## Titel der Arbeit

Timing performance studies of the hadronic calorimeter of the Scattering and Neutrino Detector at the LHC and reconstruction of neutrino inelastic scattering events

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# Abstract

The SND@LHC experiment was approved in 2021 and subsequently underwent a test-beam campaign before installation in the TI18 service tunnel 480 m away from the ATLAS interaction point. Its purpose is to exploit the intense neutrino flux in the far-forward region of this interaction point. Along with FASER$\nu$, SND@LHC constitutes the first experimental effort to study all three flavours of neutrino at LHC energies. These neutrinos offer unique physics insights, including the ability to probe the low-$x$ gluon parton density function and measure the neutrino production cross-sections at LHC energies.

This thesis focuses on the upstream section of the muon system (US) of SND@LHC. Presented first are commissioning studies to correct the unwanted time-walk effect; align the timing response of the US read-out channels to one another and other electronic subsystems of the detector; the implementation of a timing model of the US system in the simulation framework of the experiment. Building upon this commissioning, algorithms for producing spatial measurements using the aligned timing of the US are constructed and validated using experimental data and the timing model in simulation. The spatial measurements are then incorporated in machine learning methods to present the first physics study utilising the US muon system and provide a platform for further analyses using the methods and results of this thesis work.

# Kurzfassung

Das SND@LHC-Experiment wurde 2021 genehmigt und durchlief anschließend eine Teststrahlkampagne, bevor es im TI18-Servicetunnel, 480 m vom ATLAS- Wechselwirkungspunkt entfernt, installiert wurde. Ziel ist es, den intensiven Neutrinofluss im weit vorwärts gerichteten Bereich dieses Wechselwirkungspunkts zu nutzen. Zusammen mit FASER$\nu$ gehört SND@LHC zu den ersten Experimenten, die alle drei Neutrinoflavours bei LHC-Energien untersuchen. Diese Neutrinos ermöglichen einzigartige physikalische Untersuchungen, darunter die Bestimmung der Gluon-Partondichtefunktion bei kleinen $x$-Werten und die Messung von Neutrino-Produktionswirkungsquerschnitten bei LHC-Energien.

Diese Arbeit konzentriert sich auf den Upstream-Bereich des Myonsystems (US) von SND@LHC. Zunächst werden Inbetriebnahmestudien zur Korrektur des unerwünschten Time-Walk-Effekts, zur zeitlichen Angleichung der US-Auslesekanäle untereinander und an andere elektronische Subsysteme des Detektors sowie zur Implementierung eines Timing-Modells des US-Systems in die Simulationsumgebung des Experiments vorgestellt. Aufbauend auf diesen Inbetriebnahmearbeiten werden Algorithmen zur Gewinnung räumlicher Messgrößen aus den zeitlich ausgerichteten Signalen des US-Systems entwickelt und anhand experimenteller Daten sowie des Timing-Modells in der Simulation validiert. Anschließend werden diese räumlichen Messgrößen in Verfahren des maschinellen Lernens integriert, um die erste physikalische Analyse unter Verwendung des Upstream-Myonsystems durchzuführen und eine Grundlage für weiterführende Analysen mit den in dieser Arbeit entwickelten Methoden und erzielten Ergebnissen zu schaffen.

# Dedication

I dedicate this thesis to my parents, Frances and Anthony, who always emphasised education and encouraged me to be ambitious. My father consistently demonstrated his passion for the natural world. My late mother instilled in me the importance of education, often accompanied by the message from her grandmother:

> *Get yourself an education, it's the one thing they can't take away from you.*
> - Advice passed from my great grandmother to my grandmother and my mother, then passed on to me.

# Acknowledgements

Firstly, I would like to acknowledge **Prof. Dr. Heiko Markus Lacker** for continuously displaying exemplary tuition, leadership and research skills. Heiko, your contagious passion for good physics, methodical process and wealth of knowledge drove me on to work hard. A consideration of: "what would Heiko do?" helped to guide my next steps.

Several HU Berlin colleagues and friends **Anupama Reghunath**, **Alessia Brignoli**, **Doramas Jimeno**, **Lucian Fasselt**, **Jakob Schmidt**, **Dr. Christian Appelt**, **Eduard Ursov** and others have given me invaluable advice and friendship.

I am extremely lucky to have profited from so many highly talented and motivated scientists throughout HU Berlin and the SND@LHC experiment including but not limited to **Dr. Christian Scharf**, **Dr. Christopher Betancourt**, **Dr. Ettore Zaffaroni**, **Dr. Antonia Di Crezcenco**, **Dr. Thomas Ruf**, **Dr. Simona Ileva**, **Dr. Cristovao Villela**. Special thanks go to my ever-supportive and patient partner, **Charlotte Hacking**, whose positive affirmations about my progress helped me to push on in the difficult stages.

Nobody is self-made and without the examples and support of countless individuals throughout my life I would not have the privilege of studying fundamental physics. Albert Einstein is sometimes referred to as the most frequently misquoted man in modern history, but regardless I think the following is apt:

> *As our circle of knowledge expands, so does the circumference of darkness surrounding it.*

# Contents

**1 Introduction** **9**

**2 Supporting physics** **11**
2.1 The Standard Model of particle physics . . . 11
2.2 Neutrinos and The Standard Model . . . 14
2.3 Neutrino interactions with matter . . . 21
2.3.1 Elastic and quasi-elastic scattering . . . 24
2.3.2 Resonant production scattering . . . 25
2.3.3 Deep inelastic scattering . . . 25
2.3.4 Nucleon structure functions . . . 28
2.4 Neutrino physics at the LHC . . . 31
2.5 New physics searches at SND@LHC . . . 35
2.5.1 Scattering off nucleons . . . 35
2.5.2 Scattering off electrons . . . 37
2.6 Charged particle interactions with matter . . . 38

**3 The SND@LHC experiment** **41**
3.1 Overview/concept . . . 41
3.2 The SND@LHC detector design . . . 42
3.2.1 Target tracker and electromagnetic calorimeter . . . 43
3.2.2 Target and vertex detector . . . 43
3.2.3 Upstream muon system . . . 45
3.2.4 Downstream muon system . . . 48
3.2.5 Veto system . . . 48
3.2.6 Plastic scintillator: properties and physics . . . 51
3.2.7 Silicon photomultipliers . . . 53
3.2.8 The time-walk effect . . . 64
3.3 Software and electronics . . . 65
3.3.1 Front-end and data acquisition boards . . . 65
3.3.2 Event building and readout software . . . 66
3.3.3 Time alignment of the experiment . . . 67
3.3.4 Experiment Control System . . . 69
3.3.5 Online and offline data formats . . . 70
3.3.6 Track fitting algorithms . . . 71
3.4 Simulated datasets . . . 72
3.5 Measurements with SND@LHC . . . 74
3.5.1 Measurement of the muon flux in SND@LHC . . . 74
3.5.2 Observation of Collider Muon Neutrinos with the SND@LHC Experiment . . . 76
3.5.3 Observation of collider neutrinos without final state muons . . . 78

**4 US time alignment** **82**
4.1 Method . . . 82
4.1.1 Selection criteria applied to single muon events . . . 82
4.1.2 Selection criteria on digitised hits . . . 86
4.1.3 Determination of the effective speed of light measured by the SiPM . . . 87
4.1.4 Time-walk parameterisation and fitting . . . 90
4.1.5 Determination of time resolution . . . 92
4.1.6 Time-walk corrected data . . . 93
4.1.7 Fine alignment . . . 93
4.1.8 System alignment . . . 94
4.2 Laser measurements . . . 96
4.2.1 Laser calibration . . . 97
4.2.2 Pre-processing . . . 101
4.2.3 Data analysis . . . 102
4.3 Results . . . 105
4.3.1 Event selection . . . 105
4.3.2 Effective signal speed in the scintillator . . . 106
4.3.3 Time resolution . . . 115
4.3.4 Time-walk correction procedure . . . 119
4.3.5 Fine alignment parameters . . . 124
4.3.6 Scintillator timing measurements . . . 127
4.3.7 Laser Measurements . . . 137

**5 Spatial measurements using the aligned Upstream system** **146**
5.1 Method . . . 146
5.1.1 Simulation implementation . . . 146
5.1.2 Spatial measurements in the US system . . . 149
5.1.3 Angular measurements . . . 152
5.1.4 Kinematic studies of the high purity selection of muon neutrino charged current events . . . 155
5.2 Results . . . 158
5.2.1 Digitisation threshold study . . . 158
5.2.2 Spatial measurement validation . . . 161
5.2.3 Data quality of the $\nu_\mu$ CC events . . . 173

**6 Multivariate analysis** **182**
6.1 Gradient Boosted Decision Trees . . . 182
6.2 $\nu 0\mu$ BDT application . . . 185

**7 Conclusion** **200**

**List of Figures** **203**

**List of Tables** **214**

**A Appendices** **229**
A.1 Energy deposition in the US from muons . . . 229
A.2 Emission spectrum of EJ-200 . . . 229
A.3 Aluminium housing structure for the US and DS . . . 231

A.4 Typical data informing how small SiPM parameters vary with over-voltage . . . 231
A.5 Fired US bar multiplicity in selected muon events . . . 232
A.6 Evaluating the uncertainty on the muon system reference time . . . 233
A.7 Outlier uncorrected time resolution . . . 235
A.8 All US alignment parameter sets . . . 236
A.9 Bar-side time resolution derivation . . . 237
A.10 Correlation between US SiPMs and muon system reference time . . . 238
A.11 Laser intensity SiPM measuring points . . . 238
A.12 Groups of bars illuminated in multi-bar illumination modes . . . 239
A.13 Example SiPM response curves . . . 240
A.14 Example small SiPM delay - QDC correlations . . . 241
A.15 Signal velocity from Maxwell's Equations . . . 242
A.16 Energy spectrum of muons in SND@LHC . . . 243
A.17 Variation in Wacker Silgel refractive index . . . 244
A.18 $x$-barycentre data-simulation comparison . . . 244
A.19 . . . 245
A.20 Energy deposition $x-$width . . . 245
A.21 Run 5132 plane displays . . . 246
A.22 Spatial dependence of the DS system event-wise reference time . . . 246
A.23 Firing SiPMs in high QDC muon events . . . 246
A.24 Variance of spatial measurements . . . 248
A.25 Multiplicity comparison for $\nu_\mu$ CCDIS data . . . 249
A.26 $\nu 0\mu$ BDT features . . . 251

**Glossary** **252**

# Chapter 1

# Introduction

As we approach the era of the High Luminosity LHC (HL-LHC), set to begin by the mid-2030s [1, 2], the LHC will leave a rich and varied legacy that is undoubtedly fronted by the long anticipated discovery of the Higgs boson in July 2012 [3]. Beyond the Higgs boson, the LHC physics program has explored a wide range of topics, including flavour physics through measurements like mixing in the $D$-meson system [4], and QCD dynamics via dijet cross sections used to test perturbative QCD predictions [5], along with a multitude of searches for physics beyond the Standard Model (BSM) that aim to address some of the field's unresolved questions.

With the vast majority of the incredible work performed at the LHC, the common thread with the work has been that the results are in line with theory prediction. Try as we might to break The Standard Model of particle physics (SM), it consistently resists in spite of the many unanswered questions that we have. Many of the unanswered questions we have come from astrophysical and cosmological observations. The WMAP and Planck satellite mapping studies of the Cosmic Microwave Background [6] inform us that baryonic matter comprises only $\approx 5\%$ of the energy density in the Universe. Or one can look at the work of Vera Rubin in the 1970s, where Rubin observed galactic rotation curves that could not be explained by the observed mass in the vicinity, which resulted in the phenomena of dark matter that still goes unidentified [7].

The baryon asymmetry of the universe (BAU), the observed fact that the Universe contains "little or no primordial antimatter" [8] remains one of the great problems in physics. In cosmic rays we observe anti-protons on the level of about one per ten thousand protons, which is consistent with the expectation from $p + p \rightarrow 3p + \bar{p}$. One of the Sakharov conditions by which the baryon asymmetry of the Universe can occur is $C$ and $CP$ violating processes. $CP$ violation is allowed within the standard model and has most famously been observed in the quark sector. The BaBar experiment at the PEP2 asymmetric B-factory in California operated at the $\Upsilon(4S)$ resonance and collided a 9 GeV electron beam with a 3.1 GeV positron beam; this boosts the $\Upsilon(4S)$ resonance along the beam axis allowing for sufficient separation to measure the B-mesons resulting from the decay [9]. Whilst CP violation in the neutrino sector has not been observed, an upper limit on the rate is set through measurements such as T2K studying the appearance of $\nu_e$ in a $\nu_\mu$ beam [10]. From these two sources of CP violation allowed within the SM, the community has concluded that these are insufficient to explain the observed degree of baryon asymmetry in the Universe [9].

The minimal SM provided a zero mass for neutrinos, however as will be expanded on in 2.2, experimental observations by several collaborations have concluded

that at least two of the three neutrino mass eigenstates must in-fact be non-zero. The scientific community has an understanding of the mass differences between the eigenstates, $\Delta m^2_{12}$ and $\Delta m^2_{32}$ [11]. The possible orderings of these mass eigenstates; normal or inverted hierarchy, is not known, but it an analysis question informing the design of next generation neutrino experiments like DUNE and JUNO [12].

The SND@LHC experiment is one of two experiments currently sensitive to neutrino interactions of all flavours at the LHC. The main physics goal of SND@LHC is provide a measurement on the neutrino cross section at LHC energies, which will help to develop nuclear models and to constrain uncertainties in precision neutrino experiments. Furthermore, as is outlined in more detail in section 2.4, measuring the electronic neutrino flux at SND@LHC allows for a probe of the low-$x$ gluon PDF due to the fact that the electron (anti-)neutrinos in the SND@LHC acceptance are dominantly produced by the decays of charm hadrons.

In order to produce such physics measurements, since the experiment was installed between November 2021 and early 2022 there have been ongoing calibration and commissioning studies of all detector subsystems. This thesis focuses on calibration and commissioning performed on the upstream muon system (US) of the detector in order to optimise the physics capabilities of the subsystem. These calibration and commissioning studies form the bulk of chapter 4, with the remainder of the chapter covering complementary studies performed on the silicon photo-multiplier PCB using a calibrated laser. The results of this chapter leave the timing behaviour of the US well aligned throughout the system and with the other electronic subsystems of SND@LHC. Furthermore, the time alignment of chapter 4 leaves the timing behaviour of the US able to be replicated in simulation studies, which comprise a large proportion of the remaining two analysis chapters. The time alignment of the US in conjunction with the measurements of the effective signal speed inside the plastic scintillators of the US allow for spatial measurements to be constructed from the timing data in the US. The model for these spatial measurements is outlined and validated in chapter 5 before being used, alongside all other forms of US data, to study the high purity selection of events in the detector identified as muon neutrino charged current deep inelastic scattering. The final chapter of this thesis uses the breadth of US data to aid the search for $0\mu$ final state neutrino interactions (all neutrino interactions not containing a muon in the final state). The US data is used to train a BDT to identify events in the majority class of background event - events containing a final state muon, where the muon system is unable to identify the muon. Before concluding, the state of the experiment is discussed along with the future upgrade of the experiment.

# Chapter 2

# Supporting physics

## 2.1 The Standard Model of particle physics

A large part of the discussion in this subsection draws upon references [9, 13] throughout. Reference [14] being important for the electroweak unification, reference [15] being important for the discussion of the Higgs boson.

Through the 1960s and 70s, The Standard Model of particle physics (SM) was established through complementary experimental and theoretical particle physics. The SM describes the strong, electromagnetic and weak interactions, unifying the latter two of these into the electroweak interaction, whilst not attempting to include gravity. To describe these forces, the SM contains a set of fundamental particles, depicted in figure 2.1, where fermions are particles of a half-integer spin and bosons have integer spin. Further to what this figure shows, each fermion also has its accompanying anti-particle which differs only by a factor of -1 to the electronic charge. Quarks and anti-quarks combine into doublets (mesons) such as pions, or triplets (baryons) such as the proton and neutron [9].

The three forces of the SM are described through the product gauge groups:

$$SU(3)_C \times SU(2)_L \times U(1)_Y \tag{2.1.1}$$

The components of the gauge group are local symmetry groups. Each local symmetry group describes a fundamental force and must remain invariant under local symmetry transformations; in order to do so, the gauge bosons arise. The $SU(3)_C$ symmetry group describes the strong interaction and introduces eight gauge bosons, the gluons, to propagate the force. The conserved quantum number is colour charge which must be non-zero for a particle to undergo the strong interaction. Quarks carry a colour charge, anti-quarks carry an anti-colour charge while gluons carry a combination of colour and anti-colour charge. Next, the $SU(2)_L$ symmetry group which describes the weak interaction, which is maximally parity violating. This is reflected in the sub-script $L$ in equation 2.1.1 showing that the interaction couples only to left-handed chiral components of fermions and right-handed chiral components of anti-fermions. This symmetry group has three associated gauge fields: $W^1_\mu, W^2_\mu, W^3_\mu$. In the weak interaction the two conserved quantum numbers are weak isospin, $T$, and its third-component, $T_3$. Lastly, the $U(1)_Y$ symmetry group describes the conserved quantum number hypercharge, $Y$, and its associated gauge field, $B_\mu$.

The unified weak and electromagnetic theory described by the gauge group $SU(2)_L \times U(1)_Y$ is spontaneously broken by the Brout–Englert–Higgs mechanism, which requires the introduction of the Higgs field, $\phi$, a weak isospin doublet composed of two complex scalar fields:

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix} \tag{2.1.2}$$

Where its Lagrangian can be written as:

$$\mathcal{L} = (\partial_\mu \phi)^\dagger (\partial^\mu \phi) - V(\phi) \tag{2.1.3}$$

This Lagrangian possesses a global $U(1)$ symmetry due to its invariance under a transformation $\phi \to \phi' = e^{i\alpha}\phi$, because $\phi'^*\phi' = \phi^*\phi$. The Higgs potential is defined as:

$$V(\phi) = \mu^2 \phi^\dagger \phi + \lambda(\phi^\dagger \phi)^2 \tag{2.1.4}$$

Where $\lambda > 0$ is the self interaction coupling constant. When $\mu^2 < 0$, the potential - often referred to as the "Mexican hat" potential - has a set of non-zero minimum defined by:

$$\phi^\dagger \phi = \frac{1}{2}\left(\phi_1^2 + \phi_2^2 + \phi_3^2 + \phi_4^2\right) = \frac{v^2}{2} = -\frac{\mu^2}{2\lambda}$$

Where $v$ is referred to as the vacuum expectation value (VEV). The electroweak symmetry, $SU(2)_L \times U(1)_Y$, is spontaneously broken when the Higgs field acquires a non-zero VEV. It can be shown that electroweak symmetry breaking causes the four massless gauge bosons of the $SU(2)_L \times U(1)_Y$ group to become the observable massive $W^\pm$ and $Z-$bosons and the massless photon, $\gamma$. The charged combinations $W^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2)$ are the physical $W^\pm$ bosons. The gauge fields $W_\mu^3$ and $B^0$ undergo electroweak mixing to produce the neutral $Z-$boson and the massless photon [9, 13, 16]. This mixing is mediated by the Weinberg mixing angle in the rotation matrix:

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix} \tag{2.1.5}$$

Where $A_\mu$ and $Z_\mu$ are the mass eigenstates of the massless photon and $Z-$boson [9]. After spontaneous symmetry breaking, a gauge choice is made which simplifies the calculations:

$$\langle 0|\phi|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} \tag{2.1.6}$$

The fields can be expanded around this minimum and expressed, with reference to the original definition of the Higgs doublet of equation 2.1.2:

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1(x) + i\phi_2(x) \\ v + \eta(x) + i\phi_4(x) \end{pmatrix} \tag{2.1.7}$$

Where $\phi_1(x)$, $\phi_2(x)$ and $\phi_3(x)$ correspond to fluctuations of each constituent scalar field: the gauge choice of equation 2.1.6 shows that the charged complex scalar field, $\phi^+$, is zero; the neutral complex scalar field, $\phi^0$, is given the VEV. Therefore $\eta(x)$ corresponds to the fluctuation of the VEV. The crucial difference between the fluctuations of the constituent fields and the fluctuation around the VEV is that fluctuations around the VEV are in the direction where the potential is quadratic to first order, whereas fluctuations in the scalar fields are in the direction where the potential does not change [9, 16]. The massless scalar particles that arise as a

consequence of the Lagrangian formed with these scalar field fluctuations are referred to as *Goldstone bosons*. By an appropriate $U(1)$ symmetric gauge transformation of the Lagrangian, these Goldstone bosons are 'gauged away'. The transformation resulting in this elimination is referred to as unitary gauge.

Now the Higgs doublet can be written in the unitary gauge, where the perturbation of the VEV is now written as $h(x)$ to highlight the use of unitary gauge.

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} \tag{2.1.8}$$

The Lagrangian resulting from this unitary gauge choice is known as the Salam-Weinberg model and the gauge boson masses can be identified by writing the Lagrangian such that the $SU(2)_L \times U(1)_Y$ local gauge symmetry is held. A thorough derivation of the gauge boson masses is given in references [9, 13, 16]. From the spontaneous symmetry breaking and the Glashow-Salam-Weinberg model of electro-weak unification, the masses of the vector bosons $W$ and $Z$ are predicted to satisfy the relation[9, 13]:

$$\frac{m_W}{m_Z} = \cos\theta_W \tag{2.1.9}$$

As well as giving mass to gauge bosons, spontaneous breaking of the $SU(2)_L \times U(1)_Y$ symmetry also generates the fermion masses through Yukawa couplings. The down-type quarks ($d$, $s$, $b$) acquire mass via direct Yukawa couplings, the up-type quarks ($u$, $c$, $t$) acquire mass through Yukawa coupling to the charge-conjugated Higgs field, $\tilde{\phi}$ [9, 13]. Left-handed chiral fermion components are placed in $SU(2)_L$ doublets and right-handed chiral fermion components in $SU(2)_L$ singlets, the first generation of fermions are shown:

$$L_L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}, \quad Q_L = \begin{pmatrix} u_L \\ d_L \end{pmatrix}$$
$$e_R \text{ (lepton singlet)}; \quad u_R,\, d_R \text{ (quark singlets)}$$

The minimal standard model accounts for neutrinos as a left-handed Weyl fermion[1], not containing a right-handed chiral neutrino as it would be expected to be sterile: it would not interact weakly, strongly or electromagnetically, only gravitationally [17, 9]. Accounting for these sterile neutrinos in the SM, a Dirac mass term can be added to the Lagrangian without breaking gauge invariance. Similarly, the neutrino could be considered a Majorana particle, satisfying the Majorana condition $\nu = \nu^C \equiv C\bar{\nu}^T$ where C is the charge conjugation matrix [18] and the particle is its own antiparticle. This can also be incorporated into the SM without breaking its gauge invariance [19, 20].

While providing an exhaustive and precise account of fundamental physics, the SM is still incomplete. Many issues exist, like those outlined briefly in the introduction such as dark matter and dark energy. Further issues such as the neutrino masses and neutrino flavour oscillations are discussed more in the coming section.

[1] Where a Weyl fermion arises from the Dirac Hamiltonian of a massless particle [9].

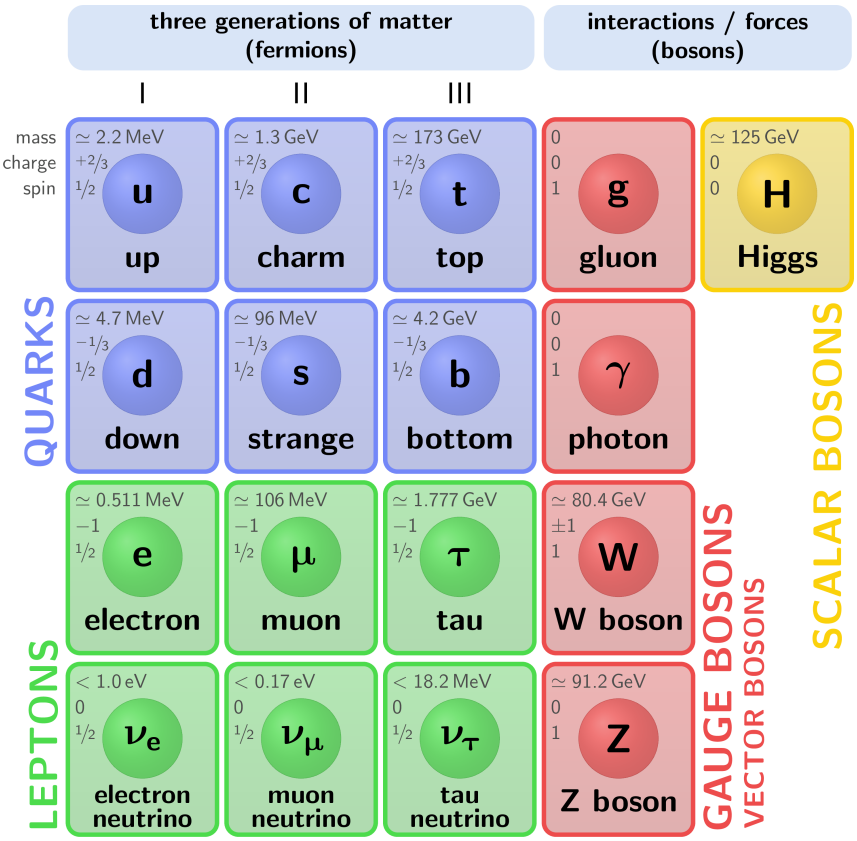



Figure 2.1: The fermions and bosons of the Standard Model. With each particle is shown the spin, mass and charge. The neutrino mass limits here are not up to date.

## 2.2 Neutrinos and The Standard Model

In 1927, the measurement of the final state electron energy distribution from Radium beta decay appeared to violate energy conservation. Three years later Wolfgang Pauli postulated the presence of a third particle in the final state in order to explain the continuous distribution, shown in figure 2.2. This new particle must be electrically neutral and small in mass, giving rise to the name neutrino, first jovially used by Enrico Fermi meaning ”little neutral one” [21].

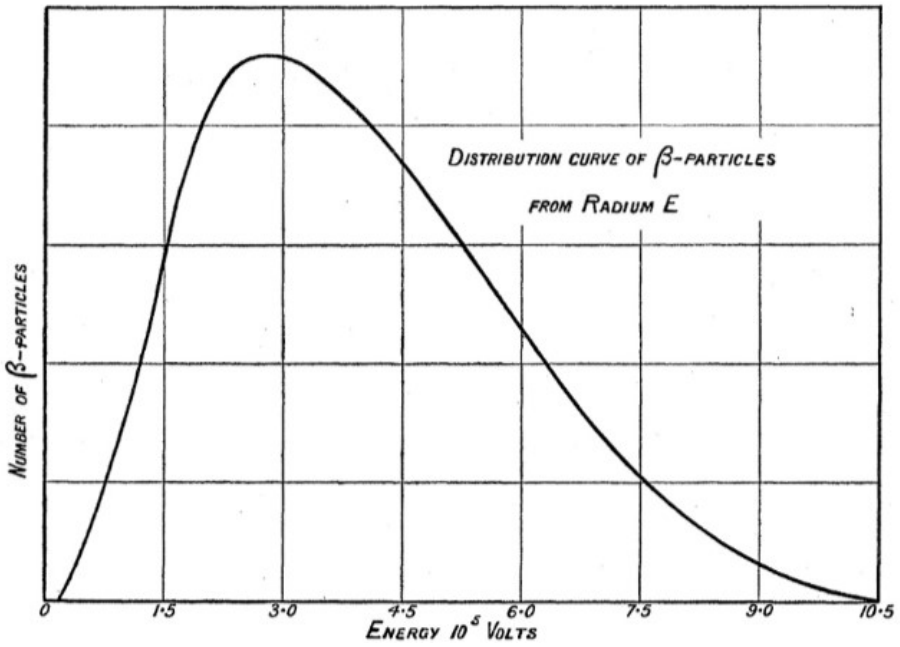



Figure 2.2: Continuous decay spectrum from 'Radium E' [22].

The electrically neutral and spin-1/2 neutrino, conserves both energy and angular momentum in the three-body nuclear decay:

$$M(A,Z) \rightarrow D(A,Z+1) + e^{-} + \bar{\nu}_e \tag{2.2.1}$$

Where $M(A,Z)$ and $D(A,Z+1)$ are the mother and daughter nuclei with mass numbers and atomic numbers given by A and Z (Z+1).

Due to the neutral charge and miniscule mass Wolfgang Pauli later said [23]:

"I have postulated a particle that cannot be detected."

To try and overcome these difficulties, Cowan and Reines attempted for years to detect neutrinos through the inverse beta decay in an experiment known as Project Poltergeist [21].

$$p^{+} + \bar{\nu}_e \rightarrow n^0 + e^{+} \tag{2.2.2}$$

While the predicted cross-section for this process was extremely small, thanks to the enormous neutrino flux caused by the nuclear reactor used as a source, Project Poltergeist could operate. The experiment employed two small tanks of $CdCl_2$ dissolved in water which separated three larger tanks of liquid scintillator. The experimental signature of inverse beta decay, equation 2.2.2, in this system comprised the prompt measurement of two gamma rays from the positron annihilation followed by a delayed gamma ray produced as a result of the neutron capture by a Cadmium nucleus[2], shown in equation 2.2.3.

$$n + {}^{108}\mathrm{Cd} \rightarrow {}^{109}\mathrm{Cd} + \gamma \tag{2.2.3}$$

[2]The gamma rays produced by the annihilation reaction and the neutron capture were not directly measured by the photon detectors but in fact that the Compton scattering those gamma rays would undergo. The gamma ray from the neutron capture would be sufficiently high energy to also induce pair production in the water.

The delayed coincidence of these photons provides the unique experimental signature that Cowan and Reines would rely upon to make their discovery. Project Poltergeist was successful under Reines and Cowan and observed the anti-electron neutrino in 1956 [24]. A key piece of the neutrino story following Project Poltergeist is the Solar Neutrino Problem. In 1966, the Homestake mine neutrino observatory relied upon neutrino capture through [21]:

$$\nu_e + {}^{37}_{17}\mathrm{Cl} \rightarrow e^- + {}^{37}_{18}\mathrm{Ar} \tag{2.2.4}$$

The neutrino energy threshold for equation 2.2.4 is 0.814 MeV which, as the Standard Solar Model shows in figure 2.3, made Homestake mainly sensitive to electron neutrinos produced by the $\beta$-decay of $^8$B [25]:

$${}^{8}_{5}\mathrm{B} \rightarrow \nu_e + e^+ + {}^{8}_{4}\mathrm{Be} \tag{2.2.5}$$

The measured neutrino flux, led by R. Davis, did not agree with the theoretical predictions, led by J. Bahcall, using the Standard Solar Model. The problem at the time was assumed to be either with the experiment, or with the theory describing the conditions in the Sun.

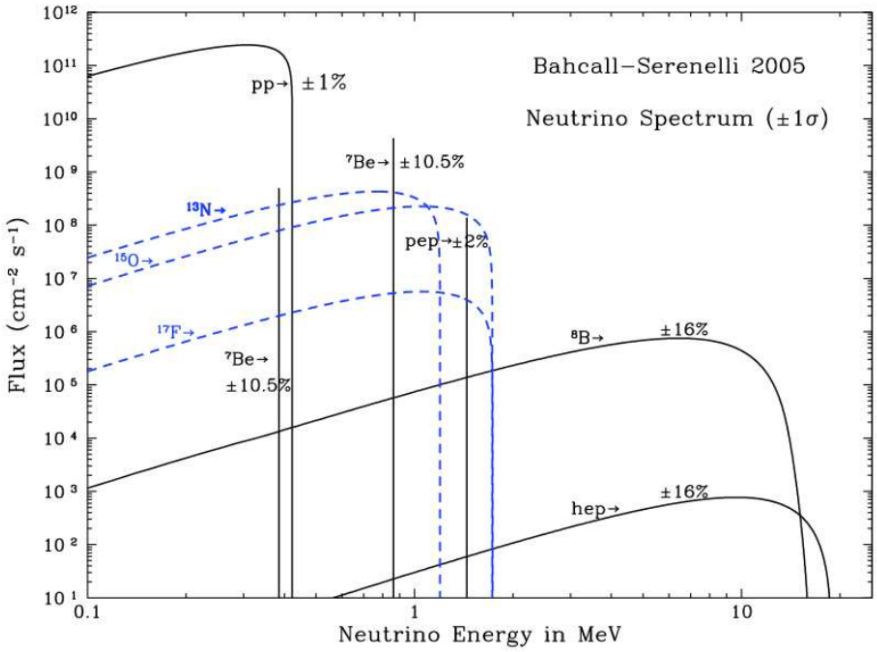



Figure 2.3: Energy spectrum of neutrinos produced by mechanisms [26] determined by the Standard Solar Model.

To resolve the Solar neutrino problem required decades of work in the neutrino sector to realise that the problem truly lay in the understanding of neutrinos. In the following decades, the work of Davis and Bahcall was supported by results at Kamioka, by the Kamiokande collaboration [27] and latterly the Super-Kamiokande (SK) collaboration [28]. It was the work of the Sudbury Neutrino Observatory (SNO), a Cherenkov detector using a 1000-tonnes of ultra-purified deuterium dioxide ($D_2O$) outfitted with 9438 photomultiplier tubes capable of measuring neutrino interactions from all three flavours of neutrino [29] that solved the Solar neutrino problem in 2001. SNO could measure the $v_e$ flux coming from $^8$B whilst measuring the total neutrino flux through:

$$\nu_e + d^+ \rightarrow p^+ + p^+ + e^- \quad \text{(CC)}$$
$$\nu_\ell + d^+ \rightarrow p^+ + n^0 + \nu_\ell \quad \text{(NC)}$$
$$\nu_\ell + e^- \rightarrow \nu_\ell + e^- \quad \text{(ES)}$$

Where $d^+$ refers to a deuterium nucleus, and $\ell$ may be $e$, $\mu$ or $\tau$. Charged current scattering is sensitive to only the electron neutrino. Neutral current is sensitive to all flavours of neutrino as is the elastic scattering however the sensitivity is lower for the $\nu_\mu$ and $\nu_\tau$. In the second run of SNO, NaCl was added to the $D_2O$ to improve to neutral-current sensitivity. With all three runs of SNO combined, the collaboration's measurement of the NC all-flavour neutrino flux was in strong agreement with the SSM theory, and with a lower uncertainty. Furthermore, the collaboration's measurements of the flux from all flavours individually gave the fraction of $v_e / v_\ell = 34\%$, which reflected the existing results beginning with Davis and Bahcall in Homestake [30].

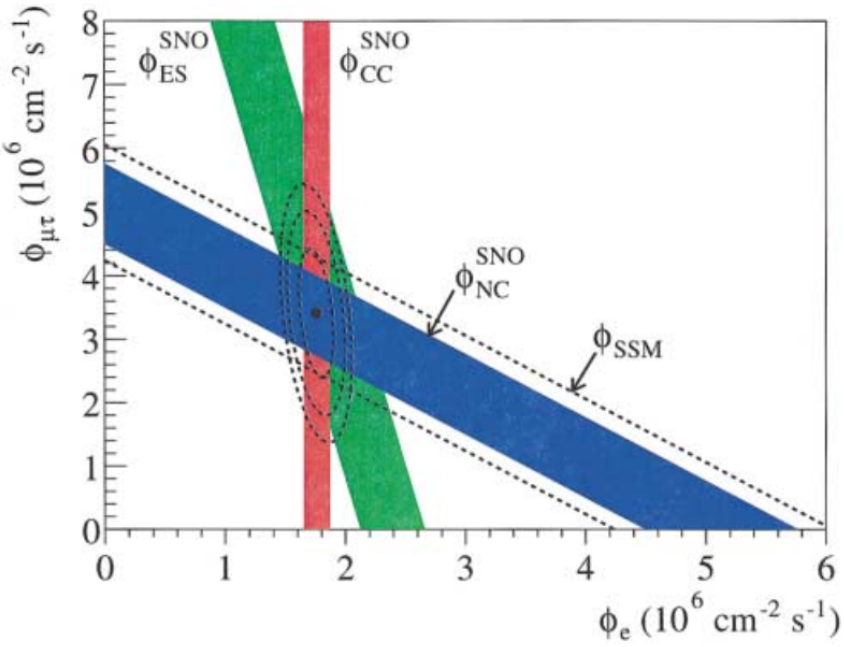



Figure 2.4: Comparison of $^8$B solar $\nu_\mu + \nu_\tau$ flux vs $\nu_e$ flux measured by SNO and SK collaborations. The diagonal bands show the predicted all-flavour $^8$B solar neutrino flux from SNO and what is predicted by the SSM [30].

This result is often misattributed as direct evidence of neutrino flavour oscillations[3]: what SNO actually observed was adiabatic flavour transformations which are described through the Mikheyev-Smirnov-Wolfenstein (MSW) effect [32].The first true experimental observation of neutrino flavour oscillations came from the SK and its study of the atmospheric neutrino problem. The atmospheric neutrino problem refers to how the ratio of measured neutrino flavours, $N_\mu/N_e$, differs between experiment and theory. This ratio was predicted to be $\approx 2$ because atmospheric neutrinos primarily come through a $\pi \rightarrow \mu$ decay chain. The so-called double-ratio of muon to electron atmospheric neutrino rates is formed by comparing data to simulation ratios:

[3]Including in the justification for the 2015 physics Nobel prize [31]

$$R = \frac{(N_\mu/N_e)_{\text{data}}}{(N_\mu/N_e)_{\text{MC}}}$$

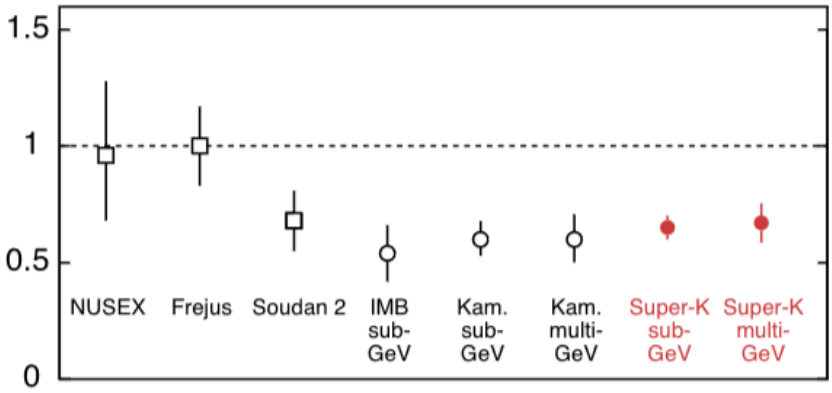



Figure 2.5: Measurements of the double-ratio of muon to electron atmospheric neutrino rates by various experiments. Central values depicted with a square refer to iron-based tracking detectors, while circular central values are water-based Cherenkov detectors [33].

As shown in figure 2.5, the SK collaboration released two papers, where the analysis was split between sub and multi-GeV neutrinos. The SK results in the figure differ from unity by more than $5\,\sigma$, showing a clear preference against the null hypothesis that the ratios are the same. SK used the $\nu_\mu \rightarrow \nu_\tau$ oscillation to show that the data obtained was in good agreement with the two-neutrino oscillation hypothesis [33]:

$$P_{a\rightarrow b} = \sin^2(2\theta)\sin^2\left(\frac{1.27\,\Delta m^2[\text{eV}^2]\,L[\text{km}]}{E_\nu[\text{GeV}]}\right) \tag{2.2.6}$$

Where $a$ and $b$ and the initial and final neutrino flavour states, $\theta$ is the mixing angle between the two states, $\Delta m^2$ is the squared mass splitting between the mass eigenstates and $L$ is the distance travelled [34]. As outlined in section 2.1, neutrinos are massless in the SM because right-handed neutrinos do not interact[4], but results such as the SK result discussed above, KamLAND 2002 [36], K2K 2006 [37] and OPERA 2012 [38] consolidated neutrino flavour oscillations, which equation 2.2.6 shows necessitates mass eigenstates of neutrinos have differences and thus are non-zero.

With three flavour eigenstates of neutrino, $|\nu_l\rangle$, $l = e, \mu, \tau$, and three mass eigenstates $|\nu_i\rangle$, $i = 1, 2, 3$: each set completing an orthonormal basis. The experimental observations that showed neutrino oscillations conclude that these orthonormal bases do not have a one-to-one correspondence. Equation 2.2.6 highlights that the oscillation probability depends on the difference of squared eigenstates masses ($\Delta m^2_{ij} = m^2_i - m^2_j$) - not the absolute mass of either eigenstate. Therefore oscillation measurements are only able to produce estimates on the relative eigenstate masses as there has to date been no absolute measurement of a neutrino eigenstate mass. The relative mass eigenstate splittings have been measured by several experiments such as SNO, KamLAND and T2K. Together, measurements have shown the masses

[4]It was noticed by Bruno Pontecorvo that quantum mechanics allowed for neutrino flavour oscillations if neutrinos had mass [35].

of the three eigenstates are such that two are relatively close together in mass, and the other differs to a far larger degree. The two possible arrangements which satisfy these observations are called the "normal mass hierarchy" and the "inverted mass hierarchy" - these are depicted in figure 2.6 and example values of the mass eigenstate splitting are given in table 2.1.

| Mass splitting | Estimate |
|---|---|
| $\Delta m^2_{21}$ | $7.53 \pm 0.18$ x$10^{-5}$ eV$^2$ |
| $\Delta m^2_{32}$ (normal hierarchy) | $2.455 \pm 0.028$ x$10^{-3}$ eV$^2$ |
| $\Delta m^2_{32}$ (inverted hierarchy) | $-2.529 \pm 0.029$ x$10^{-3}$ eV$^2$ |

Table 2.1: Values for the squared differences between mass eigenstates. All values are taken from reference [11]. More up-to date values may exist but these are here to contextualise the two different possible mass hierarchies.

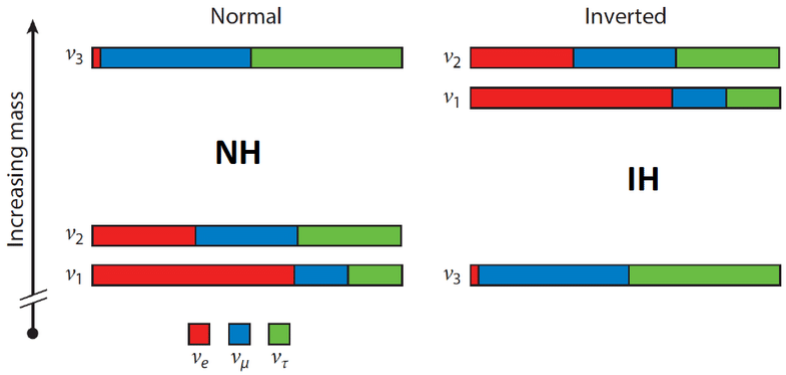



Figure 2.6: The two possible mass hierarchies given the measurements of $\Delta m^2_{ij}$. The flavour sharing in each mass eigenstate is also depicted. Taken from [39].

Each flavour eigenstate can be written as a combination of the three mass eigenstates. The formalism which encodes this information while assuming Dirac neutrinos[5] is the unitary[6] $3 \times 3$ Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix [40]:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} \tag{2.2.7}$$

Written as the product of three matrices:

$$U_{\text{PMNS}} = R_{23}(\theta_{23}) \cdot U_{13}(\theta_{13}, \delta_{CP}) \cdot R_{12}(\theta_{12}) \tag{2.2.8}$$

where the component matrices are:

$$R_{23}(\theta_{23}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_{23} & \sin\theta_{23} \\ 0 & -\sin\theta_{23} & \cos\theta_{23} \end{pmatrix}$$

$$U_{13}(\theta_{13}, \delta_{CP}) = \begin{pmatrix} \cos\theta_{13} & 0 & \sin\theta_{13} e^{-i\delta_{CP}} \\ 0 & 1 & 0 \\ -\sin\theta_{13} e^{i\delta_{CP}} & 0 & \cos\theta_{13} \end{pmatrix}$$

[5]The Dirac neutrino condition is that neutrinos have there own anti-particles.
[6]The unitarity condition is: $U^\dagger \equiv (U^*)^T$

$$R_{12}(\theta_{12}) = \begin{pmatrix} \cos\theta_{12} & \sin\theta_{12} & 0 \\ -\sin\theta_{12} & \cos\theta_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The parameters of this matrix are the three mixing angles $\theta_{12}, \theta_{23}, \theta_{13}$ and the $CP$ violating phase $\delta_{CP}$, where $CP$ violation is an asymmetry in the rate of a process occurring under the transformation through charge conjugation $C$ and parity conjugation $P$. References [20, 41] contain more thorough explanations of $CP$ violation in the neutrino sector which is briefly summarised here. Upper limits on the effective electron neutrino mass have been set by the KATRIN experiment which performs precision spectroscopy on the kinematic end point of the beta decay of tritium. The effective electron neutrino mass may be expressed as:

$$m_\nu = \sqrt{\sum_{i=3}^{3} |U_{ei}|^2 m_i^2}$$

As of April 2025 the experiment has set an upper limit of $m_{\nu_e} < 0.4$ eV at a 90% CL [42].

As briefly outlined in chapter 1, $CP$ violation has been observed in the weak sector through the kaon and beauty-meson systems but not to the extent required to explain the baryon asymmetry of the Universe. $CP$ violation in the neutrino sector has been searched for with the T2K and NO$\nu$A experiments [43, 44] however not conclusively observed. In the case of T2K, where the search for $CP$ violation has been done through comparison of the rates of muon neutrino to electron neutrino oscillations:

$$\nu_e \to \nu_\mu \quad \xrightarrow{CP} \quad \bar{\nu}_e \to \bar{\nu}_\mu \tag{2.2.9}$$

Where $C$ transforms the neutrino into its antineutrino and $P$ swaps the chirality of the neutrino, ensuring that under $CP$ the left-handed neutrino becomes a right-handed antineutrino which can undergo the weak interaction. $CP$ violation in this pair of interactions would be observed as the oscillation probabilities of these two interactions not being equal, $P(\nu_e \to \nu_\mu) \neq P(\bar{\nu}_e \to \bar{\nu}_\beta)$. Neutrino oscillations would be said to be maximally $CP$ violating if $\delta_{CP} = \pm\pi/2$. The result from the T2K measurement rules out both the $CP$ conserving values of $\delta_{CP} = 0$ and $\delta_{CP} = \pi$ with a confidence level of 95% [43]. Further measurements with T2K and other experiments will be able to further pin down the magnitude of $CP$ violation in the neutrino sector.

A final and important point on neutrino oscillations and $CP$ violation is its potential role in explaining one of the great problems in cosmology: the baryon asymmetry of the Universe (BAU). Firstly, the conditions for the BAU set by Andrei Sakharov in 1967 are [45, 46]:

1. Baryon number violation
2. $C$ and $CP$ violation
3. Interactions outwith from thermal equilibrium

The understanding of $CP$ violation in the quark sector, parameterised by the Cabibbo-Kobayashi-Maskawa matrix, does not provide sufficient $CP$ violation for the required baryon asymmetry in the early Universe [46]. Thus, other sources

of CP violation are required by Sakharov to explain BAU. Several models have been proposed for this such as Grand-Unified-Theory baryogenesis and Electroweak baryogenesis. For the purposes of relevance to the field of neutrino physics only Leptogenesis is discussed. An important foundation of this model is the Type-I Seesaw mechanism [47, 48, 49], itself one of the minimal extensions to the SM which allows the incorporation of neutrino masses through the addition of right-handed Majorana neutrinos with masses far larger than the left-handed SM neutrinos, $\nu$: $m_N \gg m_\nu$. Majorana particles give rise to lepton number violating processes like neutrino-less double beta decay. Leptogenesis then builds upon this idea. In the early Universe, the heavy right-handed neutrinos decay rapidly due to their large masses and violate lepton number conservation due to their Majorana nature, $\Delta L \neq 0$. CP violation in the decays of these heavy Majorana neutrinos ensures that leptons and anti-leptons are produced in unequal amounts, generating a net lepton asymmetry. This excess lepton number is then converted into an excess baryon number by sphaleron processes which violate $B + L$[7]. Thus, the sphaleron processes allow Leptogenesis a route to convert the lepton number violation associated with Majorana particles, to the baryon number violation required for BAU. The KAMLAND-Zen experiment, among others, are looking for neutrino-less double beta decay. The observation of such a process would be proof that the neutrino mass matrix with both Dirac and Majorana components, implying neutrinos are their own antiparticles to some extent.

## 2.3 Neutrino interactions with matter

To begin with, a general two-body interaction is considered in figure 2.7 to define kinematic variables that aid with discussion:

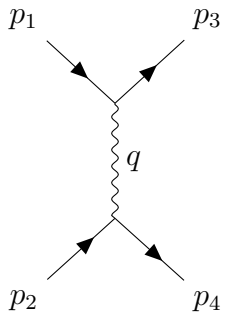



Figure 2.7: Two-body interaction with mediator four-momentum $q$.

Where $p_1$ and $p_2$ are the initial-state four-momenta; $p_2$ and $p_4$ are the final-state with the mediator carrying four-momentum $q$. The negative square of the mediator four-momentum is defined as $Q^2 = -q^2$ and can be written in four-vector algebra as $Q^2 = -(p_1 - p_3)^2$. The Mandelstam variable $s$ is defined as the square of the sum of the initial four-momenta, in the centre-of-mass frame where the interacting particles four-momenta can be written as $p_1 = (E_1, \vec{p})$ and $p_1 = (E_2, -\vec{p})$:

$$s = (p_1 + p_2)^2 = (E_1^* + E_2^*)^2 \tag{2.3.1}$$

[7]Such processes occur in extreme conditions such as the early Universe, where the temperature is above the electroweak phase transition ($T > 100$ GeV). While $B + L$ is violated, $B - L$ is conserved [46].

$s$ is a scalar product of four-vector and thus is Lorentz invariant, the centre-of-mass available energy is defined as $\sqrt{s}$. In neutrino-nucleon experiments including SND@LHC, the target is at rest, meaning that the energy of the nucleon is simply given in natural units by its rest mass, $m_N$. This allows $s$ to be written as:

$$s = 2m_N E_\nu + m_N^2 \tag{2.3.2}$$

Where it is clear that an increase in neutrino energy increases the available centre-of-mass is proportional to the incoming neutrino energy, $E_\nu$, the consequences of this for neutrino scattering is discussed in section 2.3.3, where the dominant neutrino interaction method, deep inelastic scattering (DIS), is discussed.

In quantum field theory, the propagator behaves according to the momentum transfer, $q^2$, and virtual particle, $X$, with a decay rate[8] $\Gamma_X = 1/\tau_X$ and mass, $m_X$ [9]:

$$\frac{1}{q^2 - m_X^2 + i\Gamma_X m_X} \tag{2.3.3}$$

As neutrinos do not carry a colour charge or an electric charge they cannot interact strongly or electromagnetically, leaving only the possibility of a weak interaction. As the weak force is mediated by the $W$ and $Z$ bosons of masses $m_W \approx 80\,\text{GeV}$ and $m_Z \approx 91\,\text{GeV}$, weak cross sections are suppressed at low energies where the momentum transfer is limited to lower values. The cross-section for neutrinos and anti-neutrinos scattering from quarks or antiquarks is shown in figure 2.8 to be proportional to $s$ up to several hundreds of GeV [9, 50]. The functional form will be covered in subsection 2.3.4. Given this proportionality to $s$, the LHC provides a strong neutrino research opportunity which is the fundamental motivation for the SND@LHC experiment.

The mechanisms by which neutrinos interact with matter[9] is dependent on the incoming neutrino energy, as one sees in figure 2.8. The contributions to this figure are from quasi-elastic (QE), resonance production scattering (RES) and deep inelastic scattering (DIS). While in SND@LHC, the dominant interaction mode is DIS due to the LHC energy, all three of these interaction mechanisms will be discussed for completeness. SND@LHC and FASER$\nu$ will both contribute to the understanding of neutrino interaction cross-sections in the range of $\mathcal{O}$(100)GeV - $\mathcal{O}$(100)TeV, as highlighted in figure 2.9. The region where SND@LHC and FASER$\nu$ will both contribute is often referred in literature, such as references [51, 52], as the 'energy gap'. Only the neutrino interaction components shown in figure 2.8 will be discussed, but they are not the sole neutrino scattering mechanisms. Some notable exceptions to the discussion include coherent pion production, where the transferred momentum of the neutrino induces no nuclear recoil, but a pion is emitted from the nucleus; and kaon production which is of interest purely as a background to proton decay searches [53].

[8] The decay rate is given by the inverse of the particle life-time, $\tau_X$.

[9] Here we only consider the interactions between neutrinos and nucleons.

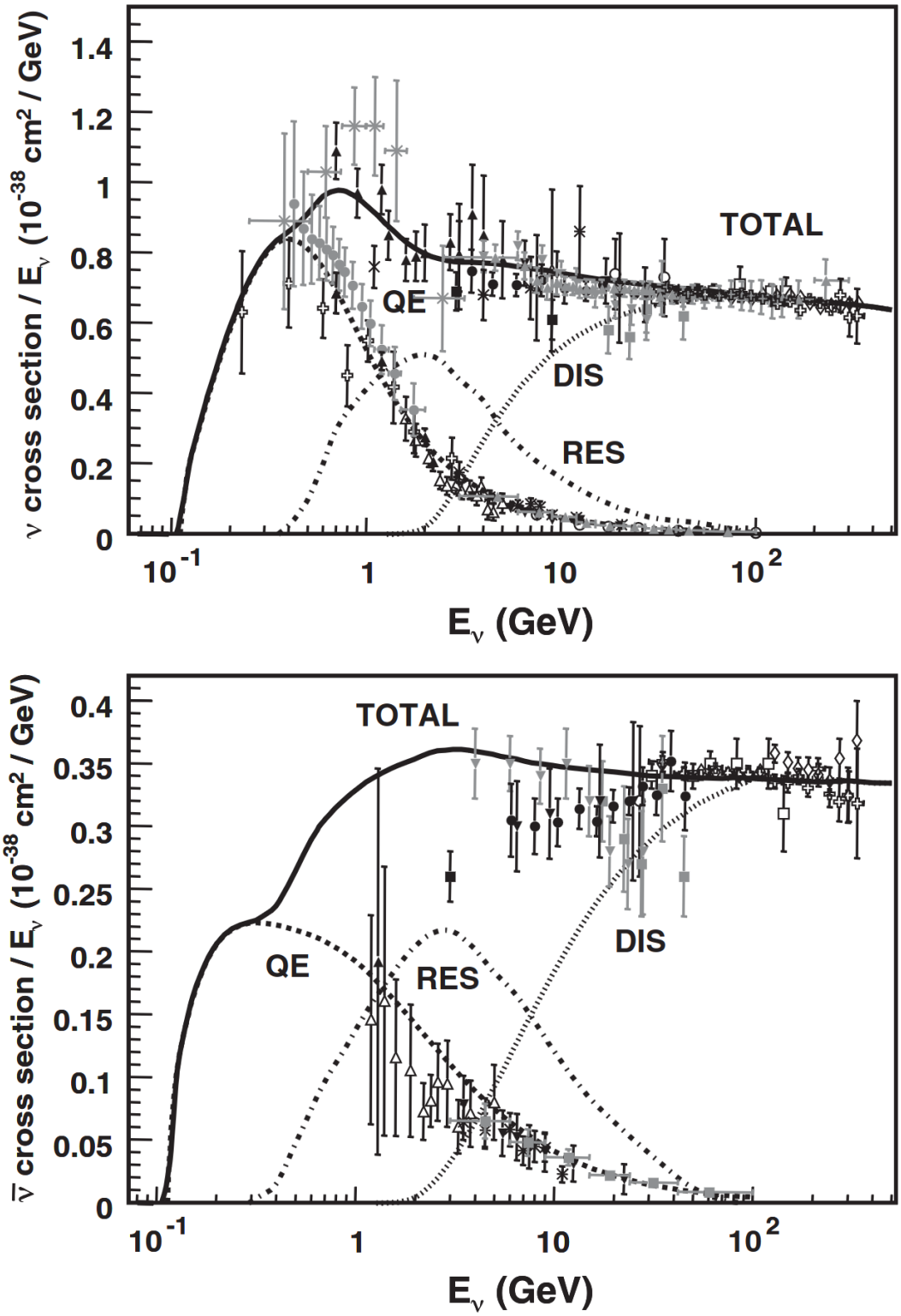



Figure 2.8: Total neutrino and anti-neutrino per nucleon charged current cross sections (isoscalar target) divided by neutrino energy as a function of neutrino energy. The total (anti-)neutrino per nucleon cross section is shown as well as the three contributions: quasielastic scattering (dashed), resonance production (dot-dashed) and DIS (dashed). Taken from [53].

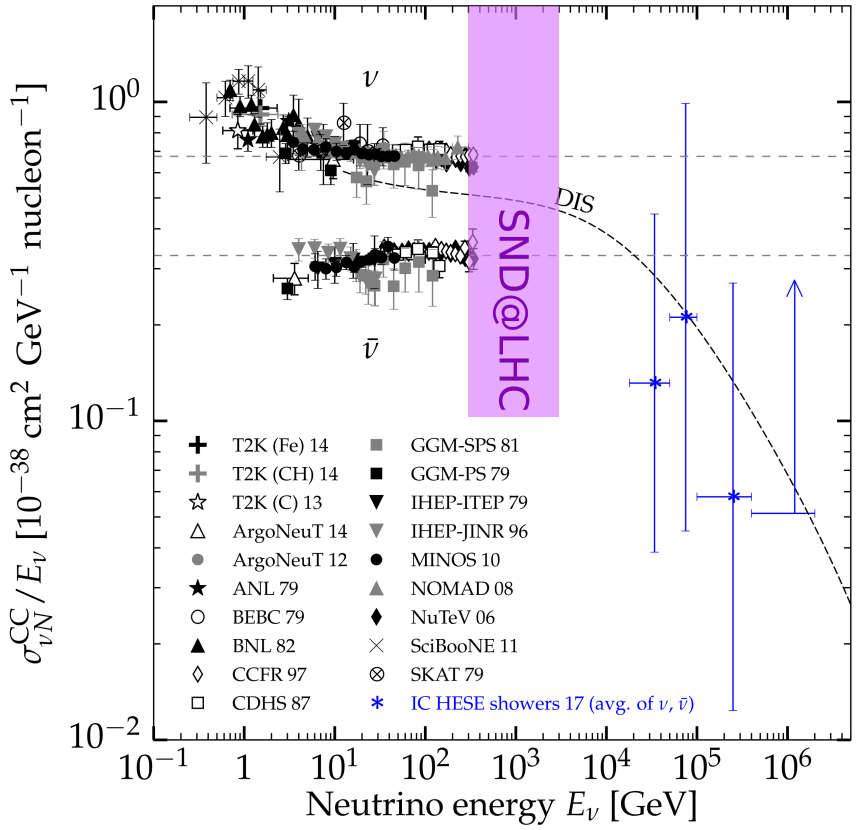



Figure 2.9: Experimental measurements of charged current neutrino interaction cross-section in the energy range $\mathcal{O}(1)$GeV - $\mathcal{O}(100)$GeV. The approximate energy range of LHC neutrinos passing through SND@LHC is shown in the magenta rectangle. Adapted from [54].

### 2.3.1 Elastic and quasi-elastic scattering

At energies of $\approx 1$ GeV and below, neutrinos interacting will predominantly scatter off the nucleon as a whole, rather than a constituent parton [53].

#### Quasi elastic scattering

The QE regime of scattering refers to instances where the momentum transfers are very low and there is no new particle creation, although minor nuclear excitations may occur[10]. QE takes one of two forms:

$$\nu_\ell + n^0 \rightarrow \ell^- + p^+ , \quad \bar{\nu}_\ell + p^+ \rightarrow \ell^+ + n^0 \tag{2.3.4}$$

Where $\ell = e, \mu, \tau$. The large mass of the tau lepton of 1777 MeV [11] means that for the majority of the energy range where QE dominates, the $\tau$ cannot be produced.

[10] Also a charged current interaction cannot strictly speaking be elastic as the internal states of the initial and final state particles are not the same.

### Elastic scattering

$$\nu_\ell + n^0 \rightarrow \nu_\ell + n^0\,, \quad \nu_\ell + p^+ \rightarrow \nu_\ell + p^+ \qquad \text{(NC)} \qquad (2.3.5)$$

$$\bar{\nu}_\ell + n^0 \rightarrow \nu_\ell + n^0\,, \quad \bar{\nu}_\ell + p^+ \rightarrow \bar{\nu}_\ell + p^+ \qquad \text{(NC)} \qquad (2.3.6)$$

Without the transmutation of a neutrino into its associated lepton, the flavour of incoming neutrino cannot be deduced.

## 2.3.2 Resonant production scattering

As the neutrino energy increases beyond the regime of QE and EL scattering, the contribution from resonance production scattering increases. The RES cross section varies between neutrino and anti-neutrino, but generally as neutrino energies increase to 1 GeV, RES becomes a key contributor until around 10 GeV, which will be discussed as an important region of the incoming neutrino energy spectrum in section 2.4. The neutrino interacts with a constituent nucleon of the target atom, transferring sufficient energy to excite the nucleon to a resonance state - such as a $\Delta$ baryon - which then typically decays into a pion and a nucleon. A neutrino resonance production scattering Feynman diagram is shown below in figure 2.10:

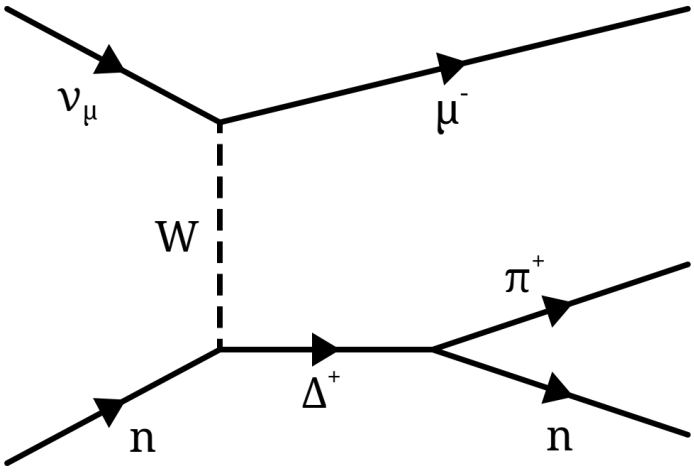



Figure 2.10: Charged current resonance production neutrino scattering where a $\Delta^+$ is produced before decaying back to a neutron ground state through the emission of a $\pi^+$ through the strong interaction [55]. The decay to $\pi^0 p^+$ is also allowed.

## 2.3.3 Deep inelastic scattering

At neutrino energies above several GeV, DIS interactions become the key interaction mechanism, and so at LHC energies this is the dominant interaction mechanism and thus of the greatest interest to the SND@LHC experiment. DIS and its uses get to the core of the SND@LHC physics program and will be referred to later in this thesis. Much of the kinematics analysis here is taken from Thomson's Modern Particle Physics [9], which the reader may consult for a more complete analysis:

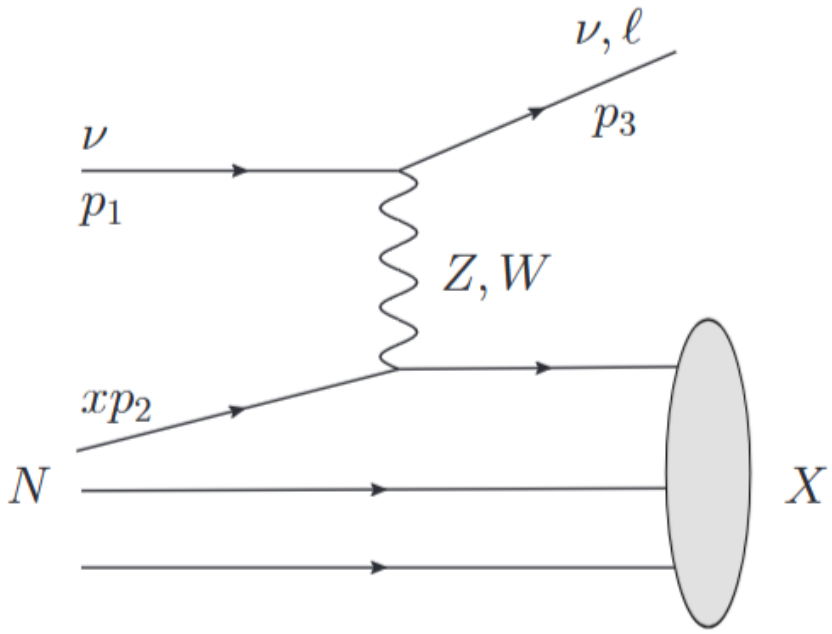



Figure 2.11: Lowest-order Feynman diagram for deep inelastic neutrino scattering [56].

Figure 2.11 shows the lowest-order Feynman diagram for neutrino nucleon charged current DIS (CCDIS) where a (anti-)neutrino of incoming energy $E_\nu$, interacts with a parton of a resting target nucleon through the exchange of a $W^{+(-)}$ boson of four-momentum $q$. DIS is allowed in both CC and NC interations but the CCDIS interactions are easier to detect due to the outgoing charged lepton, in this section the focus will be on the CCDIS:

$$\begin{aligned} \nu_\ell + N \to \ell^- + X\,, \quad \bar{\nu}_\ell + N \to \ell^+ + X \quad (CC) \\ \nu_\ell + N \to \nu_\ell + X\,, \quad \bar{\nu}_\ell + N \to \bar{\nu}_\ell + X \quad (NC) \end{aligned}$$

Where $X$ is the hadronic final state produced by the DIS interaction. The kinematics of the interaction can be studied by writing the four-momenta of the initial and final state products in the laboratory frame including the fact that the target is at rest[11]:

$$\begin{aligned} p_1 &= (E_\nu, \vec{p}_\nu), \quad p_3 = (E_\mu, \vec{p}_\mu) \\ p_2 &= (m_N, \vec{0}), \quad p_4 = (E_X, \vec{p}_X) \end{aligned}$$

Where $\vec{p}_\nu$ is the incoming neutrino three-momentum. From these vectors, one can define several Lorentz-invariant quantities:

$$\nu = \frac{p_2 \cdot q}{m_N} \quad \text{(energy transfer)} \tag{2.3.7}$$

$$y = \frac{p_2 \cdot q}{p_2 \cdot p_1} \quad \text{(inelasticity)} \tag{2.3.8}$$

$$Q^2 = -q^2 \quad \text{(negative squared momentum transfer)} \tag{2.3.9}$$

$$x = Q^2 \,/\, 2m_N\nu \quad \text{(Bjorken-x)} \tag{2.3.10}$$

[11] At rest, the four-momentum of the nucleon reduces to $(m_N, \vec{0})$ where in $\vec{0}$ means $(0, 0, 0)$.

When the four-vectors are written in the proton rest frame, $\nu$ can be shown to be equal to the energy lost by the neutrino, $E_1 - E_3$, and thus referred to as the energy transfer:

$$q = p_1 - p_3$$
$$\nu = \frac{p_2 \cdot q}{m_N}$$
$$\nu = E_\nu - E_\mu > 0$$

While discussing DIS, it will prove useful in the following section to discuss the quark-parton model. In the quark-parton model, hadrons are considered to be comprised of point-like constituents. DIS of a fundamental particle, like an electron or neutrino, and a proton is considered to be the quasi-elastic scattering of the fundamental particle from a spin-1/2 (anti-)quark constituent of the proton. The quark-parton model allows the electric and magnetic contributions in the scattering process to be related by the magnetic moment of a Dirac particle, $\vec{\mu} = g\frac{1}{2m}\vec{S}$ [9]. DIS in the quark-parton model may be formulated in the reference frame where the proton has an energy far greater than its rest mass energy, referred to the infinite momentum frame. Dealing with a struck quark in the infinite momentum frame, any component of the quark's momentum transverse to the beam axis is neglected, allowing the quark four-momentum, $p_q$, to be written as a fraction, $\xi$, of the proton four-momentum, $p_2$:

$$p_q = \xi p_2 = (\xi E_2,\, 0,\, 0,\, \xi E_2)$$

Following the quasi-elastic scattering, the momentum of the struck quark is simply $\xi p_2 + q$. Squaring this final state four-momentum gives the quark mass, $m_q^2$:

$$(\xi p_2 + q)^2 = \xi^2 p_2^2 + 2\xi p_2 \cdot q + q^2 = m_q^2$$

Given that $\xi^2 p_2^2$ is also the quark invariant mass, $m_q$, one can say that $2\xi p_2 \cdot q + q^2 = 0$

$$\xi = \frac{-q^2}{2p_2 \cdot q} = \frac{Q^2}{2p_2 \cdot q} \equiv x \tag{2.3.11}$$

So in the quark-parton model, Bjorken-x is equivalent to the proton momentum fraction carried by the struck quark and therefore bounded such that $0 \leq x \leq 1$.

Having demonstrated that $\nu = E_\nu - E_\mu$ and that $0 \leq x \leq 1$, equation 2.3.10 can be written as:

$$Q^2 = 2m_N x(E_\nu - E_\mu)$$

Clearing demonstrating the $Q^2 > 0$. At the start of section 2.3, the centre of mass energy is defined in equation 2.3.1 and then expressed in terms of the nucleon mass and incoming neutrino energy, from equation 2.3.1 and the definitions of the kinematic variables in equations 2.3.7 - 2.3.10, the negative squared momentum transfer, $Q^2$, can be shown to be given by [9]:

$$Q^2 = (s - m_N^2)xy \tag{2.3.12}$$

The parton model is used above to show that $0 \leq x \leq 1$. The inelasticity, $y$, defined in equation 2.3.8, can also be shown to be bounded such that $0 \leq y \leq 1$ by recalling $q = (p_1 - p_3)$. This allows the negative squared momentum transfer in the above expression to be simplified as $Q^2 \leq (s - m_N^2)$. Now equation 2.3.2 can be used in conjunction with equation 2.3.12 to produce the following inequality for the negative squared momentum transfer [9]:

$$Q^2 \leq 2m_N E_\nu \tag{2.3.13}$$

Neutrino DIS may result in the production of charm particles through three different mechanisms. Firstly, figure 2.12 shows how neutrino DIS can produce charmonium ($c\bar{c}$) through boson-gluon fusion, only allowed through NC and gluon bremsstrahlung, allowed in both NC and CC. Charmonium production from neutrino interactions has been studied at the CHORUS experiment [57, 58]. Secondly there is (anti-)neutrino induced single charm production. One technique for isolating single charm production looks for cases where the charmed hadron decays semi-leptonically, leaving two opposite sign leptons in the final state, as is shown in equation 2.3.15. Charm production from the DIS of an (anti-)neutrino from a nucleon allows for parton distribution functions to be probed. Primarily, the strange-quark parton distribution function can be isolated from the (anti-)neutrino DIS because it is the most likely quark to transition into charm. This probability is governed by the Cabibbo-Kobayashi-Maskawa (CKM) matrix, a $3 \times 3$, nearly-diagonal matrix which is not of direct interest in this thesis, however more information can be found in reference [9], for example.

$$\begin{aligned} \nu_\mu \ + \ \mathrm{N} \ \longrightarrow \ \mu^- \ + \ c \ + \ \mathrm{X} \\ \hookrightarrow s \ + \ l^+ \ + \ \nu_l \end{aligned} \tag{2.3.14}$$

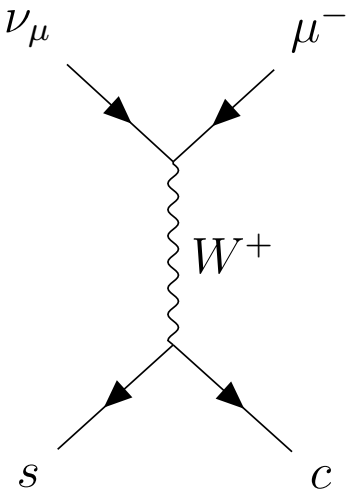
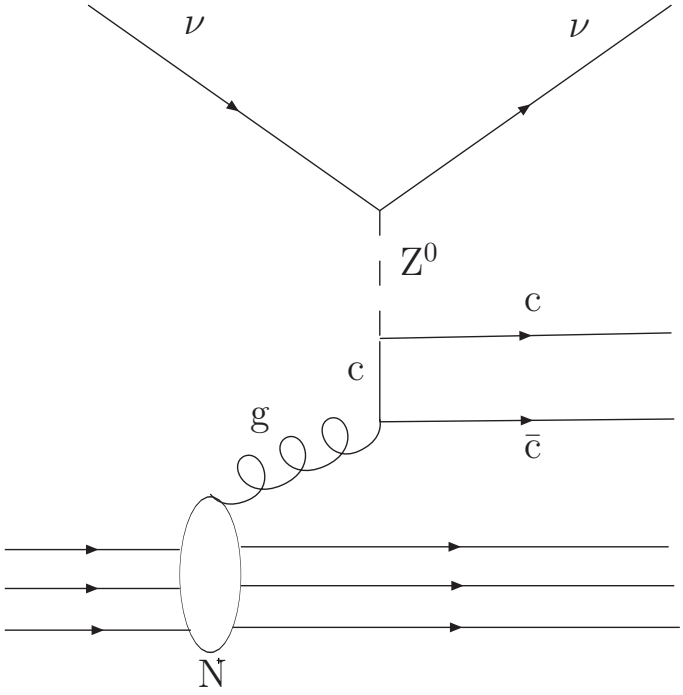


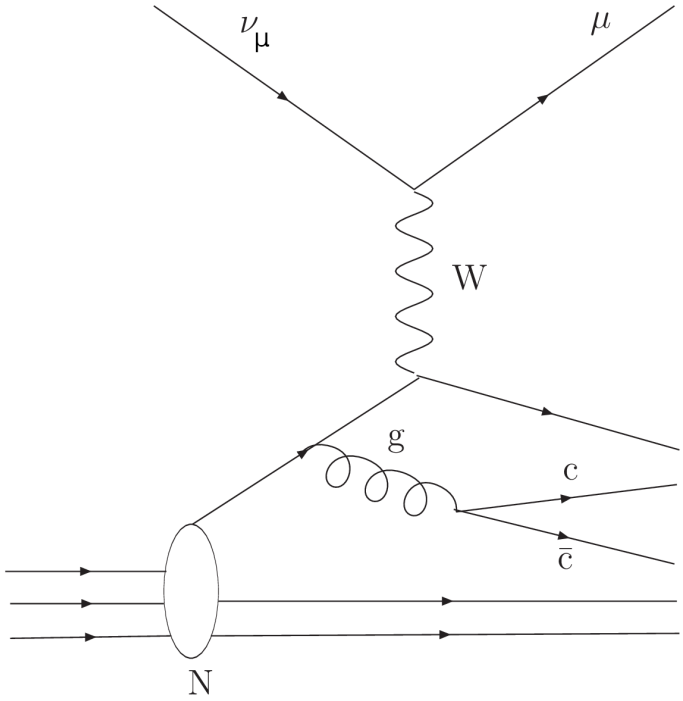

Figure 2.13: Charged-current tree-level diagram showing single charm production from a neutrino interaction.

### 2.3.4 Nucleon structure functions

Here the theory leading to the nuclear structure functions is reviewed with aid of [9, 51, 52]. The *Rosenbluth* formula gives the most general Lorentz-invariant form




(a) Boson-gluon fusion creating charmonium [59].




(b) Gluon bremsstrahlung creating charmonium [59].

Figure 2.12: Two rare processes by which neutrino DIS can produce charmonium ($c\bar{c}$).

for electron-proton scattering in the proton rest frame via the exchange of a single photon:

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E_1^2 \sin^4(\theta/2)} \frac{E_3}{E_1} \left( \frac{G_E^2 + \tau G_M^2}{1+\tau} \cos^2 \frac{\theta}{2} + 2\tau G_M^2 \sin^2 \frac{\theta}{2} \right) \tag{2.3.15}$$

Where $E_1$ and $E_3$ are the initial and final electron energies; $G_E$ and $G_M$ are the charge and magnetic moment distributions in the proton, which are dependent on $Q^2$; $\theta$ is the angle between the outgoing electron and the beam axis; $\tau = Q^2/4m_p^2$.

The Rosenbluth formula can be re-written using the definitions of $Q^2$ and $y$; expressing the cross-section in terms of two independent kinematic variables, $Q^2$ and $x$ being the negative squared four-momentum of the mediating boson and Bjorken-x respectively. Then generalising to the deep inelastic scattering regime:

$$\frac{d^2\sigma}{dxdQ^2} \approx \frac{4\pi\alpha^2}{Q^4} \left[ (1-y) \frac{F_2(x,Q^2)}{x} + y^2 F_1(x,Q^2) \right] \tag{2.3.16}$$

Here, the form factors $G_E$ and $G_M$, and $\tau$ have been replaced by two structure functions, $F_1(x,Q^2)$ and $F_2(x,Q^2)$, where $F_1(x,Q^2)$ is associated purely with the magnetic interaction while $F_2(x,Q^2)$ has components relating to both electric and magnetic interactions. These functions are not simply Fourier transforms of the proton charge and magnetic moment distribution because they are functions of both $Q^2$ and $x$[12] [9]. The Stanford Linear Accelerator Centre (SLAC) experiments were the first to perform studies of the structure functions in inelastic electron-proton studies. Incoming electron energies were selected with a magnetic field and the differential cross sections could be measured through a range of energies. The first of two key observations from these data was that the structure functions only have a slight dependence on $Q^2$, suggestive of scattering from point-like constituents in the proton, this phenomena is referred to as *Bjorken scaling* [9]. Secondly, the structure functions were observed to be dependent quantities in the deep inelastic scattering regime of $Q^2 \gtrsim 2\,\text{GeV}$; following the *Callan-Gross* relation[13]:

$$F_2(x,Q^2) = 2xF_1(x,Q^2) \tag{2.3.17}$$

The quark-parton model allows for the *Parton Distribution Functions* to be defined. For example, $u^{\text{p}}(x)\delta x$, represents the probability of an up-quark within the proton carrying a fraction of the total proton momentum between $x$ and $x+\delta x$ [9]. Moving from the electron-proton scattering system to the neutrino-proton scattering system which is of relevance to the SND@LHC experiment. While the neutrino may not wholly meet the condition of a Dirac particle, the Callan Gross relation approximately holds[14]. Now expressing the differential cross section in terms of $x$ and $y$, the corresponding general Lorentz invariant expressions for (anti-)neutrino charged-current DIS, allowing parity violation, are written as [51, 52]:

[12] In the low $Q^2$ limit, one can reduce the $G_E$ and $G_M$ to these Fourier transforms [9].

[13] The Callan-Gross is also expected *a priori* from the quark-parton model which is briefly outlined below.

[14] The Dirac equation requires particles to have their own distinct anti-particle, which in the Seesaw mechanism would not be correct.

$$\frac{d^2\sigma^{\nu/\bar{\nu}}}{dx\,dy} = \frac{G_F^2 m_N E_\nu}{\pi}\left(\frac{M_W^2}{Q^2+M_W^2}\right)^2\Big[y^2 xF_1(x,Q^2) + (1-y)F_2(x,Q^2) \pm y\Big(1-\frac{y}{2}\Big)\,xF_3(x,Q^2)\Big], \tag{2.3.18}$$

Where $G_F$ is the Fermi coupling constant $G_F/\sqrt{2} = g_W^2/8m_W^2$; $m_N$ is the nucleon mass; $E_\nu$ is the incident neutrino energy. The $\pm$ in-front of the final term is positive for neutrino and negative for anti-neutrinos. $F_3(x,Q^2)$ is introduced and can be explained by the different bosons involved: electron-proton DIS is mediated by the photon, which as a vector current does not violate parity; the neutrino-proton scattering however is mediated by the weak interaction which does violate parity [60]. The parity violating weak interaction results in a third structure function being needed to describe the difference in cross-section between neutrinos and quarks, and neutrinos and anti-quarks. Reference [50] reports that the structure function $xF_3(x,Q^2)$ for neutrinos and anti-neutrinos on protons following a leading order calculation with four active quark flavours (neglecting top and bottom flavours) and assuming a diagonal CKM matrix:

$$xF_3^{\nu p}(x,Q^2) = 2x\left(-u(x) + d(x) + s(x) - \bar{c}(x)\right)$$
$$xF_3^{\bar{\nu}p}(x,Q^2) = 2x\left(u(x) - \bar{d}(x) - \bar{s}(x) + c(x)\right)$$

The structure functions can be measured by studying the $y-$dependence of the cross section at set values of Bjorken-x. The NuTeV experiment at Fermilab, with a sampling calorimeter of iron interleaved with liquid scintillator detectors and drift chambers, performed measurements of the $F_2(x,Q^2)$ and $xF_3(x,Q^2)$ structure functions in the ranges $0.015 < x < 0.75$ and $1\,\text{GeV} \lesssim Q^2 \lesssim 200\,\text{GeV}$ [61]. Several other experiments have performed measurements of the nucleon structure functions such as HERA [62] and CHORUS [63].

## 2.4 Neutrino physics at the LHC

The idea of using the LHC as a place to study neutrinos has been around since the predecessor accelerator, the Large Electron Positron Collider (LEP) [64], and there are several reasons that a high energy collider is such an enticing place to do neutrino physics. First, the physics behind the production of the neutrinos at the LHC will be discussed at which point the qualities of these neutrino interactions will be highlighted. As stated in section 3.3, SND@LHC is capable of identifying all three flavours of (anti-)neutrino, without distinguishing between neutrino and anti-neutrino, so the production mechanisms leading to the flux of these neutrinos in SND@LHC should be elucidated, and what particular use these neutrinos have for probing the upstream interactions. The neutrino production mechanisms for all three flavours of neutrino as a function of energy are shown for both FASER$\nu$[15] and SND@LHC in figure 2.14.

[15]FASER$\nu$ operates in the complementary pseudo-rapidity region to SND@LHC of $\eta \gtrsim 8.2$ [65]

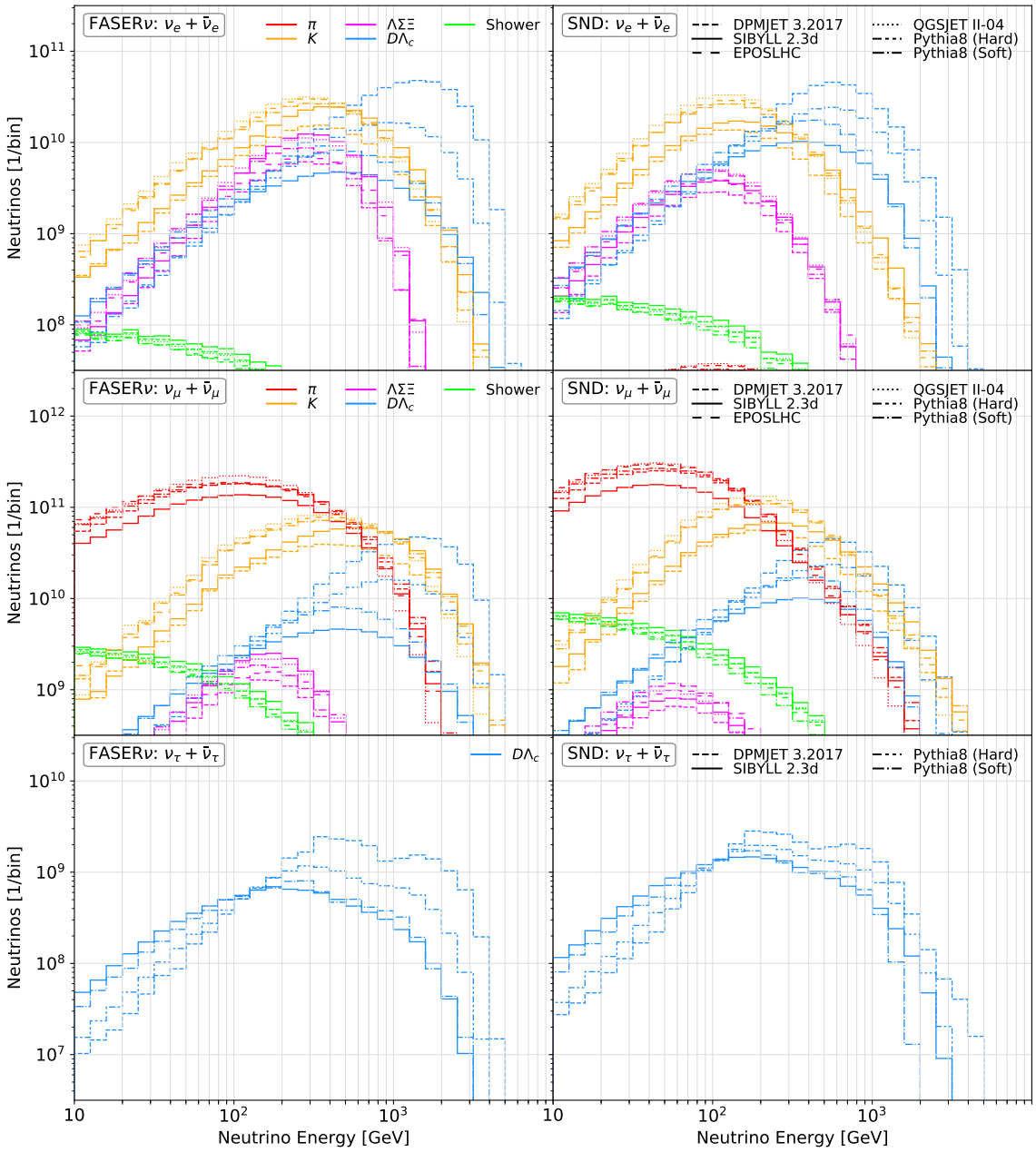



Figure 2.14: Incident neutrino spectrum for the SND@LHC on the right column, FASER$\nu$ on the left column. The scale of the ordinate axis represents the number of neutrinos passing through the detector's cross-sectional area for 150 fb$^{-1}$ of integrated luminosity. The neutrino flux is shown for each flavour, split by production method and estimated using up to six different event generators depicted with unique line styles. Taken from [65].

The semi-leptonic and fully leptonic decays of both charged and neutral kaons and ground state charmed hadrons; $D^{+/-}$, $D_s^{+/-}$ and $\Lambda_c^+$ baryons account for $\approx$ 90% of the electron neutrinos and anti-neutrinos in the acceptance of SND@LHC. From this, SND@LHC builds a charm physics programme using the flux of electron neutrinos and anti-neutrinos to probe the forward charm production at the ATLAS interaction point in a two stage process[16]: first, measuring the $pp \rightarrow \nu_e X$ cross section; second, using this to derive the charmed hadron yield [66]. The collaboration

[16]This is a *précis* of the more detailed description of the program which can be found at [66].

aims to measure the energy spectrum of electron neutrinos and anti-neutrinos, which are indistinguishable in SND@LHC. By unfolding the measured energy distribution for detector effects and assuming SM predictions for $\nu_e$ and $\bar{\nu}_e$ CC cross sections, a measurement on the total yield of $\nu_e$ and $\bar{\nu}_e$ from LHC $pp$ collisions in the pseudo-rapidity range $7.2 \leq \eta \leq 8.6$ can be made. The uncertainty on this measurement is expected to be dominated by the process of unfolding for detector effects, and with an component from the statistics. Turning now to the measurement of the charm hadron yield in the SND@LHC $\eta-$range, the first step taken is to estimate the number of $\nu_e$ and $\bar{\nu}_e$ incident through the SND@LHC $\eta-$range by statistically subtracting the electron neutrinos expected from pion and kaon decays. In this far forward region, the pseudo-rapidity of the charmed meson and its daughter electron neutrino are shown to be well correlated in figure 2.15, through a simulation with POWHEG and PYTHIA8:

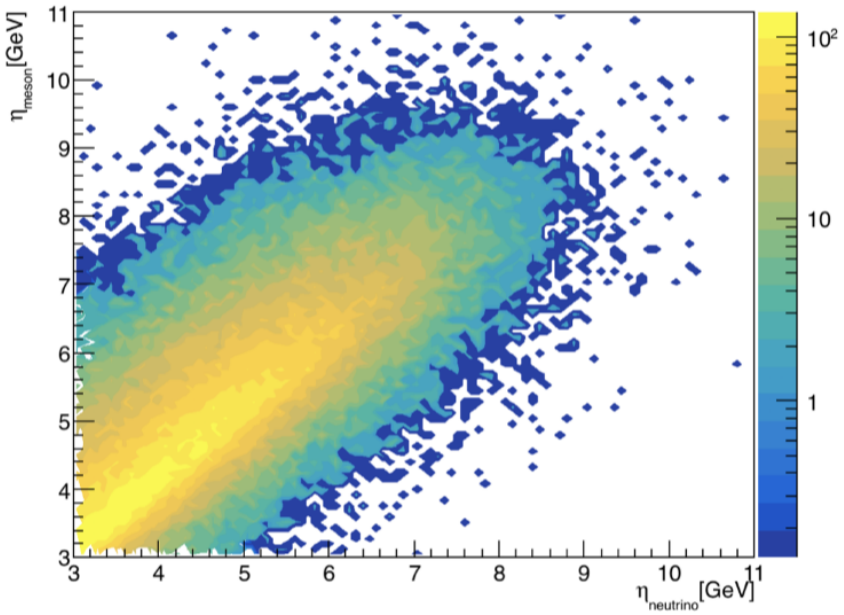



Figure 2.15: Correlation between the pseudo-rapidity of the charm meson, $\eta_{\text{meson}}$, and the electron neutrino, $\eta_{\text{neutrino}}$. Data obtained from simulations with POWHEG and PYTHIA8 from [66].

Sub-regions of figure 2.15 can then be studied to determine quantities such as, for example, the fraction of charmed mesons with $\eta > 5$ that produce neutrinos in the $\eta$-range of SND@LHC. Then the number of charmed hadrons fulfilling this criteria can be written as:

$$N(c-\text{hadron}) = N(\nu_e + \bar{\nu}_e)^{\text{charm}} \times \Gamma \times \frac{1}{Br(c \to \nu_e)} \tag{2.4.1}$$

Where $\Gamma$ is the fraction of events where the meson pseudo-rapidity is greater than 5, and the daughter electron neutrino is in the pseudo-rapidity range of SND@LHC, and $Br(c \to \nu_e)$ is the branching ratio for charm to electron neutrino final states [66]. Charm hadrons at the LHC are largely produced through gluon-gluon scattering. A simulation of gluon-gluon scattering at the LHC where the final state neutrino is in the SND@LHC acceptance is performed and figure 2.16 shows the correlation between the momentum fractions of scattering gluons: $x_1$ and $x_2$. The simulation shows that the average of the lower momentum fractions of these scattering gluons, is $\approx 10^{-6}$. At such low $x$ values, the gluon parton distribution function is rarely studied

[17]. The collaboration looks to constrain the gluon parton distribution function through the ratio to differential cross section measurements at different energies at a $\eta$-range:

$$R = \frac{d\sigma/d\eta(13\text{ TeV})}{d\sigma/d\eta_{\text{ref}}(7\text{ TeV})}$$

Where $4 < \eta_{\text{ref}} < 4.5$. The right panel of figure 2.17 shows that by making this ratio to the reference value, the scale uncertainty (red) is reduced and the PDF can be constrained by data [66].

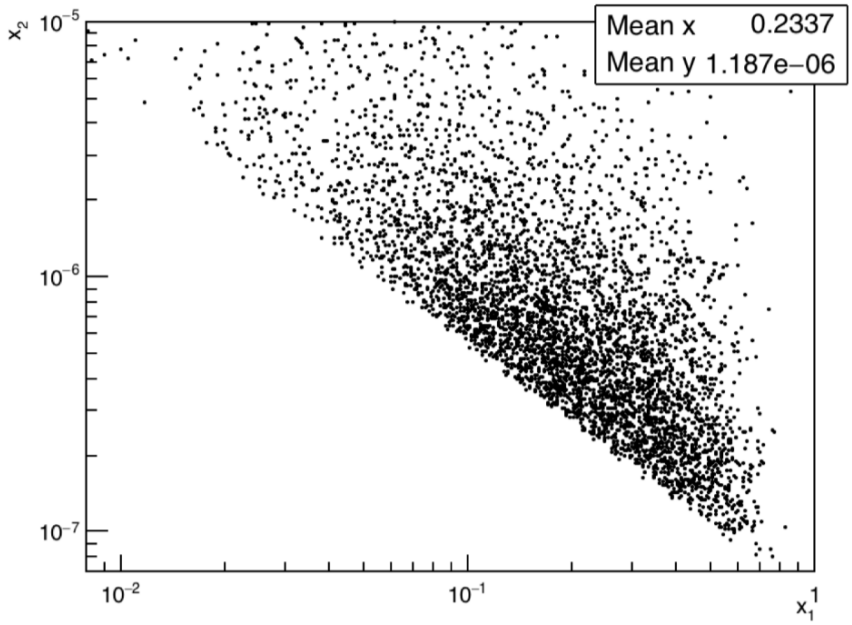



Figure 2.16: Correlation of $x_1$, $x_2$, the proton momentum fractions carried by the scattering gluons involved in open charm production at the LHC in events resulting in a neutrino in the SND@LHC $\eta$-range. Taken from [66].

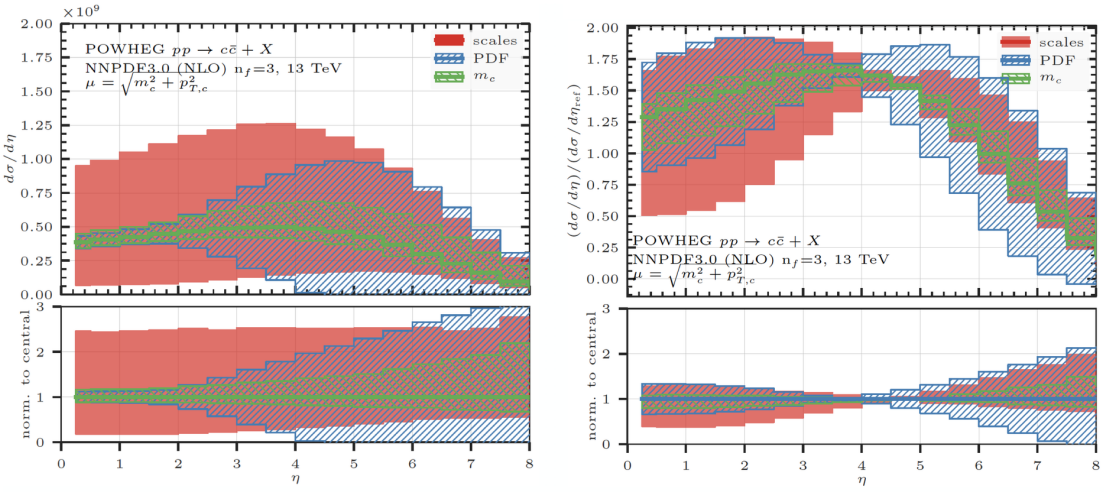



Figure 2.17: Left: differential cross section for charm production at 13 TeV. Right: ratio of differential cross section to the reference differential cross section at 7 TeV, $4 < \eta < 4.5$. Taken from [66].

[17]Such low $x$-values have been studied through $J/\psi$ [67] and open charm production of $D$-mesons at LHCb [68].

Understanding the production rate of electron neutrinos in the far-forward region of the LHC interaction points is of particular benefit to the IceCube experiment. The IceCube experiment is a high energy neutrino detector spanning a cubic kilometre of ice at the South Pole [69]. The primary research goal of the experiment is to record neutrinos from cosmic processes like gamma-ray bursts and super-novae. A background for these cosmic neutrino searches are neutrinos produced in the atmosphere through the decays of charmed mesons, themselves early-stage products in cosmic air-showers [70]. This is one of the largest backgrounds in the IceCube experiment and is more prevalent for higher energy ranges [70]. The charmed mesons which produce these atmospheric neutrinos come from the charged particles interacting with nuclei in the atmosphere at rest, thus producing very forward charm mesons. Measurements of forward charm production at the LHC will improve the estimates on prompt atmospheric neutrino production at IceCube [71].

The flux of muon neutrinos and anti-neutrinos in SND@LHC is mainly from the decays of pions and kaons. The contribution to the electron neutrino flux from pion and kaon decay is negligible due to helicity suppression (see section 11.6 of reference [9]). The tauonic neutrino flux is solely due to the decay of $D$ and $\Lambda_c$ baryons, an example of the $D_s$-meson decaying leptonically to tau is shown in figure 2.18. There is also a considerable component of muonic neutrinos produced by these meson decays for the highest energy neutrinos.

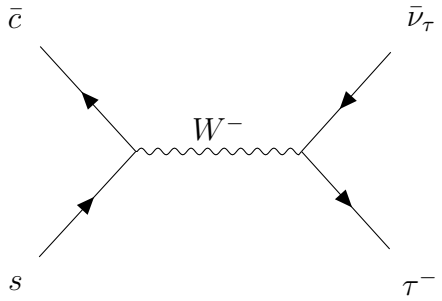



Figure 2.18: Leptonic decay of a $D_s^-$ meson to a tau final state.

## 2.5 New physics searches at SND@LHC

Worldwide research in dark matter occurs through both terrestrial direct experiments or astrophysical and cosmological indirect searches. As well as the role of SND@LHC as a neutrino scattering experiment, there is also a sensitivity in Beyond the Standard Model (BSM) physics through a search for light dark matter (LDM). SND@LHC can contribute here by performing model independent searches for Feebly Interacting Particles (FIPs) through their scattering off either electrons or nucleons and coupling this with a time-of-flight (TOF) measurement which lets the experiment reduce the background from neutrino interactions.

### 2.5.1 Scattering off nucleons

A scalar $\chi$ particle may couple through an interaction with a vector mediator $V$ with the baryon current $J_\mu^B$ to the SM via a leptophobic portal, mainly interacting with quarks, with Lagrangian $\mathcal{L}_{\text{lp}}$ [72]:

$$\mathcal{L}_{\text{lp}} = -g_B V^\mu J^B_\mu + g_B V^\mu (\partial_\mu \chi^\dagger \chi + \chi^\dagger \partial_\mu \chi), \quad J^B_\mu = \frac{1}{3}\sum_q \bar{q}\gamma_\mu q \tag{2.5.1}$$

In $pp$ collisions, the vector mediator may be produced through the decays of light mesons, $\pi \to V + \gamma$, $\eta \to V + \gamma$; proton bremsstrahlung, $p + p \to V + X$ or Drell-Yan process, $q + \bar{q} \to V + X$ [66]. The existing limits on the production of the vector mediator $V$ are from light meson decays, mono-jet searches, invisible decays of $J/\psi$, $\Upsilon$ or kaons [66, 72].

SND@LHC evaluates the sensitivity of the experiment to the leptophobic portal for both elastic and DIS scattering of $\chi$ from protons. The collaboration adopts the description of the elastic process, $\chi p \to \chi p$, from reference [73]; while for inelastic scattering, $\chi N \to \chi X$, the parton model from reference [74] is used. Only the elastic scattering from a proton is considered in the study since this the final state proton will result a charged track, giving an experimental signature. In contrast, for DIS whether the interaction is on a proton or neutron is irrelevant, so the reaction is written for a nucleon, $N$. The polar angle-energy distributions of $\pi$ and $\eta$-mesons are used to obtain the spectra of $V$ and $\chi$ particles coming from light-meson decays using an approach detailed in reference [75], the energy distributions of $V$ and $\chi$ from proton bremsstrahlung is similarly obtained. Equation 2.5.1 was used to produce the energy spectrum coming from Drell-Yan production. In the acceptance of SND@LHC, the dominant production mode of $V$ and $\chi$ was found to be light-meson ($\eta$) decay for $m_V < m_\eta$. Proton bremsstrahlung is dominant for the region $m_\eta < m_V \lesssim 3\,\text{GeV}$ while Drell-Yan dominating for masses $m_V \gtrsim 3\,\text{GeV}$. The total number of scattering events in the detector acceptance can be estimated as:

$$N_{\text{events}} = 2N_\chi^{\text{SND}} n_{\text{target}} (Z\sigma_{\text{scatt}}^{\text{elastic}} \langle E_\chi \rangle + A\sigma_{\text{scatt}}^{\text{DIS}} \langle E_\chi \rangle) l_{\text{target}} \tag{2.5.2}$$

Where $N_\chi^{\text{SND}}$ is the number of $\chi$ and $\bar{\chi}$ particles in the SND acceptance; $n_{\text{target}}$ is the atomic number density (atoms / cm$^3$) of the target material; $Z$ and $A$ are the atomic number and mass number of the target respectively, which in SND@LHC is tungsten; $\sigma_{\text{scatt}}^{\text{elastic/DIS}}$ corresponds to the relevant scattering cross-sections; $\langle E_\chi \rangle$ is the expected energy of the incident $\chi$ particle and $l_{\text{target}}$ is the length of the target region. The background to this search is found to be dominated by neutrino NC DIS and NC RES, which the collaboration believe will be reduced to negligible levels through emulsion chamber calibration and kinematics studies [66].

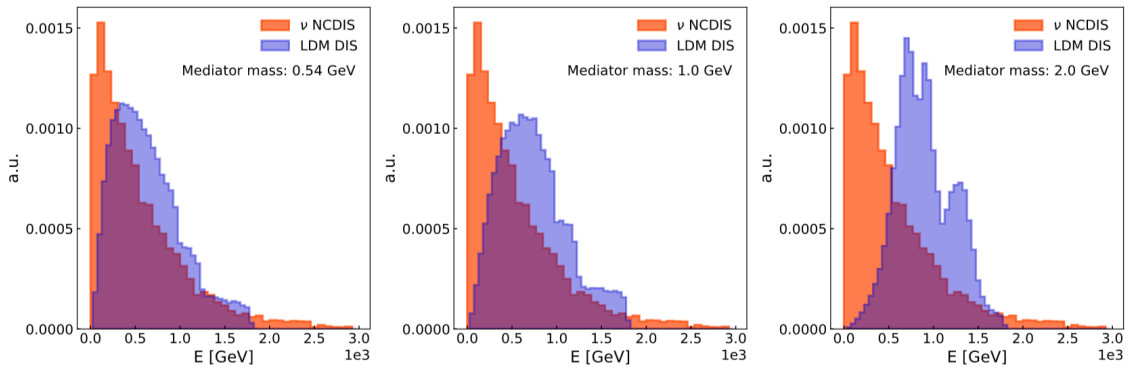



Figure 2.19: Normalised energy spectrum of final state NC DIS neutrino interactions and dark matter DIS interaction for three possible mediator masses, $m_V$. Taken from [66].

Figure 2.19 shows that SND@LHC improves its separation power on the dark matter candidate for higher mediator masses. For deriving the sensitivity of the experiment, the experiment assumes a significance of $3\sigma$; leading to figure 2.20, where the experiment shows its capacity to expand on existing limits in a broad region of the parameter space for the baryonic fine structure constant, $\alpha_B = g_B^2/4\pi$, where $g_B$ is the gauge coupling of the $U(1)_B$ leptophobic symmetry.

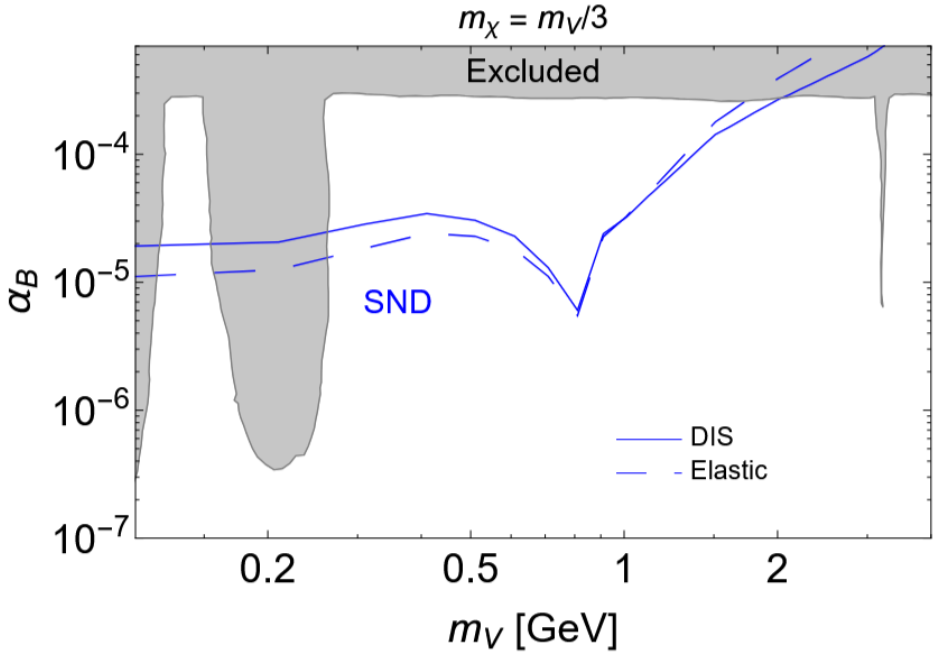



Figure 2.20: Projected sensitivity of the SND@LHC experiment to the leptophobic $U(1)_B$ dark matter portal by scattering off protons. Elastic scattering is shown in dashed blue, and DIS scattering in solid blue. Taken from [66].

### 2.5.2 Scattering off electrons

In the minimal case, the symmetry groups of the SM are extended with a new Abelian $U(1)$ group, similar to that of electromagnetism. This extension to the SM necessitates a new boson, called the dark photon (DP), $\mathcal{A}'$. Kinetic mixing arises between the dark photon and photon, due to the new $U(1)$ symmetry, similarly to the mixing of neutrino mass and flavour eigenstates [76]. The new component to the SM Lagrangian can be written as [66, 76][18]:

$$\mathcal{L}_{\mathcal{A}'} = -\frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} + \frac{m^2_{\mathcal{A}'}}{2}A'^{\mu}A'_{\mu} - \frac{\epsilon}{2}F'_{\mu\nu}F^{\mu\nu}$$

To study this, SND@LHC considers the prompt decays of a dark photon into a pair of light dark matter candidates, $\chi$, where $m_\chi \sim \mathcal{O}(1\,\text{GeV})$. The light dark matter candidate may have either half-integer or integer spin [66]:

$$\mathcal{L}_\chi = g_D A'^{\mu} \times \begin{cases} \bar{\psi}_\chi \gamma_\mu \psi_\chi \\ i\left[(\partial_\mu \phi^\dagger_\chi)\phi_\chi - \phi^\dagger \partial_\mu \phi_\mu\right] \end{cases}$$

[18] Whether the dark photon couples to SM particles through the electric charge or hypercharge, has an effect on the physical consequences for a massive dark photon. For the more thorough treatment of this one may read reference [77].

Where the top and bottom cases of the equation correspond to the vector and scalar cases respectively, and $g_D$ is the dark sector $U'$ gauge coupling. SND@LHC considers the regime where $m_{\mathcal{A}'} > 2m_\chi$ and $\alpha_D = g_D^2/4\pi \gg \epsilon e$ and assumes the branching fraction of $\mathcal{A}' \to \chi\chi^\dagger \approx 1$ [66]. For dark photon masses $m_{\mathcal{A}'} < 1\,\text{GeV}$, the primary two production modes are meson decays and proton bremsstrahlung: interacting protons at the LHC radiate DPs with a very small angular range [66].

$$\pi^0,\, \eta,\, \eta' \to \gamma\mathcal{A}',$$
$$\omega \to \pi^0\mathcal{A}'$$

To attempt and detect the light dark matter candidate, the experimental signature considered is elastic scattering with atomic electrons:

$$\chi e^- \to \chi e^-$$

Atomic electrons struck in the target region of the detector would cause electromagnetic showers in the target region which could be reconstructed with the SciFi tracker with its fine granularity and spatial resolution (see section 3.2.1)[19]. Background sources of neutrino-electron CC/NC $\nu_\ell\, e \to \nu_\ell\, e$, electron neutrino CCDIS, CCRES and CCQE with soft tracks at the vertex were investigated by the SND@LHC collaboration as part of the experiment proposal and found that the only statistically significant background could come from electron neutrino CCDIS and sufficiently soft interactions to be background would not be detectable by the emulsion system so the experiment assumes a zero-background rate for this search, as is explained with more detail in [66].

Beyond this, a complementary time-of-flight study at SND@LHC will leverage the measurable flight time from IP1 to SND@LHC.

## 2.6 Charged particle interactions with matter

Having earlier outlined the principles by which neutrinos interact with matter, this section will outline the principles by which the interaction of a neutrino is inferred by the physics interactions of the final state interaction products with the active detection media in SND@LHC. Covering this topic will contextualise the role of the US muon system of SND@LHC and how the calibration and commissioning studies are motivated. Chapter 4 of this thesis is primarily concerned with the behaviour of the detector to muons traversing the detector. Charged particles traversing a medium will lose energy by interacting electromagnetically with the atomic electrons or ionising atoms. The average rate of energy loss, $\langle dE/dx \rangle$, is described quantitively by the *Bethe equation* [79]:

$$\langle -\frac{dE}{dx} \rangle = Kz^2 \frac{Z}{A} \frac{1}{\beta^2} \left[ \frac{1}{2} \ln \frac{2m_e c^2 \beta^2 \gamma^2 W_{\max}}{I_e^2} - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right] \tag{2.6.1}$$

The symbols in this equation are as follows: $K = 4\pi N_A r_e^2 m_e c^2$; $N_A$ being Avogadro's constant; classical electron radius $r_e = e^2/4\pi\epsilon_0 m_e c^2$; $m_e$ the electron mass; $c$ the speed of light in vacuum; $z$ the charge number of the incident particle; atomic number $Z$; atomic mass of the absorber $A$; $\beta = v/c$ for the relativistic

[19] Since the radiation length of tungsten is 3.5 mm [78] and the tungsten plates in the target are 1 mm, this will not be straightforward.

particle of velocity $v$; Lorentz factor $\gamma = 1/\sqrt{1-\beta^2}$; effective ionisation potential $I_e$[20]; $W_{\max}$ is the maximum kinetic energy transferred to an electron and $\delta(\beta\gamma)$ is a density effect correction to the ionisation loss [79].

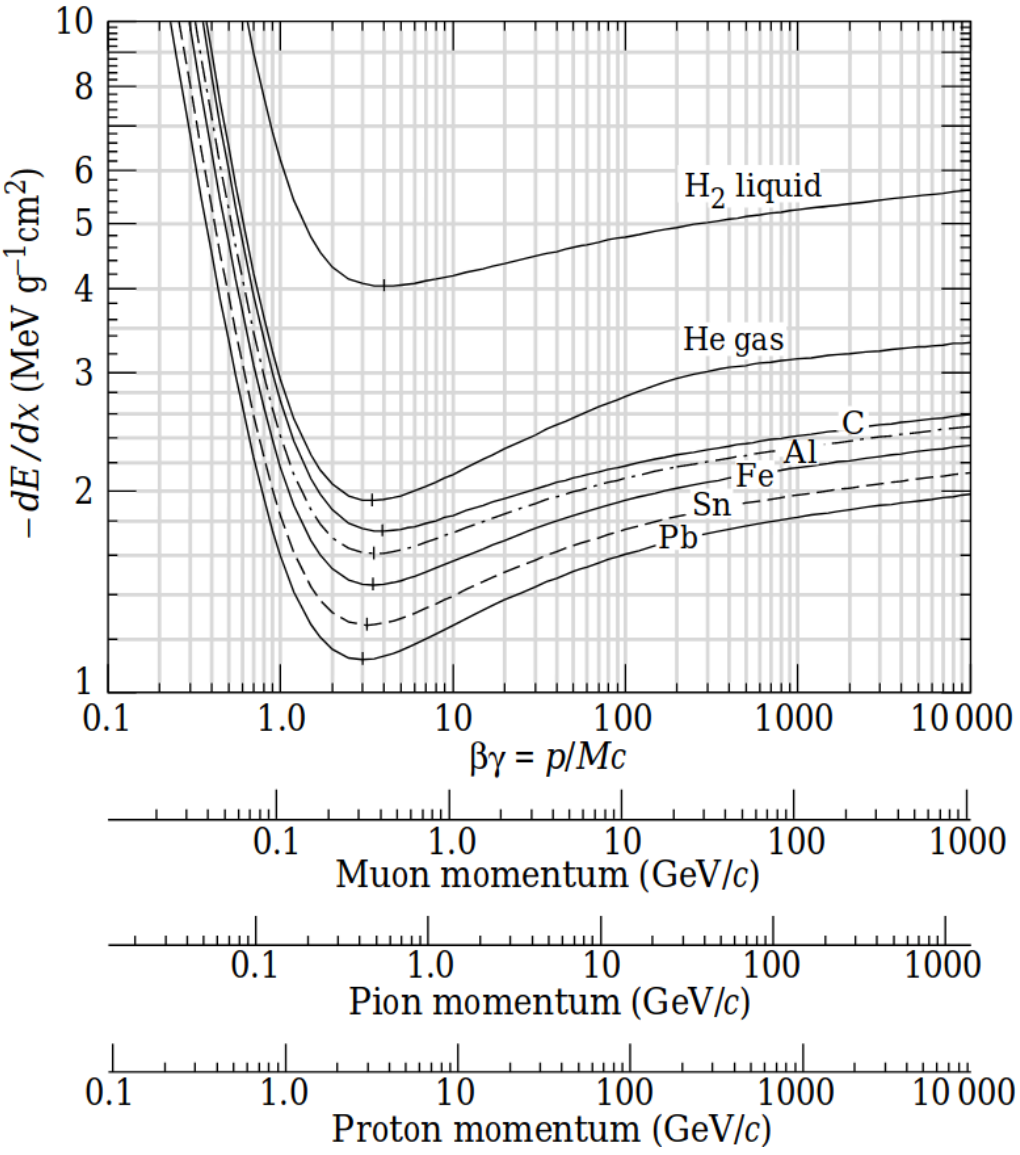



Figure 2.21: Mass stopping power for muons, pions and protons in various materials, expressed as $-1/\rho\langle dE/dx\rangle$. Taken from [79].

The Bethe equation holds for $0.1 \lesssim \beta\gamma \lesssim 1000$. The quantity, $-1/\rho\langle dE/dx\rangle$, is referred to as the *mass stopping power*, describing the rate that a charged particle loses energy in a medium relative to its density. The factor of $1/\rho$ makes for easier comparison between media. One sees in figure 2.21 the "minimum ionisation" region in the energy range of $\mathcal{O}(100)$ MeV $< |\mathbf{p}_\mu| <$ $\mathcal{O}(100)$ GeV.

The majority event case in SND@LHC are muons produced at the ATLAS IP and passing through the detector leaving an isolated track. These muons are the principle source of *in situ* calibration and commissioning work for the detector and are used heavily in this thesis, thus the nature of their interaction with the detector is paramount. The scintillators used in the SND@LHC US muon system are based on polyvinyltoluene (PVT) [80], meaning that one can approximate the mass stopping

[20]The effective ionisation potential is averaged over all atomic electrons.

power curve by the trend shown for carbon in figure 2.21, where the global minimum is around 1.8 MeV $g^{-1}$ $cm^2$. The energy spectrum of these muons at SND@LHC, normalised to unit integral, is shown in figure 2.22. From figure 2.22, 47.5% of the muons reaching SND@LHC are below 1 TeV: at an energy of 1 TeV the mass stopping power in carbon is approximately 2.6 MeV, 44% greater than that of a truly minimum ionising particle. The assumption that the mass stopping power curve for carbon best reflects the scintillator used in the US muon system is supported by how PVT is implemented in the ROOT geometry package, where one sees that the mean value of energy deposited per centimetre by the muons traversing the US muon system is $\approx$ 2.6 MeV (see appendix A.1).

So while the majority of muons at SND@LHC are considerably above the $\beta\gamma$ coefficient to be considered a truly minimum ionising particle, the impact on deposited energy is modest qualitatively well understood.

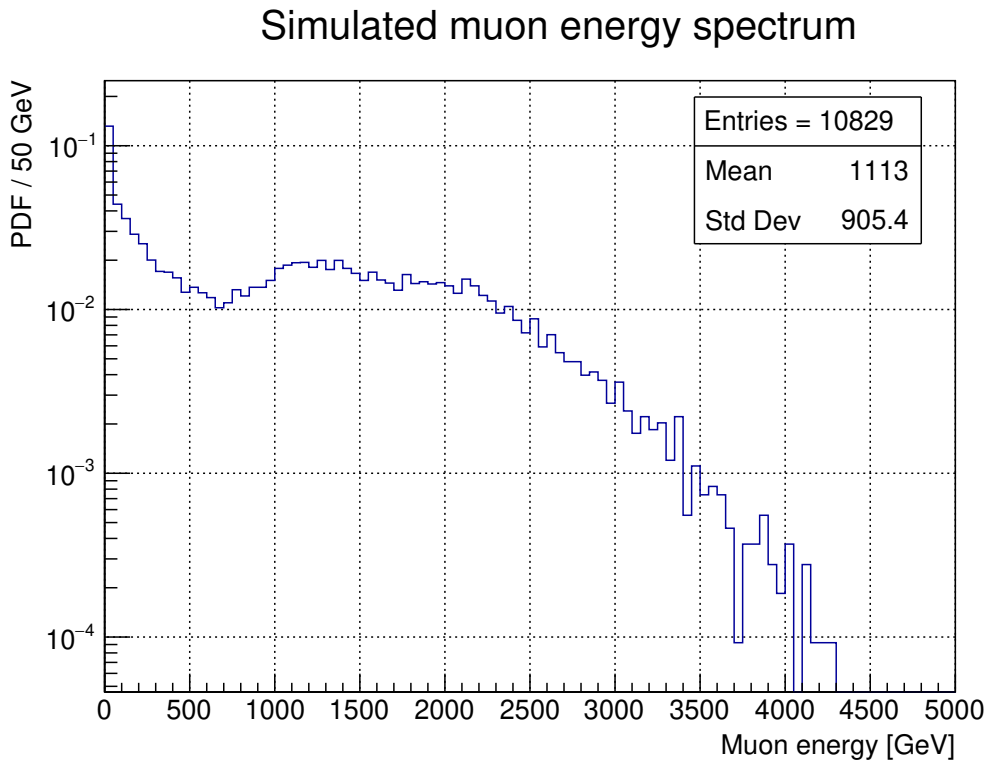



Figure 2.22: Initial energy spectrum of muons reaching SND@LHC measured from simulation and normalised to unit integral.

The charged particles traversing the US muon system deposit energy in the scintillators by exciting the atomic electrons which promptly release the deposited energy by the emission of visible light through fluorescence[21]. The fluorescence principles fundamental to the scintillator based systems at SND@LHC are discussed more in section 3.2.6.

[21]The rapid ($\mathcal{O}(ns)$ pulse width) release of the excitation energy through electromagnetic radiation is a distinguishing characteristic between fluorescence and phosphorescence [81].

# Chapter 3

# The SND@LHC experiment

This chapter outlines the design of the Scattering and Neutrino Detector at the Large Hadron Collider (SND@LHC) experiment, with a particular focus on the hadronic calorimeter and its constituent technologies. The broader experimental apparatus such as the experiment control software and event building software are also discussed, before finally outlining the data formats used by the experiment for Monte Carlo and LHC data in different analysis stages that are relevant to the work of the thesis.

## 3.1 Overview/concept

SND@LHC is a compact neutrino detector lying 480 $m$ downstream of the proton-proton ($pp$) interaction point used by the ATLAS experiment at CERN, IP1. The space found in the CERN accelerator complex is in the previously unused TI18 tunnel. SND@LHC covers an unexplored pseudo-rapidity region of $7.2 < \eta < 8.6$ which will measure all three flavours of neutrino in a mutually complementary fashion to the experiment FASER$\nu$ which exists in a complementary higher pseudo-rapidity region of $\eta > 9.2$ [82]. A diagram showing the placement of SND@LHC and FASER relative to the ATLAS IP can be found in figure 3.1.

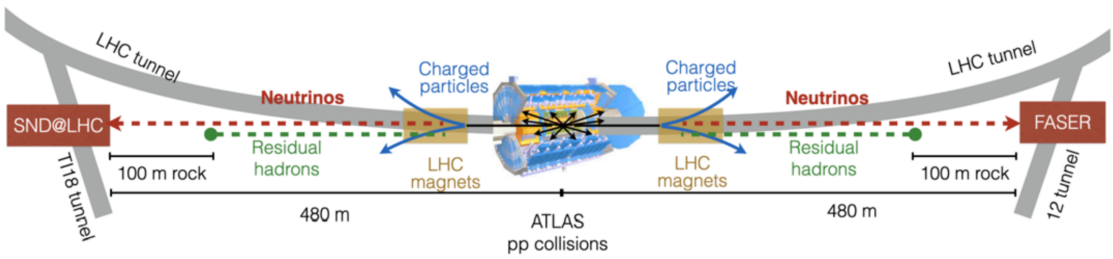



Figure 3.1: A schematic illustrating the position of both FASER and SND@LHC relative to ATLAS, and how the LHC magnets and rock removes the majority of charged particles and hadrons from impinging upon the experiments. Figure taken from reference [83].

Aside from the location of SND@LHC relative to IP1, the experimental coordinate system used is also of importance in this thesis and is shown in the top left and right corners of figure 3.2. A key point of the coordinate system that will be restated throughout the thesis is that the $x-$axis points to the left to ensure the coordinate system is right-handed. As a result of this, greater values of $x-$position refer to positions further left relative to an observer looking down the beam line, $z-$axis. This

subtle point will be periodically re-emphasised for clarity to the reader. To minimise confusion, left and right will be discussed in relative to an observer looking down the beam axis, this means that left and right are flipped relative to plots showing $x-$position.

Also shown in this figure is the collision axis and the nominal collision point of ATLAS, demonstrating that SND@LHC is an 'off-axis' experiment. The stated pseudo-rapidity range is determined for particles hitting the upper right and lower left corners of the emulsion target, evidenced by the concentric circles around the collision point in the $xy$ plane. A additional point to be noted in relation to the location of SND@LHC in the CERN accelerator complex is the direction of beam 2, that is the counter-clockwise travelling beam passing SND@LHC on its way to the ATLAS interaction point, beam 2 produces background in the experiment that will be discussed later on.

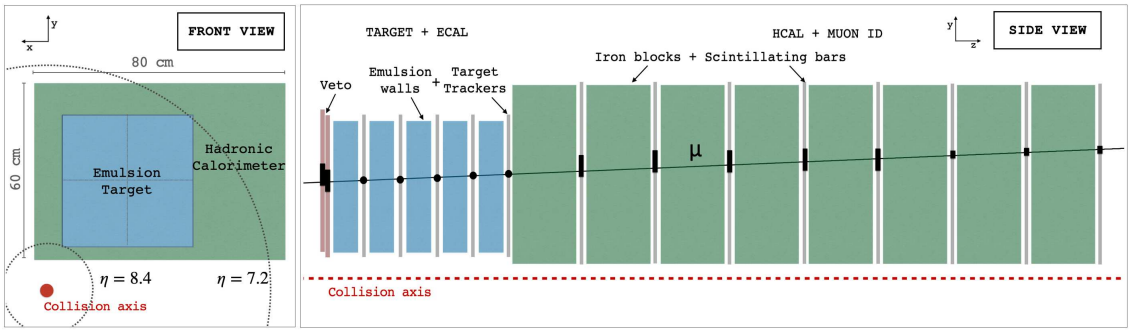



Figure 3.2: Layout of the SND@LHC detector. The coordinate systems of the $xy$ and $yz$ planes are shown in the top left and right corners respectively. Taken from [84].

## 3.2 The SND@LHC detector design

The overarching physics goal of the experiment is to measure neutrinos of all three flavours, the detector is fundamentally designed around this.

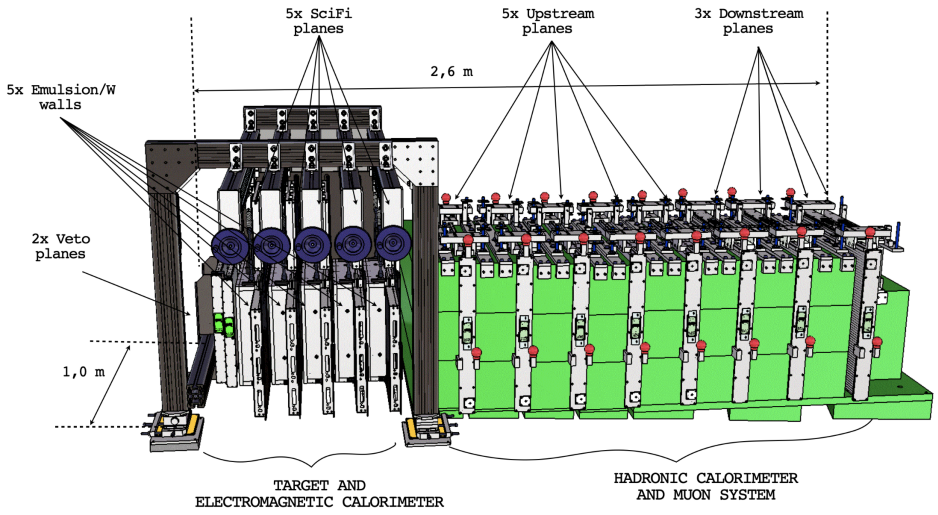



Figure 3.3: The SND@LHC detector. Particles traversing the experiment from left to right. Taken from [85].

The main sections of the detector will be discussed in detail, with particular attention to the subsystems of primary interest to this thesis work, which are the upstream muon system (US) and downstream muon system (DS).

### 3.2.1 Target tracker and electromagnetic calorimeter

The first of the two sub-detectors comprising the target region that are described in this chapter is the electromagnetic calorimeter. Situated immediately downstream of the veto system are five stations of scintillating fibre (SciFi) detectors interleaved with the five target walls described in subsection 3.2.2. The primary functions of the SciFi are firstly to timestamp neutrino interactions reconstructed in the emulsion target walls and to perform calorimetry of electromagnetic showers. Furthermore, the SciFi and muon system together act as a heterogeneous hadronic calorimeter for determining the energy of the hadronic shower caused by a neutrino interaction. Each of the five stations of the SciFi are comprised of two mats. Per station, one mat is comprised of horizontally aligned scintillating fibres and the other of vertically aligned fibres and all mats are read out by SiPMs. The aligned fibres are stacked in an overlapping fashion to remove any dead area, as shown in figure 3.4, and are stacked six layers thick and glued with a substance to suppress cross-talk [86].

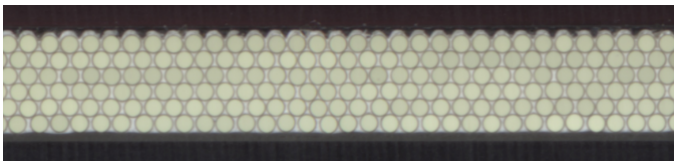


Figure 3.4: SND@LHC uses 250 $\mu$m diameter double-cladded polystyrene based scintillating fibres from Kuraray [86].

### 3.2.2 Target and vertex detector

The SND@LHC collaboration utilises a depth of knowledge in emulsion detector physics from the OPERA collaboration to achieve sub micro-metric vertex resolution. The whole emulsion target is made of five walls. Each wall is in turn made of four bricks. A brick is formed by alternating 60 emulsion films with 59 1 mm thick tungsten plates. The total material budget of one target brick is $\sim 17X_0$ [85].

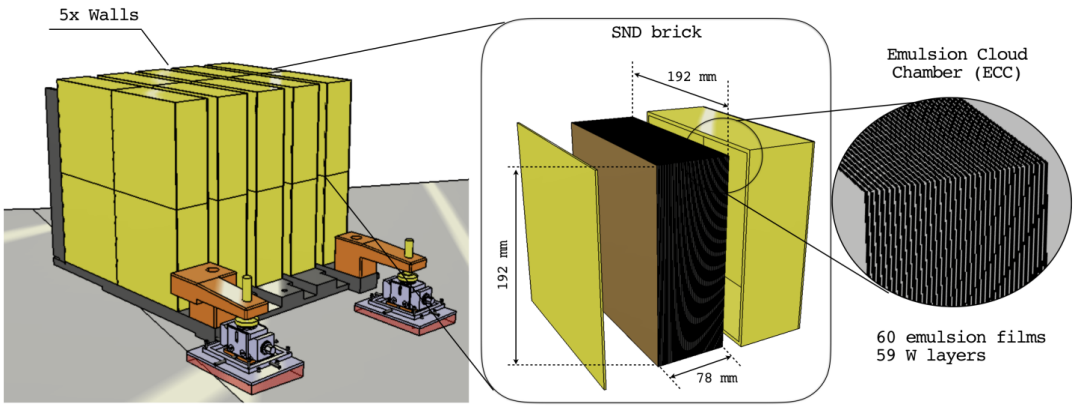



Figure 3.5: Schematic displaying the five walls of the emulsion target.

The technique of interleaving nuclear emulsion films with passive absorber layers is referred to as the Emulsion Cloud Chamber technique (ECC) and was shown by the OPERA experiment to be capable of detecting $\tau$ leptons by disentangling their production and decay vertices [87, 86]. The design goals of the target and vertex detector is to fulfil the following: cover the desired pseudo-rapidity range, maximise azimuthal angular acceptance, sufficient emulsion surface area to minimise neutrino interaction event leakage, and a minimal number of bricks per wall so that the dead area per wall is minimal. Nuclear emulsion films offer the most compact and lightest three-dimensional tracking detectors that can provide sub-micrometric and milliradian angular resolution [86]. As well as the collaboration benefiting greatly from learned experience with the OPERA experiment, the choice of emulsion films over comparably precise silicon pixel vertex detection is far cheaper and is far more spatially efficient. The usage of silicon pixel vertexing would require the installation of active cooling as well as a large number of readout cables and the hardware associated with the cabling. The downside of the emulsion is that given a finite occupancy per set of emulsion bricks, the emulsion must be manually exchanged every $\sim 25$ fb$^{-1}$ [86].

The lifespan of installed ECC walls is enhanced by optimising the temperature and humidity of the emulsion as well as minimising the neutron induced 'fog' uniformly contaminating the emulsion films. Proton-beam-gas interactions from LHC beam 2, which runs circles anti-clockwise, produce low-energy neutrons existing in an energy range of meV to $\mathcal{O}$(100 MeV). The solution to this is to construct an environment neutron shield around the target region. The neutron shield is made of 4 cm thick 30% borated polyethylene and 5 cm thick acrylic layers that result in a background reduction of $10^{-7}$ [86]. Inside the neutron shield, for the benefit of the emulsion, a temperature of $(15 \pm 1)°C$ and relative humidity of between 40% and 50% is held at all times.

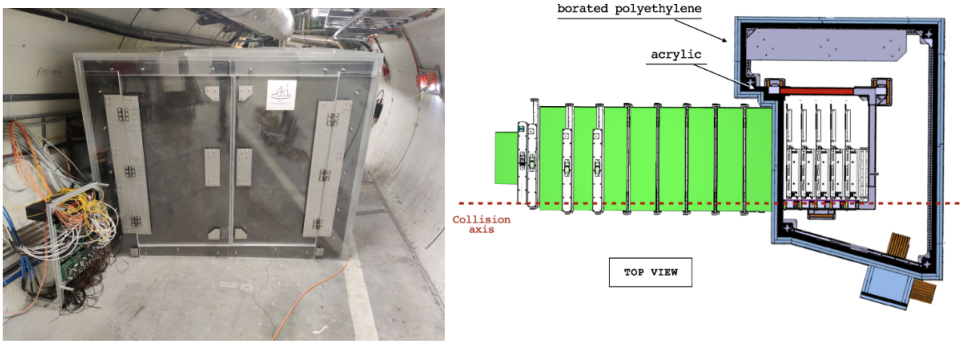


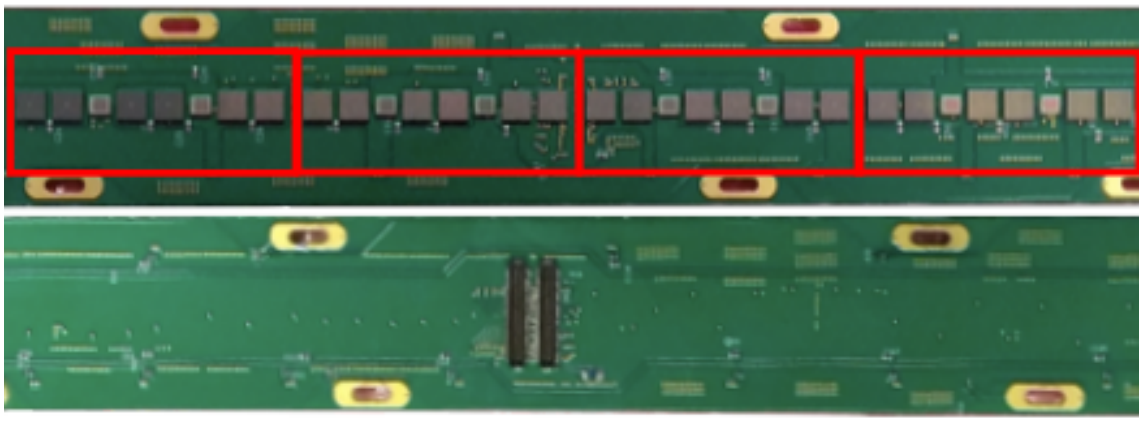

Figure 3.6: Photograph and schematic of finished neutron shielding cold box.

Once emulsion walls have collected the required luminosity they are removed from TI18 and are developed on-site at CERN meaning they are no longer sensitive to charged particles. Developed emulsions are then scanned using microscopes located at CERN, Napoli, Bologna and Zurich. Scanned emulsion data is uploaded to the EOS data storage system for offline analysis by the collaboration. Digitised emulsion data can be assigned a timestamp using the timing data from the electromagnetic calorimeter that is described in section 3.2.1.

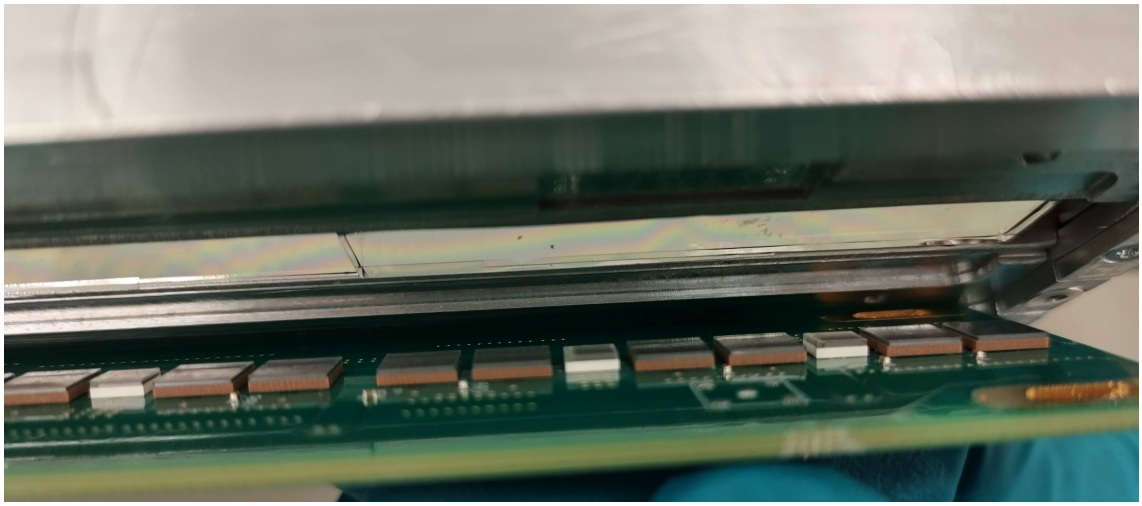
### 3.2.3 Upstream muon system

Downstream of the target system lies the subsystem of principal study in this thesis, the upstream (US) muon system. This forms the latter part of the non-homogeneous hadronic calorimeter alongside the SciFi. The US is formed of five planes of ten Eljen EJ-200 plastic scintillator cuboids with dimensions of $82.5 \times 6 \times 1\,\mathrm{cm}^3$. Each plane of scintillators is separated by a 20 cm thick iron wall resulting in the US acting as a course sampling US ($\sim 9.5\lambda_{\mathrm{int}}$ of iron in the US system). As the US, Veto, and DS systems all employ the same EJ-200 plastic scintillator, the properties of the scintillator and the physics of scintillation are deferred to section 3.2.6 to avoid repetition. The scintillators in each plane are aligned with the long axis of the scintillator parallel to the $x$-axis of the experiment, henceforth referred to as horizontally aligned. The ten scintillators per plane are individually wrapped in a aluminised Mylar foil for optical isolation and then fixed in an aluminium housing with the optical readout installed on both sides. A schematic for the aluminium housings and light shielding used for the US and DS muon systems are shown in appendix A.3.

Both end-faces of the scintillators are coupled to eight silicon photo-multipliers (SiPMs) with the group of eight SiPMs being comprised of six Hamamatsu S14160-6050HS and two Hamamatsu S14160-3010PS [88]. The 80 SiPMs per side of a plane are all connected to a printed circuit board (PCB), thus two PCBs per plane. Figures 3.7a 3.7b show how these groups of SiPMs are arranged on a printed circuit board (PCB) and coupled to the scintillators.



(a) Zoomed-in image of the US PCB. The top half shows the SiPMs that are coupled to the scintillator-ends with the red rectangles enclosing how the SiPMs are grouped to a scintillator. The bottom half shows the connector used to readout the raw SiPM data.



(b) The process of aligning a US PCB with the wrapped scintillators that are fixed in place in the aluminium housing. The Mylar foil can be seen around the bars as well as the again seeing the grouping of the SiPMs to each scintillator.

Figure 3.7

The smaller area of the two types are the Hamamatsu S14160-3010PS while the larger area are the Hamamatsu S14160-6050HS [88], these are henceforth referred to as the small area and large area SiPMs respectively. Two distinct types of SiPM are used in order to cover a larger domain of deposited energies. The large area SiPMs will provide information for energy calibration in all but the instances of highest energy deposition, where they will saturate. In such events, the small area SiPMs were installed to step in for energy reconstruction. The differences in these SiPMs leading to these different behaviours is explained in section 3.2.7. A final note on the SiPMs in the US muon system is that one of the large area SiPMs does not work, leaving 599 operation large area SiPMs.

The aluminised Mylar foil, a specular reflector, not only provides light isolation from the adjacent bars, without which, data collected by SiPMs could not solely be attributed to a single scintillator. Furthermore, the Mylar foil reflects photons which reach the edge of the scintillator while not satisfying the total internal reflection (TIR) condition of scintillator and air. The refractive index of the scintillator is quoted as 1.58 [80] which gives a critical angle of:

$$\theta_c = \sin^{-1}(1/n) \approx 39^\circ$$

Photons hitting the scintillator-air boundary with an angle greater than the critical angle, will result in TIR and the angle of the reflected ray will be equal to the angle of the incident photon [89]. Thus, a small air gap between the scintillator and the foil is important and part of the design of the US.

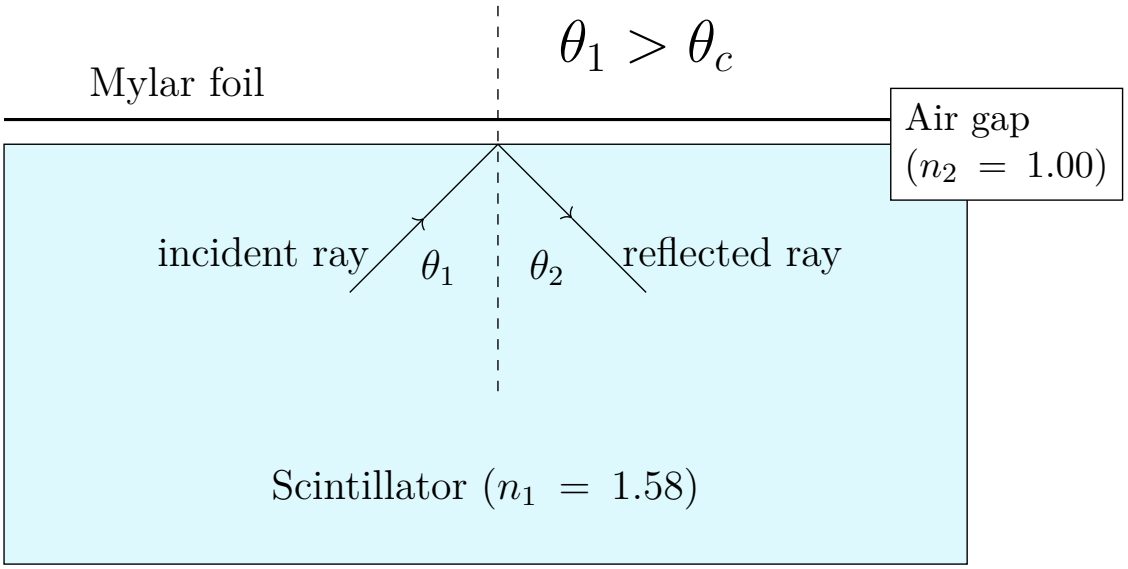



Figure 3.8: Diagram illustrating total internal reflection (TIR) in a scintillator wrapped with Mylar foil, showing the critical angle condition and refractive indices.

An important point in the construction of the US system is the use of Wacker SilGel 612 [90], hence referred to as silgel. It is impossible to avoid any air-gap between the scintillators and the SiPMs of the system, in order to minimise the air-gap and resulting signal loss at the optical interface, silgel was used to couple the SiPMs to the plastic scintillators. However, the minimum thickness of silgel which could be produced was found to be $\sim 3\,\text{mm}$, approximately a factor of two greater than the expected air-gap, and so applying silgel to both sides was not feasible. For this reason, silgel was applied to only one side of the scintillators, the scintillators were pressed firmly against the SiPMs on the non-gel side allowing the compressible silgel to minimise the air-gap on the other side of the scintillators. Wacker SilGel features only in the US system. Below is a table detailing where silgel is used in the US.

| Plane number | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Gel side | Right | Left | Right | Left | Left |

Table 3.1: Tabulation of what side gel is placed on for each US plane

The silgel material is formed from two separate viscous liquids which when mix together, vulcanises at room temperature into the final solid product. The manufacturer guidance on producing the silgel suggests the viscous liquid precursors A and B should be mixed in equal quantities (1:1). However, in laboratory tests this ratio was found unsuitable for the small thickness of $\sim 1.5\,\text{mm}$ being manufactured while a 2:1 ratio of A to B produced a more applicable final product.

### 3.2.4 Downstream muon system

The DS design overlaps greatly with US. The system is comprised of a series of planes of Eljen EJ-200 scintillators, wrapped in aluminised Mylar foil and readout on the ends by the same Hamamatsu S14160-6050HS as is used in the US. The muon system has three stations each comprised of two planes placed following one another with no iron between them. The first of these planes uses 60 scintillators of dimensions $81 \times 1 \times 1\,\mathrm{cm}^3$ aligned with the long axis parallel to the $x-$axis and readout by a single SiPM on each end, these are henceforth referred to as horizontally aligned planes. Immediately following each horizontally aligned plane is a plane of 60 shorter scintillators, $60 \times 1 \times 1\,\mathrm{cm}^3$, with the long axis parallel to the $y-$axis and readout only on the top of the bar by a single SiPM, these are henceforth referred to as vertically aligned planes. Similarly to the US, the 60 SiPMs per side are connected to a shared PCB. Then there is a final fourth station comprised only of a vertically aligned plane. All of the stations of the DS are separated by the same iron blocks that are used in the US with the exception of the space between the third station and the fourth station. The lack of an iron block between the final two DS stations is visible in figure 3.9. In this thesis what is referred to as a plane is a stack of scintillators with a common orientation, while a station is the combination of planes in-between iron blocks. In the DS system a station is not equivalent to a plane. Figure 3.9 allows one to see the combination of a horizontally aligned plane and vertically aligned plane to form a station. The fine granularity is required in order to reconstruct the track formed by a muon exiting the experiment. The algorithms used to identify a muon are discussed in the latter stages of subsection 3.3.

The vertically aligned planes of the DS do not cover the same $x$-region as the US or the horizontally aligned bars of the DS. Muons must produce data in both vertically and horizontally aligned planes for muon identification. Therefore, the region in which muons can be identified is limited by the $x$-coverage of the vertically aligned planes in the DS. One can see this under-lap in coverage in figure 3.9, the highlighted side of the experiment (right-side when looking down the beam axis). On the left-side of the experiment, against the cavern wall, the vertically aligned planes of the DS line up with the US and horizontally aligned planes of the DS. The right-edge of the vertically aligned scintillators, where a muon track can be constructed is defined as $x = L_\mu$, this presents a soft limit on the distribution of muon track positions in the $x$-axis because at least three of four vertical planes must record hits in order to reconstruct a muon track.

### 3.2.5 Veto system

Now turning to the most upstream subsystem of the experiment is the Veto system. To veto the charged particles that survive the ATLAS focusing magnets and the 480 m distance from the IP, SND@LHC began with two planes of seven Eljen EJ-200 plastic scintillators, wrapped in aluminised Mylar foil for light-tightness and readout by eight Hamamatsu S14160-6050HS SiPMs. As with the US and DS, the 56 SiPMs per side per plane are connected to a PCB. The small area SiPMs of the US are not required here because the veto is upstream of the target, so the high energy depositions expected following a neutrino or muon interaction are not expected, also the Veto system performs no calorimetry so saturation is not an issue. These planes are aligned horizontally with SiPM readout on both sides. A key design point of the Veto horizontal planes is a staggering of the planes in the y-dimension. This is done to offset the gaps between bars and ensure that the highly boosted charged

Figure 3.9: A photograph of SND@LHC, where one can see the final two vertically aligned planes and the final horizontally aligned plane of the DS system. The vertically aligned planes have a smaller length in $x$ than the horizontally aligned planes, the last of which is labelled 'DS3H'.

particles could not pass through undetected and mimic a neutrino interaction. This offset can be seen in the left sub-figure of figure 3.10.

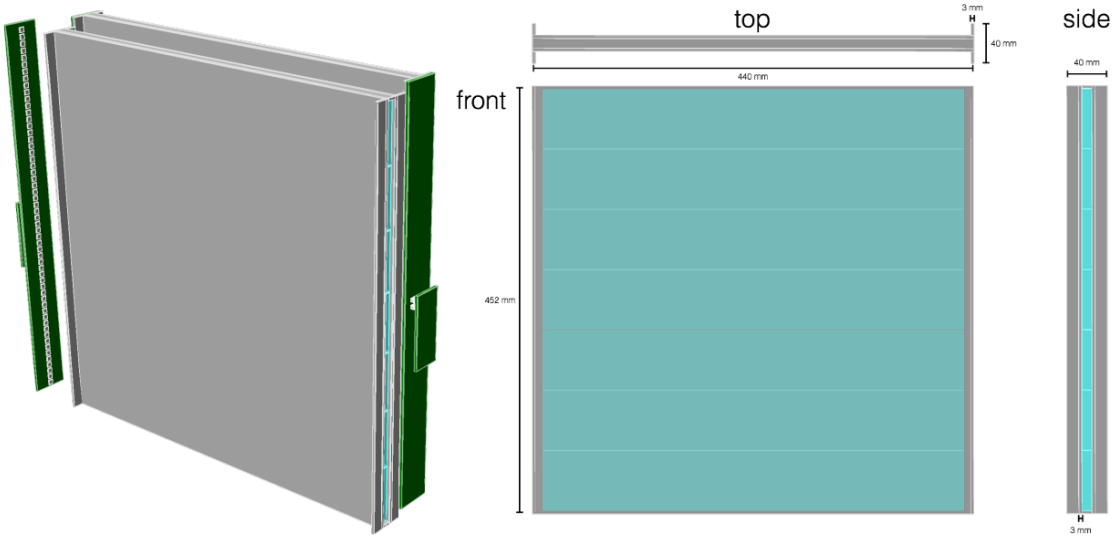



Figure 3.10: Schematic diagram of the veto bi-plane from all sides.

The efficiency of the Veto system is a key feature for the neutrino physics goals of the experiment, if too many charged particles traverse the Veto system undetected, the background to neutrino searches will greatly diminish the discriminatory power of the experiment. During the period 2022-2023, the measured Veto system inefficiency was $7.8 \times 10^{-8}$ [91]. However, as is discussed in section 3.5.2, this does

not provide sufficient rejection power for the experiment's charged current muon neutrino search. The high inefficiency of the Veto was diagnosed as originating at the ground level of the experiment. This was found by isolating events in data where a track can be fitted in the SciFi tracker while there was low activity in the Veto system (less than 13 SiPMs [91]). These fitted tracks are then extrapolated to the $z-$positions of the Veto planes and the transverse positions are measured. Plots of these positions given those criteria are shown in figure 3.11 where it is clear that the Veto inefficiency is dominated by tracks in the low-$y$ region.

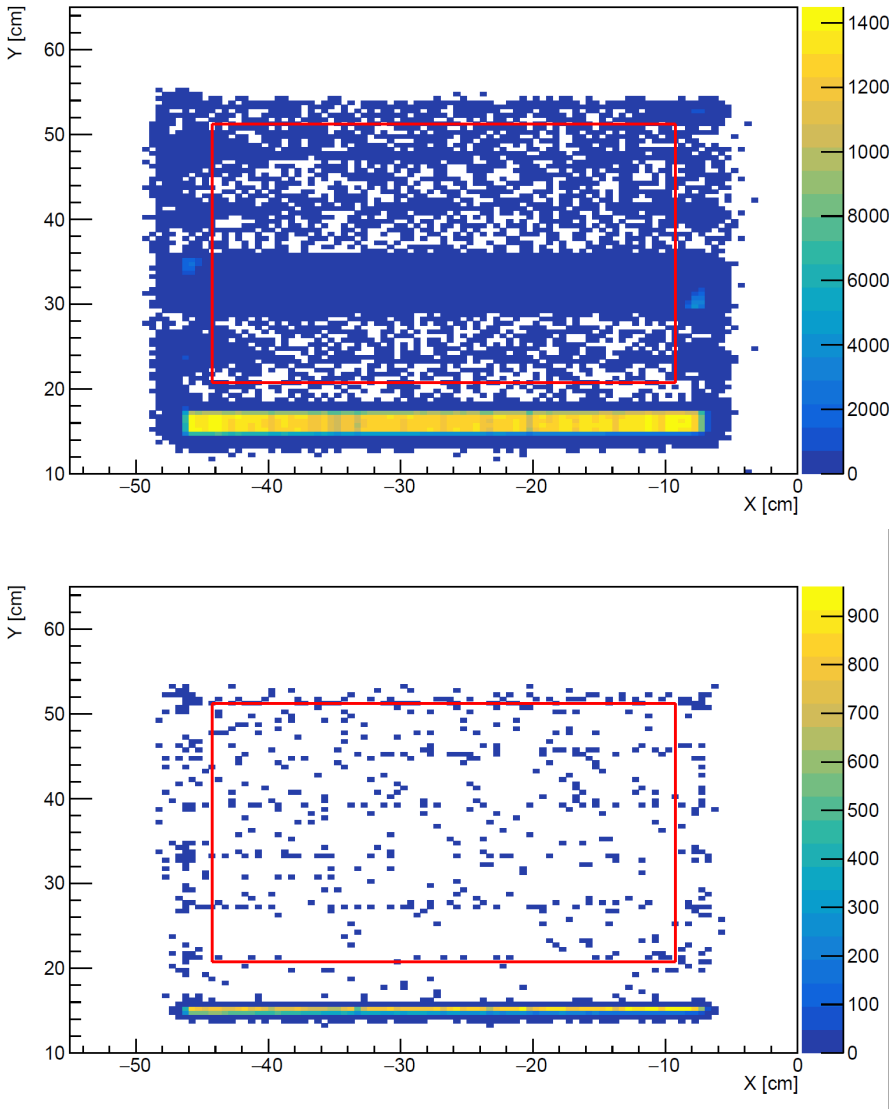



Figure 3.11: Positions of SciFi tracks extrapolated to the planes of the Veto system for events with less than 13 SiPMs firing in the Veto system. The first plane is shown on the left and the second plane on the right. Taken from [91].

The solution to this problem was the installation of a third plane, with the scintillators aligned vertically. The third plane was added at the start of 2024 to reduce the inefficiency of the system. The impact of the third plane on the Veto efficiency is well documented in reference [91]. The third plane and its impact on the experiment is of low impact to this thesis as the data used for chapter 4 was collected prior to the 3rd plane installation. However, the Veto efficiency does return

as a topic of discussion in later chapters.

Before continuing, table 3.2 summarises some of the key features for each of these three similar subsystems:

| Subsystem | Planes | Bars / plane | SiPMs / bar end |
|---|---|---|---|
| Veto | 2 | 7 | 8 |
| Upstream | 5 | 10 | 6 large area + 2 small area |
| Downstream horizontal | 3 | 60 | 1 |
| Downstream vertical | 4 | 60 | 1 (SiPM only on top) |

Table 3.2: Scintillator-SiPM configurations of veto and US subsystems. Note that a third Veto plane is added at the end of 2023 [91].

### 3.2.6 Plastic scintillator: properties and physics

The Veto, US and DS systems all use the Eljen EJ-200 polyvinyltoluene (PVT) based organic scintillator [80]. Organic scintillators such as EJ-200 are typically comprised of a primary fluorescing molecule suspended in a plastic base. The aromatic rings present in PVT provide the primary fluor with the polymer base producing negligible light yield [92]. The charged particles traversing the scintillators deposit energy by exciting the delocalised $\pi$-electron system in the aromatic ring of toluene. The excited electrons promptly release the deposited energy by the emission of ultra-violet (UV) light through fluorescence[1][81]. Following the primary UV photon emission in the aromatic ring, the photons are absorbed and re-emitted by wavelength shifting compounds in the scintillator. The nature of the wavelength shifters is proprietary and may be done by multiple compounds until the light output reaches the desired spectral peak, which for Eljen EJ-200 is quoted as 425 nm and is shown in appendix A.2.

[1]The rapid ($\mathcal{O}(ns)$ pulse width FWHM) release of the excitation energy through electromagnetic radiation is a distinguishing characteristic between fluorescence and phosphorescence [81, 93].

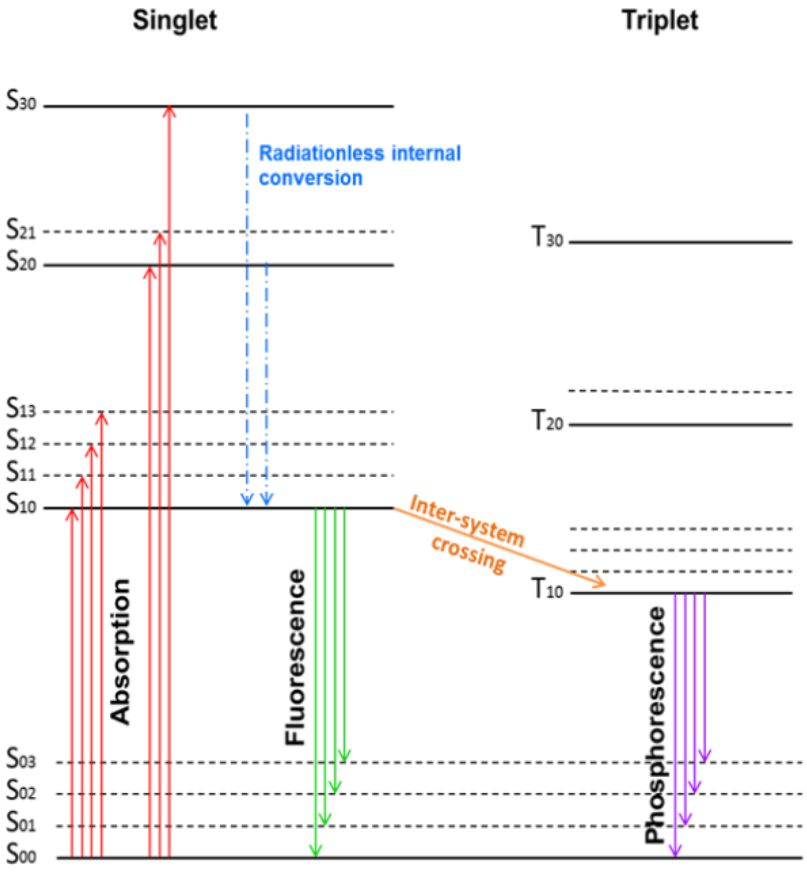



Figure 3.12: Energy level diagram of an organic molecule with $\pi$-electron structure. From [93].

In figure 3.12, the set of singlet states with spin 0 are labelled as $S_{ab}$ where $a, b$ are excitation and vibrational energy levels respectively, while the triplet states of spin 1 are similarly labelled with notation $T_{ab}$. The separation between any of these states is not to scale in figure 3.12. Typically, the spacing between adjacent excitation levels is $\sim$ 0.15 eV which is considerably larger than the typical thermal energy of an electron at room temperature from equipartition theorem ($E = 3/2 k_B T \approx 40\,\text{meV}$), most $\pi$-electrons in the aromatic ring are found in the ground excitation state. Excitations to second or third order excitation states, $S_{2b}$, $S_{3b}$, rapidly de-excite through radiationless internal transfer. Electrons on non-zero vibrational energy levels will then also rapidly de-excite to reach thermal equilibrium with their neighbours. Intersystem crossing, a forbidden spin transition as $\Delta S = T_1 \rightarrow S_0 + h\nu = 1$, can also occur but as the emission is far slower, acting on microsecond to second timescales, the impact is negligible on the timing data measured in the experiment. Thus, the majority effect of exciting electrons in the aromatic ring of PVT molecules is fluorescence through $S_{10} \rightarrow S_{0b}$ where $b$ is the vibrational state of the electron after having emitted a photon. Reference [81] lists desirable properties that should be addressed in the choice of scintillator material for a project (the following is lifted verbatim from reference [81]). The criteria cannot all be optimised, but trade-offs considering project-specific knowledge are required:

1. It should convert the kinetic energy of charged particles into detectable light with a high scintillation efficiency

2. This conversion should be linear - the light yield should be proportional to

deposited energy over as wide a range as possible.

3. The medium should be transparent to the wavelength of its own emission for good light collection.

4. The decay time of the induced luminescence should be short so that fast signal pulses can be generated.

5. The material should be of good optical quality and subject to manufacture in sizes large enough to be of interest as a practical detector.

6. Its index of refraction should be near that of glass (≈1.5) to permit efficient coupling of the scintillation light to a photomultiplier tube or other light sensor.

The Eljen EJ-200 used in the US has a light output equal to 64% of anthracene, a reference scintillator. While this may appear low, the benefits over anthracene itself are the low price of EJ-200, the relative strength of the material and the isotropic light-output [80]. The linearity of the light-output, the relationship between the number of photons produced and the deposited energy, for the scintillator was shown to hold true for low energy protons (50 keV to 5 MeV) in reference [94, 81], which is well below the energy range expected in SND@LHC. One expects the linearity of the light yield to break down for high energy depositions from heavy charged particles like alphas and protons due to effects such as Birks' law [81], while for light particles such as electrons, gammas and muons the linearity remains far better intact up to GeV scale. Eljen quote a scintillation efficiency of $10^4$ photons emitted following the deposition of 1 MeV by an electron [80], this specific property will be further discussed in section 5.2.2. The third criterion of good transparency to photons in the emission range of the fluor molecule which is true for the EJ200 and is shown in appendix A.2. The decay time of the EJ-200 is quoted as 2.1 ns by Eljen Technology [80] which is sufficient to be part of a sub-nanosecond timing system, but the decay time of the scintillator is only a factor of this design criteria of the US. The fifth criteria is two-fold: the optical quality of the scintillator has not been directly measured by the experiment and will be subject to variation in the manufacturing process; a benefit of many plastic scintillators, including the EJ-200 is the ease and flexibility of the manufacturing process meaning that the 82.5 cm length of the US scintillators is produced without difficulty and with an attenuation length substantially larger (3.5 m). Finally, the refractive index is quoted as 1.58 meaning that it very effectively satisfies the sixth criteria of being compatible with the refractive index of the light sensor, the Hamamatsu S14160-6050HS, of 1.57 [88].

### 3.2.7 Silicon photomultipliers

Silicon photomultiplier is a term used to describe a solid state photodetector comprised of an array of single-photon avalanche diodes (SPADs), which can be referred to as pixels or microcells [95, 96] and shall be referred to as pixels in this thesis. Each pixel is a a unit based on a pn-junction, which is the connection of a negatively doped semiconductor (n-doped) with a positively doped semiconductor (p-doped). The connection between these substances is called the metallurgic junction. At the construction of a pn-junction, an electric potential formed by connecting an n-doped semiconductor material (right of figure 3.13) with a p-doped semiconductor material

(left of figure 3.13)[2]. Free electrons from the n-doped region are accelerated into the p-doped region and occupy valence-band vacancies associated to acceptor atoms near the junction. This diffusion leaves behind immobile acceptor ions, creating a region depleted of free carriers — hence the name depletion zone. In this depletion zone, on the p-doped side there are negatively ionised donor ions; on the n-doped side there are positively charged acceptor ions. This resulting separation of charges constructs an inbuilt electric field in the medium that is strongest close to the metallurgic junction and dissipates in magnitude outwards towards the neutrally charged regions. The diffusion of free charge carriers continues until thermal equilibrium is reached [97].

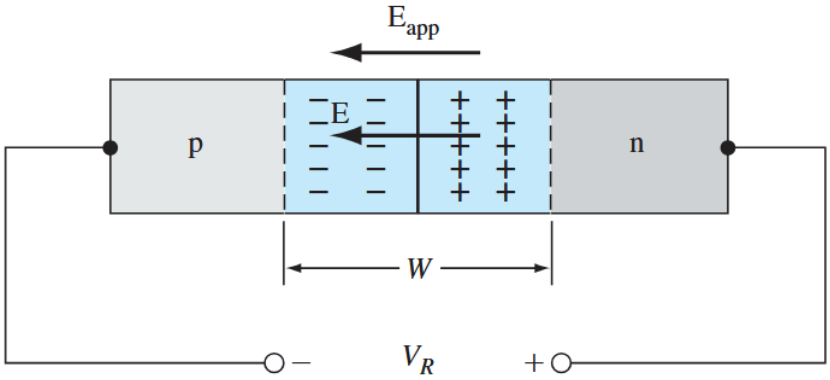



Figure 3.13: Schematic of a pn-junction powered in reverse bias with voltage, $V_R$. $E$ denotes the electric field within the depletion zone and $E_{\text{app}}$ denotes the electric field from the external potential. The direction of these fields indicate the direction that a test positive charge would accelerate. $W$ indicates the width of the depletion zone. Taken from [97].

A pn-junction with a reverse bias voltage is shown in figure 3.13. The width of the depletion zone can be shown to be square-root of the applied reverse bias voltage, $W \propto \sqrt{V_R}$, also the strength of the electric field increases with the size of the applied voltage [97]. The application of a reverse bias will accelerate the free charge carriers. As the reverse bias voltage and electric field strength increases, free charge carriers in the depletion zone undergo greater accelerations until they may perform impact ionisation which releases a new free charge carrier pair (an electron-hole pair) which increases the current across the diode. A chain of impact ionisations is referred to as an avalanche. Figure 3.14 shows there are three regimes of applied reverse bias to the pixel. At low reverse biases, there is no impact ionisation. In avalanche photodiode mode (APD), the voltage threshold allowing for impact ionisation for electrons is surpassed but the kinematics are still such that each electron is expected to free less than one further electron, therefore the avalanche dies out. At higher reverse biases, the current is observed to exponentially increase - known as Geiger mode. This corresponds to a chain reaction of impact ionisation that is self-sustaining. Both electrons and holes can now also perform impact ionisation and each freed charge carrier is expected to release more than one charge carrier. The voltage above which the pixel operates in Geiger mode is known as the breakdown voltage, $V_{\text{bd}}$ [96, 95, 98].

[2]To note, the p- and n-doped regions are initially electrically neutral.

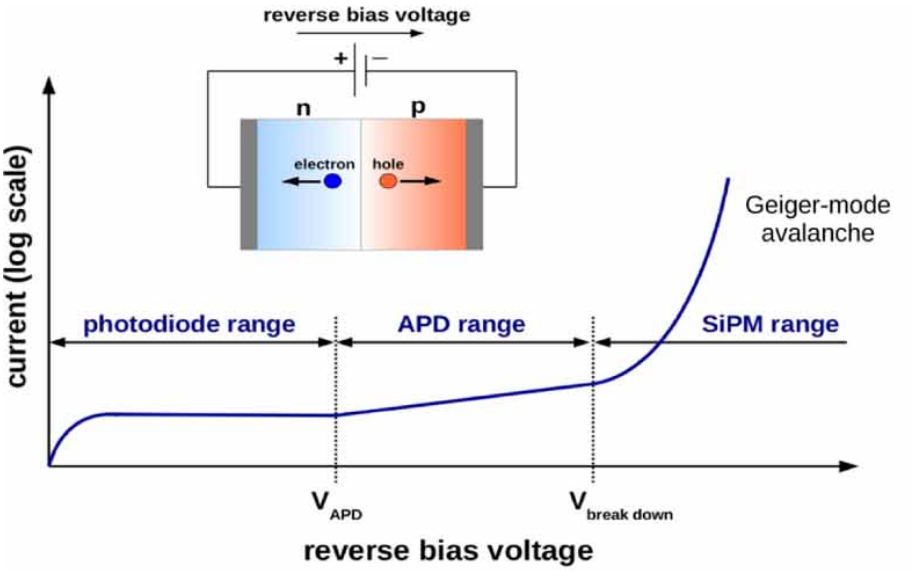



Figure 3.14: Depiction of how the current across the photodiode increases with the applied reverse bias. Taken from [96].

In a SiPM these pixels are connected in parallel, each have a reverse bias voltage, $V_{\text{bias}}$, which is above the breakdown voltage, $V_{\text{bd}}$, putting the diodes into Geiger mode. This excess voltage above breakdown is called the over-voltage, $\Delta V = V_{\text{bias}} - V_{\text{bd}}$. When a photon releases an electron-hole pair in the Geiger mode pixel, the electron and hole accelerate in opposite directions, each with a probability of each to create an avalanche. Their respective probabilities are not equal. The probability of an electron to trigger an avalanche is known to be far higher [99].

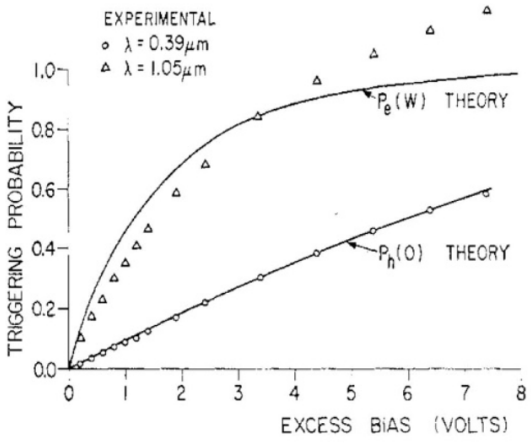



Figure 3.15: Avalanche triggering probability of electrons and holes in a pixel. Taken from [96].

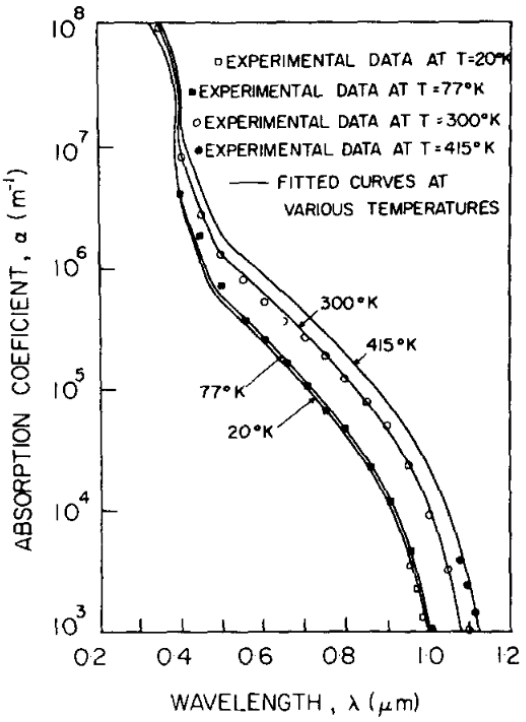



Figure 3.16: The absorption coefficient of photons in silicon as a function of photon wavelength for different temperatures between 40 K and 415 K. Taken from [100].

Furthermore, it is known that the absorption coefficient of light in silicon is far greater at lower wavelengths than longer wavelengths [100]. The different avalanche probability and absorption coefficient data inform the design structure of these pixels and lead to what is referred to as the p-on-n structure. The Hamamatsu SiPMs used in SND@LHC are made with a p-on-n structure, as is evidenced by the photon detection efficiency curve in figure 3.18. The reason for this is that in a p-on-n structure pixel shorter wavelength photons will be absorbed closer to the surface due to the higher absorption coefficient. The generated electron will then accelerate towards the n-doped region, due to the reverse bias, and potentially cause an avalanche. In the case of longer wavelength photons, the electron-hole pair will likely be created deeper in the pixel, where the electric field is weaker and thus the energy gained by the electron is lesser which reduces the probability of generating an avalanche.

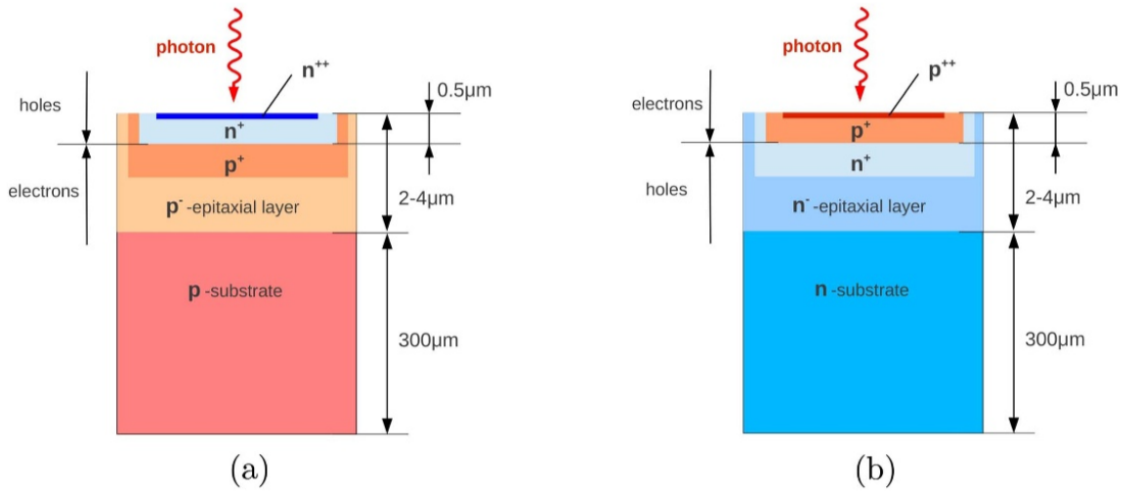



Figure 3.17: a.) n-on-p pixel structure, more sensitive in the red region of light. b.) p-on-n pixel structure, more sensitive in the blue, near-UV region of light. Figure taken from [96].

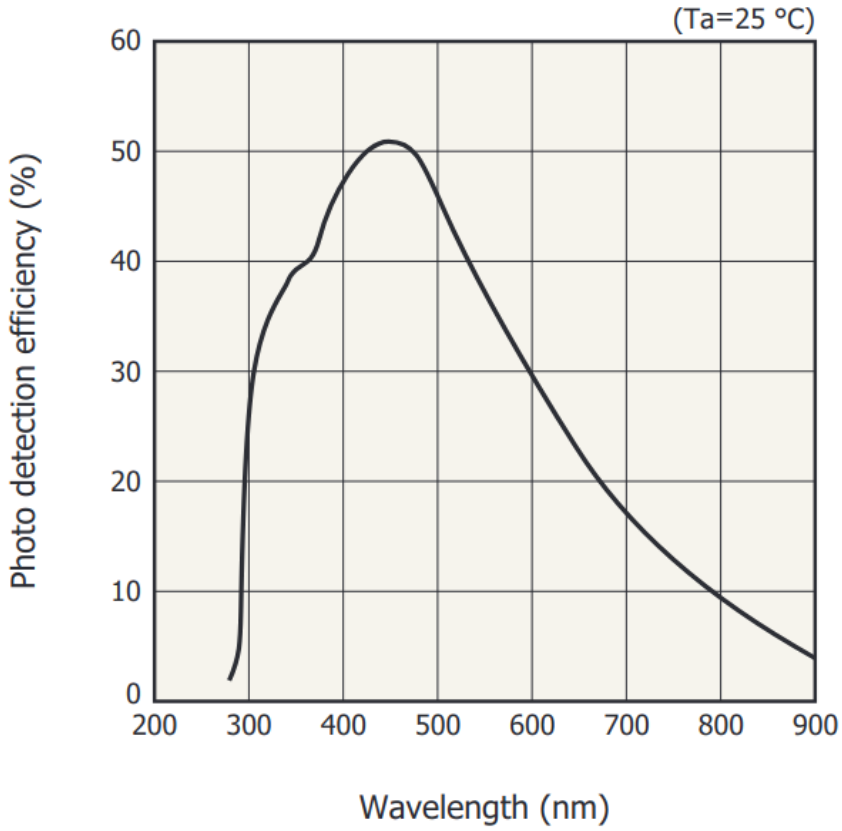



Figure 3.18: Typical photon detection efficiency curve for the Hamamatsu S14160/S14161 series SiPMs. The nature of the data varies between SiPM type and even among SiPMs of the same type. This figure demonstrates the typical data for these series of SiPM, the peak PDE is quoted as 50% at $\Delta V = 2.7\,\mathrm{V}$ for all SiPMs in this series [88].

In Geiger mode, the avalanche will grow exponentially and therefore require external termination. This is done by the quenching resistor, connected in series, to cease the SiPM from being damaged, to reset and register future data. The large current pulse created by the avalanche causes an increased voltage across the quenching resistor, which in turn causes a voltage drop across the SiPM until the voltage is no longer above breakdown, stopping the avalanche. At this point the SiPM must reset back to its ready-state over the recovery time. An electrical model (equivalent circuit) of this system is shown in figure 3.19. The model consists of the pixel and its depletion layer capacitance in parallel, $C_d$; the quenching circuit is comprised of the quenching resistor and a parasitic capacitance, $C_q$[3], in parallel to the quenching resistor. The recovery time of the pixel, $\tau_r$, can then be written as $\tau_r = R_q(C_q + C_d)$ [96].

[3]Parasitic capacitance refers to the unwanted capacitance between two different conductors that arises due to their proximity to one another and different voltages.

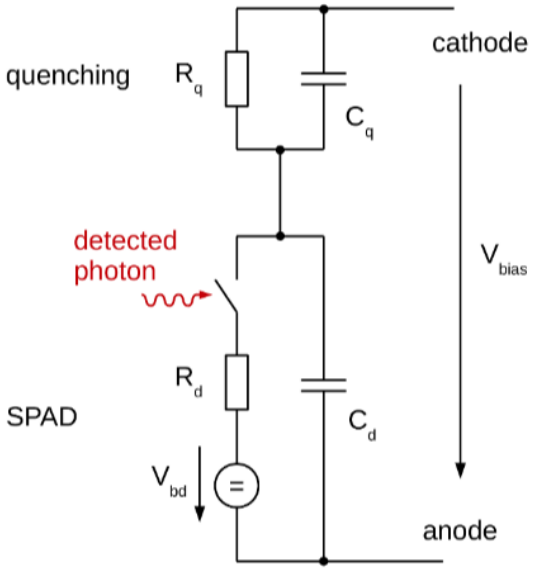



Figure 3.19: Equivalent circuit for a SiPM pixel and quenching resistor. The resistance of the pixel during an avalanche is denoted as $R_d$. Taken from [96]

The recovery times of pixels are typically on the order of several hundreds of nanoseconds and thus far after the physics event triggering a pixel is over. When a pixel detects a photon, that pixel is rendered insensitive until the SiPM has quenched and recovered back to the bias voltage. This features makes a SiPM an inherently non-linear detector. The number of fired pixels, $N_{\text{fired}}$, can be modelled by assuming a uniform light distribution over the pixel array. The expected number of fired pixels expressed as a function of the number of incident photons, $N_\gamma$, can be derived. Denoting the total number of pixels in the SiPM as $N_{\text{pixels}}$ and starting from the probability of a given pixel not being hit after a total of $N_\gamma$ photons:

$$P(\text{pixel does not fire}) = \left(1 - \frac{1}{N_{\text{pixels}}}\right)^{N_\gamma}$$
$$P(\text{pixel does fire}) = 1 - \left(1 - \frac{1}{N_{\text{pixels}}}\right)^{N_\gamma}$$

Thus the expected number of unique fired pixels, $\langle N_{\text{hit}} \rangle$, is simply obtained:

$$\langle N_{\text{hit}} \rangle = N_{\text{pixels}} \left[1 - \left(1 - \frac{1}{N_{\text{pixels}}}\right)^{N_\gamma}\right] \tag{3.2.1}$$

In the limit of a large number of pixels on the SiPM, the following limit can be used to express equation 3.2.1 as a decaying exponential:

$$\lim_{N\to\infty} \left(1 - \frac{1}{N}\right)^{Nx} = e^{-x}$$

Let $x = N_\gamma / N_{\text{pixels}}$:

$$\langle N_{\text{hit}} \rangle = N_{\text{pixels}} \left[1 - e^{-\frac{N_\gamma}{N_{\text{pixels}}}}\right] \tag{3.2.2}$$

The width of a pixel is referred to as the pixel pitch and the percentage of this pixel area that is sensitive to light is called the geometric fill factor. The number

of pixels, pixel pitch and geometric fill factor of the SiPMs used for the US and DS muon system are shown in table 3.3:

| | SiPM type | |
|---|---|---|
| | S14160-6050HS (large area) | S14160-3010PS (small area) |
| Number of pixels | 14331 | 89984 |
| Pixel pitch | 50 | 10 |
| Geometric fill factor | 74 | 31 |

Table 3.3: The number of pixels in each type of SiPM, width of the pixels (pitch) and the geometric fill factor. Values taken from the respectively data sheets for the SiPM models, found at reference [88] for the large area SiPMs and reference [101] for the small area SiPMs.

The SiPM gain, $G_{\text{SiPM}}$, is defined as the number of secondary charge carriers produced by each primary charge carrier. This value provides an upper limit for how a realistic pixel will perform, the effective gain is the number of secondary charge carriers which reach the cathode or anode of the pixel within the duration of the avalanche. The secondary charge carriers which fail to do so may subsequently reach the bias terminals in what is referred to as afterpulsing. This afterpulsing probability is dependent on several factors including temperature, over-voltage and the pixel recovery time. In the absence of signal, diodes in Geiger mode are experience dark counts. Dark counts occur when thermally produced charge carriers in the diode cause avalanches. The dark count rate (DCR) is highly dependent on the sensor temperature as well as the quality of the manufacturing process and materials used [95]. The underlying physics which causes dark counts are discussed in reference [98]. A strong quality of the SiPM, and a quality that has improved considerably in recent years, is the photon detection efficiency (PDE) which can be written as the product the quantum efficiency (QE) the geometric fill factor, ($G_{\text{ff}}$) and the probability that a photoelectron the instantiate an avalanche, $P_{\text{bd}}$ [98, 95]:

$$\text{PDE}(\lambda,\, U,\, T) \;=\; \text{QE}(\lambda) \;\cdot\; G_{\text{ff}} \cdot\; P_{\text{bd}}(\lambda,\, U,\, T) \tag{3.2.3}$$

The geometric fill factor simply corresponds to the fraction of the SiPM area that is photosensitive. The arguments given earlier in this section showing why a p-on-n pixel structure is used, will clearly enhance $P_{\text{bd}}$, and the geometric fill factors of SiPMs have improved from 20-30% [95]. It is improvements like these that give modern-era SiPMs such strong PDEs and thus enhanced usage in particle physics experiments.

SiPMs also provide very good time resolutions, with components to the resolution coming from electrical noise, dark current, photon statistics and other components that depend on the implementation, such as components from the clock of the DAQ system. These components can be added in quadrature [102]:

$$\sigma_t \;=\; \sigma_t^{\text{ele}} \;\oplus\; \sigma_t^{\text{DCR}} \;\oplus\; \sigma_t^{\text{photo}} \;\oplus\; \sigma_t^{\text{clock}} \tag{3.2.4}$$

Where $\sigma^{\text{ele}}$ is the component from electronic noise which depends on the steepness, $dI/dt$, of the signal pulses at the triggering threshold, this can be expressed as [102, 96, 98]:

$$\sigma_t^{\text{ele}} = \frac{\sigma_{\text{noise}}}{dI/dt} \oplus \sigma_t^{\text{TDC}}$$

Where $\sigma_{\text{noise}}$ is the baseline noise when no signal is present, $\sigma_{\text{TDC}}$ is the jitter of the time-to-digital converter in the readout. The component from photon statistics at zero DCR is due to fluctuations in the arrival time of the photons and can be expressed as [96, 102]:

$$\sigma_t^{\text{photo}} \propto 1/\sqrt{N_{\text{pe}}} \tag{3.2.5}$$

The term $\sigma_t^{\text{DCR}}$ accounts for the time resolution component coming from the dark count rate. The DCR degrades the time resolution by polluting the signal with charge carriers thermally generated in other pixels of the SiPM. This component of the resolution can be expressed as [102]:

$$\sigma_t^{\text{DCR}} \propto \frac{\sqrt{\text{DCR}}}{N_{\text{pe}}}$$

With the nature of this proportionality arising from dark counts being well approximated by Poissonian statistics. Finally, the contribution to the time resolution from the clock is denoted $\sigma_t^{\text{clock}}$.

Another property of SiPMs to be covered is optical cross-talk. Optical cross-talk (OCT) is a form of correlated noise, means that OCT usually follows a previous event. The three types that are discussed here are prompt, delayed and external. Firstly, the underlying point behind OCT is that during an avalanche in a SiPM, photons can be emitted as the freed charge carriers interact with lattice atoms en-route to the cathode/anode, these photons are referred to as cross-talk photons [103]. If these photons travel into a neighbouring pixel on a SiPM, they have a chance to instance an avalanche. For prompt and delayed OCT, a photon produced in the avalanche will travel into the silicon of a neighbouring pixel and free a charge carrier pair. The difference between prompt and delayed OCT refers to what section of the neighbouring pixel the cross-talk photon frees a charge carrier pair. If this occurs in a region of the neighbouring pixel where the electric field is large, then the charge carriers are rapidly accelerated and may initiate an avalanche, this is prompt OCT and is depicted in the left image of figure 3.20. If, however, the charge carriers are freed in a region of low electric field strength, then the charge carriers may diffuse through the pixel and reach the high electric field strength region where they accelerate and instance an avalanche. The diffusion process may provide a substantial time delay resulting in that pixel not polluting the initial event which produced the cross-talk photon, one sees delayed optical OCT in the middle pane of figure 3.20. Finally, external OCT refers to the case where the cross-talk photon exits the silicon but is then reflected upon an optical boundary condition, such as the SiPM protective window, and re-enters a different pixel of the SiPM where it instances an avalanche, shown in the right pane of figure 3.20.

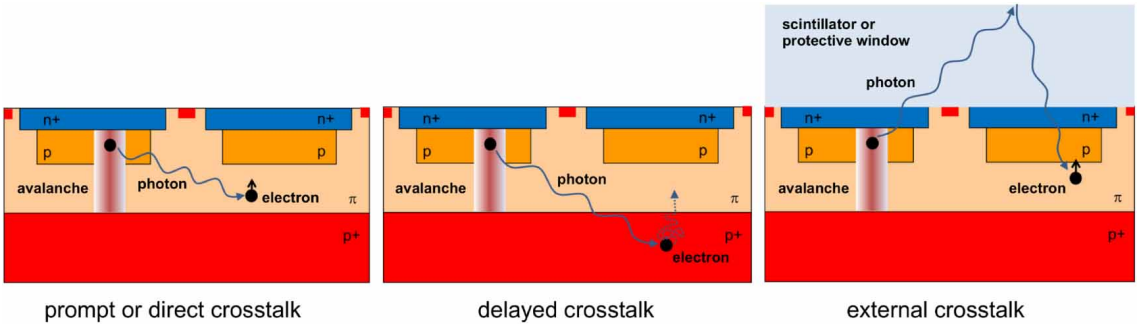



Figure 3.20: Depiction of left: prompt, middle: delayed and right: external optical cross-talk. Taken from [98].

Both prompt and delayed OCT can be reduced by the additional of optical trenches: these are cuts into the silicon between pixels that are filled with reflective layers [98]. Optical trenches typically provide shielding between adjacent depletion zones, minimising prompt OCT, so deeper optical trenches can be used to also reduce the delayed OCT rate. Further methods for reducing the OCT rates can be found in reference [98].

Many SiPM properties are dependent on the over-voltage. These include the gain, DCR, PDE and OCT probability. The stated breakdown voltages and selected over-voltages for the small area and large area SiPMs of the US and DS muon systems are shown in table 3.4.

| | SiPM type | |
|---|---|---|
| | S14160-6050HS (large area) | S14160-3010PS (small area) |
| Breakdown voltage, $V_{\text{bd}}$ $(V)$ | 38 | 38 |
| Bias voltage, $V_{\text{bias}}$ $(V)$ | 42.5 | 48 |
| Over-voltage, $\Delta V$ $(V)$ | 4.5 | 10 |

Table 3.4: The breakdown voltages are taken from the respectively data sheets for the SiPM models, found at reference [88] for the large area SiPMs and reference [101] for the small area SiPMs.

Some key performance metrics of the SiPM at these over-voltage can be estimated from the data sheets which show the typical trends of the quantities with respect to over-voltage. Figure 3.21 shows gain, OCT probability and PDE as a function of over-voltage for a typical SiPM in the Hamamatsu S14160/S14161 series:

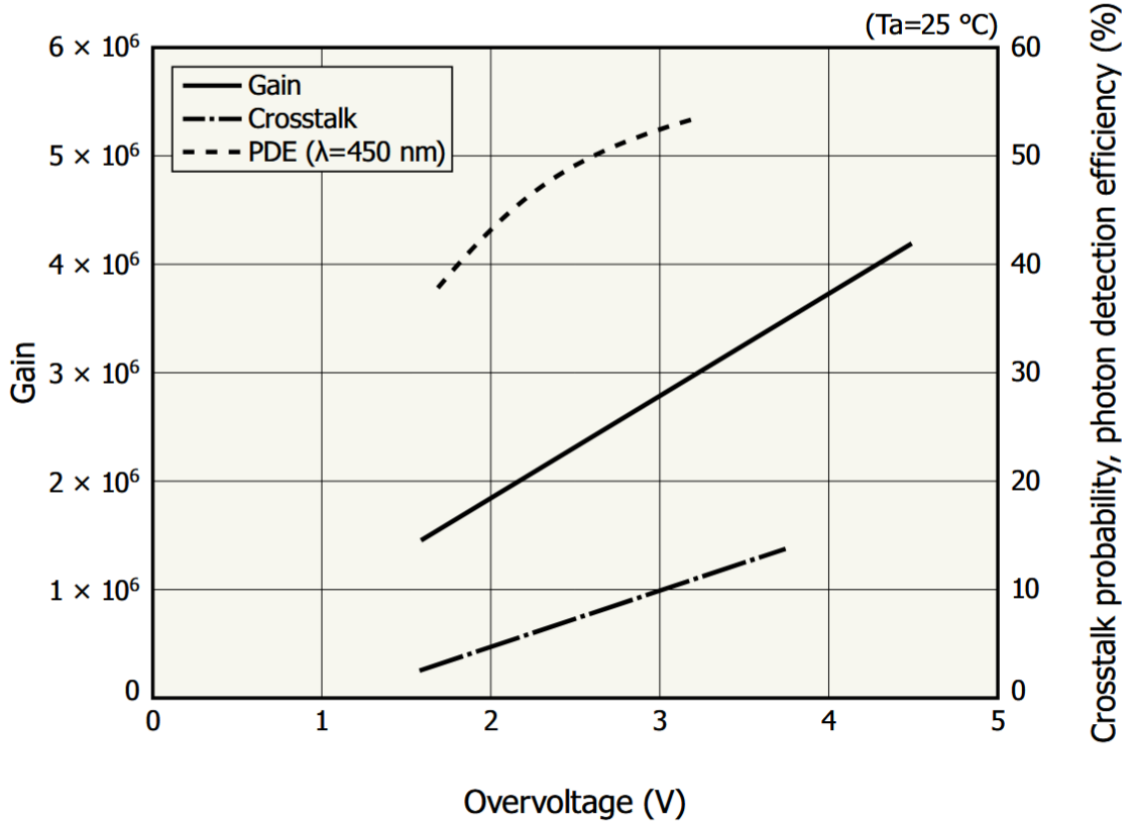



Figure 3.21: Data for a typical SiPM in the Hamamatsu S14160/S14161 series for how gain, OCT probability and PDE($\lambda = 450\,\text{nm}$) varies with the applied over-voltage. Figure taken from [88].

A similar figure for the small SiPMs, of lesser interest to this thesis, can be found in appendix A.4. The values of these parameters at the operated voltage in the experiment can only be estimated. The trends for OCT probability and gain may be assumed to remain on a linear trajectory, while the PDE is clearly plateauing as one would expect from plots of this trend from other SiPMs in literature such as reference [104]. Estimates from this figure are summarised in table 3.5:

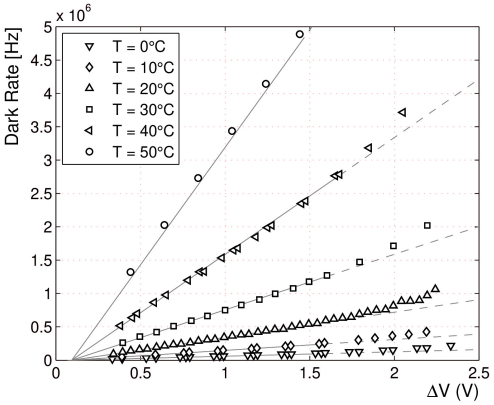
| | Parameter estimate at $\Delta V = 4.5\,V$ |
|---|---|
| Gain [dimensionless] | $4 \times 10^6$ |
| PDE($\lambda = 450\,\text{nm}$) [%] | 50 |
| Cross-talk probability [%] | 20 |

Table 3.5: Estimated Hamamatsu S14160-6050HS SiPM performance parameters at the SND@LHC chosen over-voltage of $4.5\,V$.

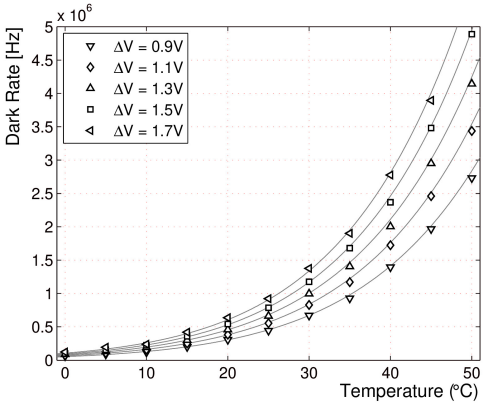
A further trade-off with over-voltage not shown in figure A.4 or table 3.5 is the DCR, which is also strongly dependent on temperature. The dependence on over-voltage and temperature is shown for a Hamamatsu S10362-11-050C in figure 3.22.

The equation used to fit the DCR in sub-figure 3.22b has a power law term, $T^{3/2}$, and an exponential decay term[4]. So reducing the operating temperature of the SiPMs is always of great importance. SND@LHC has a temperature and humidity

[4]The function is found to empirically describe the data from reference [105] so the precise functional form is not of interest.




(a) Solid lines show the result of a linear fit in the range $\Delta V \leq 1.6\,\mathrm{V}$ and DCR $\leq$ 5 MHz. The dashed lines are extrapolations from the fit.




(b) Each data set with a fixed temperature is fitted using all data points.

Figure 3.22: Measurements of the DCR for various over-voltages with linear fits for a range of temperatures. This data is collected for a Hamamatsu S10362-11-050C. The Taken from [105].

monitoring system that will be discussed in section 3.3. While figure 3.22a shows a proportionality between DCR and $\Delta V$, this is not universally true. Reference [102] studies how the DCR varies with over-voltage for SiPMs which have been exposed to different levels of radiation, an example of this is shown in figure 3.23.

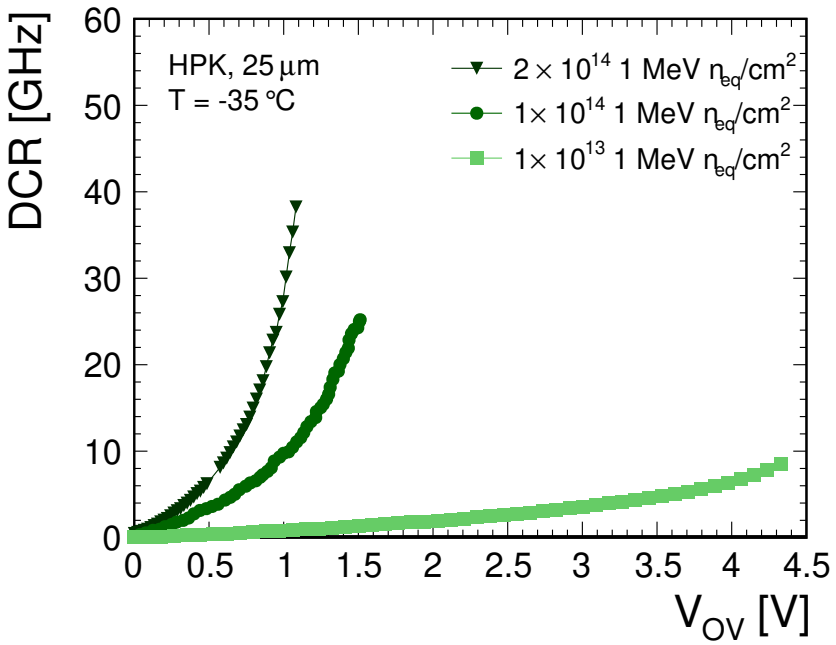



Figure 3.23: DCR measurements at a wide range of over-voltages (labelled $V_{OV}$ in the figure but referred to as $\Delta V$ in this thesis) for SiPMs irradiated to different levels. The SiPMs used in this study are from Hamamatsu (labelled HPK in the upper-left of the plot) with a pitch of 25 µm. Taken from [102].

Part of the work performed in reference [102] was to compare the performance of SiPMs irradiated to different levels which are expected to be reached during the

operation of CMS at the HL-LHC. The irradiation level of $2 \times 10^{14}$ 1 MeV $n_{eq}/cm^2$ corresponds to what is expected in MIP timing detector barrel sensor of CMS after an integrated luminosity of 3000 $fb^{-1}$. The irradiation was performed at the JSI neutron reactor[5]. It is important to note that the irradiance expected at SND@LHC is orders of magnitude lower than what is expected at CMS during the HL-LHC. Reference [106] presents a thorough review of radiation damage in SiPMs and the methods of studying the impact of radiation damage. The radiation effects on other performance metrics like gain and PDE are far less effected than DCR [106].

### 3.2.8 The time-walk effect

A key part of the first chapter of this thesis is the treatment of the observed time-walk effect in the SiPMs of the US. Here, the time-walk effect will be defined and its causes outlined. The time-walk effect is a trait inherent to the architecture of a SiPM. Each firing pixel produces a current and the slope of the current rising edge is related to the number of concurrently firing pixels on the SiPM [107, 108, 109]. In particle physics detectors, where photon fluxes through the SiPM can be small due to minimum ionising particles, or large due to either electromagnetic or hadronic showers, these differences in signal amplitudes are sufficient for the time-walk effect to cause substantial shifts in the recorded signal time. Data acquisition systems that employ a fixed threshold such as SND@LHC are particularly vulnerable to this as a typical method of minimising time-walk is to use a constant fraction discriminator.

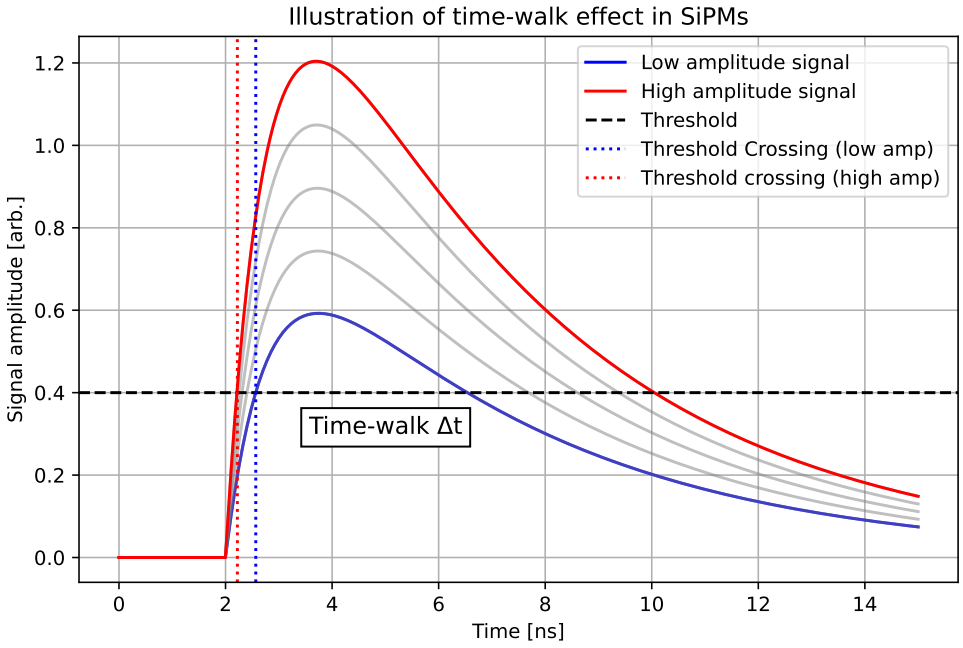



Figure 3.24: Illustration of how two signals of different signal shapes will be affected by the time-walk effect when a fixed threshold is applied.

Figure 3.24 shows the impact of the time-walk effect through two simulated SiPM signals. The $\Delta t$ induced by the time-walk effect broadens the time resolution of the system. Chapter 4 section 4.1.4 highlights the extent of the time-walk effect

[5]Radiation exposure of detector elements is often quoted in terms of the equivalent number of 1 MeV neutrons. See chapter 2 of reference [81] for a discussion of this.

observed in the US of SND@LHC and how this effect is then parametrised and corrected for.

## 3.3 Software and electronics

### 3.3.1 Front-end and data acquisition boards

All electronic subsystems of SND@LHC; Veto, SciFi, US and muon system, are all read out using the same data acquisition (DAQ) system of front-end (FE) board and DAQ board. The SiPM PCBs of Veto, SciFi and US are directly connected to the FE boards of SND@LHC, shown in figure 3.25, utilise the TOFPET2 application specific integrated circuit (ASIC) by PETsys [110]. The FE board digitises the SiPM data and sends the data to the DAQ board. The FE board contains two TOFPET2 ASICs at 64 channels per ASIC and the ASIC runs on a 160 MHz clock. The purpose of the ASIC is to combine signal amplification, discriminators, charge integrators, analog-to-digital converters (ADCs), time-to-digital converters (TDC) as well as temperature monitoring. The ADCs in the TOFPET2 ASIC are actually charge-to-digital converters (QDCs) which are tested to respond linearly to an input charge up to 1500 pC. The gain of the QDC branch can be set between 1.00, 2.50 and 3.65 to optimise the response depending on the scale of the SiPM signals received by the ASIC [110]. In SND@LHC this QDC branch gain is set to 2.50. Each channel of the ASIC has three discriminators; T1, T2 and E, each set to their own trans-impedance amplifier (TIA). T1 is the lowest threshold channel which is used to set the time of the hit. The higher threshold T2 discriminator is to begin the charge integration while the E discriminator is used to discriminate against dark counts [110, 111, 112].

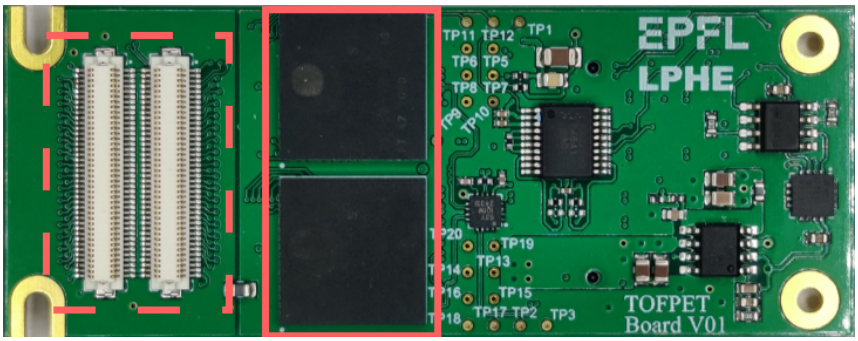



Figure 3.25: Photo of the SND@LHC FE board. The dashed red box shows the dual connector strips which connect to the SiPM PCB via a small interposer PCB. The solid red box shows the two TOFPET2 ASICs. Taken from [85].

The calibration stage of the ASIC requires three steps. Firstly, the input level to TIA for each channel in the ASIC are adjusted to be above the electronic noise. Following this, the TDCs and QDCs of the ASIC are calibrated using input pulses with a defined duration and phase with respect to the FPGA clock. This calibration step is performed while the SiPMs are powered below breakdown voltage. Finally, the SiPMs are powered to the operating voltage and the dark count rate of each SiPM is measured as a function of the threshold setting on the three discriminators in the TOFPET2. The results from the calibration are stored in configuration files which are used in data-taking. Data and commands from/to the DAQ boards are transmitted through the 1 Gb Ethernet port on the board. The DAQ board is

equipped with an Altera Cyclone V FPGA and transmits the data for four FE boards (512 channels) to the DAQ server located above ground.

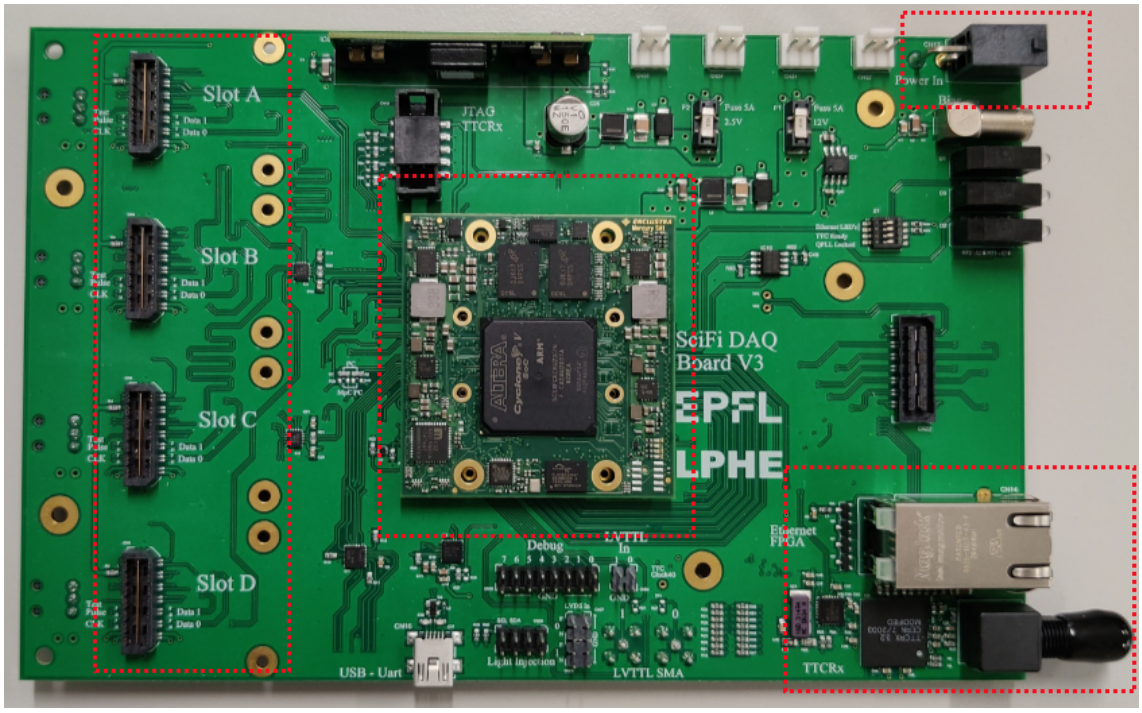



Figure 3.26: Photo of the SND@LHC DAQ board. The dotted rectangle on the left shows the four connector slots for reading in digitised SiPM data from the FE boards. The central dotted rectangle shows the Enclustra Mercury SA 1 module including the FPGA. On the top right a rectangle highlights the power socket and on the bottom right the Ethernet and TTCrx ports. Taken from [85].

Now the usage of FE and DAQ boards are summarised throughout the Veto, US and DS systems. All PCBs have one FE board per. In the Veto and the US, a DAQ board covers two stations, there are four connector slots on a DAQ board, each slot capable of reading in the data from one FE board. Since both the Veto and US have two FE boards per station, a single DAQ board is capable of serving two planes. Whereas in the DS system, since a horizontal plane uses two FE boards and a vertical plane uses one, a DAQ board can accommodate a single DS composite station (horizontal plane and vertical plane) leaving one slot unused. The final vertical-only station of the DS is served with its own DAQ board thus leaving three empty slots.

### 3.3.2 Event building and readout software

SND@LHC does not operate with triggers, so all hits recorded by the detector are transmitted to the DAQ server for event building, filtering and saving. Firstly, the hits sent to the server are grouped into 25 ns windows, this is the timespan of an event in the DAQ and this step corresponds to the step labelled 1 in figure 3.27. With events formed, noise filtering is applied. SND@LHC employs a two-stage noise filter to optimise computing time. The fast noise filter is a series of statements applied with conditional logic, where each statement looks for the event to pass a threshold on the number of individual hits or DAQ boards active in each subsystem i.e. if any of the Veto, SciFi, US or DS muon system meet a threshold in either the

number of hits or the number of active DAQ boards then the event passes the fast filter. The second noise filter then looks more at the topology of the events by asking how many planes of the individual subsystems are active, again in conditional logic. If any subsystem is active beyond the threshold set for that subsystem, the event passes the second noise filter. The application of the two-stage noise filter is the step labelled 2 in figure 3.27. Events passing these filtering stages are then saved to disk. The data are subsequently converted from the online data format into the offline analysis format and saved in permanent storage. Key elements of the offline analysis format and the online data format are covered later in this section for reference later in this thesis.

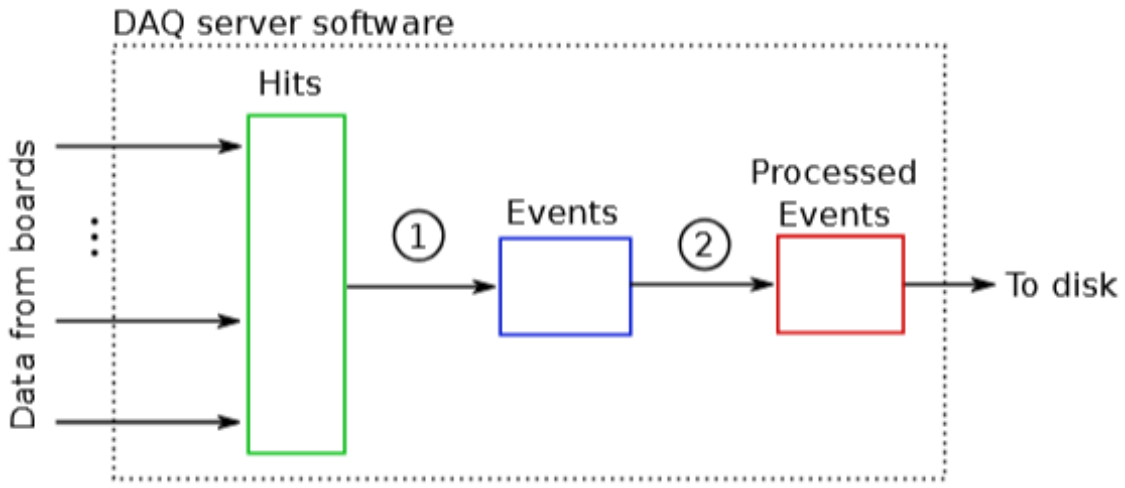



Figure 3.27: Schematic of the event building process [85]. Step 1 corresponds to the collection of hits into 25 ns events. Step 2 corresponds to the application of the two-stage noise filter. Taken from [85].

### 3.3.3 Time alignment of the experiment

The LHC clock and orbit signals from the LHC Beam Synchronous Timing (BST) system are received by a Trigger Timing Control (TTC) crate and distributed to the DAQ boards. The LHC clock (40.079 MHZ bunch crossing frequency) and orbit clock (11.245 kHz revolution frequency) signals are sent by the LHC BST system through fibre optical cables to a dedicated DAQ board at the experiment. The DAQ system itself has a time alignment which has evolved through the operation of the experiment. Knowing which DAQ time alignment was active during a particular run period is required in the analysis of timing data so that the full statistics collected can be analysed. Initially in the operation of SND@LHC, events were grouped with the earliest hit providing the event start time. Furthermore, the times recorded by the Veto system and one of the US planes were observed to be systematically lower than other detector planes[6]. This could give rise to a very dangerous type of background in the experiment: in the case of a muon traversing the experiment causing hits in the Veto and all subsequent systems, the lower time recorded by the Veto could cause the event builder to package the Veto hits into a previous event with noise hits shortly before the muon. This would then lead to a neutrino-like topology of a neutral particle interacting in the target and producing charged particle products. To combat this, the observed shift between detector planes was corrected for and

[6]First reported in an SND@LHC collaboration meeting contribution found at reference [113].

this represents the first change of time alignment mechanism in the experiment. The data collected before this could not be retroactively corrected. The next change of the time alignment mechanism in the experiment was to run the event building synchronously with the LHC. While the Veto timing offset was corrected for, the event builder still packaged events without knowledge of the LHC clock. Figure 3.28 illustrates the benefit of running the time alignment synchronously with the LHC BST clock received by a dedicated DAQ board in the experiment. The impact of the systematically earlier times in the Veto and US on the depicted physics event in figure 3.28 would be to elongate the duration of the event.

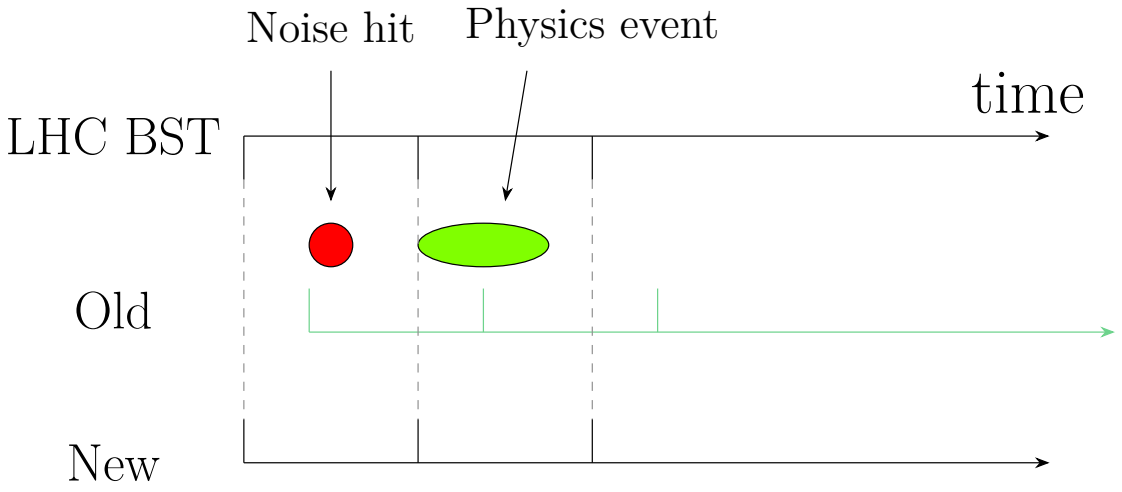



Figure 3.28: Illustration of the difference between the SND@LHC time alignment before (A) and after (B) running the event builder synchronously with the LHC BST clock. Before, a noise hit could split a physics event into two separate recorded events. With the LHC BST clock, the expected start time of an event can remove this unwanted feature.

Running the event building synchronously with the LHC-BST then allows SND@LHC to incorporate the information provided by the LHC filling scheme. Every LHC fill, a dose of protons injected into the LHC, has a defined structure which is openly documented. The filling scheme tells us what bunches of the LHC fill are crossing at the interaction points, which bunches of beam 1 (clockwise travelling beam) and beam 2 (counter-clockwise travelling beam) are not colliding [114, 115]. Bunches which are not colliding may interact with machine elements or residual matter in the beam-pipe leading to events at SND@LHC. Since SND@LHC is 480 m away from IP1 and 2 km away from IP2, there is a phase shift between the filling scheme and the respective crossing points. The phase shifts for the respective crossing points are found by maximising the overlap with the measured SND@LHC event rate. With these phase shifts in plane, the software of SND@LHC readily allows an analyst to check whether an events is tagged as originating from IP1, IP2; or from non-colliding bunches from either beam 1 or 2.

Chapter 4 requires that data for all time alignments are analysed, so knowing the time periods that the different time alignment settings were active is important. Table 3.6 shows the time periods in which different time alignments were employed.

| Time alignment status | Condition on run number, run | Time period covered |
|---|---|---|
| Veto and partial US misalignment | run $\leq$ 5116 | Start of experiment - 26/10/2022 |
| Misalignment corrected | 5117 $\leq$ run $\leq$ 5174 | 26/10/2022 - 04/11/2022 |
| Veto and partial US misalignment | 5175 $\leq$ run $\leq$ 5192 | 04/11/2022 - 07/11/2022 |
| Misalignment corrected | 5193 $\leq$ run $\leq$ 5412 | 07/11/2022 - 30/11/2022 |
| LHC synchronous alignment | run $\geq$ 5413 | 29/03/2023 - onward |

Table 3.6: As is clear, the misalignment reappeared for several days due to a technically issue. The uncovered time period from December to March 29 2023 is due to the LHC not running over these dates.

Now table 3.7 the selected run to cover each time alignment period is shown, this will be repeated in chapter 4 for clarity. The reason for selecting these runs within the corresponding time period is that they contain large numbers of events.

| | Time alignment | | |
|---|---|---|---|
| | Misalignment | No misalignment | LHC synch |
| Run number | 5097 | 5408 | 5999 |

Table 3.7: Runs used in chapter 4 to study the timing behaviour in the experiment.

### 3.3.4 Experiment Control System

The experiment control system (ECS) is bespoke software to control the entirety of the SND@LHC experiment. From here individuals may stop or start data collection runs, monitor power supplies and stop or start the various machines involved in running the experiment. The ECS is designed to operate entirely without user-input through software designed to handle any issues which may arise during data taking such as the DAQ server crashing. A user is typically assigned to monitor the experiment via the graphical user interface of the ECS to ensure no lost periods of data collection. The ECS also receives the status of the LHC which it uses to stop and start runs. For example, when the LHC beam condition changes, the ECS will stop the active run and begin a new one so that each discrete run of the SND@LHC experiment contains data at a fixed LHC run condition [85].

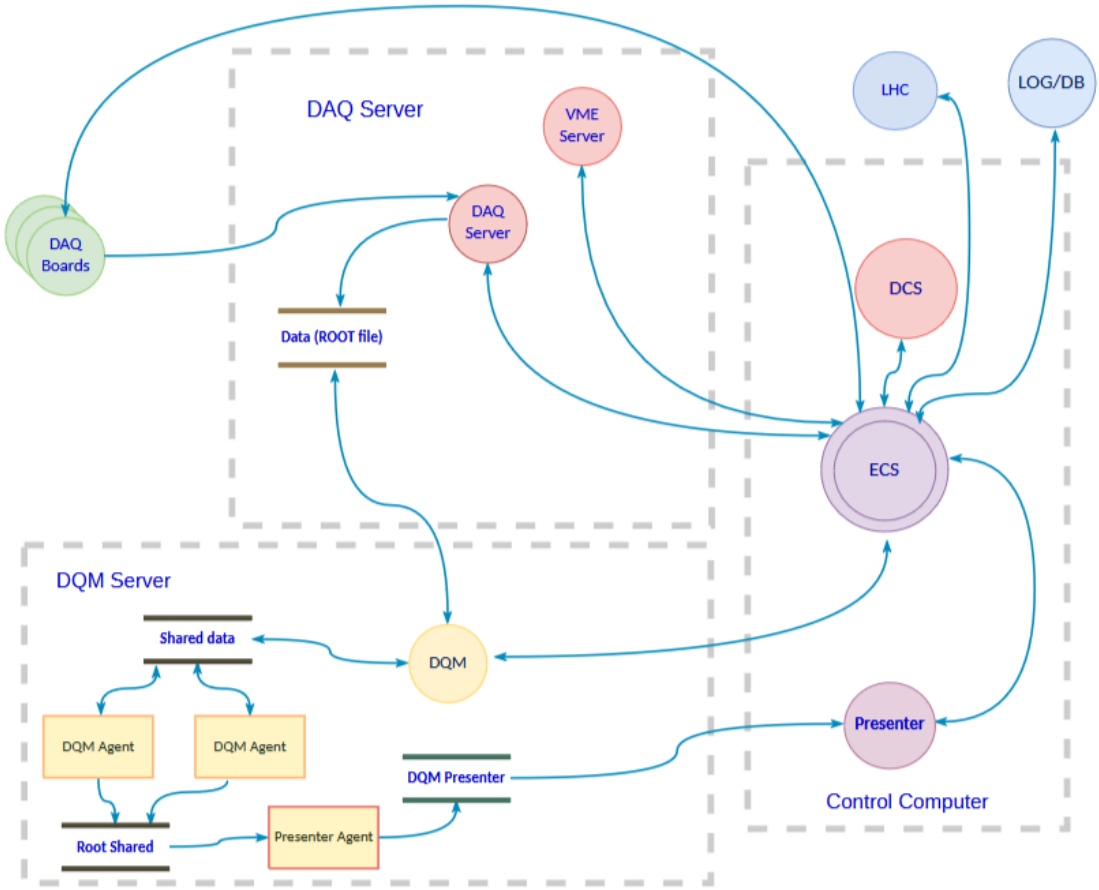



Figure 3.29: Simplified schematic of the global SND@LHC DAQ system [85].

The detector control system (DCS) monitors the power supplied to the electronics of SND@LHC as well as the cooling system and humidity sensors inside the neutron shielded cold box. Alarms are coded into the DCS such that if conditions in either the power supplies or the cold box fail to meet specified operating conditions then emails are sent to the responsible individuals. In the case of the power supplies, the DCS can also cut power supplies to the experiment if the electronics are at risk. Alarms are then deactivated through the ECS and the ECS can also reactivate the power supplied [85].

### 3.3.5 Online and offline data formats

The distinction and differences between the SND@LHC online and offline data formats should be made. The offline data format is used throughout the thesis with the exception of the work with data recorded by laser measurement described in chapter 4 section 4.2 so a brief explanation of the differences will be useful for this section. The online data format is the output of figure 3.27, the data are grouped into events and if calibration data are available, the information from each SiPM may be used to measure the time and QDC before converting to the offline analysis format[7]. The online data format has no understanding of the geometry of the experiment, so a key goal of the conversion to the offline format is to assign each fired SiPM to a detector ID which encodes the position of each scintillator in the Veto, US and DS muon system as follows:

[7]The calibration functions can be found at reference [111] but are not covered in this thesis.

$$\text{Veto, US, DS detector ID} = [\text{subsystem}]\,[\text{station}]\,[\text{bar}] \tag{3.3.1}$$

Where subsystem and station are single digit integers. The subsystem digit equals either 1,2 or 3 for the Veto, US or DS subsystem respectively. The station number begins at 0 for the furthest upstream station of each subsystem and progresses downstream. The bar number is between 0-119 to cover all 120 bars of a DS muon system station (60 horizontal bars + 60 vertical bars). A similar detector ID nomenclature is employed in the Scifi but it is not covered in this thesis. To encode this information in the offline data format, one extracts the identification number of the DAQ board which registers the data and then maps this DAQ board number to a subsystem and station; the channel of the FE board which registers the data is then similarly mapped to the SiPM number. These two steps allow the detector ID to be formulated. With SiPM hits in grouped into detector IDs, the information of these SiPMs can then be encoded into the digitised hit classes of SND@LHC. The digitised hit class of the US allows an analyst to extract the array of QDC values, timing values, perform other analysis algorithms such as finding the average time on one side of a scintillator.

Another difference to be considered is the unit of time. Both format stores time as a number of cycles of the TOFPET clock however a conversion to nanoseconds is used in the majority of this thesis. As stated earlier, the length in time of an event in the experiment is 25 ns, this time arises from the TOFPET clock of 160 MHz: the clock completes a cycle in $1/f = 6.25$ ns, multiplying by four is considered to give the experiment sufficient scope to contain all the relevant physics associated with an interaction and results in an event being an integer number of nanoseconds.

### 3.3.6 Track fitting algorithms

Fitting tracks associated with muons is of principle importance to this thesis work, so the track fitting algorithms used are covered here. Very little input to these algorithms was given by the author. The DS muon system is used for tracking in this thesis but the SciFi tracker can also be used with the same algorithms.

#### Simple Tracking

The name given to this algorithm in the collaboration is *Simple Tracking*. Firstly, the hits in the DS muon system for an event are formed into clusters, where each cluster is a container of hits that are adjacent to one another in a plane of scintillators. Events with too many clusters per plane ($n > 5$) are considered too noisy for a reliable fit. Events which do not have at least three planes per projection registering a suitable number of clusters ($5 > n \geq 1$) are not considered for track fitting to reject ghost tracks[8]. The best candidate set of clusters for representing a muon is found by first requiring that one plane per projection in the muon system must contain only one cluster, which is referred to as the seed cluster. This seed cluster is then used to make a simple linear parameterisation with clusters in an adjacent plane. This parameterisation is used to evaluate the residuals to clusters in the adjacent planes. The set of measurements in each projection with the minimal residual is then defined as set of measurements best representing a muon. This set of clusters is then passed to GENFIT [116] for track fitting using the Kalman Filter method, along with SND@LHC specific information such as the spatial resolution

[8]Ghost tracks are the random combinatorics of noise appearing like a track.

of the measurements. The spatial resolution in the DS muon system is found from the granularity. Each DS scintillator spans 1 cm in both dimensions transverse to its long axis. If a given scintillator fires, the true particle crossing point is assumed to be uniformly distributed over this 1 cm width. Using the fact that the standard deviation of a uniform distribution of width $a$ is given by $a/\sqrt{12}$, the spatial resolution from one fired scintillator, whether horizontally aligned (DSH bar) or vertically aligned (DSV bar) in the DS muon system is simply:

$$\sigma_y(\text{DSH bar}) = \sigma_x(\text{DSV bar}) = \frac{1}{\sqrt{12}}\,\text{cm} \tag{3.3.2}$$

**Hough Transform**

The is a secondary tracking algorithm implemented in the software of the experiment which utilises the Hough transform [117]. The Hough transform method is not utilised in this thesis work however its efficiency is reported during the literature review of section 3.5.1. The track fitting for the Hough transform method is also performed by the Kalman filter of GENFIT [84].

Having been passed to GENFIT, the simple underlying mechanisms behind the fitting are outlined here. The track fitting software of SND@LHC uses the Kalman Filter of GENFIT. The Kalman Filter [118] is a method of estimating the state $s$ and covariance of a system from a set of measurements with covariance matrices. The track state is parameterised with five parameters [119]:

$$\mathbf{s} = (q/p, u', v', u, v)^T \tag{3.3.3}$$

Where $q/p$ is the ratio of the charge to momentum; $u'$ and $v'$ are the track slopes in orthonormal basis vectors $\hat{u}$ and $\hat{v}$, in the case of SND@LHC these $\hat{u}$ and $\hat{v}$ vectors are the $\hat{x}$ and $\hat{y}$ vectors; $u$ and $v$ are the positions of the track measured in coordinates $\hat{u}$ and $\hat{v}$.

The Kalman Filter is an iterative approach where every successive measurement of the state of the system is used to update the estimates of the state and covariances. The process begins with initial, "seed", measurements of the state and covariance. The Kalman Filter then predicts the track parameters in the future. When a new measurement of position is provided, this measurement is compared to the prediction from the seed state and the prediction error is corrected. This cycle of prediction and correction is repeated for each subsequent measurement passed to the Kalman Filter where each successive measurement typically reduces the covariance [119, 116].

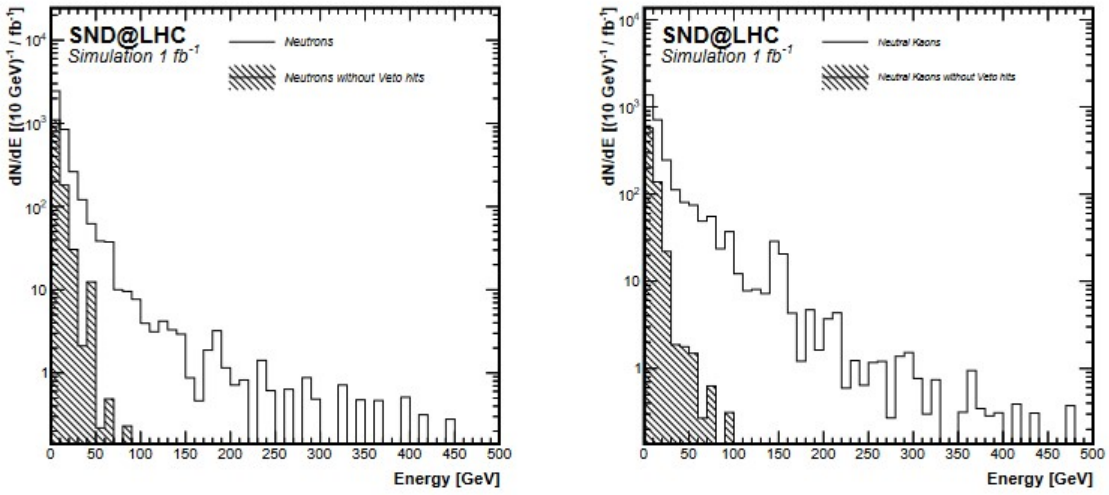
## 3.4 Simulated datasets

Precise simulated datasets of the physics processes occurring in the SND@LHC experiment are paramount to optimising the physics output of the experiment. As muons produced promptly around the ATLAS IP constitute the majority of events in SND@LHC [66, 84], understanding the kinematics of these is the natural first physics case to simulate.

DPM-JET3 (Dual Parton Model) [120] and SIBYLL [121], both hadronic interaction simulation frameworks, are used to generate the proton-proton, $pp$, interactions using LHC run 3 beam conditions. Following this, the $pp$ interaction products are propagated towards the SND@LHC experiment using the LHC model from FLUKA [122] until a $1.8 \times 1.8\text{m}^2$ scoring plane located in the rock around 60

m upstream of SND@LHC. The kinematics of $200 \times 10^6$ *pp* collision products which arrive at this scoring plane are saved. Muons from the scoring plane are propagated to the detector using GEANT4 [123]. The majority of muons from IP1 pass clean through the detector acting as near-MIPs, this class of event we call *passing muons* and provide a clean physics environment for in-situ detector calibration and commissioning studies.

As scintillation is the fundamental physics principle underlying the active detector components of the experiment, electrically neutral particles are not directly detectable in the experiment, only their hadronic interactions are. For this reason, neutral hadrons interacting in the fiducial volume of SND@LHC are a source of background in neutrino analyses and thus the kinematics of these neutral particle interactions must be well understood through simulation studies. Primary muons from the ATLAS IP interacting in the rock and concrete between ATLAS and SND@LHC may produce neutrons and $K_L^0$s. PYTHIA V6.4 uses the muon energy spectrum to simulate the interactions of $\mu^-$ and $\mu^+$ on protons and neutrons at rest. These muon interactions are placed along the muon flight path according to the material density with the final state particles of these interactions being propagated to SND@LHC by GEANT4. Each sub-figure in figure 3.30 shows the energy spectrum of neutral particle interactions within the target system of SND@LHC for cases where there is accompanying activity in the Veto system and when there is not.




(a) *Neutrons* (b) *Neutral Kaons*

Figure 3.30: Energy spectra of neutrons and neutral kaons interacting in the target system of SND@LHC shown for both with (blank) and without (shaded) accompanying hits in the Veto subsystem. Taken from [124]

All neutrino production in LHC *pp* collisions is simulated with DPMJET3 code, with FLUKA being used to simulate the particle flight toward SND@LHC, as well as incorporating the production of secondary neutrinos from particles decaying en-route to SND@LHC. The interactions of neutrinos inside the acceptance of SND@LHC is handled by GENIE with the propagation of the decay products by GEANT4 [125, 126]. In total, the statistics of neutrino interactions simulated is equivalent to an integrated luminosity of $400\text{fb}^{-1}$, corresponding to approximately 10 years of detector operation.

## 3.5 Measurements with SND@LHC

As of January 2025, SND@LHC has three physics publications; a measurement of the recorded muon flux through the detector [84], the observation of muon neutrino charged current interactions [125] and most recently an observation of neutrino interactions without a final state muon [126]. Methods and conclusions from each of these papers provide important input to this thesis, so a brief overview of these publications will be given to provide context where needed for points throughout this thesis. The author provided no contribution to these papers.

### 3.5.1 Measurement of the muon flux in SND@LHC

This publication evaluates the muon flux as measured through the SciFi tracker, the emulsion and the DS muon system of SND@LHC. As the acceptance of the SciFi and DS muon system detectors is different, the muon fluxes are quoted separately. While this paper provides the first demonstrated use of the digitised emulsion data in SND@LHC, that is not of interest in this thesis so only the results with some relevance are discussed. Both the tracking algorithms available in the SND@LHC software are utilised in this study.

Firstly reported in the muon flux measurement of the experiment are the angular distributions in the electronic detectors. While it is clear that the vast majority of reconstructible tracks traversing the experiment have slopes relative to the beam axis very close to zero, some structure around the peak can be analysed. In the distribution of fitted tracks in the SciFi tracker, a shoulder to the main peak of the angular distribution is concluded to arise from interactions and particle decays at various locations. For example, in the more detailed note on the muon flux, the location where Beam 1 is deflected by a quadrupole magnet following the straight section around IP1, is concluded to contribute to the shoulder peak of the angular distribution measured by the SciFi [127].

The tracking efficiencies of both the Simple Tracking and Hough Transform methods are studied using data and Monte Carlo. The underlying methods behind how this is done are not described in the EPJ-C[9], but in the more detailed internal note on the subject. In data, the strategy is to use the SciFi to measure the efficiency in the DS and vice versa. In data the data sets are very large and cover the entire acceptance of all subsystems well. The final tracking efficiencies that are measured in the internal note, referenced in the peer-reviewed publication and utilised in section 3.5.3 are shown in table 3.8:

| Subsystem | Tracking algorithm | Tracking efficiency in data (%) |
|---|---|---|
| SciFi | Simple Tracking | $0.868 \pm 0.009$ |
| | Hough Transform | $0.956 \pm 0.007$ |
| DS | Simple Tracking | $0.937 \pm 0.007$ |
| | Hough Transform | $0.944 \pm 0.009$ |

Table 3.8: Tracking efficiency measured in data for both SciFi and DS, each using both available tracking algorithms. Adapted from [127].

The simulation of proton-proton collisions and the transportation of interaction products to SND@LHC leads to the prediction that the flux of muons through

[9]Journal where the muon flux article was published: European Journal of Physics - C.

SND@LHC is expected to be correlated with the $y$ coordinate of the experiment due to the magnetic fields acting on charged particles as they exit IP1. How the muon flux varies with $y$ represents a more robust examination of MC-data agreement than a one-dimensional check. This MC-data comparison is shown in 3.31:

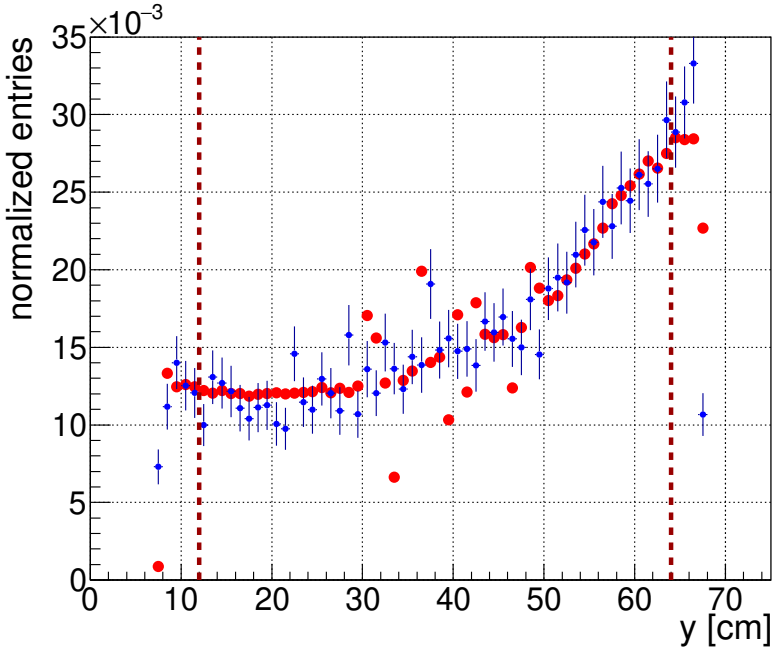



Figure 3.31: DS tracks shown in the red data series, Monte Carlo simulation in blue. The dotted lines at 12 cm and 64 cm represent the lower and upper limits on $y$ considered for the muon flux measurement. Taken from [84].

Overall, the comparison between data and simulation in figure 3.31 is very strong. The error bars for the experimental data are too small to see, as one expects given the very small deviations between successive data points especially in the $\sim 11 - 30$ cm region. The outlying data points for the experimental data are put down to inefficient bars by the analysis in this paper [84]. The larger error bars on the Monte Carlo measurements are simply due to lower statistics.

Due to the correlation between muon flux and $y-$position and the different acceptances of the SciFi and the DS, the flux per unit area in the DS is expected to be larger than what is measured in the SciFi. The fiducial area used in the analysis for the SciFi was $-42\,\text{cm} \leq x \leq -11\,\text{cm}$ and $18\,\text{cm} \leq y \leq 49\,\text{cm}$ while the acceptance area in the DS was $-54\,\text{cm} \leq x \leq -2\,\text{cm}$ and $12\,\text{cm} \leq y \leq 64\,\text{cm}$. The results for the muon fluxes measured with these independent fiducial areas is shown in table 3.9:

| Subsystem | Muon flux ($10 \times 10^4$ fb/cm$^2$) |
|---|---|
| SciFi | $2.06 \pm 0.01$ (stat.) $\pm 0.12$ (sys.) |
| DS | $2.25 \pm 0.01$ (stat.) $\pm 0.10$ (sys.) |

Table 3.9: Measured muon flux in the SciFi and DS detectors with distinct fiducial areas. Adapted from [84].

To allow for a comparison between the two subsystems, the fluxes are also quoted when both subsystems use the same fiducial area, which is the area given above for the SciFi. The results with the common fiducial area are in table 3.10:

| Subsystem | Muon flux ($10 \times 10^4$ fb/cm$^2$) *same fiducial area* |
|---|---|
| SciFi | $2.06 \pm 0.01$ (stat.) $\pm 0.12$ (sys.) |
| DS | $2.02 \pm 0.01$ (stat.) $\pm 0.08$ (sys.) |

Table 3.10: Measured muon flux in the SciFi and DS detectors using a common fiducial area. Adapted from [84].

As muons provide the most common source of events in the detector, a thorough measurement of the muon flux provides a study of paramount importance to the further physics goals of SND@LHC. Not just the flux measurement itself, but the tracking efficiency comparisons and MC-data comparisons will be regularly quoted as the experiment pursues its physics programme.

### 3.5.2 Observation of Collider Muon Neutrinos with the SND@LHC Experiment

The goal of this paper was to produce a high-purity sample of charged-current muon neutrino deep inelastic scattering events using the data collected during 2022, which the ATLAS collaboration measured to be 38.7 fb$^{-1}$. As outlined in section 2.3.3, DIS is the dominant interaction mechanism for LHC neutrinos traversing SND@LHC. As the charged current mode of DIS for muon neutrinos produces an isolated muon in the final state, this is the natural first analysis channel for the experiment. In this data collection period the experiment is expected to see $157 \pm 37$ $\nu_\mu$ CCDIS events in the target given the mass used in 2022 (792 kg) before the application of selection criteria. The uncertainty in the expected number of event is estimated by the different predictions by the DPMJET3 and SIBYLL simulations.

This analysis demanded selection criteria with a very strong background rejection power. The first section of the criteria seeks to identify events occurring in the target of the detector. Fundamental to this is vetoing any events with activity in the Veto system. Due to a higher than expected inefficiency in the Veto system, the first two planes of the target tracker were also adopted into an extended Veto system to greatly enhance the veto power against charged particles. An algorithm developed to determine the target wall in which the interaction occurs was implemented to veto events occurring in the final target wall to ensure that at least two Scifi planes are available for calorimetric sampling. Furthermore, a restriction of the average channel number firing in the horizontal and vertically aligned planes of both the Scifi and the muon system were used to remove events at the edges of the detector acceptance. These fiducial volume cuts were shown to have an efficiency of 7.5% on the neutrino interaction simulation data. These fiducial volume cuts are summarised in table

| Cut | Description | Data | Neutrino MC |
|---|---|---|---|
| A | Average SciFi channel in [200, 1200] (vert.) and [300, 1336] (hor.) | $6.54 \times 10^{-2}$ | 0.44 |
| B | Average DS bar number in [10, 50] (vert.) and [70, 105] (hor.) | 0.85 | 0.81 |
| C | No Veto hits | $3.69 \times 10^{-3}$ | 0.56 |
| D | No hits in the first SF station | 0.58 | 0.86 |
| E | No hits in the second SF station | 0.66 | 0.72 |
| F | Interaction vertex not in fifth target wall | 0.75 | 0.60 |

Table 3.11: Fiducial volume selection criteria and the fraction events for data and neutrino simulation [125] which pass each subsequent criterion.

The second section of the selection criteria uses a series of cuts to isolate events with signal-like topologies, referred to as the topological selection criteria. A final state muon track is required and events with more than 10 hits in either projection of the muon system are rejected in order to retain only events with clean muon tracks. There are then requirements on the properties of the muon track: firstly, it is required that the muon track intersects the first Scifi plane more than 5 cm away from the edge of the detector. Secondly, the sum of closest approach distances to a hit in the vertical and horizontal planes of each SciFi station must be less than 3 cm. As the $\nu_\mu$ CCDIS are expected to have large energy depositions, there are requirements on the hit multiplicity in the Scifi and on the QDC in the US: there must be greater than 35 hits in the SciFi and the US QDC must be greater than 600 (700) for data (MC). The fiducial volume selection criteria are applied first and then the topological selection criteria, so the value of total selection efficiency refers to the percentage of data/neutrino MC remaining after both the fiducial volume and the topological selection criteria are applied:

| Cut | Description | Data | Neutrino MC |
|---|---|---|---|
| G | At least two consecutive SF planes fire | 0.20 | 0.93 |
| H | All US planes must be fire | $8.8 \times 10^{-2}$ | 1.00 |
| I | Prior event at least 100 clock cycles away | 0.98 | N/A |
| J | Track fitted in DS | 0.47 | 0.90 |
| K | Latest DS time > earliest SF hit time | 0.61 | 1.00 |
| L | DS track intersects first SF plane > 5cm away from edge | $7.7 \times 10^{-2}$ | 0.78 |
| M | Sum of min(DOCA) of track to SF hits < 3cm in both projections per station | $8.99 \times 10^{-2}$ | 0.79 |
| N | More than 35 SF hits | 0.77 | 0.83 |
| O | US total QDC larger than 600 (700) | 0.92 | 0.92 |
| P | Fewer than 10 DS hits per projection | 0.67 | 0.9 |
| | Fiducial volume eff. [A-F] | $5.9 \times 10^{-5}$ | $7.5 \times 10^{-2}$ |
| | Total selection eff. [A-P] | $9.5 \times 10^{-10}$ | $2.7 \times 10^{-2}$ |

Table 3.12: $\nu_\mu$ CCDIS identification selection criteria and the fraction of events in data and neutrino simulation passing each successive criterion. The final efficiencies are tabulated in the bottom two rows. Data taken from [125].

With all of the selection criteria applied, eight $\nu_\mu$ CCDIS candidates remain. The significance of this measurement is evaluated in terms of rejecting the null hypothesis, defined by setting the neutrino signal strength $\mu$, to zero. The one-sided profile likelihood ratio test $\lambda(\mu)$ was used to quantify the rejection power against the null hypothesis. The likelihood is constructed by assuming the probability to observe signal follows Poissonian statistics, $\mathcal{P}$, and the background follows a normal distribution, $\mathcal{N}$:

$$\mathcal{L} = \mathcal{P}(n|\mu s + \beta)\,\mathcal{N}(\beta|b,\,\sigma_b) \tag{3.5.1}$$

Where $n$ is the number of observed events, $s$ and $b$ are the number of expected signal and background events, and $\beta$ is the observed number of background events which along with $\mu$ is a free parameter in the likelihood fit. RooStats [128] is then used to maximise $\mathcal{L}(\mu, \beta)$ with numerical methods to give a $p-$value of $7.15 \times 10^{-12}$ which is then used in the inverse normal cumulative distribution function to express the $p$-value as an equivalent number of standard deviations, $\sigma$:

$$Z = \Phi^{-1}(1 - p)$$
$$Z \approx 6.8\sigma$$

### 3.5.3 Observation of collider neutrinos without final state muons

This most recent physics analysis of the experiment searches for neutrinos without a final state muon, this analysis channel comprises the charged-current interactions of both $\nu_e$ and $\nu_\tau$ as well as neutral current interactions of all flavour [126]. Adopting the notation of this article, the signal events will be referred to as $\nu 0\mu$ events and background from neutrino interactions containing a single final state muon referred to as $\nu 1\mu$. This work acts as a precursor to a flavour classification with the experiment and also the experiment's first analysis of $\nu_e$ charged-current interactions which as was discussed in section 2.4 forms an important step in the broad physics goals of the experiment. The experimental data used was collected between July 2022 and October 2023, totally an integrated luminosity of 68.6fb $^{-1}$ and a data collection efficiency of 97.2%. The simulated data used in this analysis is gathered in an identical fashion to the $\nu_\mu$ CC paper discussed in section 3.5.2.

The first conditions applied to the data selection is that the event is correlated with a bunch crossing at IP1 through the method outlined in section 3.3.3 and that there is no activity in the Veto system. Following this, the events must be occurring at least 625 ns since the previous event, to reduce the veto inefficiency. It is noteworthy that the IP1 bunch crossing and requirement that the previous event was at least 625 ns earlier retain more than 99.75% of the signal. As in the $\nu_\mu$ CC paper, the average SciFi channel number both horizontally and vertically is utilised to provide a fiducial volume cut to remove events where charged particles enter the target region through the edges of the target and do not traverse the Veto system. In this analysis, the upper and lowermost two scintillators in the first two planes of the US system is also used to veto charged particles entering the detector. Following the application of these fiducial volume criteria the experimental dataset contains $7.9 \times 10^6$ events with a signal acceptance of 20%. To ensure adequate sampling in the homogeneous HCAL, it is required that two SciFi stations and two US stations are active in events, with a minimum of 35 SciFi hits and at least a "large" activity

in the US which is expected to correspond to an energy deposition of approximately 10 GeV. As discussed in the tracking algorithms section 3.3.6, fitting a track in the muon system requires three planes per projection to register data, thus for this $\nu 0\mu$ search, events reporting activity in the last two of the stations muon system are vetoed. This cut reduces the $\nu_\mu$ CC background by a factor of 14. The remaining $\nu_\mu$ CC background is mostly due to events in which the muon exits the detector before the third station of the muon system[10].

The next analysis stage requires the introduction of the discriminating variable developed for this article, the sum of SciFi hit densities. Each SciFi hit is assigned a weight that is given by counting the number of other hits in the same SciFi plane:

$$w_i = \sum_{i \neq j}^{N} H(x_j - x_i + 1\,\text{cm}) \times H(x_j - x_i - 1\,\text{cm}) \tag{3.5.2}$$

Where $N$ is the number of SciFi hits in a plane; $x$ is the hit position in centimetres and $H(x)$ is the Heaviside step function. Each plane has the sum of it's weights determined. This variable was devised as one expects a neutrino interaction to deposit a high density of hits coming from electromagnetic or hadronic showers. This sum of weights allows for many events coming from debris of muon interactions in the detector surroundings to be rejected. Furthermore, this variable allows for events with sparse hit patterns to be rejected. This is achieved by requiring that two stations have a sum of weights greater than 20 and greater than $20 \times 10^3$ in at least one station. The discriminating variable is set to be the largest sum of SciFi hit density weights found over the SciFi planes. The number of events passing all cuts is shown in table 3.13.

| | Data | $\nu 0\mu$ simulation | $\nu 1\mu$ simulation |
|---|---|---|---|
| All events | $1.1 \times 10^{10}$ | 307.7 | 496.8 |
| Fiducial volume | $7.91 \times 10^6$ | 63.6 | 111.6 |
| Hits in 2 SciFi and 2 US stations | $5.95 \times 10^5$ | 38.0 | 83.4 |
| Large SciFi activity | $6.34 \times 10^4$ | 37.1 | 74.5 |
| Large US+DS activity | $2.28 \times 10^4$ | 35.8 | 68.3 |
| No hits in last 2 DS stations | $1.47 \times 10^4$ | 27.7 | 4.9 |
| Sparse-shower removal | 25 | 16.7 | 2.0 |
| Signal region | 9 | 7.2 | 0.30 |

Table 3.13: Number of events passing each selection cut in the data, $\nu 0\mu$ simulation (signal) and $\nu 1\mu$ (background).

An important feature of this study is its link with an SND@LHC test beam and how the test beam data is utilised by the collaboration to validate the use of the sum of SciFi hit densities. In 2023, the collaboration performed a test beam at the CERN North Area with a detector consisting of four SciFi stations, an exact replica of the US system and a single replica DS station. The SciFi planes were interleaved with iron blocks each of an equivalent number of interaction lengths to a tungsten SND@LHC target wall. This test beam detector was studied with beams of pions

[10]This background will be returned to in chapter 6.

ranging in energy from 100 to 300 GeV. The validation of the discriminating variable used in the $\nu 0\mu$ paper was done by comparing how this variable was distributed for the test beam data and the test beam simulation, which is shown in figure 3.32.

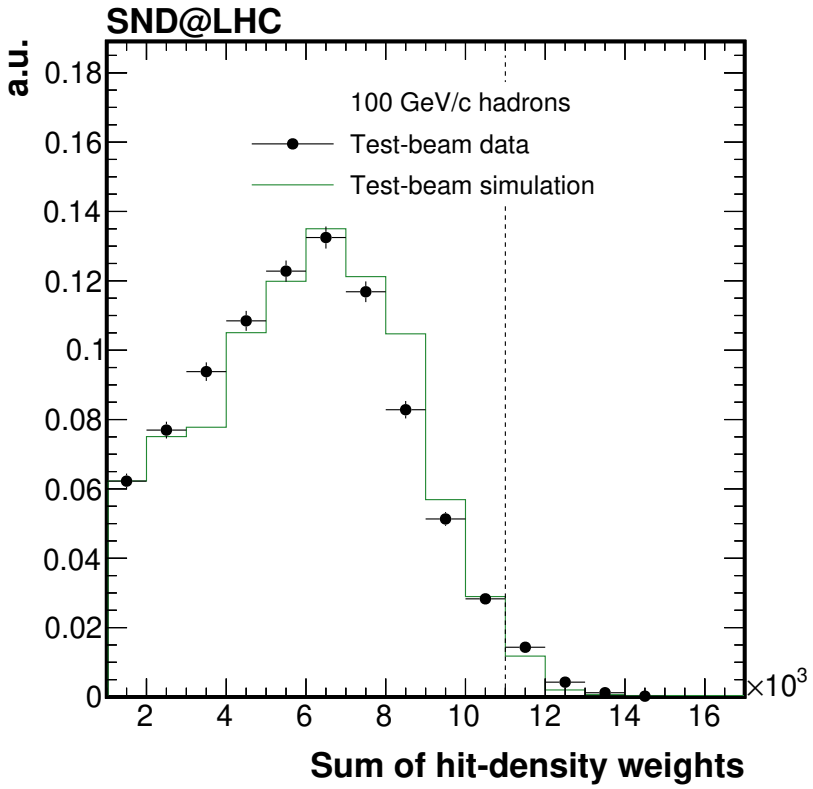



Figure 3.32: Distribution of the sum of SciFi hit density weights as measured with 100 GeV pions in the 2023 test beam campaign. The dotted vertical line is at $11 \times 10^3$. Taken from [126].

As figure 3.32 shows, most 100 GeV hadrons have a maximum sum of SciFi hit density weights below $11 \times 10^3$. Events with a sum of SciFi hit density weights less than $5 \times 10^3$ is used as the control region in which an estimate is evaluated of the neutral hadron background originating from muon DIS in the material around SND@LHC. In the control region, 3.1 neutrino interactions and 8.2 neutral hadron events from muon DIS are expected, the observation in data of ten events is inline with the 90% confidence level upper limit of this estimate. The discriminating variable is then varied between $5 \times 10^3$ and $20 \times 10^3$, evaluating the the sensitivity to $\nu 0\mu$ at incremental steps. The sensitivity is quantified in terms of rejecting the null hypothesis in a similar way to section 3.5.2: the null hypothesis is defined by setting the signal strength to zero. A one-sided profile likelihood ratio test is the test statistic. The likelihood includes a Gaussian factor to model background uncertainty. See equation 3.5.1.

Figure 3.32 shows that the agreement between simulation and data breaks down above $11 \times 10^3$, this is used to derive a 100% uncertainty in this higher sum of SciFi hit density weights region. The uncertainty on $\nu_\mu$ CC background is quoted as the statistical uncertainty in the number of events measured in the $\nu_\mu$ CC paper added in quadrature with the DS system tracking systematic uncertainty [84] which gives 18.2%. The uncertainty of $\nu_\tau$ CC interactions with a final state muon, $\tau \to \mu$, is set to 100%. The article notes that the $\nu_\mu$ CC background component is so

dominant that doubling the background uncertainties on neutral hadrons and $\nu_\tau$ CC components leaves the results unchanged.

Having performed the sensitivity scan using these background figures, the optimal threshold on the sum of SciFi hit density weights to observe the $\nu 0\mu$ signal is shown to be $11 \times 10^3$. The final results of this analysis channel yield an expectation of 7.2 signal events with a significance of $5.5\sigma$ where there is an expected background of 0.30 $\nu_\mu$ CC events and $1.5 \times 10^{-2}$ events from neutral hadrons following muon DIS in the surrounding material, finally there are $1.7 \times 10^{-3}$ expected background events from $\nu_\tau$ CC interactions producing a final state muon. The expected 7.2 signal events can be broken down into an expected 4.9 $\nu_e$ CC events, 2.2 NC events and 0.1 $\nu_\tau$ CC events with no final state muon. Nine events in total are observed in the signal region with a significance of $6.4\,\sigma$.

Also considered in the analysis is the significance to observe $\nu_e$ CC interactions. The article reports the observation of $\nu_e$ CC with a significance of $3.7\,\sigma$ and an expected significance of $2.2\,\sigma$, representing a stepping stone for a more significant $\nu_e$ CC measurement. In parallel, studies searching for $\nu_e$ CC using the emulsion detectors of SND@LHC are ongoing.

# Chapter 4

# Correction and alignment of the US SiPM timing data

*The work in this chapter was conducted entirely by the author using data collected by the experimental collaboration* ***unless stated otherwise in this preamble or in the text.*** *The tracking algorithms and offline analysis classes in the experiment software were developed by various members of the SND@LHC collaboration. The measurements of the trace-line lengths in the PCB, discussed in section 4.3.5, were performed by a colleague PhD student at HU-Berlin. The laser calibration data for section 4.3.7 was not collected by the author, but by a PhD student in EPFL. The preparation of the simulated data-sets studied in the latter section of this chapter was done by various members of the SND@LHC collaboration.*

## 4.1 Method

The first stage of the thesis is the correction of the time-walk effect in the followed by an alignment of the timing measurements. The correction and alignment of the was performed with events representing passing muons due to the clean physics environment which they present. Firstly, the method for selecting passing muon tracks is presented, before evaluating the effective signal speed of the light in the scintillator, correcting the time-walk effect, measuring the effect on the time resolution following the time-walk correction. Finally, applying a systemic time alignment to the and evaluating the subsequent timing performance.

The final section of this chapter covers lab measurements of the PCB taken by firing laser light directly at the SiPMs of the PCB. These data were studied to produce conclusions regarding the PCB that cannot be made in the TI18 setting of the full detector.

### 4.1.1 Selection criteria applied to single muon events

The first step in selecting passing muons is to apply the muon tracking code that is outlined in section 3.3. With the tracking algorithm applied, the quality of a track is ensured by a requirement that the $\chi^2$ per degree of freedom, $\chi^2_\nu$, of the track is less than 5.

The next key feature of fitted tracks that can be analysed is the slope in each projection:

$$s_x \equiv \frac{dx}{dz} = \frac{\vec{p} \cdot \hat{x}}{\vec{p} \cdot \hat{z}}, \quad s_y \equiv \frac{dy}{dz} = \frac{\vec{p} \cdot \hat{y}}{\vec{p} \cdot \hat{z}},$$

Where $s_x, s_y$ are the measured slopes in $x$ and $y$ respectively; $\hat{x}, \hat{y}, \hat{z}$ are the unit vectors of the experimental reference system; $\vec{p}$ is the momentum vector of the track to be fitted. Since SND@LHC does not have an applied external magnetic field, the momentum cannot be measured. To bypass this, the assumption is made that all the particles that will be trackable in the DS muon system are highly relativistic and boosted along the beam axis: the classical three-momentum component parallel to the beam axis is assumed to be far greater than either transverse components. This initial assumption is passed to GENFIT which then uses the measured hit positions in the DS system to return a fitted direction vector. Determining $s_x$ and $s_y$ as ratios of the fitted momentum vector components extracts the tangent of the track angle relative to the beam axis with no dependence on the arbitrary magnitude of momentum.

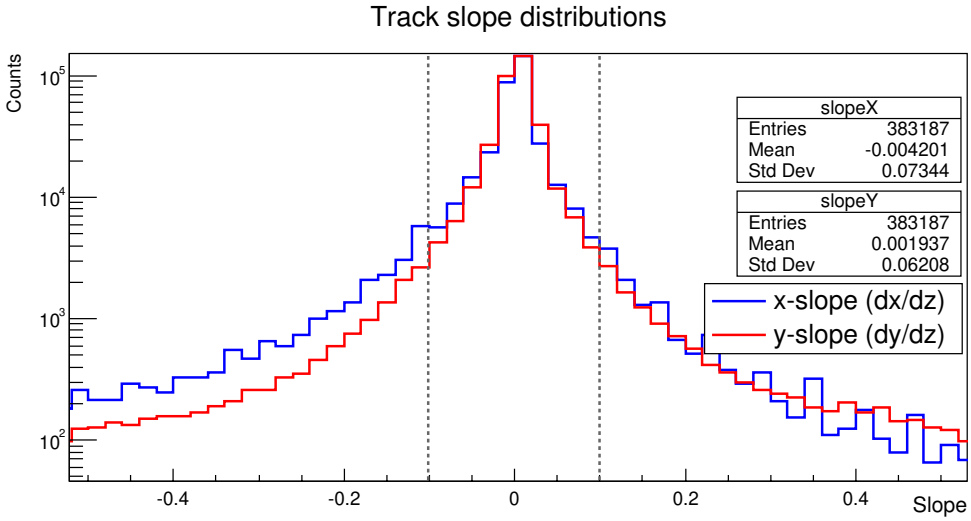



Figure 4.1: Distribution of fitted track slopes in the $x$ and $y$ projections. Data from run 5408 with the only criteria applied being the requirement of a fitted track in the muon system.

The vertical lines shown in figure 4.1 mark the slope ranges $-0.1 < s < 0.1$ which is the criteria placed on this value. The asymmetry between $-1 < s_x < -0.5$, this can be attributed to background originating in the LHC Q12 quadrupole magnet [127]. One can see the location of the Q12 quadrupole magnet in relation to SND@LHC in figure 4.2, principally that Q12 is downstream of SND@LHC on the $z$-axis. Such events contaminate timing studies as the timing chronology will be inverted due to Q12 being further downstream than SND@LHC. As particles from Q12 travel in the negative $z$ and positive $x$ directions to reach SND@LHC, it is clear that a negative value of $s_x$ is to be expected.

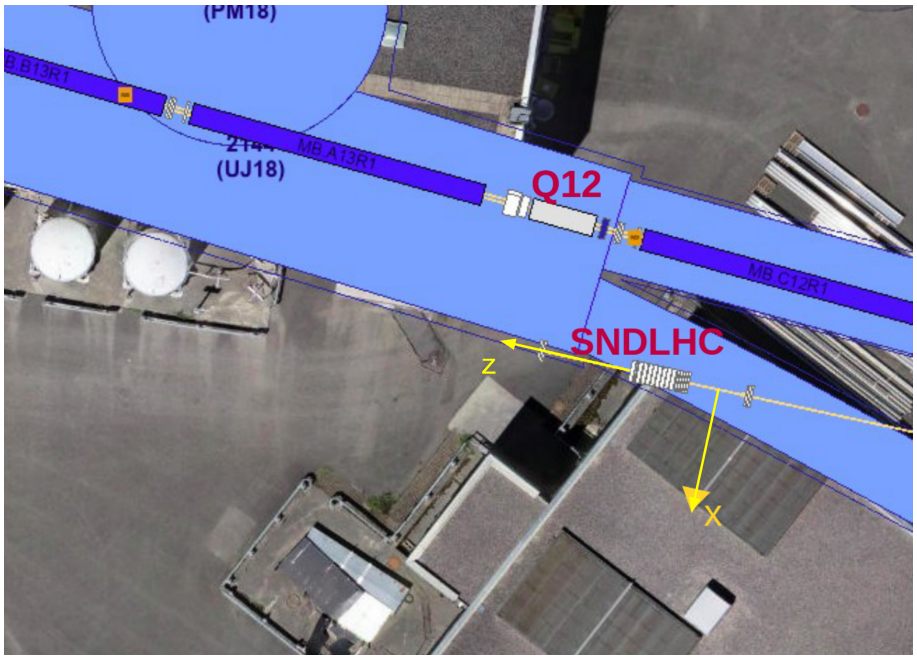



Figure 4.2: Location of Q12 quadrupole magnet and the SND@LHC positioned in TI18.

In order to study the timing response of single SiPMs, a reference event time can be used to remove the effect of the event-wise start time, $t_0$, and remove timing effects outwith the propagation of the scintillation light such as signal propagation in the read-out cables[1]. A true event start time cannot be used for this analysis because such a time would be provided by either the Veto or the Scifi subsystems; neither of these provide an appropriate solution: both the Veto and Scifi subsystems have a far smaller geometric acceptance in the $xy$ plane and due to the narrow $s_y$ distribution, the upper scintillators in the cannot be traversed by muons which also fire the Veto or Scifi system. The solution to this problem was found by taking advantage of the sharper time resolution observed in the DS muon system. Additionally, due to the horizontally aligned DS scintillators being read-out by a single SiPM on each side, the average time recorded by the SiPMs connected to a horizontally aligned scintillator is independent from the point where the particle crosses a DS scintillator. Utilising this, the event-wise reference time is determined as the average time recorded by the $N_{\text{DSH}}$ fired horizontally aligned scintillators:

$$t_0^{\text{DS}} = \frac{1}{N_{\text{DSH}}} \times \sum_{\text{DSH } i}^{N_{\text{DSH}}} \overline{t_{DSH\,i}}$$

$\overline{t_{DSH\,i}}$ is the average of the times recorded by the SiPMs on either side of the horizontally aligned scintillators:

$$\overline{t_{DSH\,i}} = \frac{1}{2} \times (t_{\text{i, left}} + t_{\text{i, right}})$$

Thus we can say that $t_0^{\text{DS}}$ is simply an average over the fired SiPMs of the DS horizontal planes.

$$t_0^{\text{DS}} = \frac{1}{N} \sum_{\text{SiPM } i}^{N} t_{\text{SiPM } i} \tag{4.1.1}$$

[1] All the electronic read-out cables in SND@LHC have the same length

This value is of central importance to this chapter, and is hence referred to as $t_0^{\mathrm{DS}}$. The times recorded by SiPMs in the US system are consistent quoted in reference to this event-wise reference time, $t_0^{\mathrm{DS}} - t_{\mathrm{SiPM}}$, this will be referred to as $dt_{\mathrm{SiPM}}$.

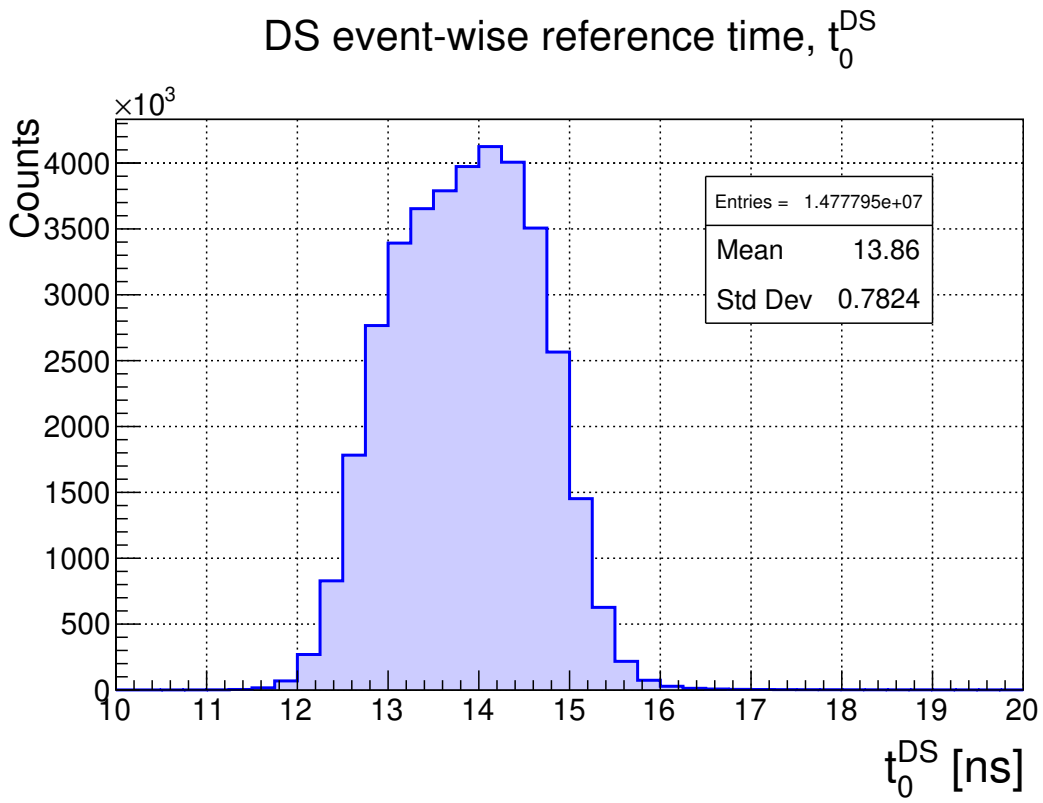



Figure 4.3: Distribution of the average time value measured by the fired horizontal muon system scintillators used in track reconstruction. This distribution is taken from run 5408.

Similarly, an event-wise reference time in the SciFi system is constructed by making a simple average of the SiPM times, $t_0^{\mathrm{SF}}$:

$$t_0^{\mathrm{SF}} = \frac{1}{N} \sum_N t_{\mathrm{SiPM}}^{\mathrm{SF}} \tag{4.1.2}$$

Where $N$ is the number of SiPMs in the SciFi registering data and $t_{\mathrm{SiPM}}^{\mathrm{SF}}$ is the time measured by individual SiPMs in the SciFi. This simple average produces a distribution that is sufficiently well defined to serve as an event-wise reference time.

### 4.1.2 Selection criteria on digitised hits

| Criterion name | Requirement |
|---|---|
| *Event selection criteria* | |
| Fitted DS track, $\chi^2_\nu$ | $\chi^2_\nu < 5$ |
| Track slopes; $s_x$, $s_y$ | $s_x < 0.1$<br>$s_y < 0.1$ |
| *Fired scintillator selection criteria* | |
| y-residual between track and bar centre, $\Delta y$ | $\Delta y \geq E[\Delta y] - 1 \times \sigma_{\mathrm{RMS}}(\Delta y)$<br>$\Delta y \leq E[\Delta y] + 1 \times \sigma_{\mathrm{RMS}}(\Delta y)$ |

Table 4.1: Selection criteria for events and fired scintillators applied to data.

On the level of the digitised hit objects, the fitted track is extrapolated to the $z-$position of the fired scintillators of the US recorded in the given event. Once extrapolated to the $z-$position of a scintillator we can compare the $y-$position of a fired scintillator, $y_{\mathrm{scint}}$, with the $y-$position of the track, $y_{\mathrm{track}}$. The distributions of residual between expected position from the DS track and the position of fired bars are made for all planes in the , such as is shown for the 5th US plane in figure 4.4. For a digitised hit in the event, the $y-$residual must fall within 1 standard deviation of the mean of the $y-$residual distribution, $E[\Delta y]$, for the plane of scintillators extrapolated to.

$$\Delta y = y_{\mathrm{scint}} - y_{\mathrm{track}} \tag{4.1.3}$$

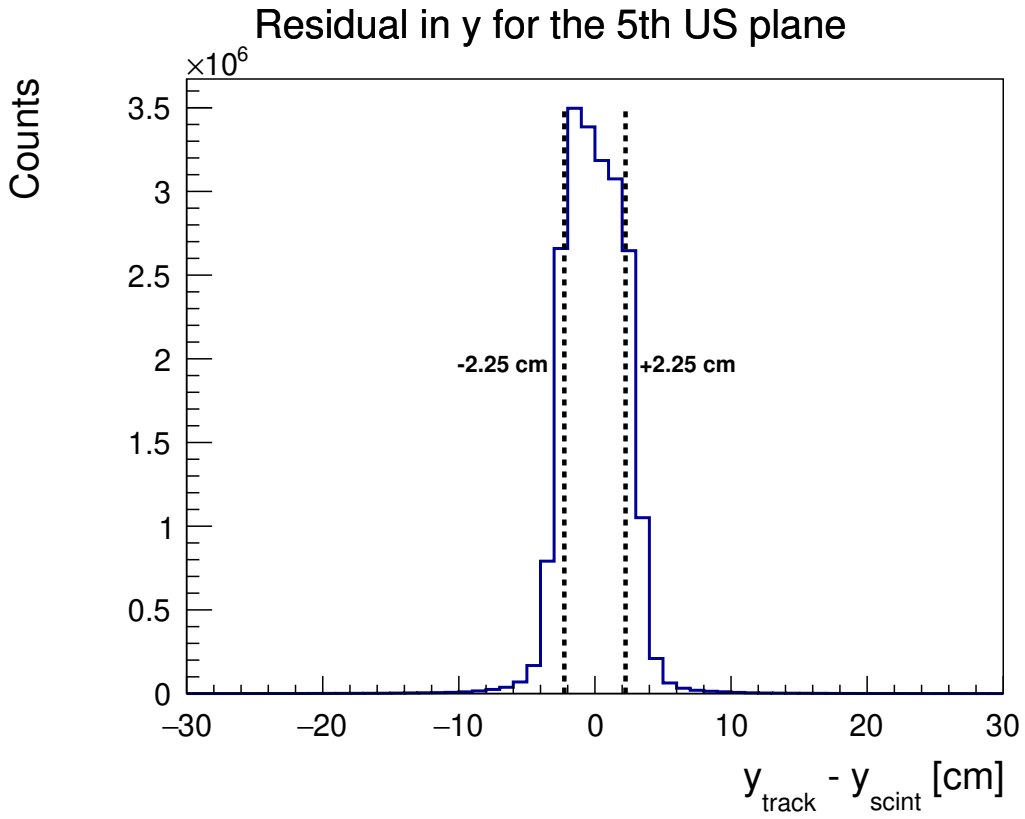



Figure 4.4: Residual between the $y$-position of the track extrapolated to the 5th US plane and the y-midpoint of a fired scintillator in the 5th US plane from run 5097.

### 4.1.3 Determination of the effective speed of light measured by the SiPM

Following the application of these selection criteria, the SiPM times are fetched from the muon system digitised hit object. By correlating the $t_0^{\mathrm{DS}}$ relative SiPM time, $dt_{\mathrm{SiPM}}$, with the $x-$position from the extrapolation of a fitted DS track, a distribution containing the effective signal speed observed inside the scintillator by the SiPM channel, $c_{\mathrm{SiPM}}$, is formed and an example is shown in figure 4.5. Here it is recalled that since the $x$-axis of SND@LHC points left, greater values of $x$ refer to positions further left-wards for an observer looking down the beam-axis.

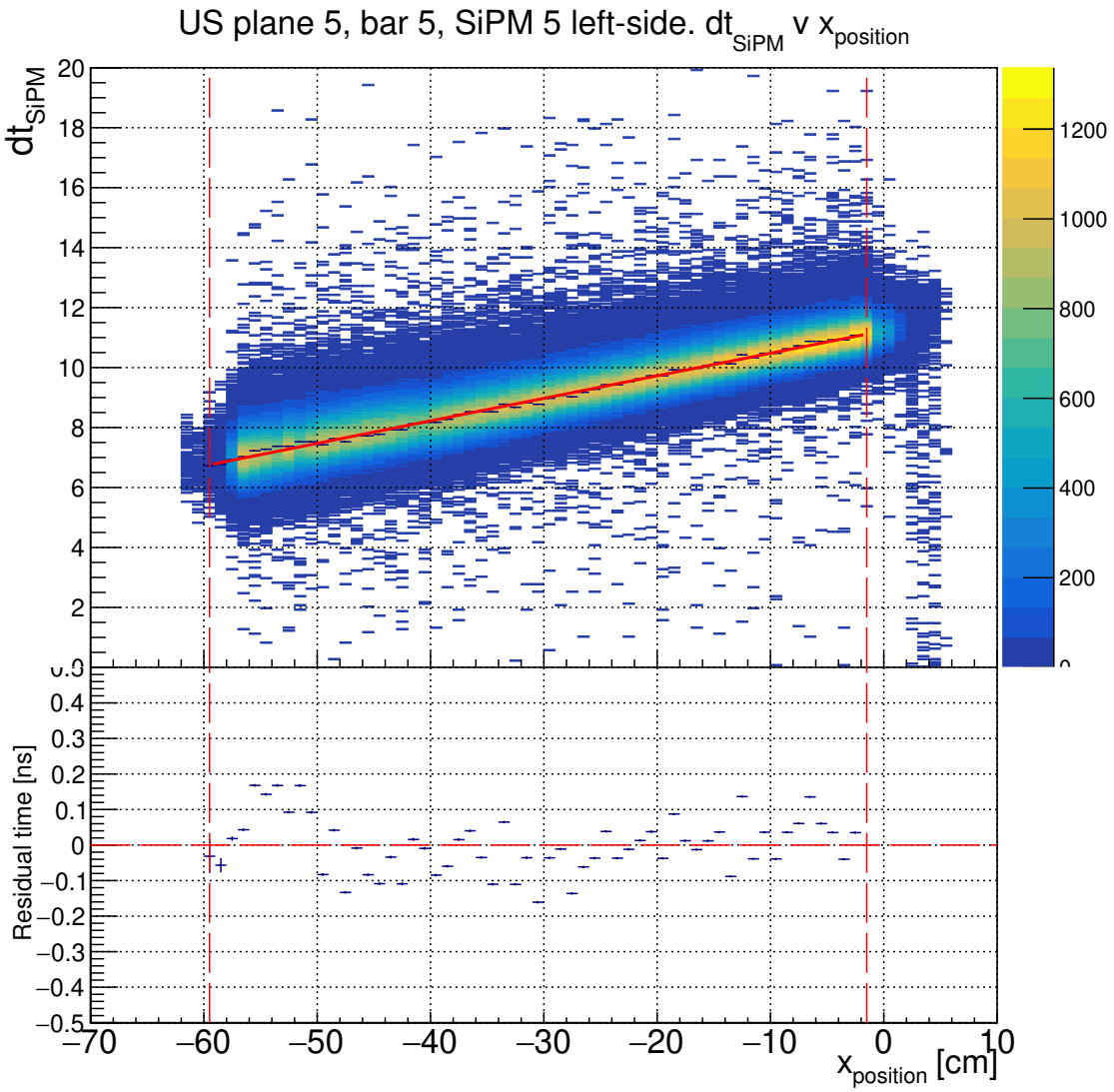



Figure 4.5: 2D histogram of $dt_{\mathrm{SiPM}}$ against the extrapolated DS $x-$value. The vertical red dashed lined indicate the fit range. Data from run 5408.

From the underlying linear relationship between $dt_{\mathrm{SiPM}}$ and $x-$position, the effective signal speed in the scintillator, $c_{\mathrm{SiPM}}$, as measured by the SiPM can be extracted. In order to do this, the 2D histogram is used to form a weighted one-dimensional histogram for fitting in the following process. The projection in the $dt_{\mathrm{SiPM}}$ axis is obtained for all bins of $x-$position in the range of the $[A - 5\,\mathrm{cm},\, L_{\mu}]$ where $A$ is the left (looking down the beam-axis) edge of the scintillator in centimetres and, as outlined in section 3.2.4, $L_{\mu}$ denotes the right edge of the vertically aligned scintillators of the DS system which presents a soft limit on where fitted tracks can be found. Figure 4.6 depicts a scintillator with the left- and right-edges labelled $A$ and $B$ respectively; the total length of the scintillator $L_{\mathrm{bar}}$ and the directions of the experiment coordinate system are shown. For each timing projection, the most-probable value of the $dt_{\mathrm{SiPM}}$ projection is taken as the central timing value for each value of $x$, with the uncertainty in time taken as the standard error on the mean.

A linear fit is then applied such that $dt_{\mathrm{SiPM}}$ is written as a function of $x-$position, $x$:

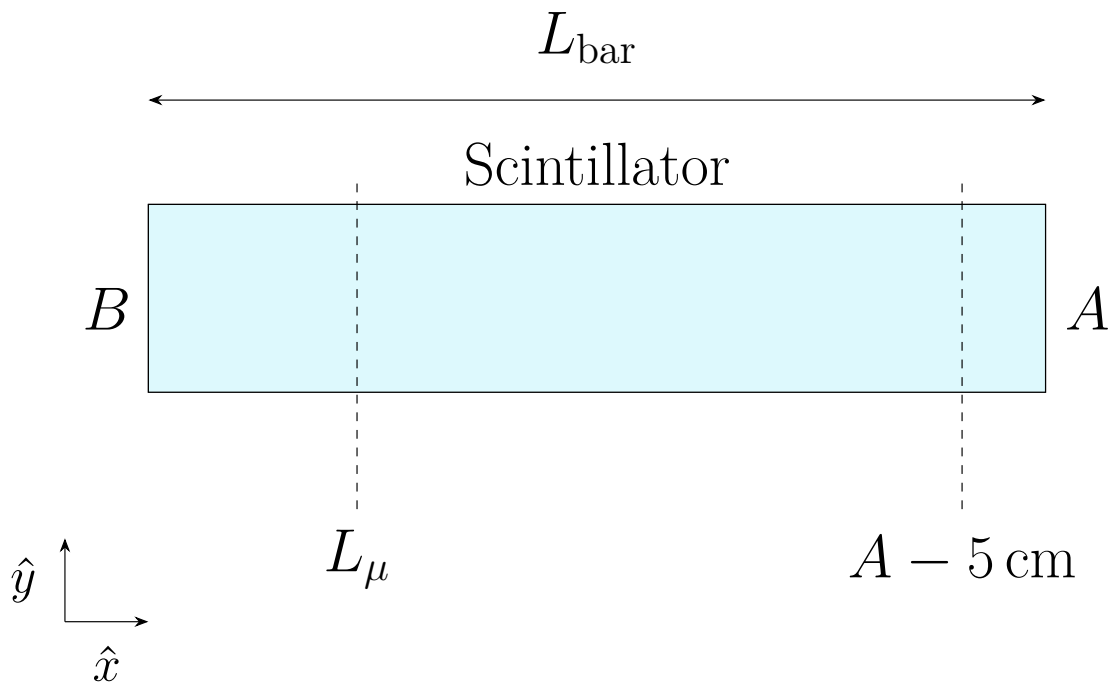



Figure 4.6: Schematic of a scintillator illustrating what the fit range used to determine the effective signal speed in the scintillator is. The figure is not to scale.

$$dt_{\text{SiPM}} = \frac{x}{c_{\text{SiPM}}} + \text{const.} \tag{4.1.4}$$

Where const is the constant offset of the fit. The effective speed of light in the scintillator is extracted as:

$$c_{\text{SiPM}} = \left\{ \begin{array}{ll} 1/m \pm \delta m/m, & \text{if SiPM on left} \\ -1/m \pm \delta m/m, & \text{if SiPM on right} \end{array} \right\} \tag{4.1.5}$$

Where $m$ and $\delta m$ are the gradient and the error on the gradient from the fit. The factor of $-1$ distinguishing SiPM channels on the left and right side is due to SiPMs on the right side where the signal travels in the negative $x$-axis giving a negative gradient. The fit residuals shown in the lower half of figure 4.5 highlight that the statistical uncertainties, taken from the standard error on the mean in each bin of $x-$position, are entirely unable to explain the observed deviation from the simple linear model. Therefore, some systematic uncertainty is unaccounted for. A result of this is that the uncertainty on the fitted parameters cannot physically represent the system. To obtain a more empirical measurement of the uncertainty on the signal speed, the $x-$range $[A-5,\, L_\mu]$ is divided into five equal-length segments, $c_{\text{SiPM}}$ is extracted, and the final uncertainty quoted as the square-root of the variance of the sub-measurement set.

$$\Delta c^{\text{SiPM}}_{\text{signal}} = \sqrt{\frac{\sum_i^N (c_i - c_\mu)^2}{N}} \tag{4.1.6}$$

Where $c_i$ is the set of effective signal speeds measured in the five segments and $c_\mu$ is the mean of this set.

### 4.1.4 Time-walk parameterisation and fitting

The presence of the time-walk effect is made clear by the correlation displayed between $dt_{\text{SiPM}}$ and $\text{QDC}_{\text{SiPM}}$, as shown in figure 4.7. To produce a histogram such as figure 4.7, one must correct for the time-of-flight of the signal photons, $t^{\gamma}_{\text{SiPM}}$, by using the value of $c_{\text{SiPM}}$ determined as described in the previous section, otherwise the time smearing coming from the different photon flight times through the scintillator would entirely dilute the time-walk effect. The signal time-of-flight is corrected to the bar-centre point, which is arbitrarily chosen but will return to the discussion in section 4.3.5. The SiPM time corrected by the scintillation photon time-of-flight (ToF), $t^{\text{ToF}}$, is given by:

$$t^{\gamma}_{\text{SiPM}} = \frac{x_{\text{track}} - x_{\text{ref}}}{c_{\text{SiPM}}} \tag{4.1.7}$$

$$t^{\text{ToF}} = \begin{cases} t_{\text{SiPM}} + t^{\gamma}_{\text{SiPM}}, & \text{for SiPMs on left,} \\ t_{\text{SiPM}} - t^{\gamma}_{\text{SiPM}}, & \text{for SiPMs on right} \end{cases} \tag{4.1.8}$$

With this photon time-of-flight correction applied to the US SiPM time, the quantity of time used to display the time-walk effect is $dt^{\text{ToF}}_{\text{SiPM}} = t^{\text{DS}}_0 - t^{\text{ToF}}_{\text{SiPM}}$. A weighted histogram from the 2D plot is constructed here in a similar manner to what was performed and outlined in section 4.1.3. A projection on the $dt^{\text{ToF}}_{\text{SiPM}}$ axis is made for each bin of $\text{QDC}_{\text{SiPM}}$ and the modal value of time is set as the bin contents with the bin error given by the standard error on the mean. With the weighted histogram formed, the time-walk correction function shown in equation 4.1.9 is fitted:

$$\Delta t^{TW}(\text{QDC}) = t_0 + \frac{\alpha}{\beta \times \text{QDC} - \text{QDC}_0} + \gamma \times \text{QDC} \tag{4.1.9}$$

Using the Desmos graphing calculator [129], parameter ranges were established and are shown in table 4.2. These parameter ranges were selected to allow sufficient flexibility through the observed $\text{QDC}_{\text{SiPM}}$ domain.

| Parameter | Lower limit | Upper limit |
|---|---|---|
| $t_0$ | $\frac{1}{2} \times t_0^{\text{estimate}}$ | $\frac{3}{2} \times t_0^{\text{estimate}}$ |
| $\alpha$ | -500 | 0 |
| $\beta$ | 0 | 30 |
| $\text{QDC}_0$ | -10 | 0 |
| $\gamma$ | 0 | 1 |

Table 4.2: Upper and lower parameter limits for fitting the time-walk function, equation 4.1.9.

The estimate for $t_0$, $t_0^{\text{estimate}}$, is made using the high-QDC data where the middle term of equation 4.1.9 will diminish; $\Delta t^{TW}$ at high-QDC will mostly reflect $t_0$ as it is assumed that $0 < \gamma < 1$. The starting values of the parameters are simply set as the midpoints of the allowed range. The fit is performed over the QDC range $[0, \text{QDC}_{\text{max}}]$, where , $\text{QDC}_{\text{max}}$ is the last bin before there are fewer than 10 entries. The choice of 10 is arbitrary and could be increased up to approximately 100 before this would have a substantial impact upon the fit range.

Only the statistical uncertainty is accounted for in this process which was found to not be sufficient, this is discussed more in section 4.3.4. Due to this, the fitting

process occurs in two stages: firstly, the time-walk function is fitted by ignoring the error bars, fitting by minimising the residuals between data and expected value, thus fitting only the shape of the time-walk effect. These resulting fit parameters are then used with the measured data to produce a maximum likelihood estimate (MLE) for the unaccounted for systematic uncertainty, $\sigma_{\text{sys}}$. This estimate is then used with the statistical uncertainty to produce a $\chi^2$ fit where the total error in each bin is given by $\sigma = \sqrt{\sigma^2_{\text{stat}} + \sigma^2_{\text{sys}}}$. The arguments behind this method, the mechanics of the MLE used to produce the estimate of the systematic uncertainty and the results of this method throughout the US system are discussed in 4.3.4. The resulting fitted parameters and systematic uncertainty estimate are then written out to file for subsequent analysis stages.

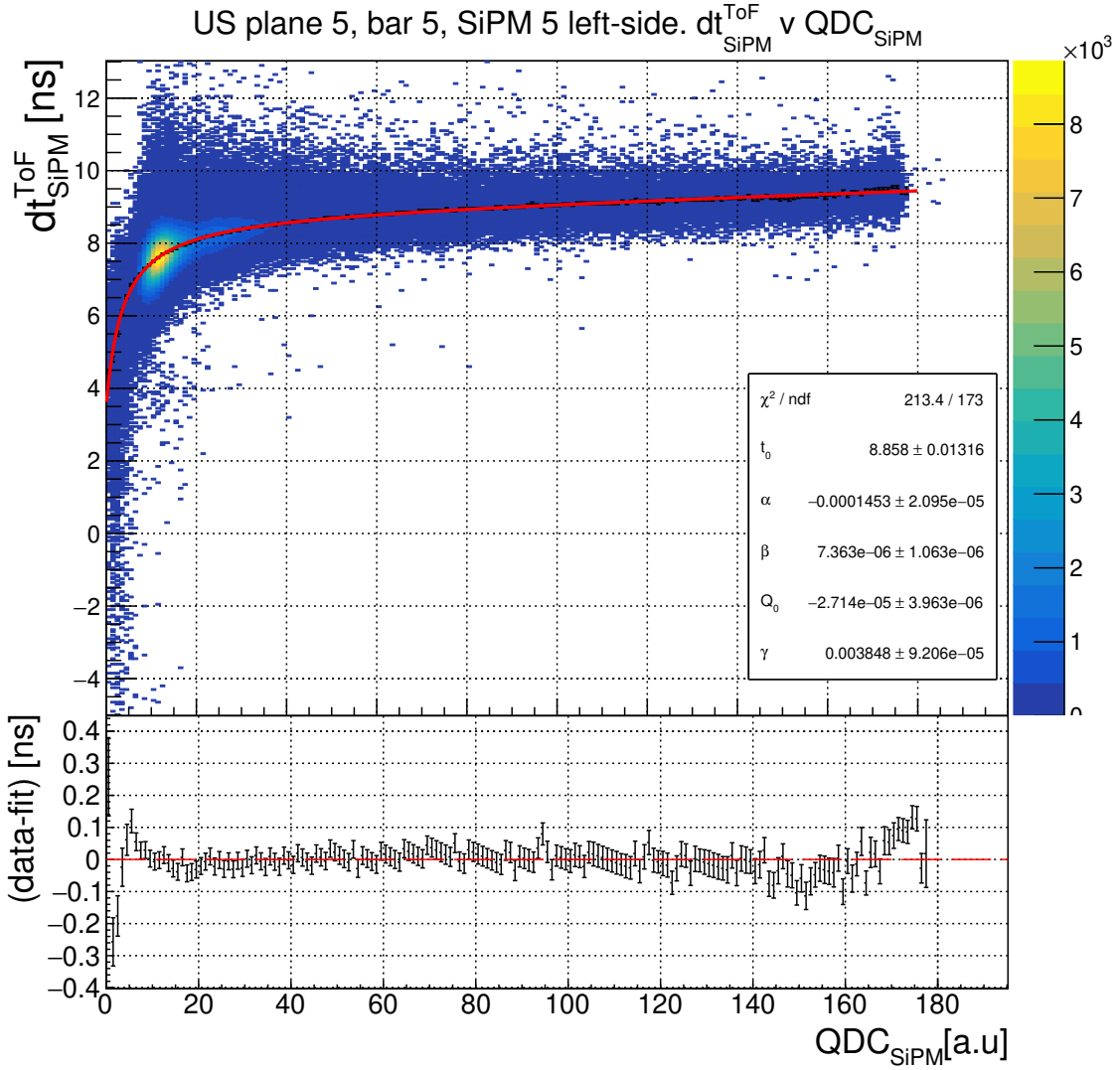



Figure 4.7: QDC of a single SiPM correlated against $dt^{\text{ToF}}_{\text{SiPM}}$. The lower pane shows the ratio of the data to the fit for each QDC point.

A final point that is evident from figure 4.7 is the insufficiency of the correction procedure at very low values of QDC: there is a very long albeit sparsely populated tail in negative values of $dt^{\text{ToF}}_{\text{SiPM}}$. The presence of this tail induces a small effect

downstream in the analysis that is returned to for discussion in section 5.2.2.

### 4.1.5 Determination of time resolution

The correlation histograms for $dt_{\text{SiPM}}$ and $\text{QDC}_{\text{SiPM}}$ such as figure 4.7 also allow for the time resolution to be evaluated. To do this, the projection of this histogram on the $dt_{\text{SiPM}}$ axis is made with the time resolution of the SiPM extracted in two steps: firstly, the full-width at half maximum (FWHM) is taken, divided by 2.355, to give the time resolution of $dt_{\text{SiPM}}$ and then the resolution component from the muon system reference time is statistically subtracted:

$$\sigma_t^{\text{SiPM}} = \sqrt{\left(\frac{\text{FWHM}}{2.355}\right)^2 - \sigma_{t_0^{DS}}^2} \tag{4.1.10}$$

Where the resolution on $t_0^{DS}$ is measured through the time differences between successive horizontally aligned planes of the muon system labelled $i$ and $j$:

$$\Lambda_{ij} = t_{\text{DSH j}} - t_{\text{DSH i}}$$
$$\sigma_{t_0^{\text{DS}}} = \frac{1}{\sqrt{6}}\sigma_{\Lambda_{ij}}$$

A full derivation of this resolution can be found in appendix A.6. The value extracted as the resolution on $t_0^{\text{DS}}$ is 132 ps. An example of the time resolution of a SiPM following the time-walk correction is shown in figure 4.8.

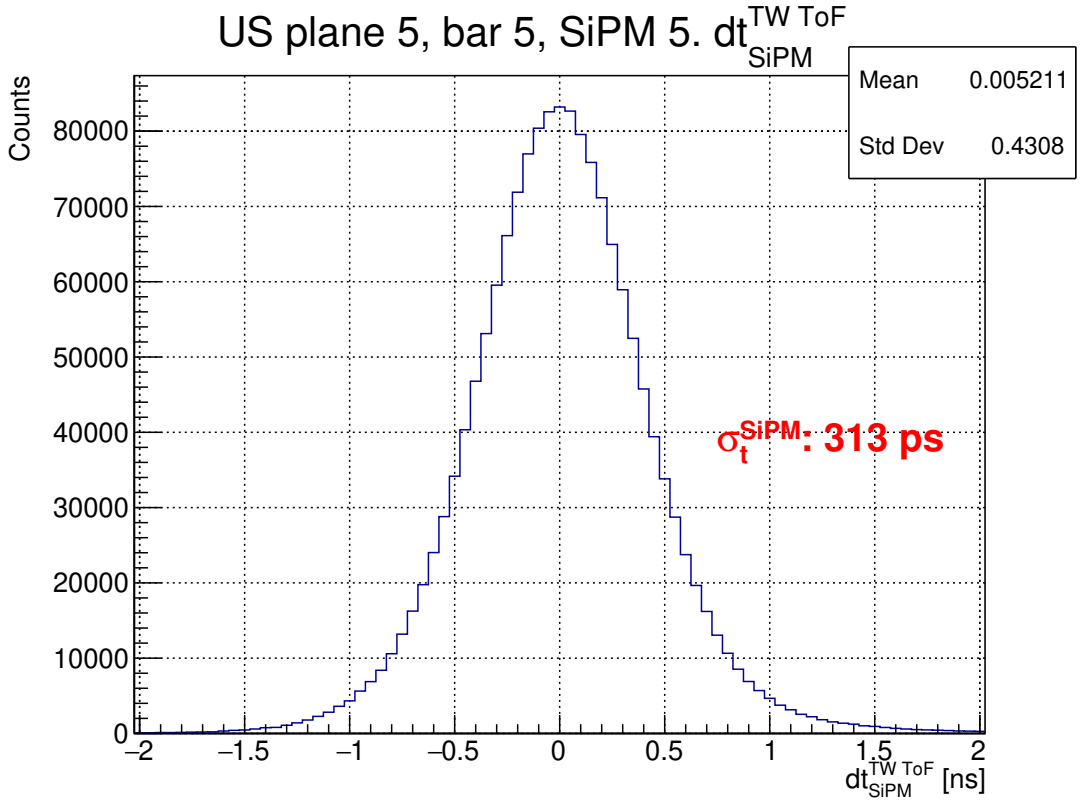



Figure 4.8: Time resolution of a SiPM with signal time-of-flight correction, time-walk correction and its fine alignment parameter applied. Time resolution extracted according to equation 4.1.10.

### 4.1.6 Time-walk corrected data

Time-walk correction allows for the $c_{\text{SiPM}}$ to be re-evaluated, and then the time-walk corrected time resolution to be measured. Both of these values are determined identically to how is described earlier in this chapter.

Equation 4.1.9 includes a parameter to allow for a shift in the $dt^{\text{ToF}}_{\text{SiPM}}$ axis, $t_0$. Once the parameters for a given channel are determined, $t_0$ is not used for the time walk correction as this parameter only to allow sufficient flexibility in the model. The time-walk corrected SiPM time, $t^{\text{TW}}_{\text{SiPM}}$, is determined as:

$$t^{\text{TW}}_{\text{SiPM}} = t_{\text{SiPM}} + \frac{\alpha}{\beta \times \text{QDC} - \text{QDC}_0} + \gamma \times \text{QDC} \tag{4.1.11}$$

The time-walk corrected time taken in reference to the DS muon system reference time is written as $dt^{\text{TW}}_{\text{SiPM}} = t^{\text{DS}}_0 - t^{\text{TW}}_{\text{SiPM}}$. The effect of the time-walk correction is visualised by correlating the values of $dt^{\text{TW ToF}}_{\text{SiPM}}$ (the superscripts TW and ToF indicate both the photon time-of-flight correction and the time-walk corrections are applied) against the $\text{QDC}_{\text{SiPM}}$ value.

### 4.1.7 Fine alignment

With all US SiPMs having a time-walk correction applied. There remains an offset in time with respect to the $t^{\text{DS}}_0$. The final alignment, henceforth referred to as the fine alignment, to correct US SiPM times to be statistically aligned with the DS muon system is required for the full timing power of a scintillator to be utilised.

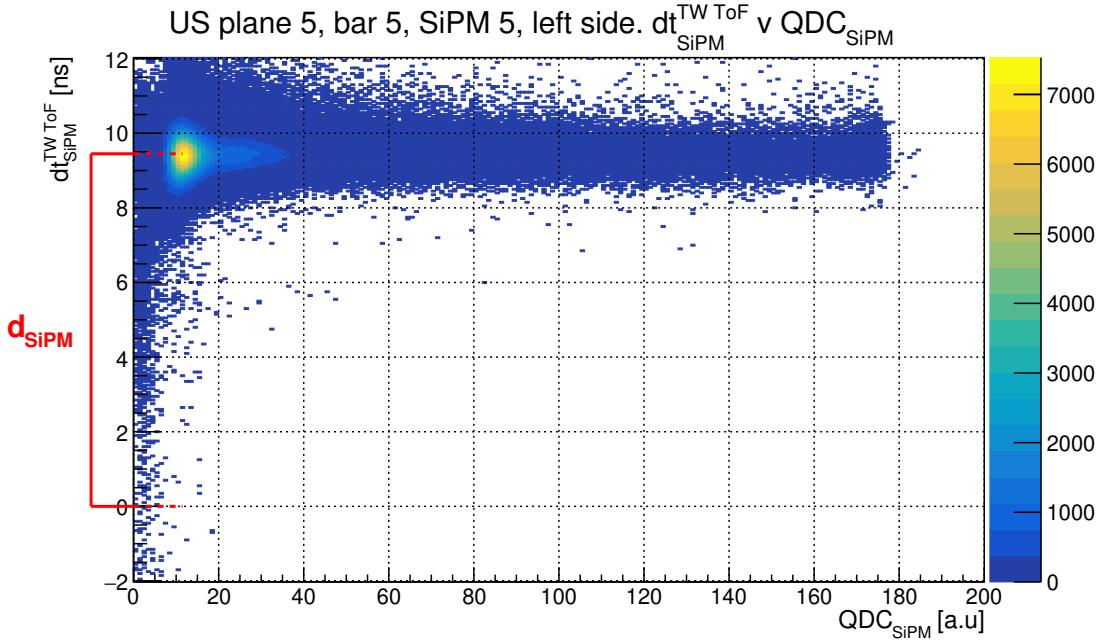



Figure 4.9: Definition of the alignment parameter. Using a centrally positioned SiPM on the left side of the 5th bar of the 5th plane of the US system as an example.

Utilising histograms of $dt^{\text{TW ToF}}_{\text{SiPM}}$ v $\text{QDC}_{\text{SiPM}}$, an example is shown in figure 4.9, the projection on the $dt^{\text{TW ToF}}_{\text{SiPM}}$ axis is taken. From this projection, a $\pm 1$ standard deviation region around the modal value of the $dt^{\text{TW ToF}}_{\text{SiPM}}$ projection is selected. The alignment parameter with respect to the DS muon system, $d_{\text{SiPM}}$, is determined as

the mean of this truncated region with an associated error given by the standard error on the mean within the truncated region. The DS alignment parameter is then subtracted event-wise from $dt_{\text{SiPM}}^{\text{TW ToF}}$, having the effect of shifting the distribution to be centred at zero:

$$E[dt_{\text{SiPM}}^{\text{TW ToF}}] - d_{\text{SiPM}} \approx 0 \tag{4.1.12}$$

Where $E[dt_{\text{SiPM}}^{\text{TW ToF}}]$ is the modal value of the distribution. Just as the timing of the US can be aligned to the muon system, the alignment can also be done to the SciFi. In order to do so, the measurement of an event-wise reference time in the SciFi given in equation 4.1.2 is used. When the US SiPM times are aligned to the SciFi and the corresponding SciFi alignment parameters are determined, these are denoted as $d_{\text{SiPM}}^{\text{SF}}$ to distinguish from the DS alignment parameters, $d_{\text{SiPM}}$. The fine alignment to the SciFi will be used in analysis in chapter 6.

Sets of SciFi alignment parameters were measured for the US in order to allow analysis where the DS muon system is not expected to be active. The method for doing so is exactly the same as described here in this section, the time-walk does not need to be re-evaluated depending on which subsystem provides the reference timing. These alignment parameters are referred to as the DS/SF alignment parameters depending on whether the DS or SciFi reference time is used.

Full scintillators, sides of scintillators of individual SiPMs timing measurements that contain the application of SiPM fine alignment constants will be henceforth referred to as aligned.

### 4.1.8 System alignment

With all SiPMs in the US aligned to the rest of the electronic subsystems of the experiment, the time resolution of a full scintillator in the detector can be used by combining the measurements. The time-walk corrected and aligned time measured by the side of a scintillator , $dt_{\text{side}}^{\text{ToF}}$, is defined as an average of the time-walk corrected SiPM times with their respective fine alignment constants applied:

$$dt_{\text{side}}^{\text{ToF}} = \frac{1}{N_{\text{side}}} \sum_{\text{SiPM}}^{N_{\text{side}}} \left( t_0^{\text{DS}} - t_{\text{SiPM}}^{\text{TW ToF}} - d_{\text{SiPM}} \right) \tag{4.1.13}$$

The superscript TW denoting the application of the time-walk correction is now to be assumed, there is no use-case of not applying to time-walk correction, whereas the application of the time-of-flight correction remains application specific and its application will remain denoted by a superscript ToF. By combining the SiPM timing measurements, the time resolution measured by each side of a scintillator will reduce by a maximum of $\frac{1}{\sqrt{N_{\text{side}}}}$ if there is no cross-talk between the SiPM channels, as one can deduce from the equations in appendix A.9.

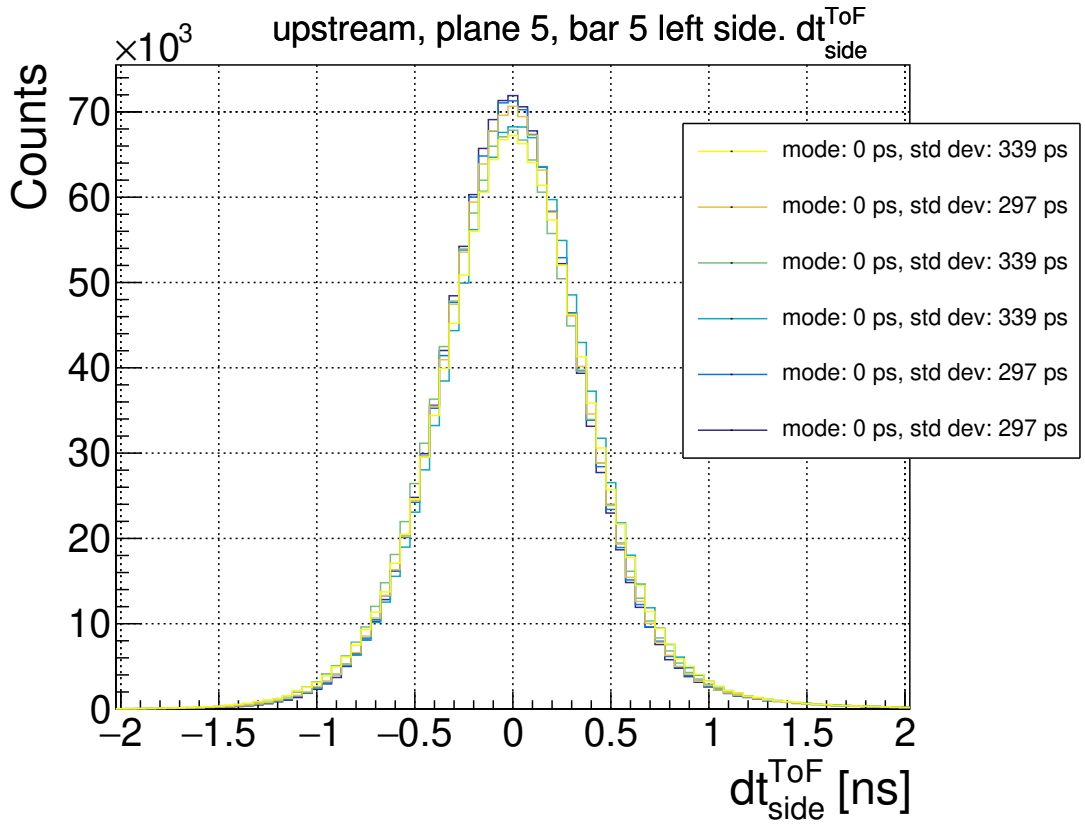



Figure 4.10: Corrected and aligned SiPMs coupled to the same side of a scintillator in the with the modal times and standard deviations shown.

Figures such as figure 4.10 show the remaining offset between SiPMs in the US system is negligible relative to the time resolution. Now the SiPMs can be treated like independent measurements and used to measure an average time which will reduce the time resolution.

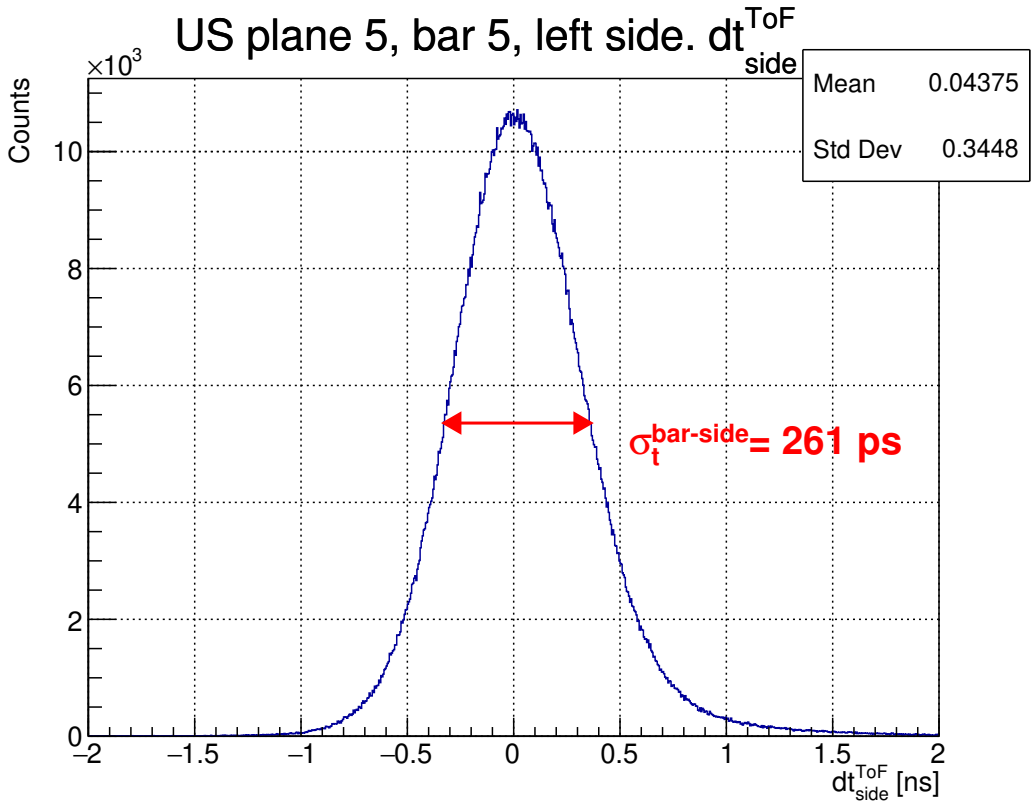



Figure 4.11: Average time measured by fired SiPMs on left side of scintillator. The time resolution is extracted according to equation 4.1.10.

Figure 4.11 shows the distribution $dt_{\text{side}}^{\text{ToF}}$. Having combined the SiPM timing measurements for the six large area SiPMs on the left side of this scintillator, the time resolution reduces by a less than one would expect from the averaging of 6 independent measurements:

$$\frac{\sigma_t^{\text{bar-side}}}{\sigma_t^{\text{SiPM}}} = \frac{261}{313} > \frac{1}{\sqrt{6}}$$

Where the time resolution of $dt_{\text{side}}^{\text{ToF}}$ is denoted $\sigma_t^{\text{bar-side}}$ and the time resolution of $dt_{\text{SiPM}}^{\text{ToF}}$ is denoted $\sigma_t^{\text{SiPM}}$. This behaviour indicates that the SiPM timing measurements cannot be considered independent and that cross-talk is present somewhere in the system. The extent and origin of this cross-talk is discussed more in section 4.3.6.

## 4.2 Laser measurements

Having observed the intrinsic time-walk effect of the SiPMs in the PCB and consequences of cross-talk, dedicated laser measurements of the SiPMs were produced in order to understand the origins of these behaviours and how they are impacted by factors such as the intensity of light incident upon the SiPM and the number of neighbouring SiPMs are being illuminated. Beyond the time-walk and alignment, open questions such as the behaviour of the small area SiPMs would be studied by use of the laser data as well. To use the PCB, the SND@LHC DAQ server must be run on a PC so that the data can be transmitted and events can be constructed using the online event builder (see section 3.3). This data was collected at École Polytechnique Fédérale de Lausanne (EPFL).

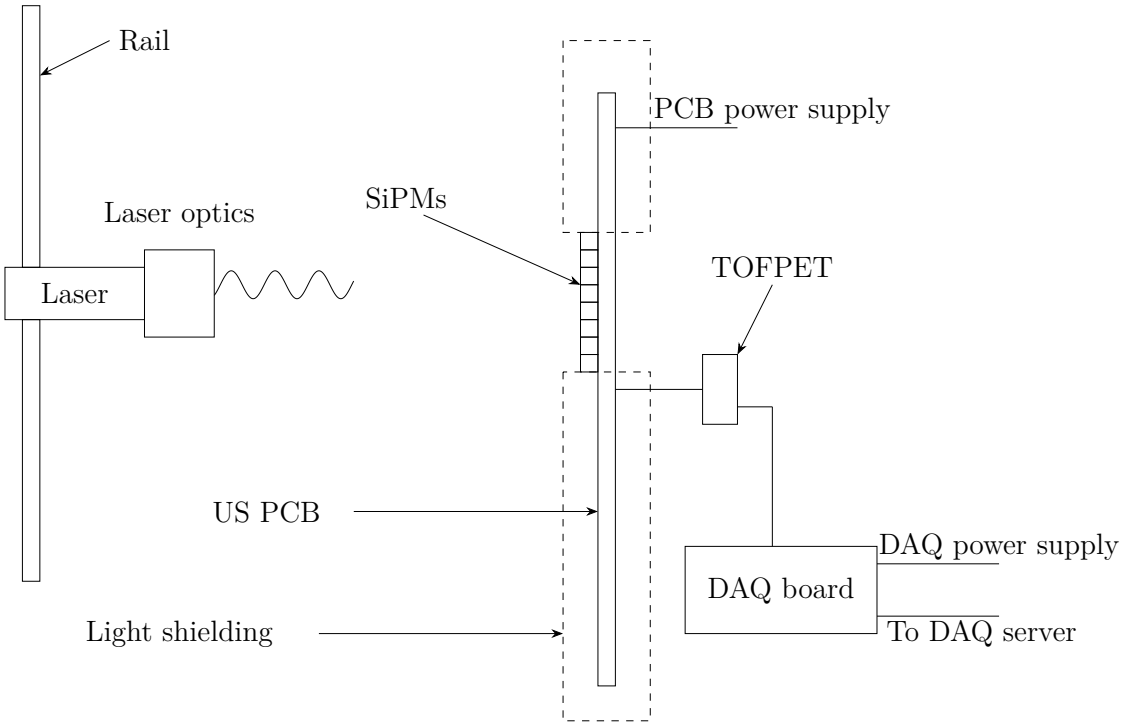



Figure 4.12: Schematic of the experimental set-up inside the dark room at EPFL. The items in this schematic are not to scale. There are eight SiPMs visible to the laser: the one-bar illumination regime.

The goals of the data-taking period where to create datasets where two main control variables were manipulated: number of fired neighbouring SiPMs and the intensity of light incident upon the SiPM(s). From data collected in TI18, it is known that when a relativistic charged particle traverses a scintillator the majority of the SiPMs coupled to that scintillator will register data[2]. For this reason, when studying the number of fired neighbouring SiPMs the natural increment of firing SiPMs is by the number coupled to each side of a scintillator, eight, how many groups of eight SiPMs are illuminated by the laser is referred to as the illumination regime. By varying the intensity of light incident upon the PCB and studying the measured time and QDC, the time-walk in the lab environment - with no scintillator - can be evaluated.

### 4.2.1 Laser calibration

*The data collection for the laser calibration was performed by a PhD student at EPFL, not the author. The calibration procedure was performed by the author and the PhD student who collected the calibration data.*

The laser produces light at 450 nm with a pulse frequency of 10 kHz. To calibrate the laser, data were taken with two SiPMs[3]: one SiPM was placed directly in-front of a laser and the other SiPM was positioned with a transverse shift, simply illustrated with the schematic in figure 4.13. The transverse shift means that SiPM 2 (as labelled in figure 4.13) will receive far less light, thus not saturating at high laser intensities.

[2]This is shown later in figure 5.24

[3]These SiPMs are unrelated to the US system of SND@LHC

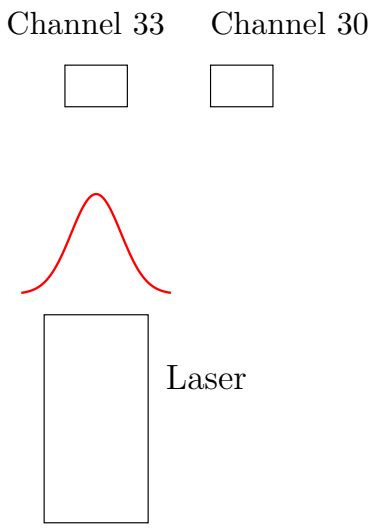
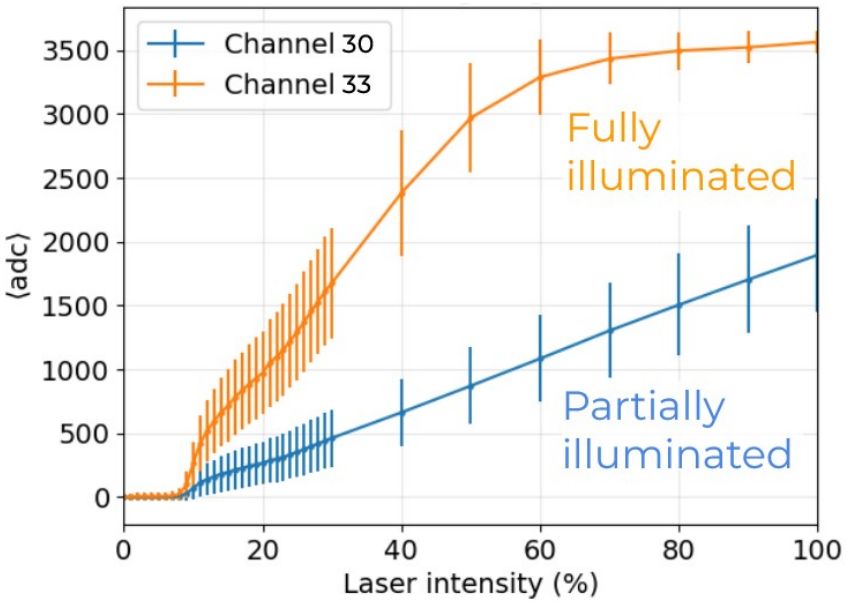



Figure 4.13: Schematic illustrating the positions of the two SiPMs used for calibrating the laser. The red Gaussian simplistically illustrates the Gaussian beam profile of the laser output.

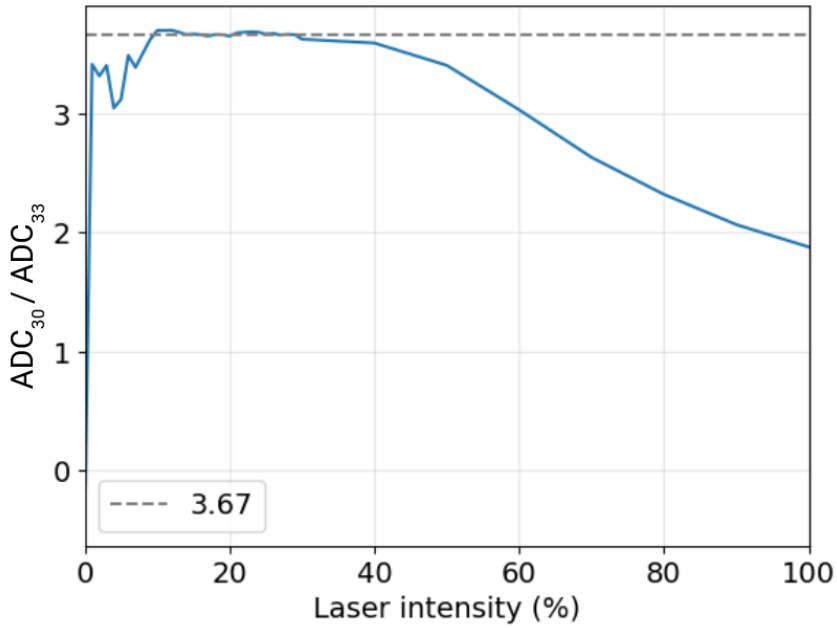
The two SiPMs in the calibration set-up do not measure QDC like the SiPMs in the SND@LHC US system, instead they measure an analogue-digital-converter (ADC) value. While QDC measures charge by estimating the integrated current with respect to time, ADC is a method of estimating the peak voltage. Both QDC and ADC are signal processing values which are proportional to the avalanche charge (proportionality notwithstanding saturation effects). The ADC values were recorded scanning through the intensity of the laser in steps of 1% from 0% to 30%, then in 10% steps from 30% to 100%. Each stage in intensity records $10^4$ events with the mean and standard deviation extracted.

There are two stages to the calibration: the low intensity fluctuations and the high intensity saturation. Both of these effects are clear in figure 4.14a. Firstly, the calibration deals with the saturation observed in channel 33 for high intensities, shown in figure 4.14a. The saturation is accounted for by measuring the ratio of ADC values of the two channels, $\mathrm{ADC}_{30}/\mathrm{ADC}_{33}$, and using this ratio to estimate what ADC would be measured by the saturating channel if the dynamic range was sufficiently large. The value of the ratio used to correct for saturation is the average ratio computed in the laser intensity range where the ADC values of both channels appear proportional, approximately between 15% and 30%. This first step was not performed by the author, the result of correcting the saturation with this method is shown in figure 4.15.

Figure 4.15 shows clearly that intensities $\lesssim$ 20% show fluctuations from the linear relation shown at larger laser intensities. These low-intensity fluctuations are corrected for by performing a linear fit in the range [20%, 100%] and then performing a linear regression using the measured ADC values to determine an estimate of laser intensity received by channel 30, this is the calibrated laser intensity. The result of this linear regression is shown in figure




(a) ADC measured on two calibration channels plotted against the relative laser intensity. The error bars here show the standard deviation.




(b) The ratio of ADCs, $ADC_{30}/ADC_{33}$. The average value of the ratio in the approximately proportional region [15%, 25%] is shown to be 3.67.

Figure 4.14: Both figures here were not produced by the author but by the PhD student who collected the data at EPFL.

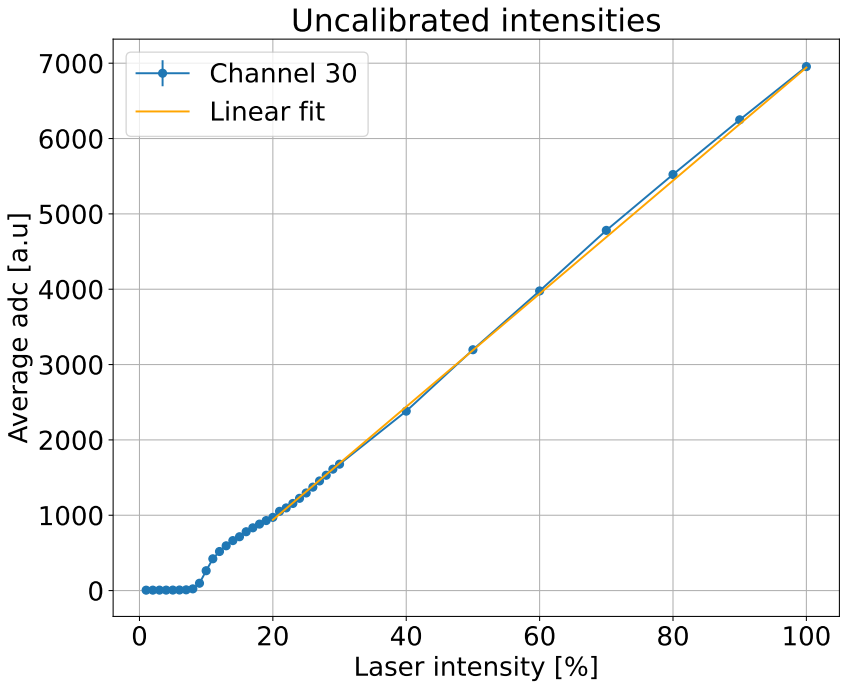



Figure 4.15: Plot of the ADC recorded by channel 30 for laser intensities between 0 and 100% following the correction of saturation effects using the ratio of ADC values between the two calibration channels. The linear fit is performed in the range [20%, 100%].

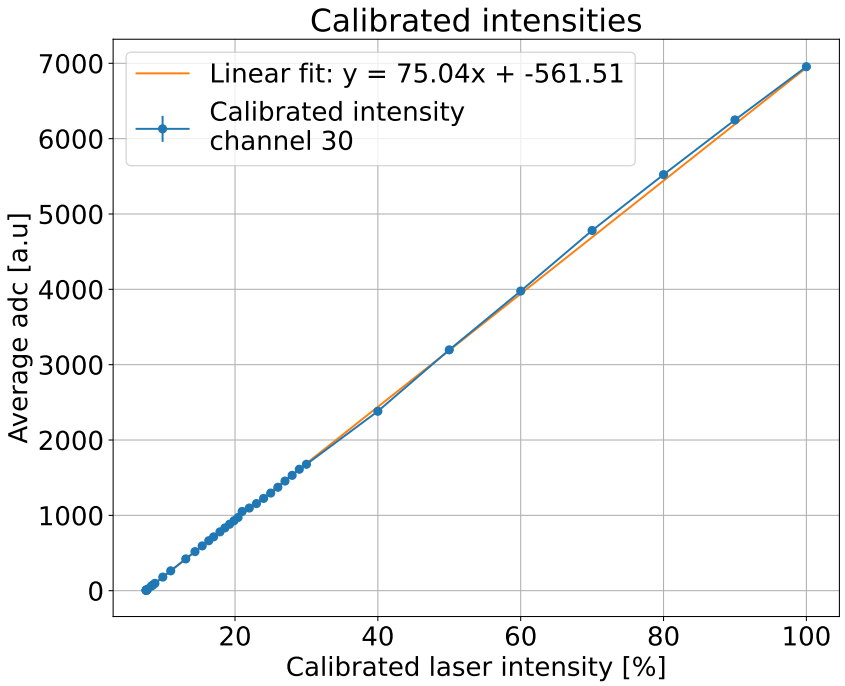



Figure 4.16: Plot of ADC recorded by channel 30 and the calibration laser intensities following the linear regression defined in the legend. The small discontinuity around 20% is an artefact of the calibration process with negligible impact on the resulting analysis.

With both stages of calibration complete, the laser intensity can be converted to a number of photons per unit area using:

$$n_\gamma/A = \mathrm{ADC}_{33} \times \frac{1}{G_{33}} \times \frac{1}{A_{33}} \times \frac{1}{\mathrm{PDE}_{33}} \tag{4.2.1}$$

Where $\mathrm{ADC}_{33}$ is the ADC measured by channel 33, the fully illuminated channel corrected for saturation. $G_{33}$, $A_{33}$ and $\mathrm{PDE}_{33}$ are the gain, area and photon detection efficiency of the calibration channel. For maximal cross-over between the laser data and TI18, it had to be ensured that the full area of the SiPM was being uniformly illuminated by the laser and that the largest range of laser intensity could be used without the SiPM entering saturation. In order to do so, neutral density and ground glass diffusion filters were attached to the end of the laser optics to ensure that the SiPMs were uniformly illuminated. The set of diffusion filters used were kept as fixed through the data taking campaign, as were multiple other parameters summarised in table 4.3.

| **Parameter name** | **Value** |
|---|---|
| Photon wavelength | 450 nm |
| Pulse frequency | 10 kHz |
| Trigger phase | 10 cc |
| Run length | 5 s |
| Optics | n=1.5, 1720 DG |

Table 4.3: Fixed parameters during laser data taking and their values.

The pulse frequency refers to the frequency of the laser, the trigger phase refers to the fixed phase difference between the DAQ clock and the signal sent to the laser. The optics configuration encodes the neutral density filter, $n$, and the total diffusion grade $DG$ of the applied ground glass filters.

With a calibrated laser, data was collected for each SiPM alone at uncalibrated relative intensity points for the SiPMs in appendix A.11. The data with multiple SiPMs firing were collected subsequently where light shields allowed the laser to illuminate one, two or three groups of eight SiPMs, henceforth referred to as one, two or three-group illuminations. The multi-SiPM data sets were studied at the same uncalibrated intensity points as the large SiPMs. All 10 of the groups of eight SiPMs were measured in one-group illumination, the two and three-group illumination combinations that were investigated can be found at appendix A.12. The multi-group combinations measured were all combinations of adjacent groups in order to have the most carry over to experimental data.

### 4.2.2 Pre-processing

At this point the data has run through the event builder, but is still in the raw data format and preprocessing is required before the final analysis stages. As each run with the laser corresponds to a fixed light intensity per event: to study how, for example, light intensity affects the recorded time on the SiPM, one must extract one data point of time and QDC from each run. To do this, the correlation of QDC and time is studied, and the modal values from each projection are extracted to form this data point. An example of the correlation histograms from which the data are extracted is shown in figure 4.17.

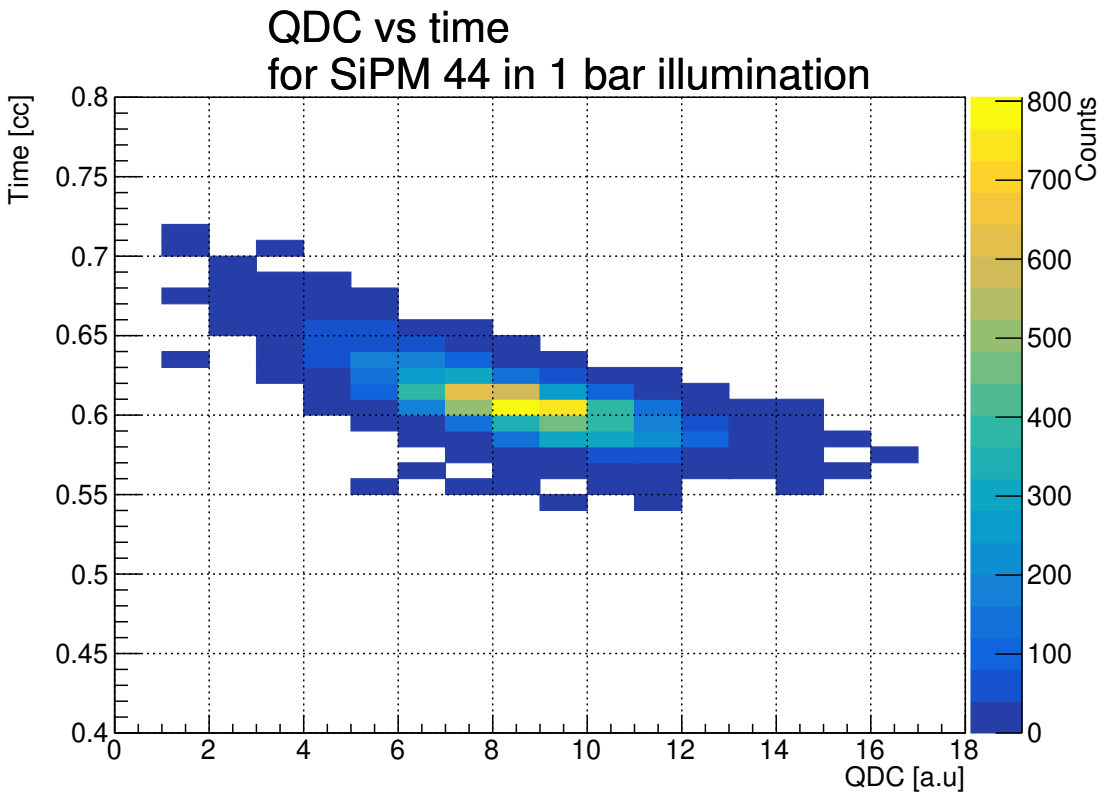



Figure 4.17: Time versus QDC for a single laser run for a single SiPM on the PCB. The unit "[cc]" refers to the fraction of the 6.25 ns clock-cycle.

### 4.2.3 Data analysis

The time-walk in the lab across the full dynamic range of the SiPM is then done by investigating the collection of data points for a given SiPM with a given illumination regime[4]. An example of how the time-walk behaviour changes with respect to the illumination regime as shown in figure 4.18. The causes of the observed offset between the single SiPM illumination and the multi-SiPM illuminations are discussed in section 4.3.7. The QDC saturation curve can be studied as shown in figure 4.19.

[4]Whether only the SiPM in question is illuminated, or how many groups of eight are illuminated.

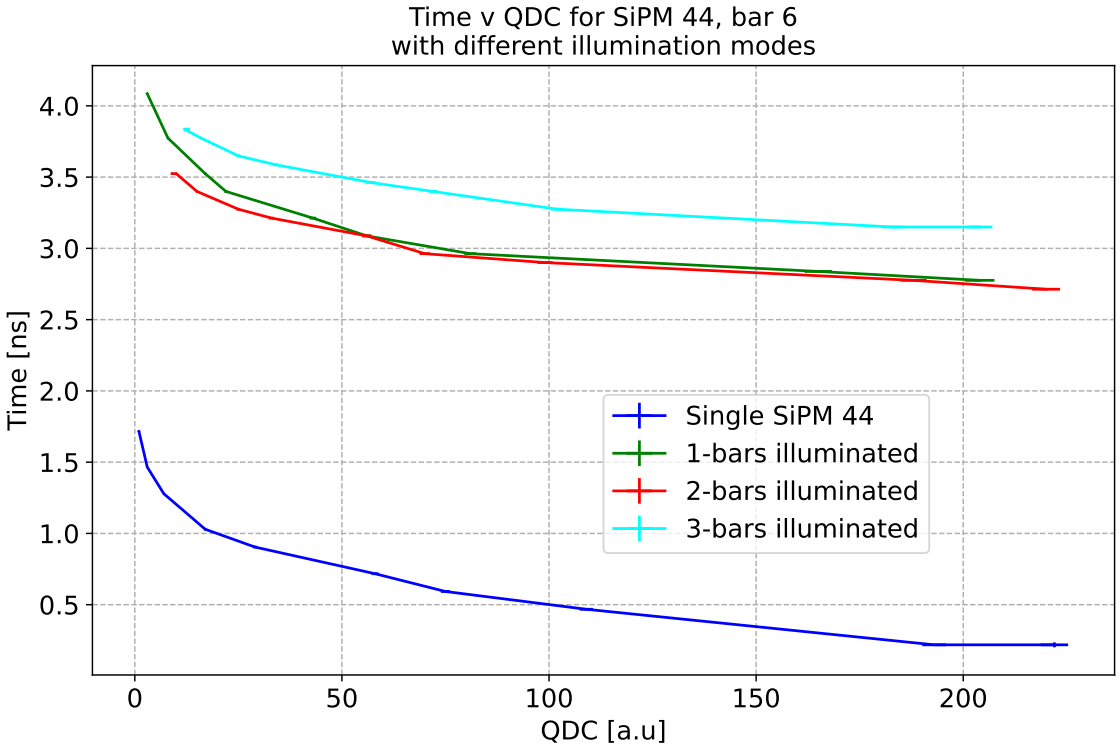
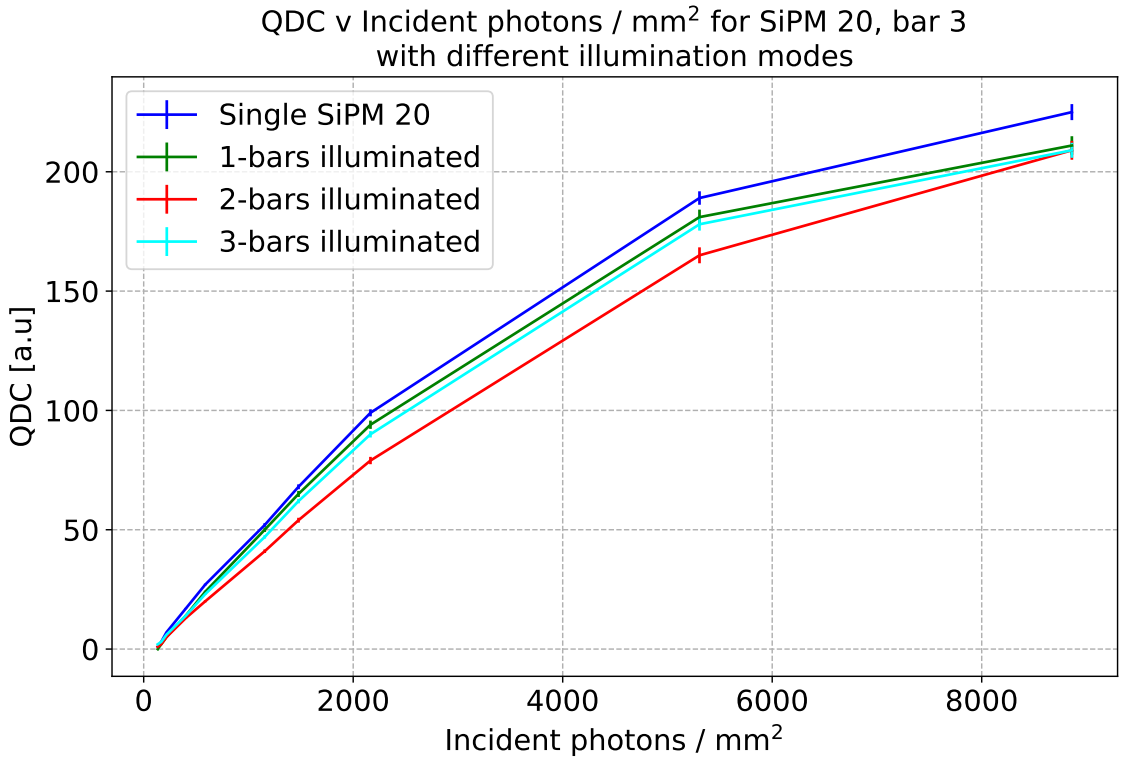
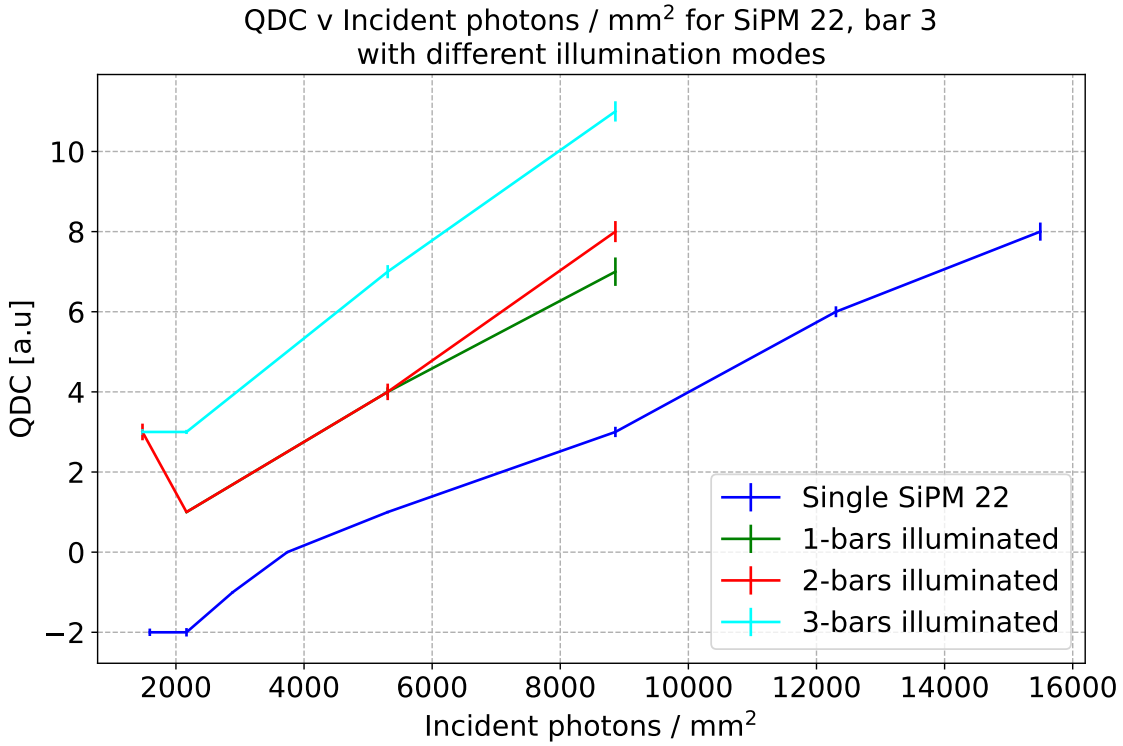



Figure 4.18: Comparison of the time-walk behaviour observed for a given SiPM in the different illumination regimes investigated.

Figure 4.19 displays the independent variable, laser intensity, in terms of the number of incident photons per square millimetre which is established through the calibration process. These tools of comparing time and QDC with different illumination modes comprise the majority of the scope of the results obtained during the dedicated laser data measurement period.




(a) Data-series showing how the QDC saturation curve is affected by the illumination regime for a large area SiPM.




(b) Comparison of how the QDC saturation curve is affected by the illumination regime for a small area SiPM.

Figure 4.19: Comparison of the different illumination modes for a large area SiPM (top) and small area SiPM (bottom).

## 4.3 Results

### 4.3.1 Event selection

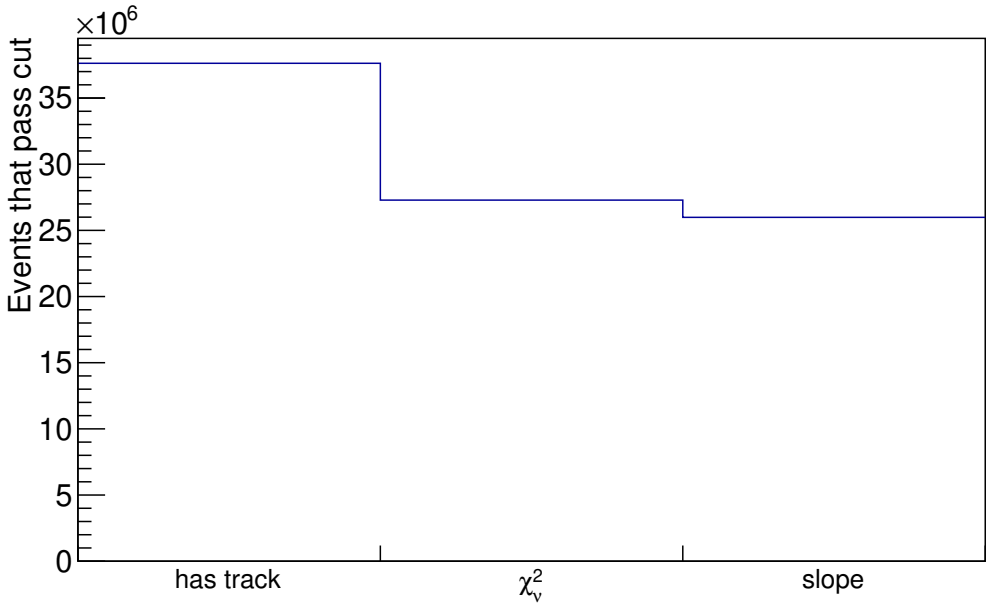



Figure 4.20: Effect of selection requirements applied on events for selecting muons

Figure 4.20 shows the impact of each criterion used in the muon selection process. The first criterion, "has track", selects events in which a track can be fitted using the simple tracking algorithm detailed in section 3.3.6, but also events where the three fired horizontal planes of the muon system also have at least one bar per horizontal plane with both SiPMs firing. The requirement for both SiPMs firing on the horizontally aligned scintillators is to ensure that $t_0^{\mathrm{DS}}$ remains spatially independent.

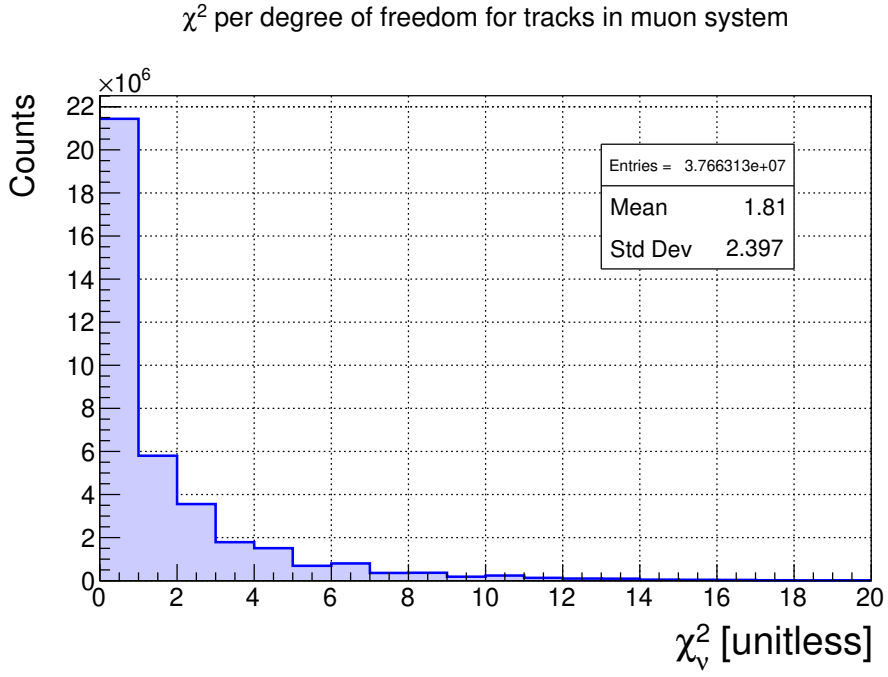



Figure 4.21: Distribution of the fitted track $\chi^2_\nu$.

Figure 4.20 shows furthermore that the requirement of $\chi^2_\nu < 5$ has only a miniscule effect on the statistics in the analysis. The example distribution shown in figure 4.21 is a $\chi^2_\nu$ distribution characteristic of a system with a low number of degrees of freedom which is correct for the fitting in the muon system. Tracks in the muon system of SND@LHC will have one measurement per fired scintillator and there are five fitted parameters, as outlined in section 3.3.6. For muons passing through the detector, there will typically be seven measurements, one per station of the muon system (three horizontal plus four vertical).

$$\text{Degrees of freedom} = N - p = 7 - 5$$

By this stage of the event selection, the final two event selection criteria are almost entirely defunct, but are kept in the analysis for the marginal improvements in data quality that they bring.

No criteria are applied to the SiPMs themselves. However, one key point realised early in the SND@LHC experiment is the absence of the small SiPMs in the data. Whilst the small SiPMs are functionally implemented to deal with the large energy deposition instances associated with showers, even in the cases of large QDC values being recorded following the selection criteria here - potentially caused by muon photo-nuclear interactions or muon bremsstrahlung - the small SiPMs were not active to any useable degree. Unless otherwise stated, the analyses in this thesis is concerned with the large area SiPMs.

### 4.3.2 Effective signal speed in the scintillator

An important metric for success in the evaluation of the effective signal speed in the scintillator is found in reference [130]. This article uses the same model of scintillator as SND@LHC and scintillators of two different cross-sections, one of which is $6 \times 1$ cm$^2$, which is also used for the Veto and US system. Due to this, the value for $c_{\text{SiPM}}$ can be compared to that reported in reference [130] of 15.5 cm/ns. The upper plot in figure 4.22 shows the distribution of $c_{\text{SiPM}}$ before time-walk correction through the system. While the mean value and standard deviation are outwith the value reported in reference [130], the value is not wholly comparable due to differences in the experimental set up. The lower plot in figure 4.22 is used to look for patterns appearing in the value of $c_{\text{SiPM}}$ through the system. The error bars used are those determined as the systematic uncertainty from the sub-ranges used to extract $c_{\text{SiPM}}$ as outlined in section 4.1.3. As shown in the legend, the data is split by left and right side, where each side of each plane corresponds to one PCB. The PCBs of the which include a Wacker SilGel interface between scintillator and SiPM are shown in table 3.1, consulting with this table highlights that there is no effect from the gel on the measured signal speed. It is helpful here to recall that the main runs used for the analysis in this chapter are 5097, 5408 and 5999.

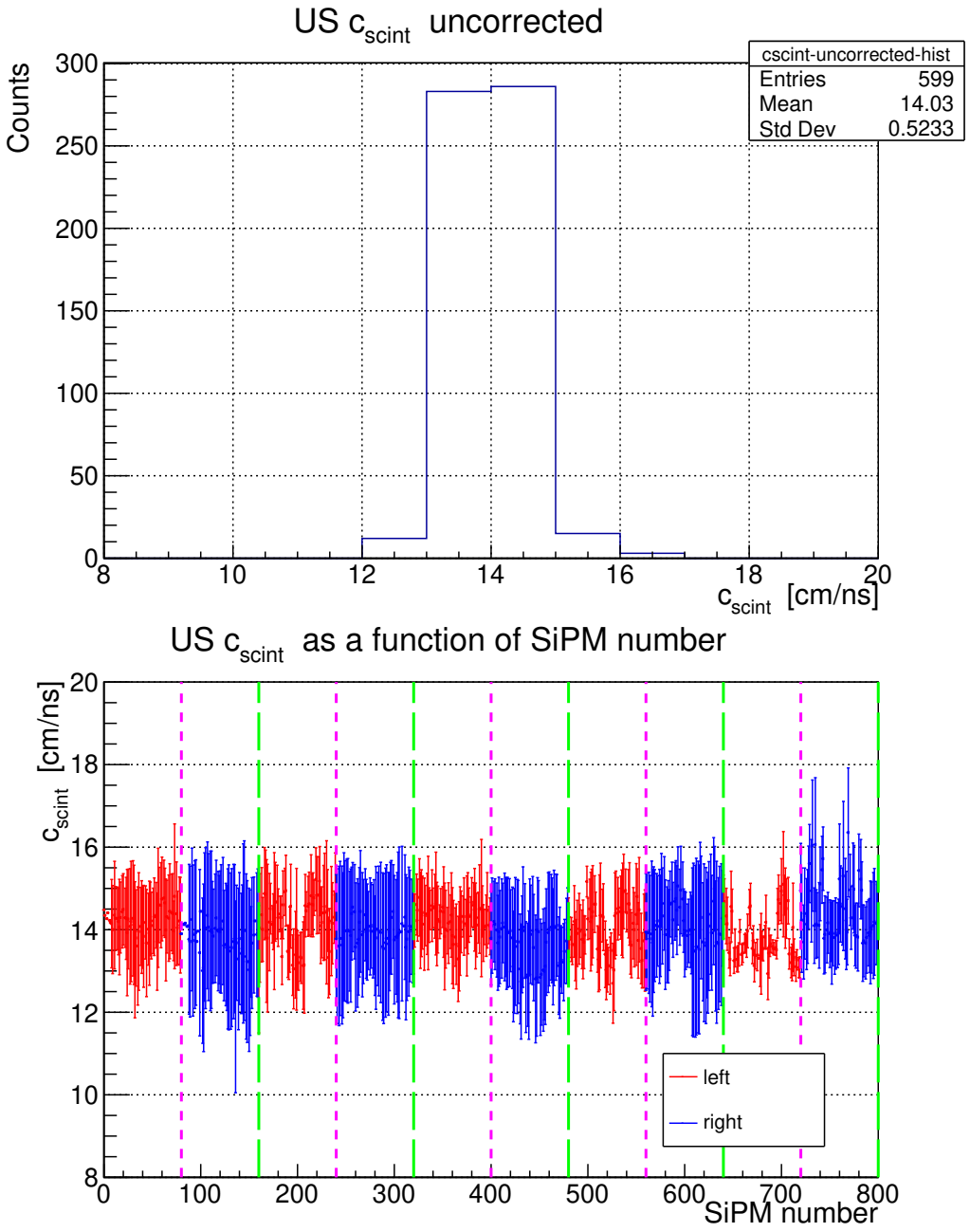



Figure 4.22: Distribution of measured effective signal speeds with no time walk correction applied for run 5408. Error bars correspond to the estimated systematic uncertainty.

The systematic uncertainty estimate described in section 4.1.3 assumes that the underlying true effective signal speed is independent of the segment of scintillator used to measure it. This assumption, and therefore the linear model, are not entirely accurate. The effective signal speed systematic uncertainty is estimated by measuring the effective signal speed in five equal-length segments of the scintillator. The systematic uncertainty estimation assumes that these five measurements in equal-length segments of the scintillator are identical and independently distributed. This assumption can be tested using the mean-centred correlation matrix [131, 132]:

$$R_{ij} = \frac{\sum_{k=1}^{N} \left(X_{k,i} - \mu_k\right)\left(X_{k,j} - \mu_k\right)}{\sqrt{\sum_{k=1}^{N} \left(X_{k,i} - \mu_k\right)^2 \sum_{k=1}^{N} \left(X_{k,j} - \mu_k\right)^2}} \tag{4.3.1}$$

Where the sum is over the number of large area SiPMs, one SiPM channel is inactive therefore the total is 599 instead of 600. The resulting matrix is shown in figure 4.23. It is by definition diagonal and satisfies the condition $R_{ii} = 1$.

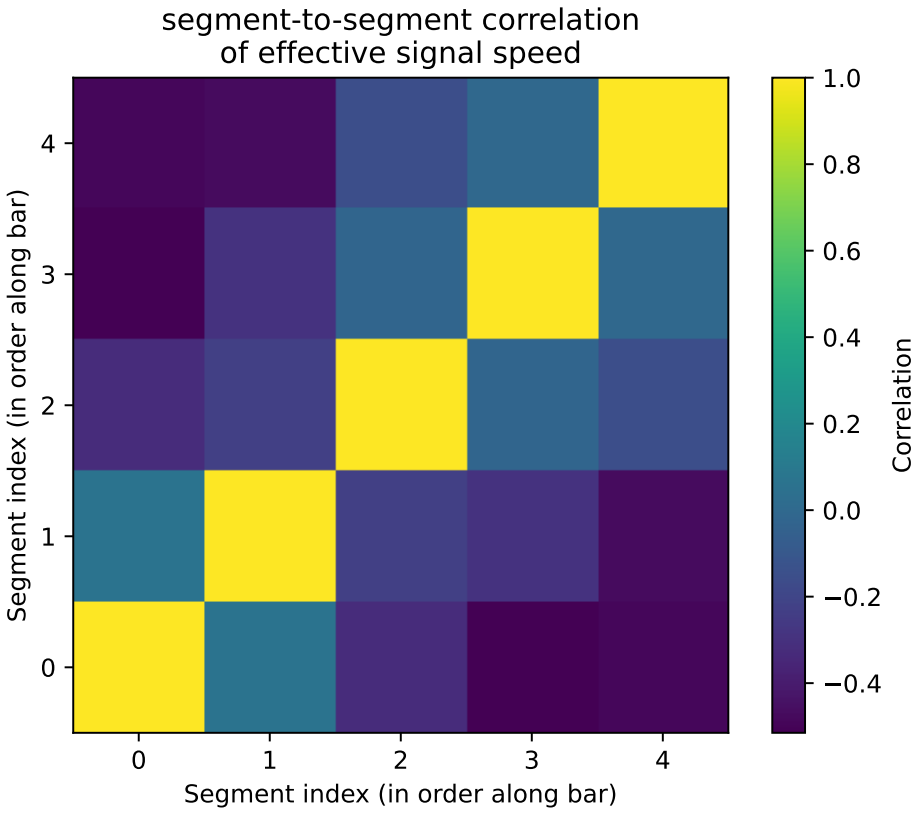



Figure 4.23: Mean centred correlation matrix of the measured effective signal speeds in the five equal-length segments throughout the scintillators. By definition of equation 4.3.1, matrix is symmetric and the main diagonal values are all unity.

The systematic uncertainty estimation assumes that the measurements of the effective signal speed are independent of position in the scintillator and identically distributed. If this is true, the off-diagonal values in the mean-centred correlation coefficient matrix should be zero, which is clearly not what is shown in figure 4.23. The correlation matrix shows that above each diagonal value there is a trend of decreasing correlation, the linear model therefore is not the ideal case for the physics of the SiPM-scintillator system. While the linear model is shown to not wholly capture the physics, the degree of its insufficiency can be quantified by the root-mean-squared residual between

One must also investigate the temporal stability of this procedure by comparing the effective speed of light measured by a SiPM over different data sets. Figure 4.24 covers a time period of October 22nd 2022 (run 5097) to May 11th 2023 (run 5999). The reproducibility of $c_{\text{SiPM}}$ is more clearly shown in figure 4.25 where one can see that the channel-to-channel variations in the determined values are maintained over different data sets when measuring the uncertainty through an estimated systematic with the approach described in 4.1.3. Such reproducibility over time allows for a constant set of $c_{\text{SiPM}}$ values to be used without reproduction.

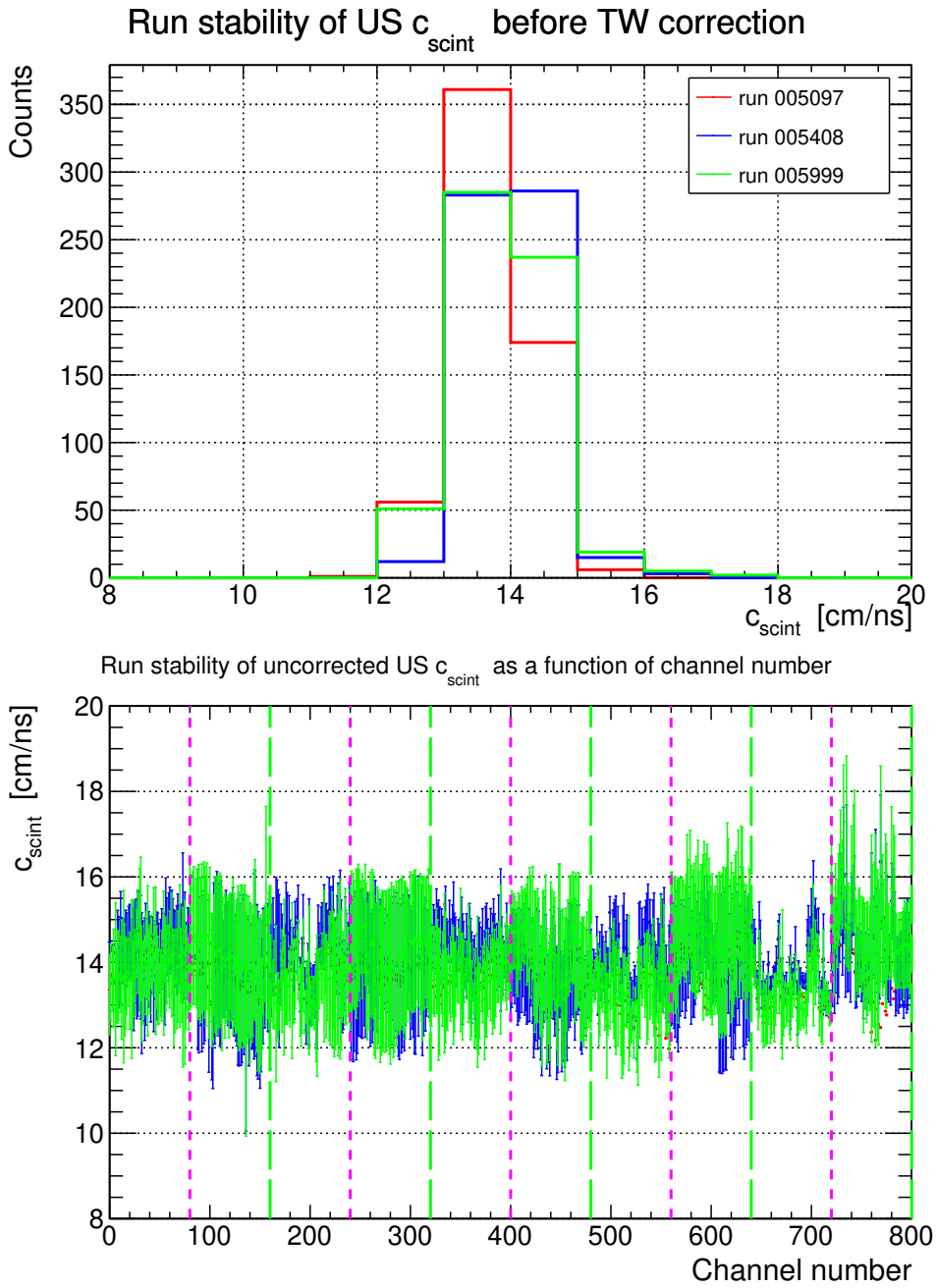



Figure 4.24: Run stability plots of the effective signal speed before time-walk correction. Overlaid histograms of the quantity on the top subplot and a channel-wise plot for the quantity on the lower plot.

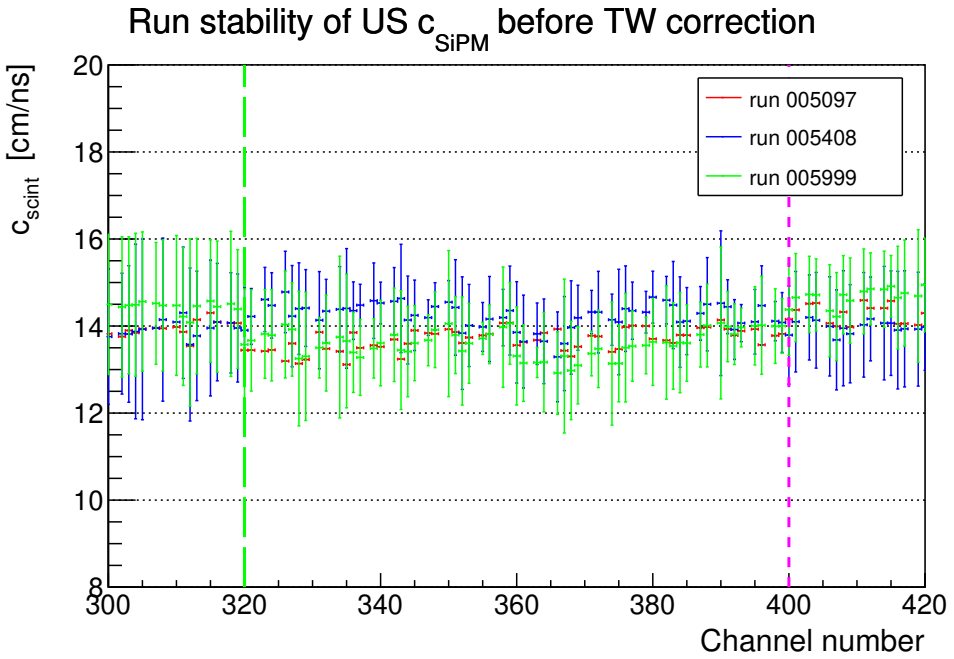



Figure 4.25: Zoom-in on run stability plot for uncorrected $c_{\mathrm{SiPM}}$ values on the left of plane 3 in the Upstream muon system.

Observing a stability in the measured value of the effective signal speed in the scintillator before the application of the time-walk correction provides a strong foundation for monitoring the stability of the time-walk corrected effective signal speed later.

With a successfully determined time walk correction, the resulting function is used to correct SiPM time values and re-evaluate the effective signal speed in the scintillator. The top sub-figure of figure 4.26, shows that the correction systematically shifts the value of $c_{\mathrm{SiPM}}$ to higher values.

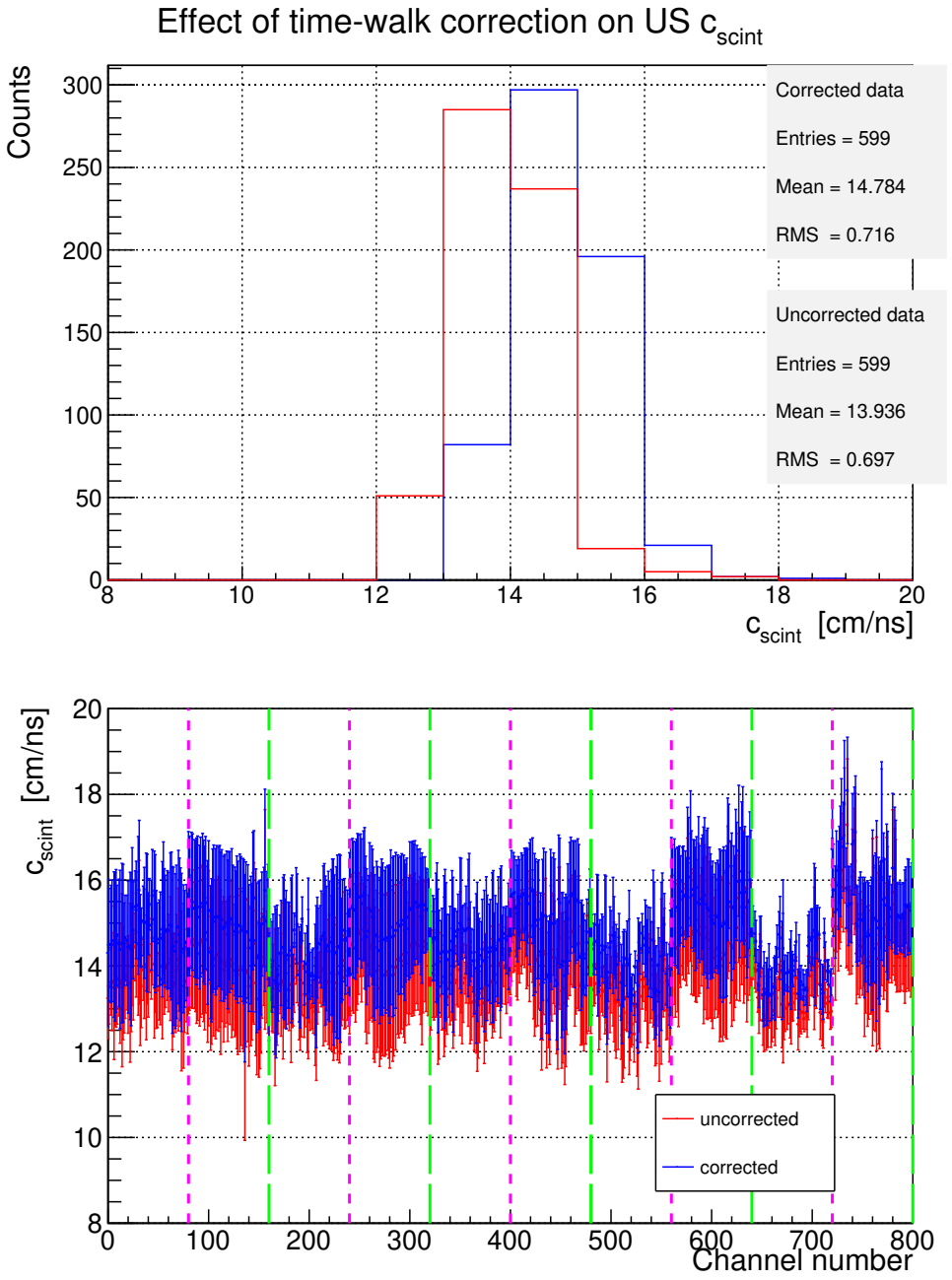



Figure 4.26: Effect on the measured effective signal speed when applying the time-walk correction function.

To explain why the effective signal speed is measured lower before time-walk correction, we first recall the time-walk effect: smaller signals fire fewer pixels of the SiPM concurrently, the slew rate of the rising edge is correlated with the number of concurrently firing pixels, meaning that signals causing greater rising edge slew rates will cross a fixed threshold earlier than a signal with a smaller rising-edge slew rate, as is depicted in figure 3.24. Signals travelling through the scintillator will have a fraction of the signal attenuated where this fraction is given by the Beer-Lambert law:

$$I(x) = I_0 \exp\left(-\frac{x}{\lambda}\right) \tag{4.3.2}$$

Where $I_0$ is the intensity of the signal at the point of signal creation; $x$ is the distance travelled by the signal from the point of creation; $\lambda$ is the attenuation

length of the material and $I(x)$ is the remaining signal intensity at distance $x$. A more attenuated signal translates into fewer concurrently firing SiPM pixels and brings into consideration the time-walk effect. The time-walk correction minimises the impact of the signal attenuation on the time recorded by the SiPM which will then increase the measured effective signal speed. This is illustrated by figure 4.27 where QDC is correlated against $x-$position. The SiPM used in this plot is on the left-side of the system, the right side of the $x-$position axis. This correlation plot shows that the most probable value of the QDC drops over the length of the bar, from the figures that illustrate the time-walk effect in the US system, such as figure 4.28, it is clear that a small change in QDC in the region around the most probable value results in a considerable change in the time recorded by the SiPM. This underpins why the time-walk correction induces an increase on the measured effective signal speed.

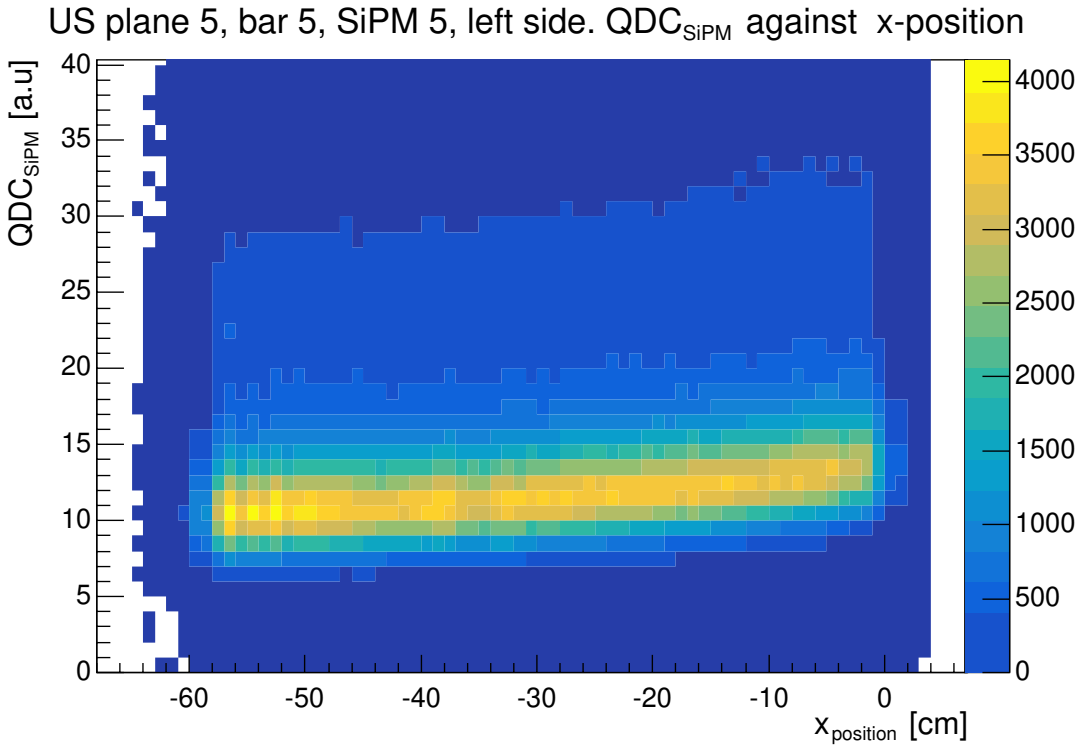



Figure 4.27: QDC value for a SiPM correlated against the x-position.

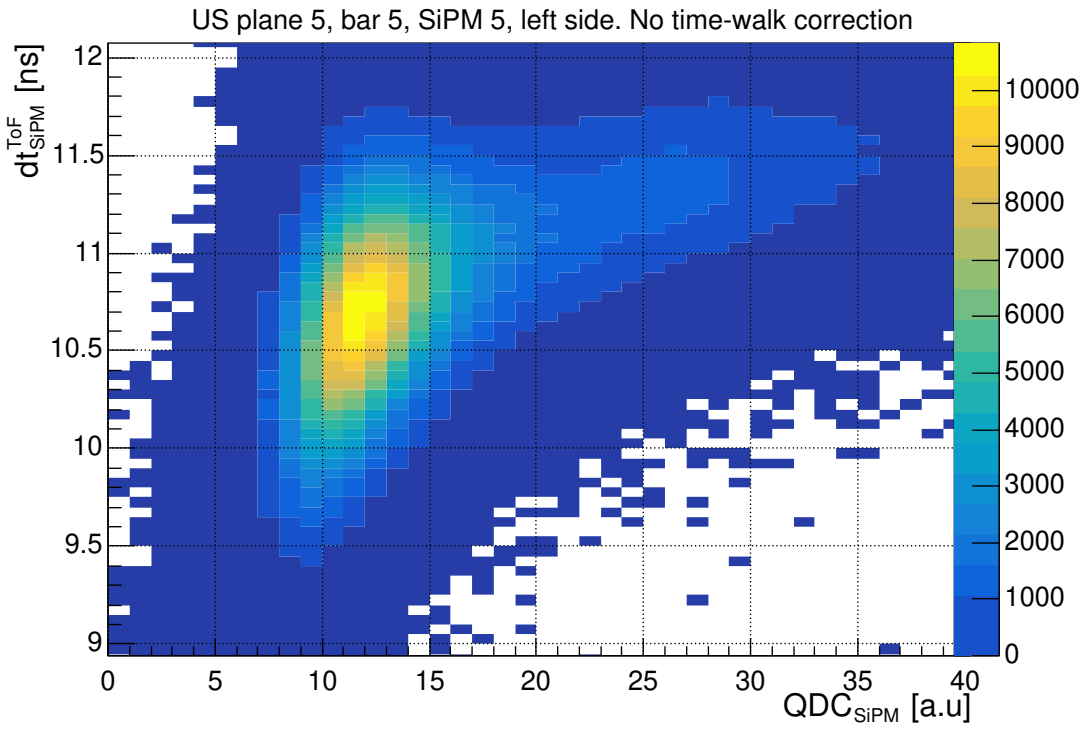



Figure 4.28: Zoom-in of $dt^{\text{ToF}}_{\text{SiPM}}$ versus $x-$position for SiPM 5, bar 5, plane 5 of the US zoomed-in around the MPV peak to illustrate what impact on time can be induced by a small QDC change around the most probable QDC value.

Just as with the effective signal speed before the time-walk correction, the run stability of the time-walk corrective effective signal speed can be studied. Figure 4.29 shows the overlaid histograms of the time-walk corrected effective signal speed in the upper half with the values for each channel shown in the lower half. The estimated systematic uncertainty is used as the uncertainty plotted in the lower half. Figure 4.30 shows a zoom-in of the SiPMs on the left side of plane 3 in the US system to demonstrate that the central values obtained for the three runs investigated are within the systematic uncertainties of one another, allowing for the conclusion that the time-walk corrected effective signal speed is well reproduced between the three runs studied.

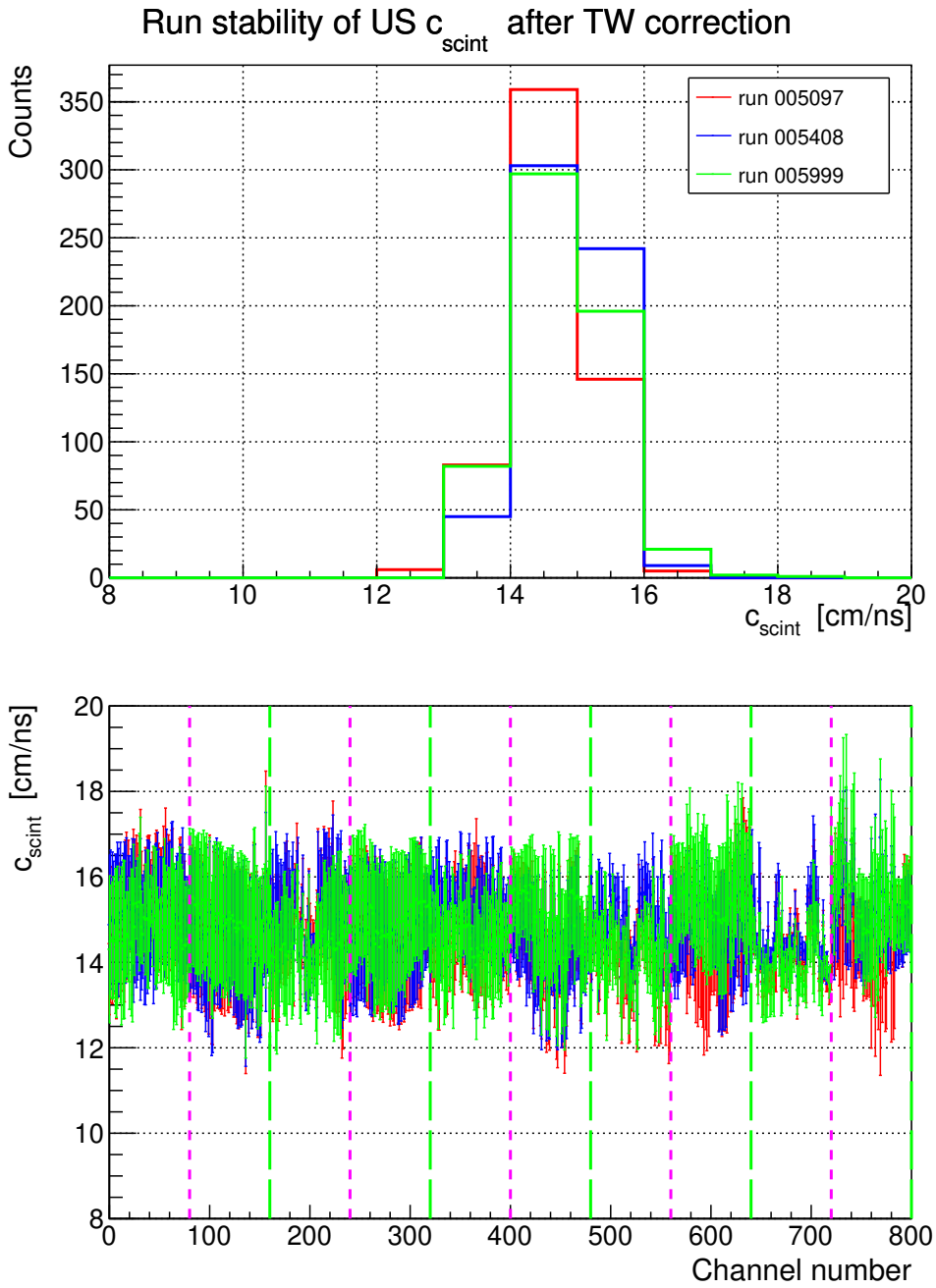



Figure 4.29: Run stability plots of the time-walk corrected effective signal speed. Overlaid histograms of the quantity on the top subplot and a channel-wise plot for the quantity on the lower plot.

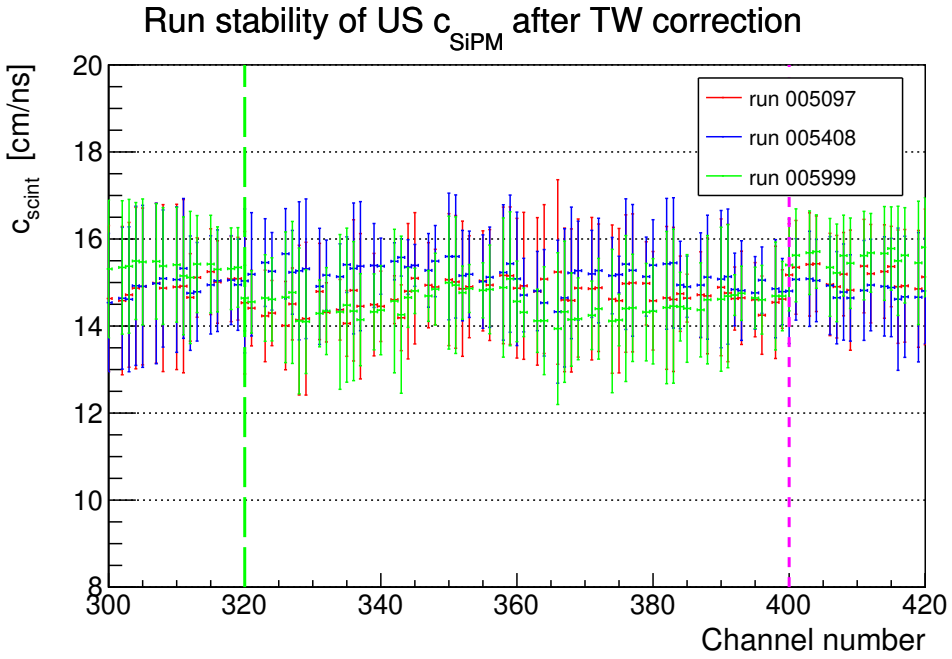



Figure 4.30: Zoom in on run stability plot for time-walk corrected effective signal speed values in the US. Channels 320 to 399 correspond to the left side of the 3rd plane of the US.

### 4.3.3 Time resolution

Following the time-walk correction, the time resolution is re-evaluated using the same method as section 4.1.5. Figure 4.31 shows that the time-walk correction has the systematic effect of improving the time resolution across the entire US system. The time-walk effect broadens the distribution of times measured by introducing a signal amplitude dependent bias, so it is clear that acting to minimise the time-walk effect would improve the time resolution. The shape of the time resolution distribution itself is also narrowed which reflects the variance in the extent of time-walk effect present across the 600 large SiPM channels and how the time-walk correction function can cope with these differing extents.

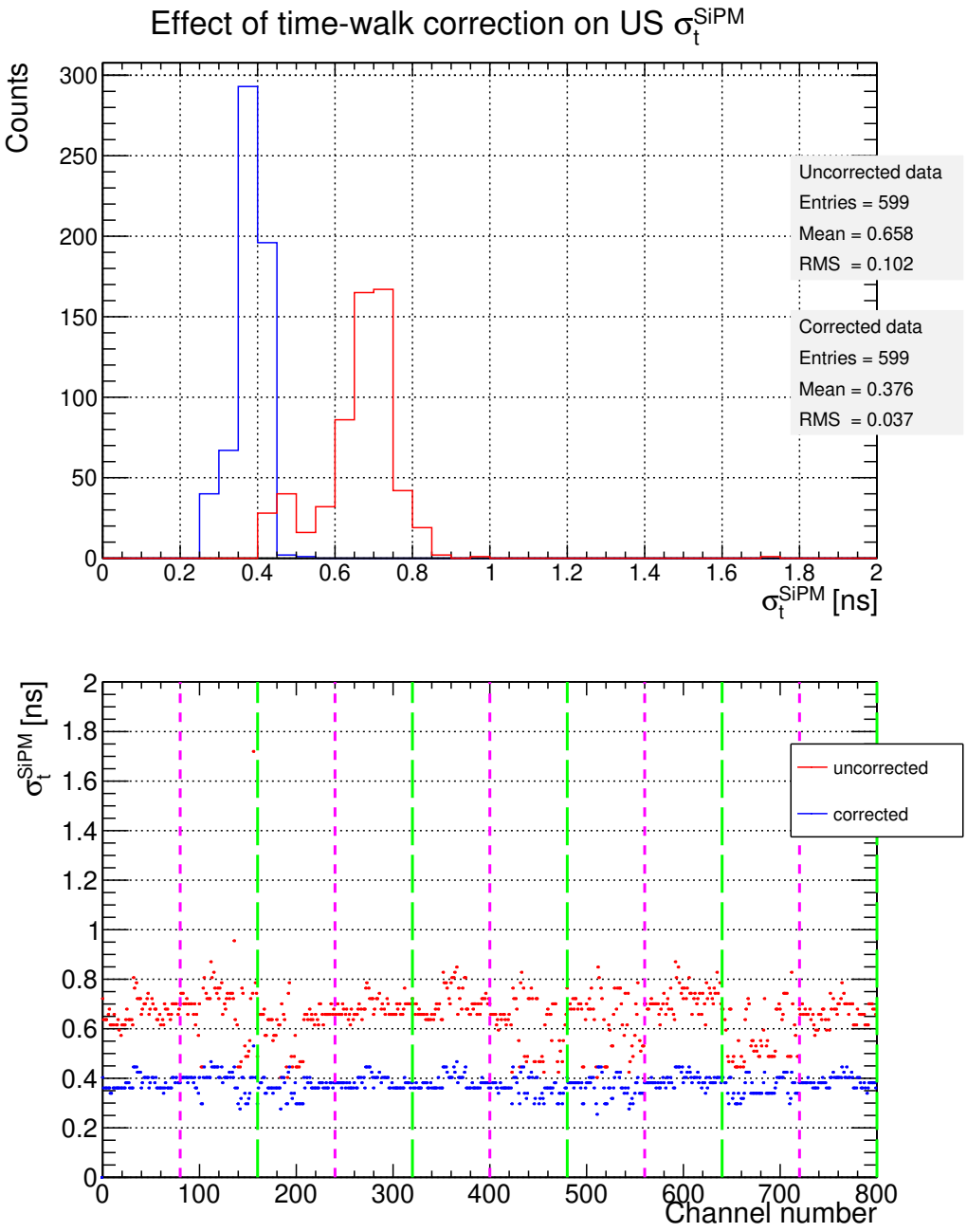



Figure 4.31: The effect on time resolution by applying the time walk correction parameters for run 5408.

The largest outlier in this dataset, with a time resolution of $\approx$ 1.7 ns is measured on a channel on the right side of the first plane in the US system. When investigating the data quality one sees that there is an issue with that particular SiPM and this is shown in appendix A.7. Figure 4.31 also shows that certain PCBs contain a greater proportion of lower values of uncorrected time resolution, which give rise to the small peak of better uncorrected time resolutions in the upper plot. The PCBs that contribute to this peak correspond perfectly to the PCBs which have gel pads used as shown in table 3.1. The SiPMs contained in this lower peak may have near ideal optical coupling between scintillator, gel and SiPM. If this is the case, then given that the majority of the SiPMs coupled to gel see no discernibly different time resolution then the gel must not be providing good optical coupling to most SiPMs[5].

[5]The author was heavily involved in the testing, production and application of the gel in the

The time resolution is expected to be best for energy depositions closest to the SiPM with the resolution worsening for energy depositions further away from the SiPM. From the Beer-Lambert equation 4.3.2 one may determine the expected signal loss from 6 cm away from the left-side of the US scintillator to the edge of the muon track acceptance[6], $L_\mu$:

$$I(L_\mu) = I_0 \exp\left(-\frac{L_\mu}{\lambda}\right), \quad I(x_{\text{left}} - 6) = I_0 \exp\left(-\frac{(x_{\text{left}} - 6)}{\lambda}\right)$$
$$\frac{I(L_\mu)}{I(x_{\text{left}} - 6)} = \exp\left(-\frac{1}{\lambda}|L_\mu - x_{\text{left}} + 6|\right)$$
$$\frac{I(L_\mu)}{I(x_{\text{left}} - 6)} = 84.8\%$$

The $x-$position of the left-side of the US scintillator is denoted as $x_{\text{left}}$; 6 cm away from the left-side is used to remove any edge-effects that could skew the results[7]. $L_\mu - x_{\text{left}} + 6$ is approximately $-50.6$ cm, the modulus of this value is taken to return what fraction of the signal is remaining after travelling 50.6 cm. This is specifically for a SiPM on the left side of the system such that $I(x_{\text{left}} - 6) > I(L_\mu)$ whereas for SiPMs on the right side of the system this quotient would be inverted. With the relative intensity only dropping by $\approx 15.2\%$ over this distance, one can use equation 3.2.5, the relationship between the time resolution of a SiPM and the number of photoelectrons, to estimate how the time resolution should change with this drop in intensity:

$$\sigma_t(x_{\text{left}} - 6) = 1/\sqrt{N_{pe}}, \quad \sigma_t(L_\mu) = 1/\sqrt{0.848\, N_{pe}}$$
$$\frac{\sigma_t(L_\mu)}{\sigma_t(x_{\text{left}} - 6)} \approx 1.08 \tag{4.3.3}$$

From these estimates, the ratio of time resolutions measured approximately 50.6 cm away should only cause an 8% rise in the time resolution. For a SiPM on the right side of the system, the ratio of time resolutions in equation 4.3.3 would be inverted, so when all SiPMs are studied in figure 4.33, the value plotted is the time resolution far from the SiPM is divided by the time resolution measured closer to the SiPM, $\sigma_t^{\text{SiPM}}(\text{far})/\sigma_t^{\text{SiPM}}(\text{near})$.

detector.

[6]The right-most (looking downstream) point that a muon track can be reconstructed is defined in $L_\mu$, see section 3.2.4, is the x-range in which muon tracks can be reconstructed.

[7]These edge-effects are discussed in section 5.2.2

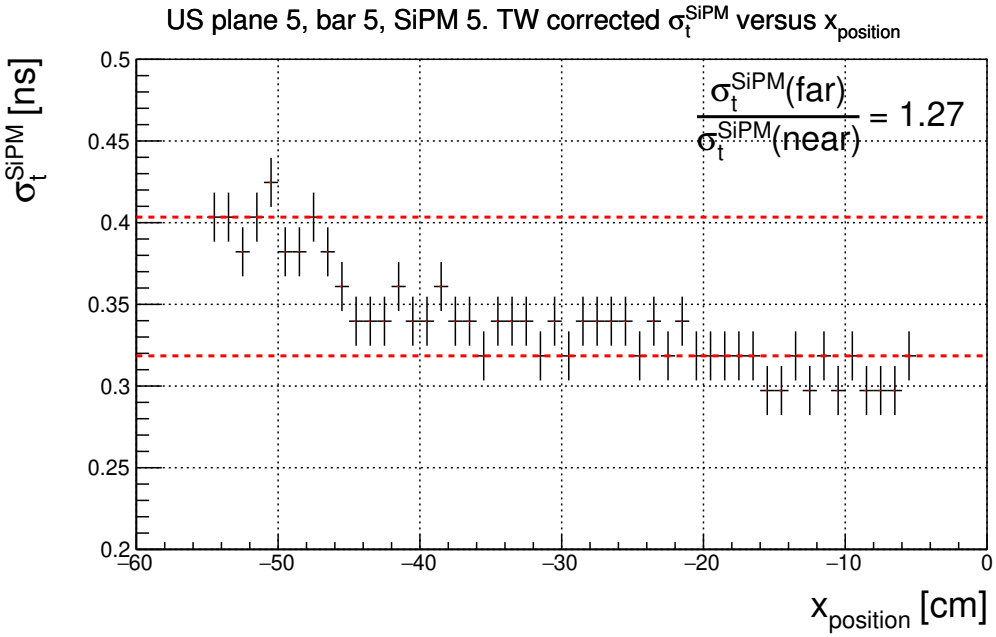



Figure 4.32: Measured time resolutions after the application of a time-walk correction for US plane 5, bar 5, SiPM 5 with data from run 5999. The time resolution at $L_\mu$ and $x_{\text{left}} - 5$ is measured and the ratio shown in the top right.

The change in time resolution measured with this same method is studied for all SiPMs in the US and the results shown in figure 4.33. The uncertainty on the value is propagated simply from $f = A/B$, $\sigma_f/f = \sqrt{(\sigma_A/A)^2 + (\sigma_B/B)^2}$. The mean value and standard error on the mean of $\sigma_t^{\text{SiPM}}(\text{far})/\sigma_t^{\text{SiPM}}(\text{near})$ are written in the lower plot of figure 4.33. It is clear in the graphs shown in the bottom half of figure 4.33 that there is a systematic difference in how the time resolution changes for the two sides of the US system. The right side of the US system shows a mean change of time resolution over 50.6 cm corresponding well to what one expects from attenuation arguments whereas the left-side SiPMs show a larger change in time resolution. Since the silgel is not applied to the same side of all planes, as documented in table 3.1, the silgel cannot be the underlying cause. Differing electronic behaviour could be the root cause although this cannot be directly concluded without experiment in TI18 itself.

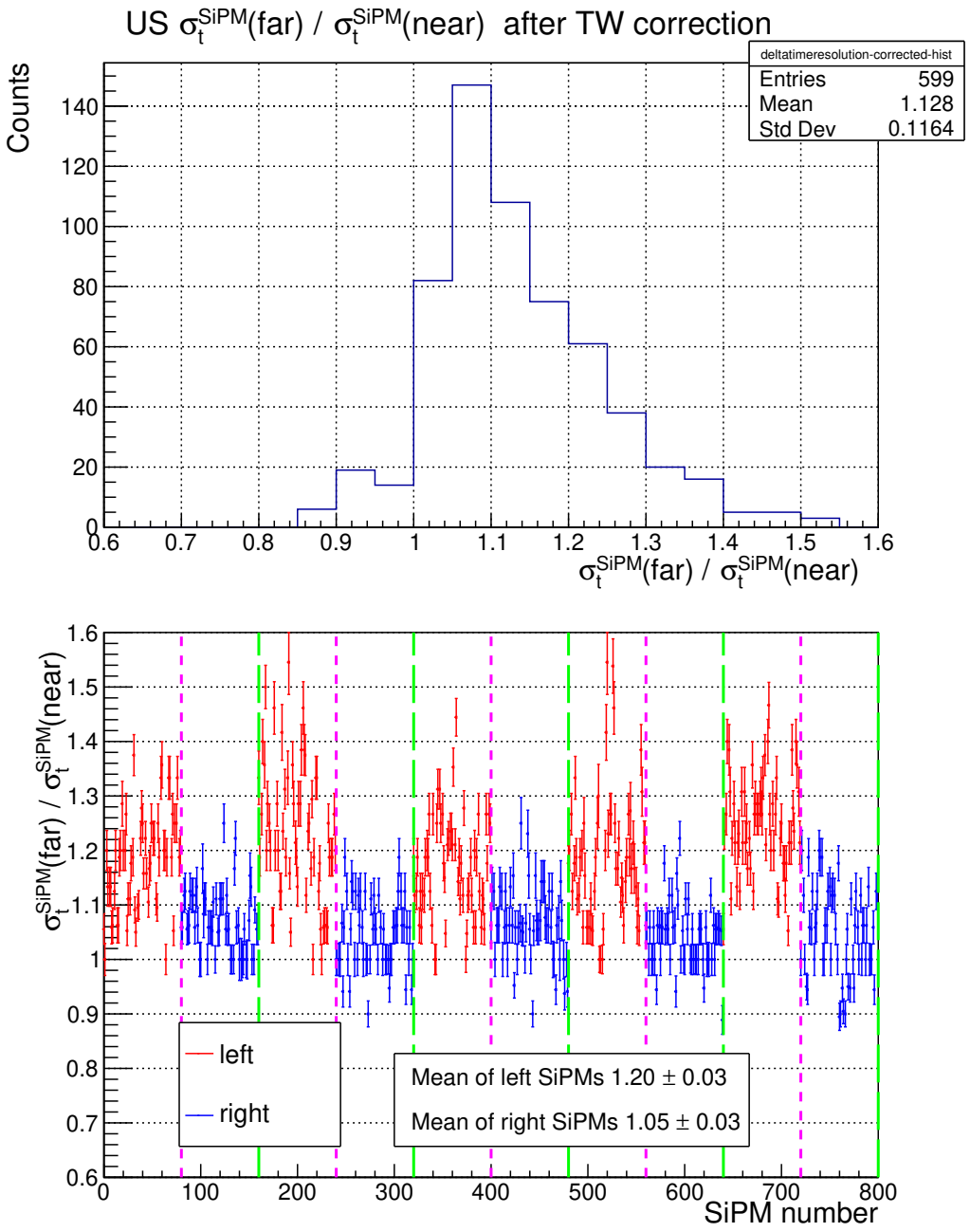



Figure 4.33: Distribution and graph showing how the measured time-walk corrected time resolution changes over a distance of approximately 50.6 cm for all the SiPMs in the US muon system.

### 4.3.4 Time-walk correction procedure

With the $dt_{\text{SiPM}}^{\text{ToF}}$ against QDC histograms created and the time-walk correction function defined with parameter limits defined in section 4.1.4, the details of the fitting process and the results can be discussed. As stated in section 4.1.4, the time-walk function is fitted with a lower limit of zero, meaning that the region between zero and the QDC most probable value, where the time-walk effect is strongest, is described by the fitted function. The upper limit of the fit is the last bin before there are fewer than 10 entries.

A key feature outlined in section 4.1.4 is that the statistical uncertainties do not account for the variance observed in the system. Visualising the $\chi^2_\nu$ for all SiPMs

using only statistical uncertainties returns values far greater than 1, where a $\chi^2_\nu$ value of 1 indicates that the extent of agreement between data and expectation is well described by the variance. The results of the $\chi^2_\nu$ calculation for all channels using statistical uncertainties are shown in figure 4.34.

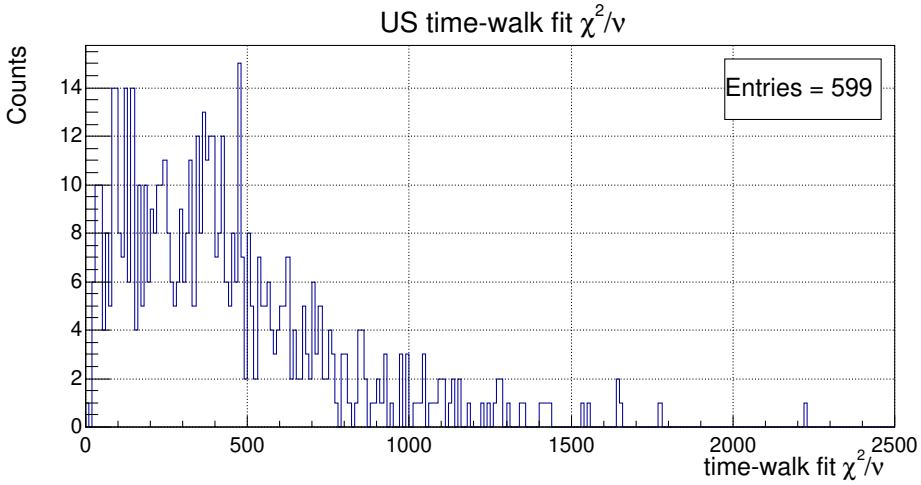



Figure 4.34: Distribution of $\chi^2_\nu$ values using only statistical uncertainties from the time-walk fits of the US SiPM data from run 5999.

As the QDC distribution follows a Landau convolved with a Gaussian, the region around the modal QDC is sharply peaked meaning that the statistical uncertainties in that region are very small. To support the hypothesis that there is an unaccounted for systematic uncertainty, a comparison between the $\chi^2_\nu$ and the root-mean-squared residual (RMS residual) of the SiPM data and their respective fitted time-walk functions is done. The RMS residual is computed as:

$$\mathrm{RMS}_{\mathrm{residual}} = \sqrt{\frac{1}{N}\sum_{i}^{N}\left(\frac{y_i}{f(x_i)} - 1\right)^2} \tag{4.3.4}$$

Where $y_i$ and $f(x_i)$ are the $i^{\mathrm{th}}$ data point and fitted function evaluated at this data-point, and $N$ is the number of data-points. This quantity represents a measurement of how far on average the ratio of data-point and expected value from the fitted function vary from unity. The distribution of this value for all US SiPMs is shown in 4.35.

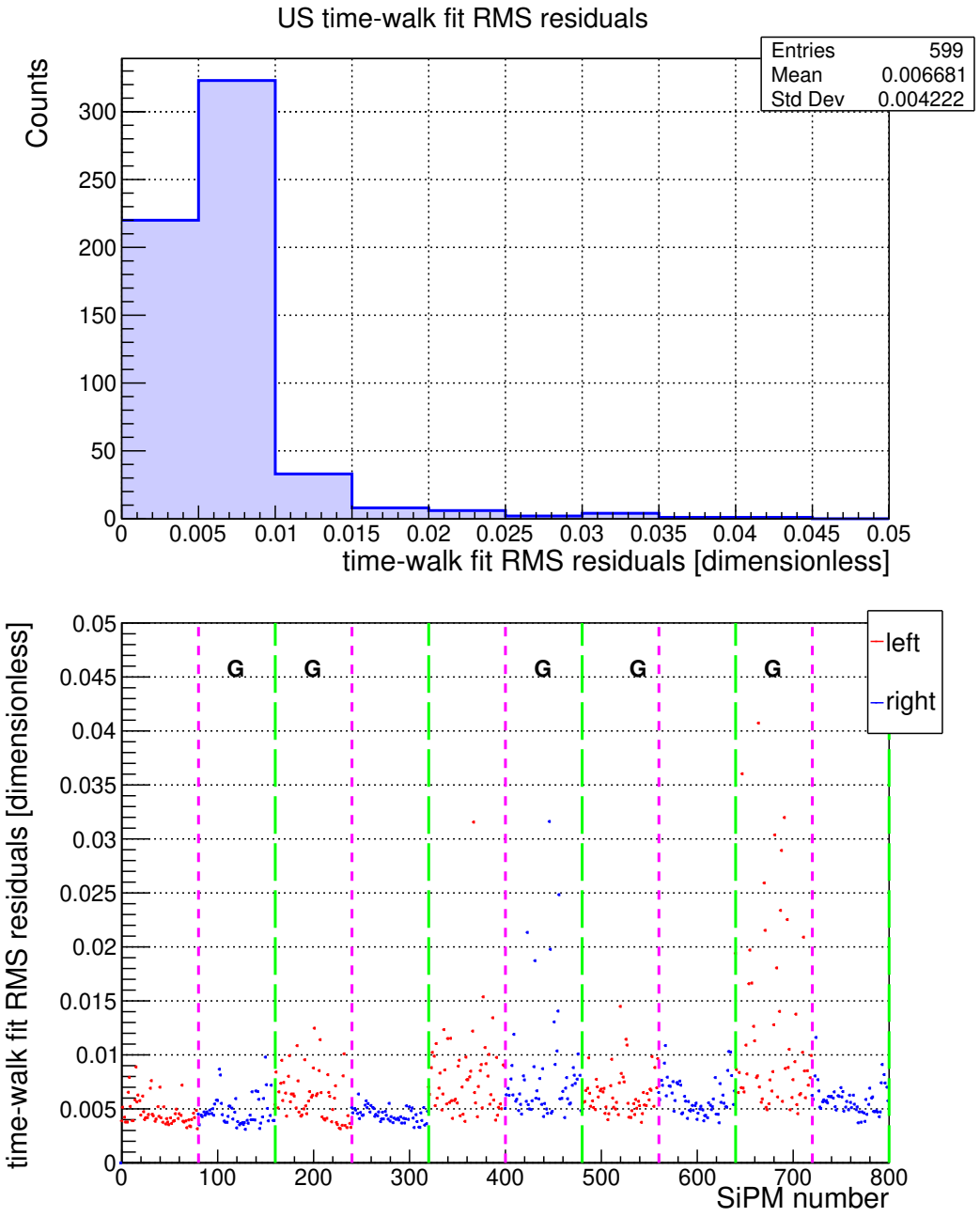



Figure 4.35: Distributions of the RMS residual for the time-walk fit on data from run 5999.

Figure 4.35 clearly shows that the vast majority of SiPMs in the US have a value of RMS residual of 0.01 or less. This means that the majority of SiPMs deviate from the fitted function by $\leq 1\%$. The $\chi^2_\nu$ and RMS residual plots together give strong evidence that the statistical errors are not fully representative of the variance observed and therefore there is an unaccounted for systematic uncertainty, $\sigma_{\text{sys}}$. This unaccounted for systematic uncertainty can be estimated if we assume that the errors in each bin of QDC are normally distributed, then one can write down the probability to observe a value $y_i$ given the time-walk function fitted with parameters $\theta$ evaluated at point $x_i$, $f(x_i;\theta)$:

$$p(x_i|\theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(y_i - f(x_i;\theta)^2}{2\sigma^2}\right] \tag{4.3.5}$$

Where one can say $\sigma^2 = \sigma^2_{\text{stat}} + \sigma^2_{\text{sys}}$. The likelihood to observe the full set of measurements $\{y_i\}$ is the product of probabilities for fitted parameters, $\theta$:

$$\mathcal{L}(\theta) = \prod_i^N p(x_i|\theta)$$

The negative log likelihood, $-\ln(\mathcal{L})$, referred to as NLL, can be shown to be given by:

$$\text{NLL}(\theta) = \frac{1}{2}\sum_i^N \frac{(y_i - f(x_i;\theta))^2}{\sigma^2_{\text{stat}} + \sigma^2_{\text{sys}}} + \ln(\sigma^2_{\text{stat}} + \sigma^2_{\text{sys}}) \tag{4.3.6}$$

The negative log likelihood is then numerically minimised using the SciPy package [133] which allows the $\chi^2_\nu$ to be re-evaluated using the optimised estimate of $\sigma_{\text{sys}}$.

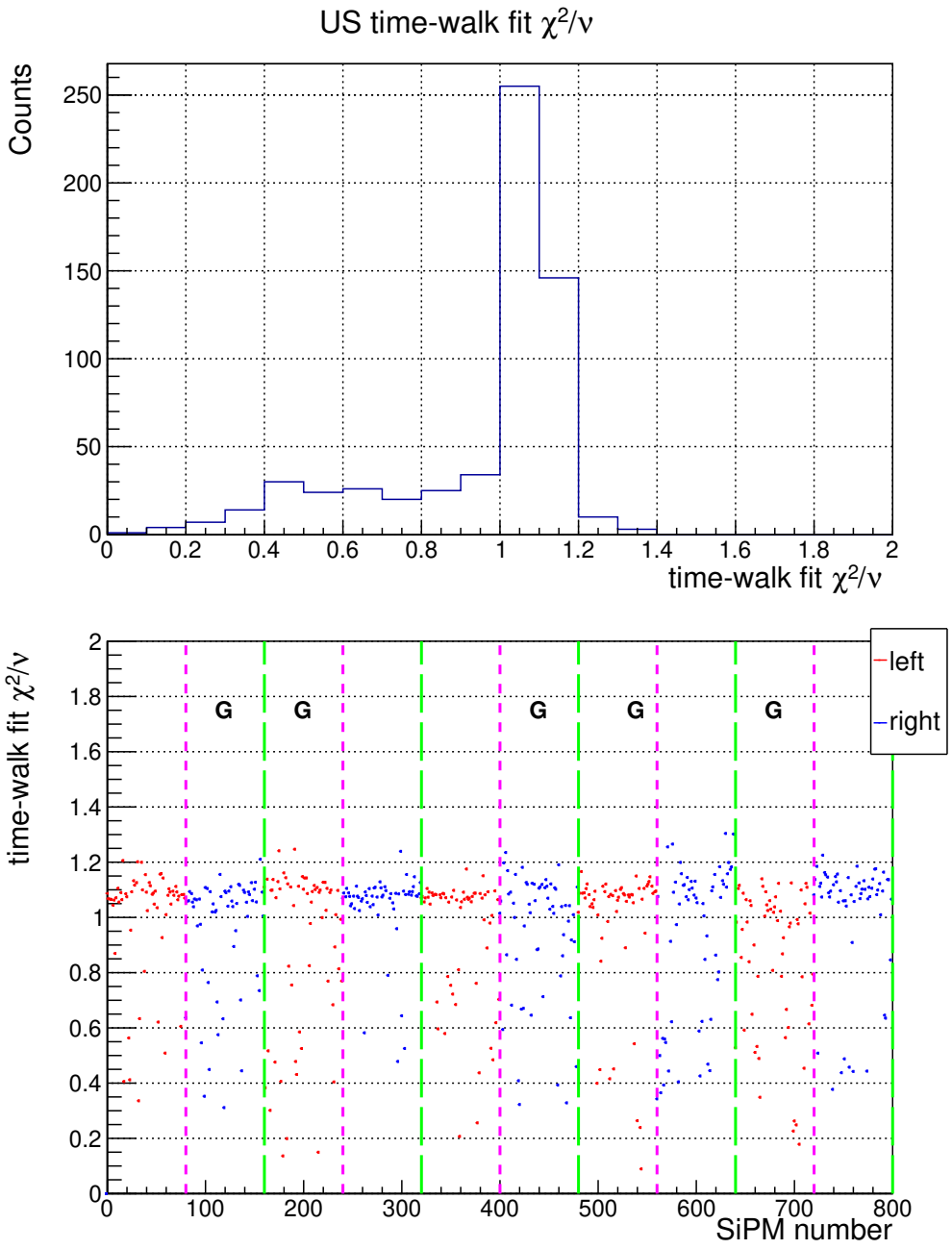



Figure 4.36: Distribution of $\chi^2_\nu$ using the quadratic sum of statistical uncertainty and numerically optimised estimate of the systematic uncertainty from the time-walk fits of the US SiPM data from run 5999.

The shape of the distribution of $\chi^2_\nu$ accounting for $\sigma_{\text{sys}}$ shows a long tail for $0 < \chi^2_\nu < 1$. This indicates that the values of $\sigma_{\text{sys}}$ are overestimated which will be caused by the low-QDC region of the domain, where the time-walk correction function begins to fail to describe the data.

The distribution for the estimated systematic uncertainty is shown in figure 4.37 where one sees a nearly symmetric

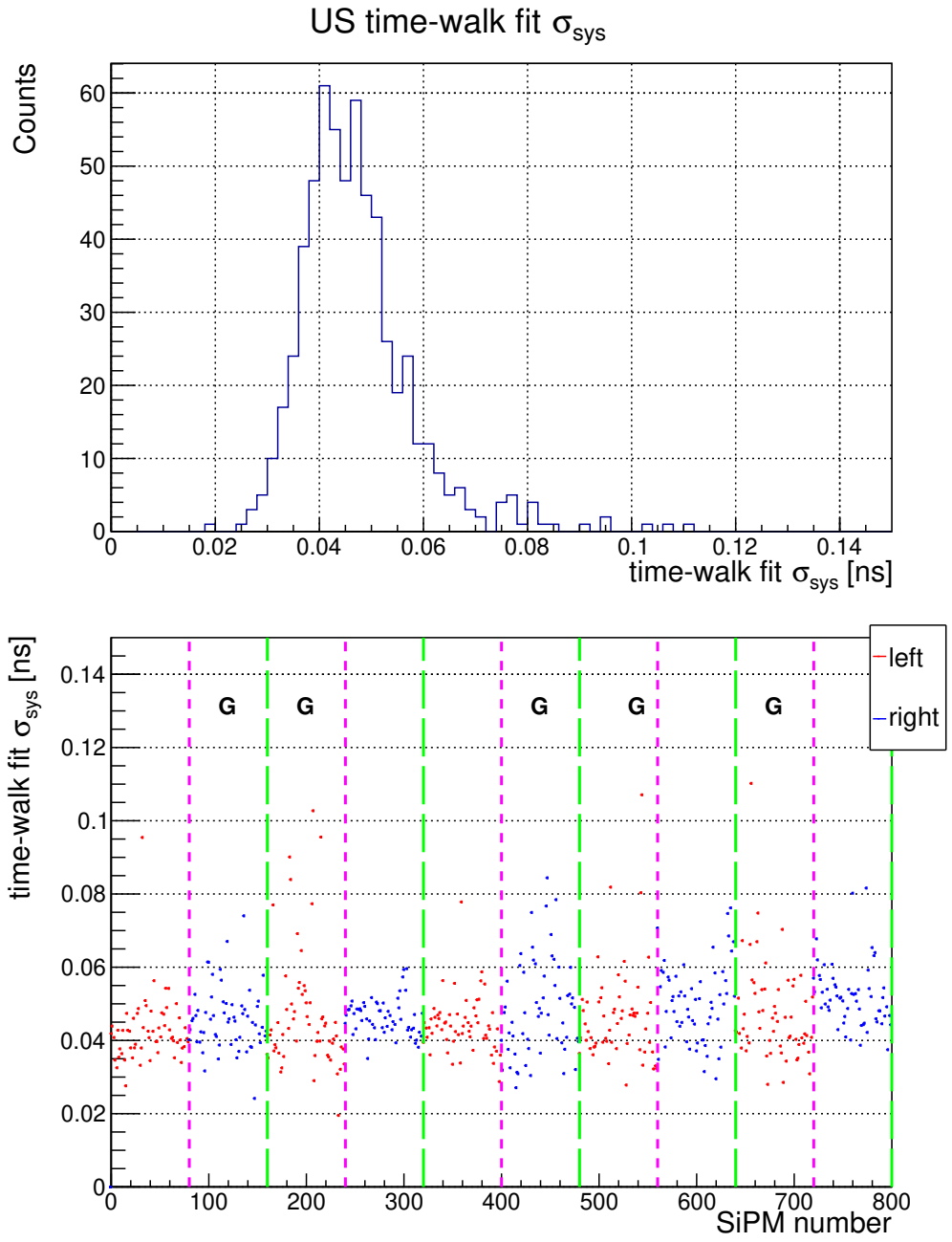



Figure 4.37: Distribution of measured $\sigma_{sys}$ using MLE.

### 4.3.5 Fine alignment parameters

The US may be aligned to either the SciFi (SF) or the muon system (DS) such that the modal time difference between any SiPM in the US and the event-wise reference time is zero. There are three time alignment settings with which data is collected (explained in section 3.3, table 3.6). To cover all of these combinations there are six sets of parameters determined. Here only the DS alignment parameters for the LHC synchronous time alignment setting[8] is presented but in appendix A.8, all six data sets are visualised just as shown in figure 4.38.

[8]This is the current alignment and covers the vast majority of the collected luminosity of the experiment.

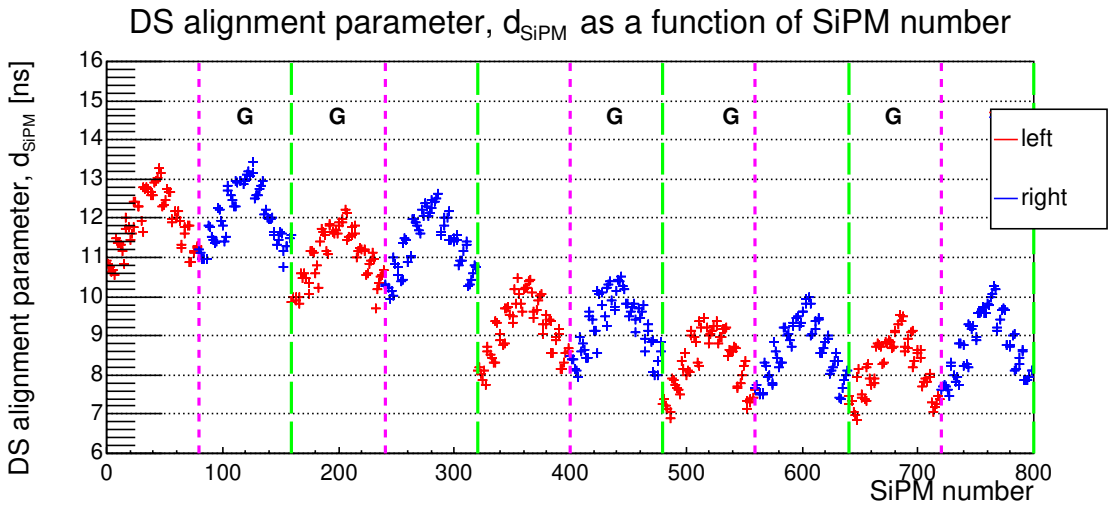



Figure 4.38: Determined DS alignment parameters from run 5999 for the US muon system. These alignment parameters are appropriate for use on any data using the LHC synchronous time alignment.

With the alignment parameters, equation 4.1.12 can be followed to align the timing response of the SiPMs and highlight that now all US SiPMs have a $dt^{\mathrm{ToF}}_{\mathrm{SiPM}} - d_{\mathrm{SiPM}}$ expectation value of 0 ns following the application of the time-walk correction of equation 4.1.9, signal time of flight correction of equation 4.1.8 and the fine alignment:

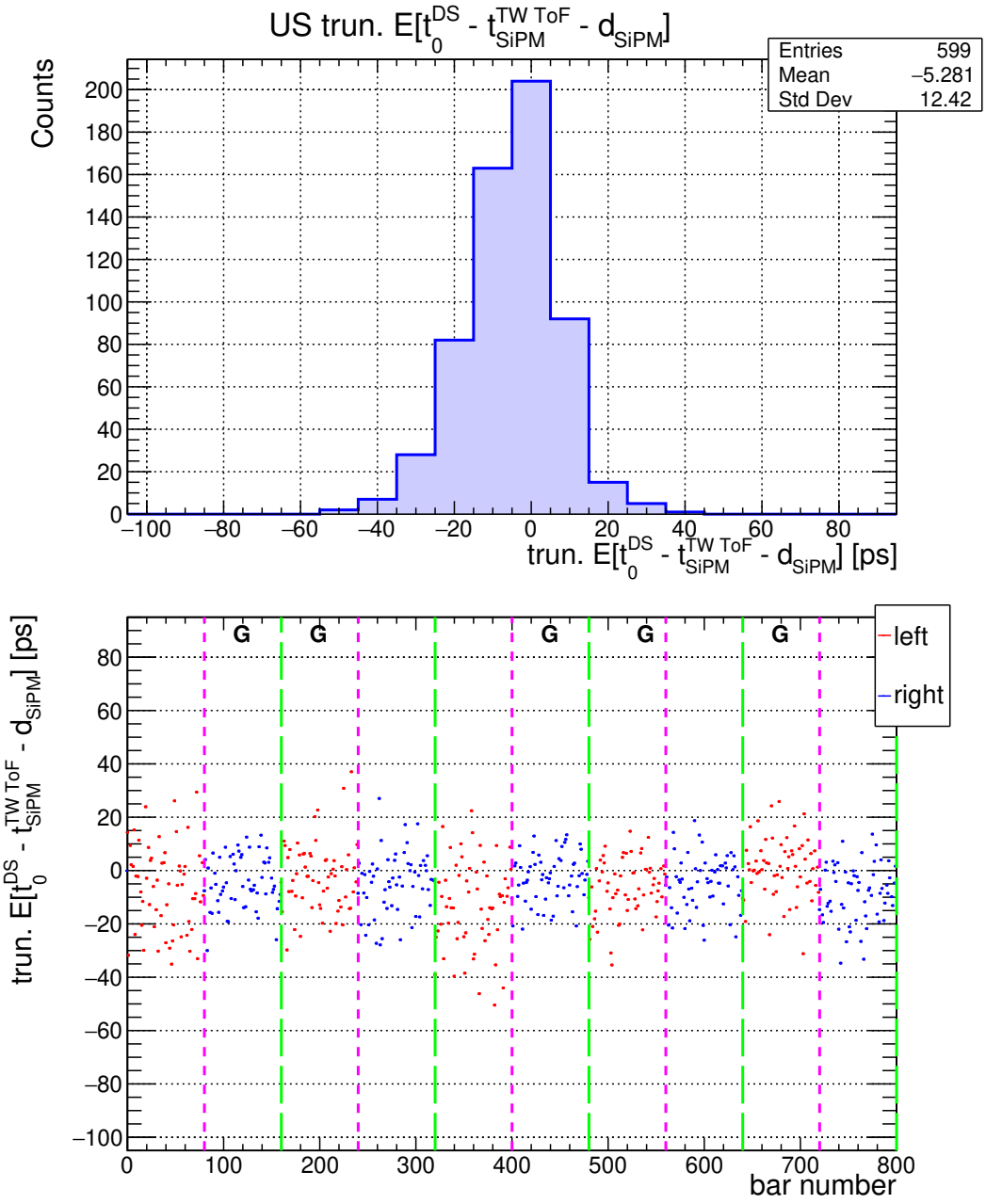



Figure 4.39: The timing value mean truncated to within $\pm 1\sigma$ measured for all US channels after applying the time-walk, signal time-of-flight corrections and alignment parameters for run 5999.

Figure 4.39 shows the mean of the aligned time-walk and signal time-of-flight corrected SiPM times taken within a $\pm\sigma$ range around the modal timing value with the error shown in figure 4.39 coming from the standard error on this truncated mean, which is not visible in the figure. For all sets of alignment parameters, the same intra-PCB structure is observed, the inverse-u shape as well as the structure observed within the inverse-us. These two structural elements are most likely due to trace-length differences in the US PCB. Trace-line measurements of the US PCB were taken by a colleague of the author and can provided upon request to the author. These measurements were done using the computer aided design (CAD) schematic of the PCB which was used in the production of the PCBs. With the measurement of the trace-lines, the timing component expected can be estimated by assuming a signal speed in the PCB trace of 14 cm/ns. This assumed signal speed

is achieved by using the commonplace expression for which a derivation is provided in appendix A.15:

$$v = \frac{c}{\sqrt{\epsilon_r}} \tag{4.3.7}$$

Where $c$ is the speed of light in vacuum; $\epsilon_r$ is the relative permittivity of the dielectric material used to construct the US PCBs, FR-4[9]. The value of $\epsilon_r$ for FR-4 is found to vary depending on the source: it is quoted as $\epsilon_r = 4.6$ in the data-sheet at reference [135], whereas it is quoted as $\epsilon_r = 4.0$ in reference [136]. Arbitrarily, the value of 4.6 is chosen as the nominal value, returning a signal velocity of 14.0 cm/ns. Larger and smaller values are then chosen to demonstrate how variation in $\epsilon_r$ will impact the timing contribution from the PCB trace-lines. The relative permittivity is impacted by several factors including the frequency of the electric field, temperature and humidity [136]. The error bars correspond to a fixed systematic uncertainty of $\pm 1$ cm.

Clearly from comparison of the figures in 4.40, the timing component from the trace-line lengths cannot account for the variation observed in the value of $d_{\text{SiPM}}$.

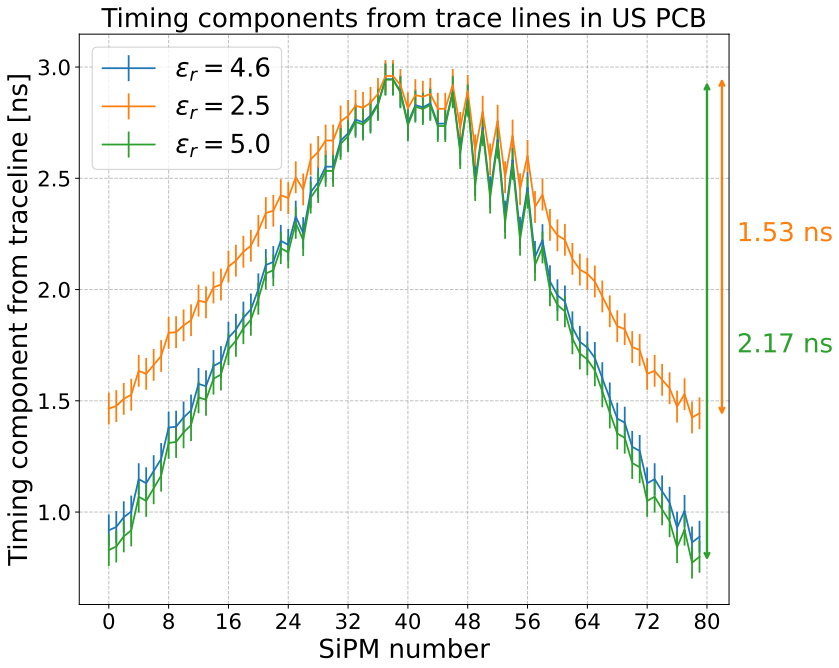
### 4.3.6 Scintillator timing measurements

The application of the time-walk correction and alignment parameters allows for the timing power of a full scintillator to be utilised. To understand how one expects using the timing capability of a scintillator to impact the time resolution, one may begin from the construction of the average aligned time measured by a bar-side with the signal time-of-flight correction applied, $dt^{\text{ToF}}_{\text{SiPM}}$:

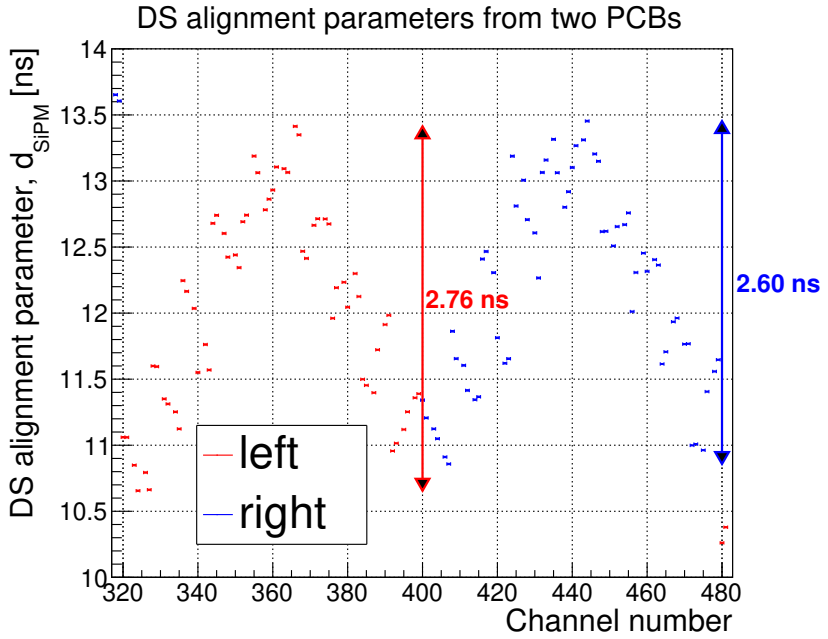
$$dt^{\text{ToF}}_{\text{side}} \;=\; \frac{1}{N}\sum_{\text{SiPM}}^{N}\left(t_0^{\text{DS}} - t^{\text{TW ToF}}_{\text{SiPM}} - d_{\text{SiPM}}\right) \tag{4.3.8}$$

Combining the measurements of the $N$ SiPMs per side, measuring the resolution and statistically removing the contribution from the muon system reference time as formulated in equation 4.1.10, is shown in figure 4.41.

[9]The use of FR-4 is not documented but was confirmed via email communication with an engineer who worked on the PCB design [134].




(a) Timing components estimated from PCB trace-length measurements. The timing ranges for $\epsilon_r = 2.5$ and $\epsilon_r = 5.0$ are shown by the arrows and text with colour-scheme provided in the legend.




(b) A zoom in on the intra-PCB structure of figure 4.38. The arrows and corresponding times correspond to the measured $d_{\text{SiPM}}$ range on a PCB.

Figure 4.40: The estimated timing component coming from the trace-line measurements supply an explanation for the observed inverse-u shape of the alignment parameters for a given PCB.

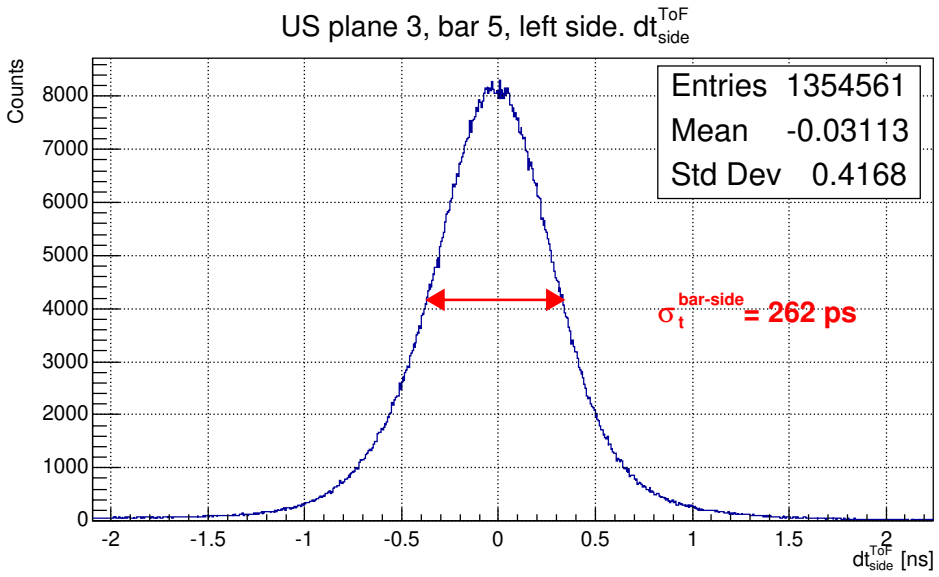



Figure 4.41: Average aligned time distribution for all the SiPMs on the left side of the 5th scintillator in the 5th plane. The FWHM extracted directly from the data and $\sigma_t^{\text{bar-side}}$ is measured by removing the resolution component from the event-wise reference time.

As with the single SiPMs, the tails present in the timing distributions created a large degree of difficulty in finding a functional form to describe the data. The quoted uncertainty on figure 4.41 is obtained as is performed for the individual SiPM channels: estimating the sigma value from the FWHM (FWHM/2.355) and then statistically removing the resolution component from the DS muon system reference time, $\sigma_t^{\text{bar-side}} = \sqrt{(\text{FWHM}/2.355)^2 - \sigma_{t_0^{\text{DS}}}^2}$.

Performing this measurement for both sides of all 50 scintillators of the US system gives the results as shown in figure 4.42.

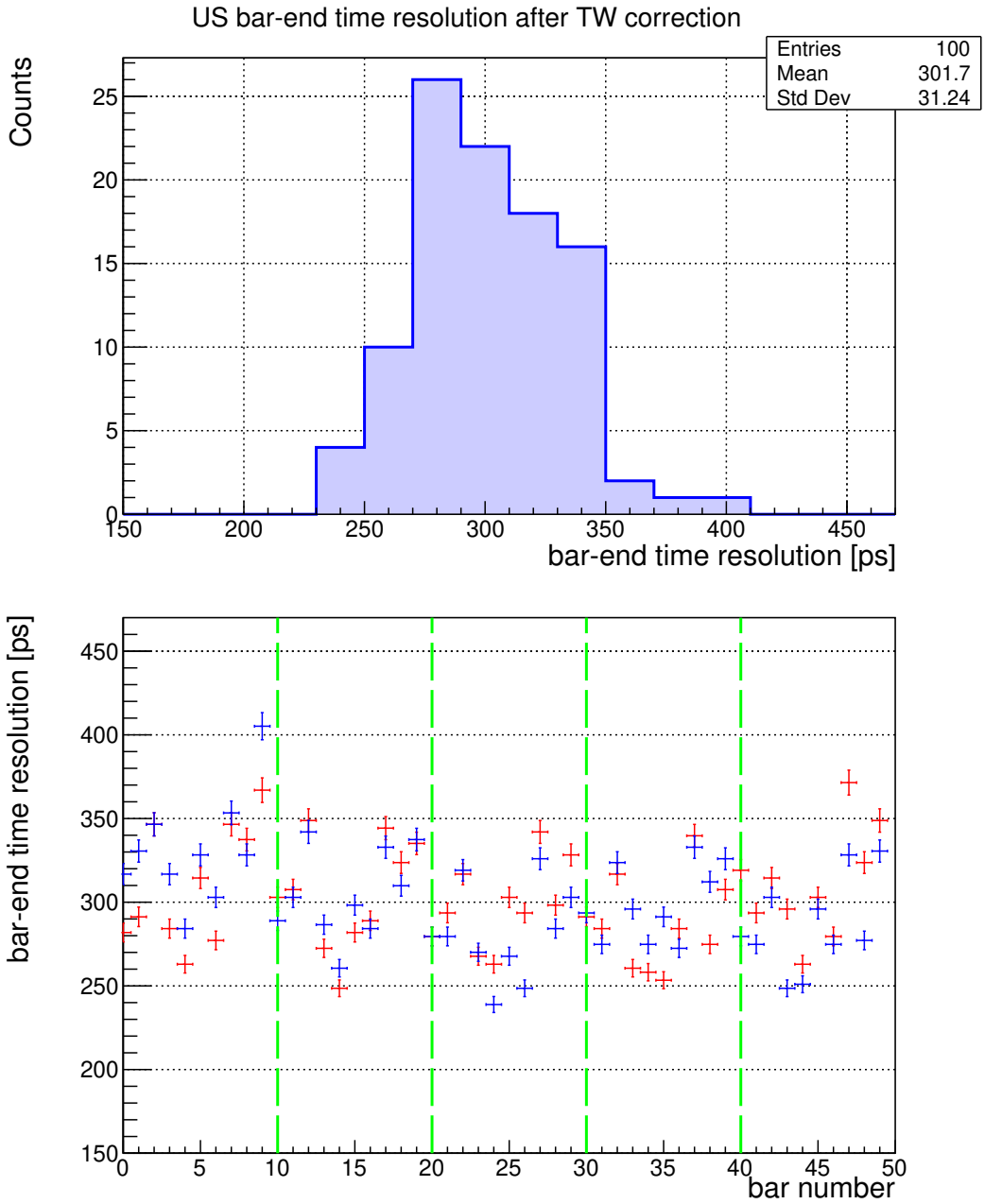



Figure 4.42: Measured bar-side time resolutions for all scintillators in the . The green vertical lines indicate the borders between planes of the .

These measured resolutions can be broken down into their constituent components using [131]:

$$\begin{aligned} \sigma_{dt^{\mathrm{ToF}}_{\mathrm{side}}} &= \sigma_y \\ \sigma_y^2 &\approx \sum_{i,j=1}^{n} \left[ \frac{\partial y}{\partial x_i} \frac{\partial y}{\partial x_j} \right]_{x=\mu} V_{ij} \end{aligned} \tag{4.3.9}$$

Where $V$ is the covariance matrix. The uncertainties in alignment parameter, $d_{\mathrm{SiPM}}$, are negligible and dropped for the construction of the expression. Equation 4.3.9 can be employed with equation 4.3.8 to give:

$$\sigma^2_{dt^{\text{ToF}}_{\text{bar side}}} - \sigma^2_{t_0^{DS}} = \frac{1}{N^2}\left[\sum_{i=1}^{N}\sigma^2_{t_i^{\text{TW ToF}}} + 2\sum_{1\leq i<j\leq N}\sigma_{t_i^{\text{TW ToF}}\, t_j^{\text{TW ToF}}}\right] \tag{4.3.10}$$

Where $\sum_{1\leq i<j\leq N}$ is the number of pair-wise SiPM combinations on a bar-side. The first term accounts for the resolutions of the individual SiPMs in the group; the second term accounts for covariance contributions between the pair-wise combinations of SiPMs in the group; the final term reflects the covariance between the US SiPMs and the event-wise reference time[10]. The factors of two are required to account for the symmetric covariance matrix, since both ordered pairs $(i, j)$ and $(j, i)$ have identical covariance contributions. The terms accounting for the covariance between SiPMs and the event-wise reference time are neglected because the US SiPMs and the DS SiPMs are electrically isolated from one another and should have no covariance. A more comprehensive derivation of equation 4.3.10 can be found in appendix A.9. In order to evaluate the accuracy of equation 4.3.10 one must study the covariance of SiPM times, and their covariance with the event-wise reference time. An example of the correlation between two same-side SiPMs is shown in figure 4.43. From these histograms, a covariance is extracted which can then be used to determine the expected bar-side time resolution.

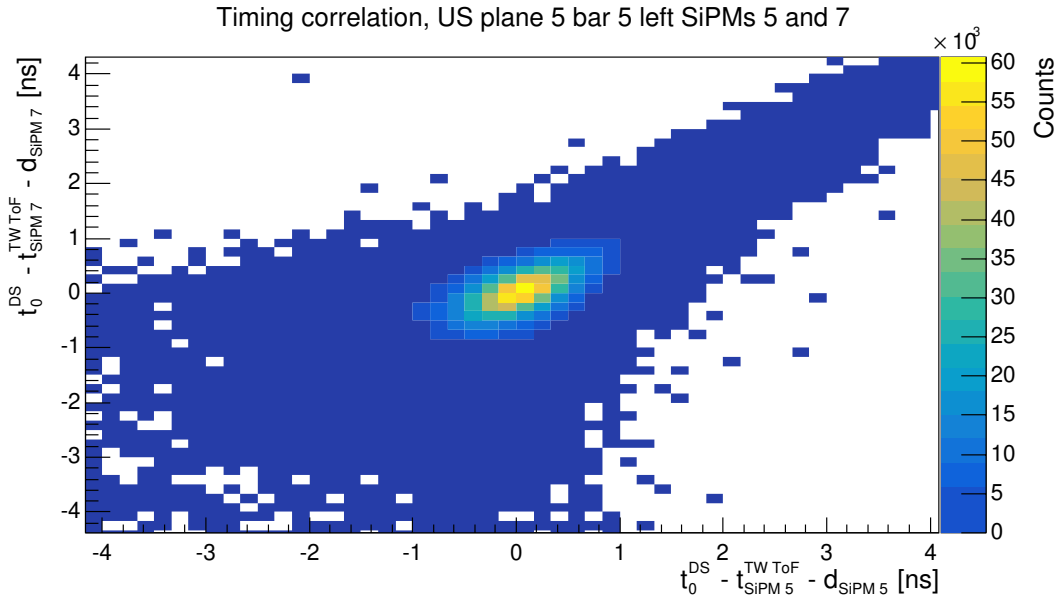



Figure 4.43: Correlation histogram between the aligned times of two SiPMs reading out the same side of a scintillator in the 3rd plane of the

The covariance of SiPM time measurements is simply obtained with the standard equation:

$$\sigma_{xy} = \text{Cov}(x, y) = 1/(n-1)\sum_{i}^{n}(\bar{x} - x_i)(\bar{y} - y_i)$$

[10]the DS muon system estimate of event-wise reference time is used here, but the SciFi reference time could equally be used.

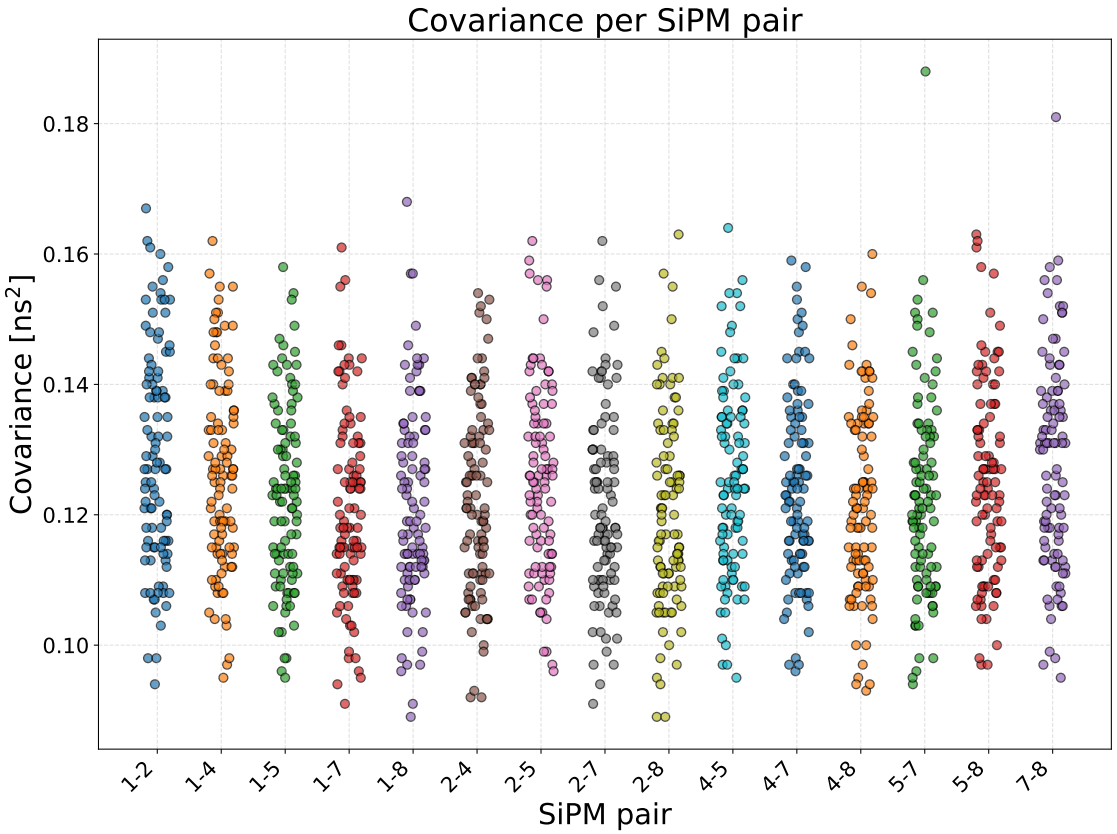



Figure 4.44: Scatter plot of the truncated covariance measured for SiPM pairs coupled to the same side of a US bar.

Each side of a scintillator has 15 permutations, $\binom{6}{2}$. To investigate whether the covariance is correlated with the distance between the SiPMs in the pair, the truncated covariances measured for each of these 15 permutations is plotted in figure 4.44 where the labels on the abscissa axis show the SiPMs in the pair. Some horizontal noise is added into the plot purely for cosmetic purposes. The range of measured truncated covariance is clearly independent of the SiPM pair.

The distribution of the residual bar-side time resolutions shown in figure 4.45 demonstrates that by considering the covariance between SiPMs coupled to the same side of a scintillator, the measured bar-side time resolution appears well reproduced. *A priori* one does not expect this distribution to be normally distributed so evaluating the distribution is not trivial. Qualitatively, one should expect a symmetric distribution centred at zero. The mean of the distribution is written on figure 4.45 as $0.4 \pm 1.4\,\text{ps}$ with the uncertainty given by the standard error on the mean. The standard error on the mean includes zero, allowing for the conclusion that the covariance contributions from SiPMs coupled to the same side of a scintillator are sufficient to reproduce the measured bar-side time resolution. The skew is computed using the method implemented in the SCIPY.STATS package [133] with the values adjusted for bias by the Fisher-Pearson standardised moment coefficient:

$$g_1 = \frac{m_3}{m_2^{3/2}}$$

where:

$$m_i = \frac{1}{N}\sum_{n}^{N}(x_i - \bar{x})^i$$

The data can be used with the D'Agostino-Pearson $K^2$ test [137], also implemented in SCIPY.STATS, to evaluate the probability that the dataset is drawn from an underlying normally distributed random variable. The statistic $K^2$ is defined as:

$$K^2 = Z_1(g_1)^2 + Z_2(g_2)^2$$

Where $g_1$ is the skew and $g_2$ is the kurtosis, $(m_4/m_2)^2 - 3$. While the distribution of the residual time resolution is not expected to be necessarily be normally distributed, as the D'Agostino-Pearson $K^2$ test deals with the skew and kurtosis, it is well suited to test the symmetry of the distribution and the size of the tails. For normally distributed data, $g_1 = g_2 = 0$, so ideally the $K^2$ is as close to zero as possible. $K^2$ is approximately $\chi^2-$distributed with two degrees of freedom, meaning that a probability value to observe such a value of $K^2$ or greater can be determined from the cumulative distribution function of the $\chi^2$ distribution for two degrees of freedom (1 degree each for skew and kurtosis).

| Parameter | Value |
|---|---|
| Skew | 0.175 |
| Kurtosis | -0.195 |
| D'Agostino-Pearson p-value | 0.743 |

Table 4.4: Measured skew and kurtosis of the residual bar-side time resolution. These values are then used with the D'Agostino-Pearson test to produce a probability that the data comes from an underlying normal distribution.

As shown in table 4.4, the D'Agostino-Pearson test shows that the data are well in agreement with the hypothesis that the residual is drawn from an underlying normal distribution: 74.3% of normal datasets would produce $K^2$ values greater than or equal to what this data reports. Again, there is no argument given that the distribution of the residuals should be normally distributed, just that the residuals should be symmetric, which is supported by the D'Agostino-Pearson test.

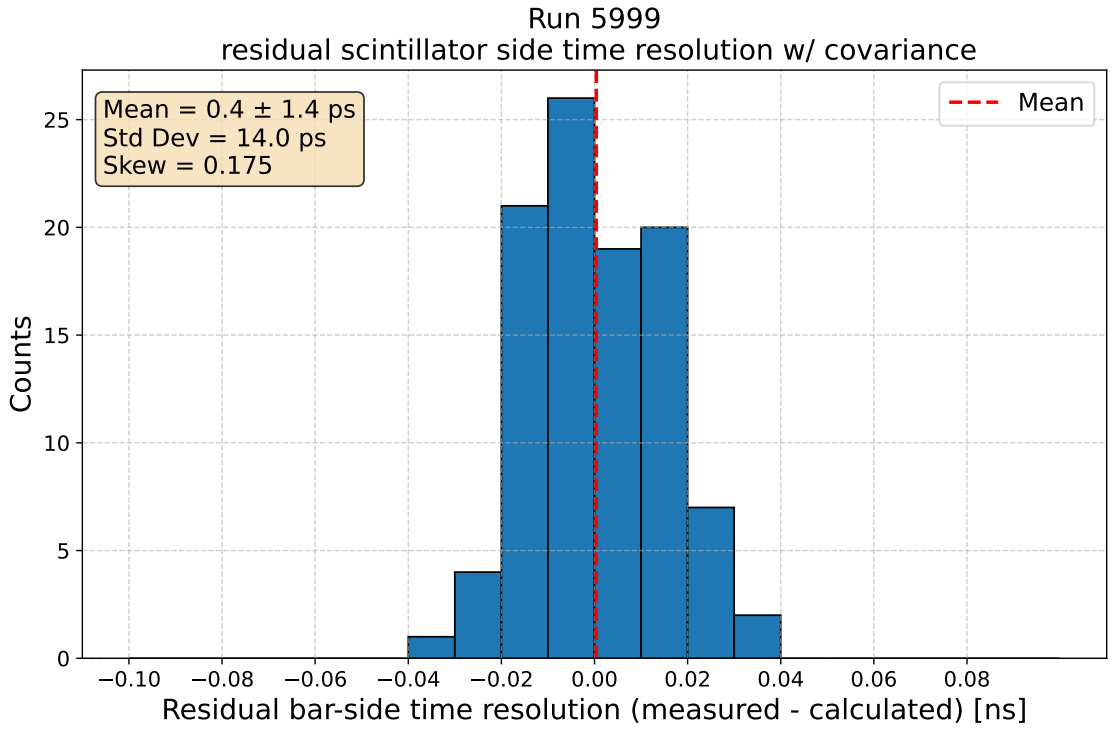



Figure 4.45: Residual in bar-side time resolution between what is directly measured and what is calculated from SiPM resolutions and covariances.

Concerning the use of Gaussian error propagation in equation 4.3.10: while the underlying distributions are known to not be well described by a Gaussian function - hence the use of the FWHM over a Gaussian fit.

The time resolution of a full scintillator can be derived beginning from the average time measured by left and right:

$$dt_{\text{bar}}^{\text{ToF}} = \frac{1}{2}\left(dt_{\text{left}}^{\text{ToF}} + dt_{\text{right}}^{\text{ToF}}\right) \tag{4.3.11}$$

Again using equation 4.3.9 the time resolution of a full scintillator can be expressed as:

$$\sigma^2_{dt_{\text{bar}}^{\text{ToF}}} = \frac{1}{4}\left(\sigma^2_{dt_{\text{left}}^{\text{ToF}}} + \sigma^2_{dt_{\text{right}}^{\text{ToF}}}\right) \tag{4.3.12}$$

The results for the measure time resolution of full scintillators in the US system are shown in figure 4.46

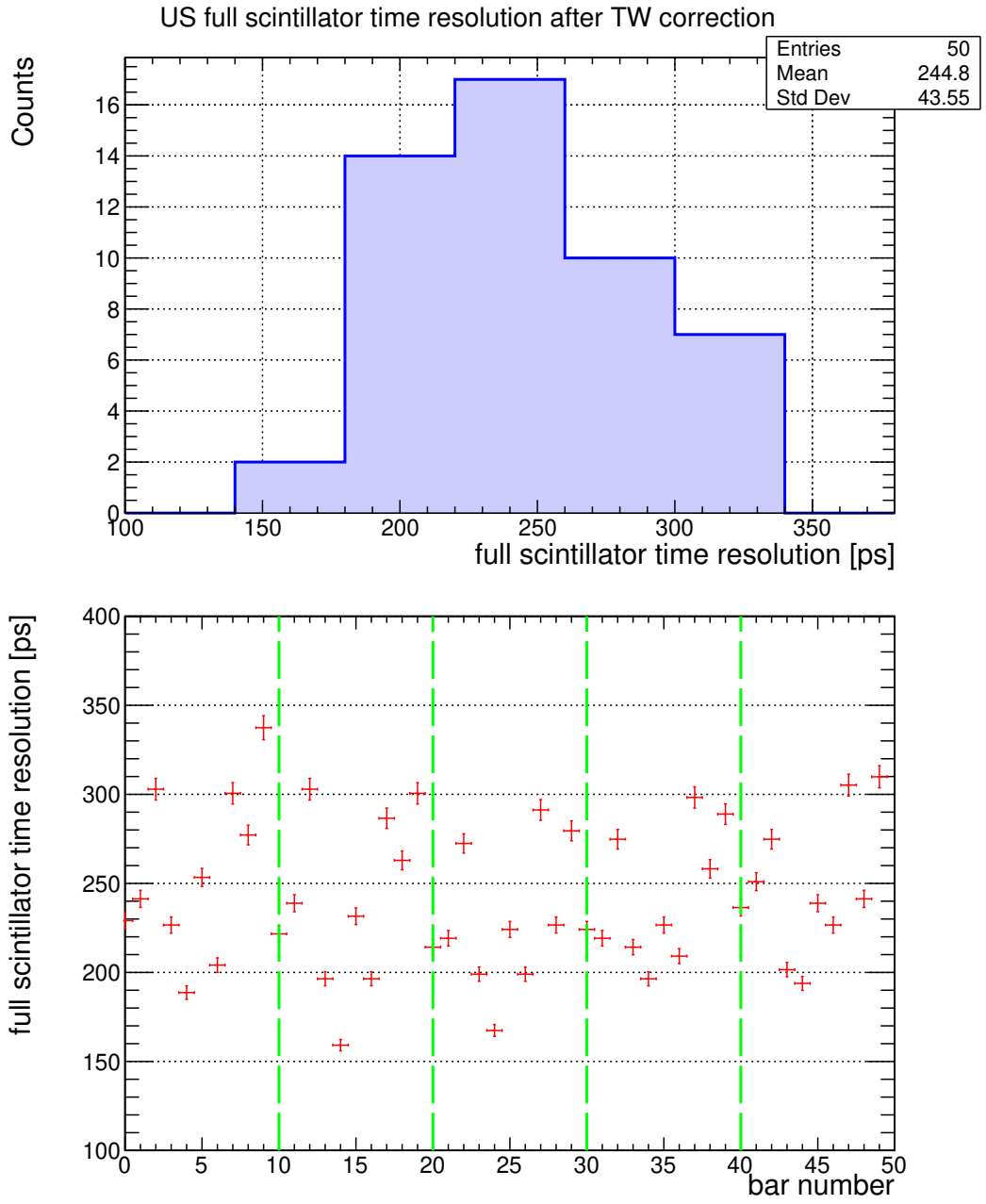



Figure 4.46: Measured time resolution of full scintillators of the US system with data taken from run 5999.

The resolution of the average time recorded by a scintillator can be compared to what is calculated using equation 4.3.12 and the difference between measured and calculated is shown in figure 4.47. The residual measurement shows that the mean and its standard error band do not include zero, indicating the presence of a systematic bias. Since the mean of the residual distribution is negative, the implication is that the calculated values are overestimated.

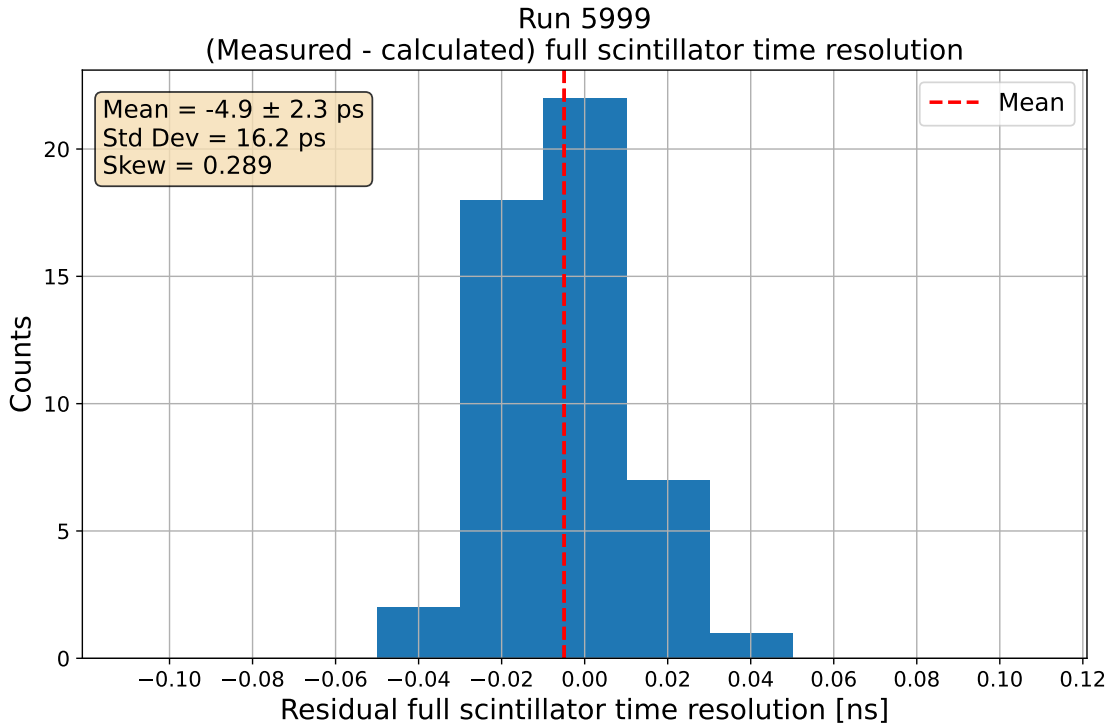



Figure 4.47: Residual (measured - calculated) full scintillator time resolution measurement for all scintillators in the US system.

Since there are only 50 data-points present, the exclusion of zero from the mean estimate was investigated using bootstrapping [138]. For each bootstrap: the dataset is random sampled with replacement to recreate a 'bootstrapped' dataset of the same length and its mean is determined. This is performed $N_{\text{boot}} = 10^3$ times to produce a distribution of bootstrapped mean values, shown in figure 4.48. The maximum confidence level upper bound for which the mean remains negative was determined to be 93% and is shown in the legend of figure 4.48. The 93% confidence level upper limit remains less than zero, indicating that with 93% confidence, the true mean of the residual time resolution is less than zero. To support this further, a p-value of 7.08% is also shown. The p-value represents the probability to observe a mean value this low if the population mean were zero.

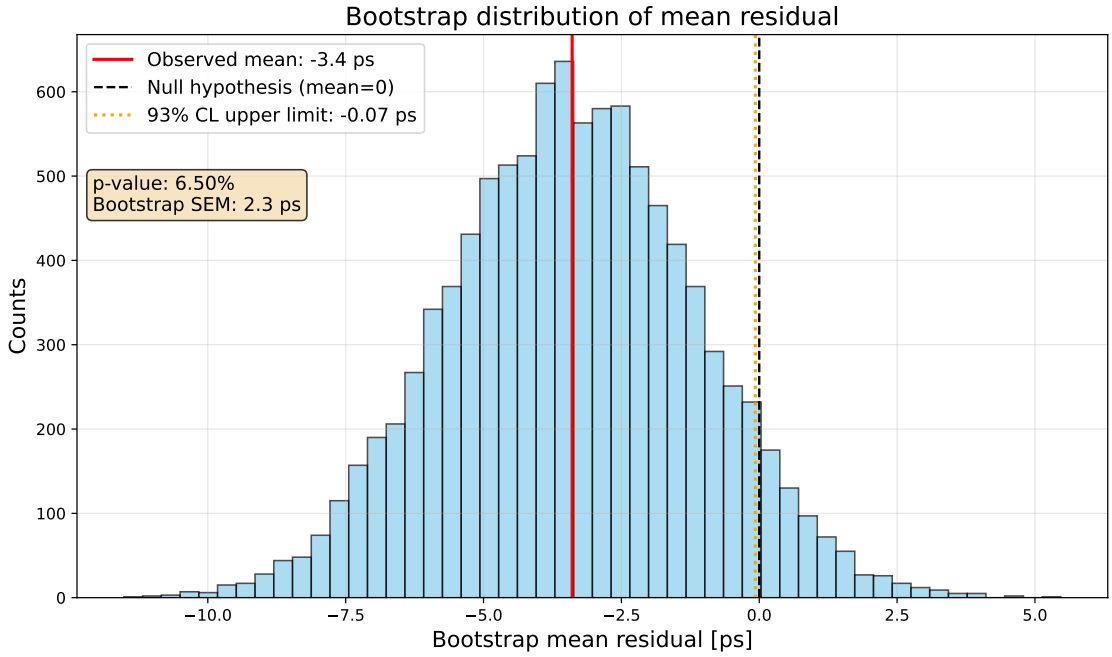



Figure 4.48: For each bootstrap (sample with replacement for the measured residual time resolutions), the mean residual is determined and these bootstrapped means are filled into this histogram.

The evidence greatly supports one of two conclusions: there is a systematic overestimate of the full scintillator time resolution when calculated from equation 4.3.12, or the time resolution is being underestimated when directly measured using equation 4.3.11. While the residual offset is statistically significant, the value is only 3.4 ps, which is far less than the time resolution of a full scintillator.

The D'Agostino-Pearson test is also carried out on the full scintillator residual time resolution data, with the results shown in table 4.5.

| Parameter | Value |
|---|---|
| Skew | 0.296 |
| Kurtosis | -0.617 |
| D'Agostino-Pearson p-value | 0.427 |

Table 4.5: Measured skew and kurtosis of the residual full scintillator time resolution. These values are then used with the D'Agostino-Pearson test to produce a probability that the data comes from an underlying normal distribution.

Once again, albeit to a lesser extent, the data are consistent with a normally distributed quantity, with normally distributed datasets producing greater than or equal values of $K^2$ 42.7% of the time due to random variation.

### 4.3.7 Laser Measurements

The fundamental goals of the dedicated laser measurements campaign were to study how two variables impact the data collected by the SiPMs mounted on the PCB. These variables are light intensity and the number of fired SiPMs, where the number of SiPMs firing is discretised into the number SiPM groups firing, with a group being a group of eight that is coupled to a scintillator in the full experiment. Firstly,

studying the QDC as a function of laser intensity, henceforth referred to as the QDC response curve gives a qualitative view of the saturation observed in the front end. Combining the laser calibration with the quoted photon detection efficiency[11] allows for the laser intensity to be expressed in terms of the number of photons detected by the SiPM. These plots give insight into the cause of non-linearity in the front-end. A simulation of the Hamamatsu S14160-6050HS was performed to understand after how many detected photons one expects SiPM saturation to occur. The result of this simulation is shown in figure 4.49. To perform the simulation, $N_\gamma^{\text{inc}}$ incident photons are fired at the pixel array, where $N_\gamma^{\text{inc}} \times \text{PDE} = N_\gamma^{\text{det}}$ of these photons are detected photons successfully firing pixels on the array, where PDE is the SiPMs photon detection efficiency at the gain of the SiPMs in US system of SND@LHC. The list of pixel indices being hit by a detected photon is used to make a count of the number of uniquely fired pixels, this is the property which makes SiPMs inherently non-linear. Values of $N_\gamma^{\text{inc}}$ from zero to $10^5$ in 200 increments are studied to see how the number of uniquely fired pixels and thus the output signal is dependent on $N_\gamma^{\text{inc}}$. The saturation effects start to become clear from approximately above 50% of the pixels firing, which will require $\approx 10^4$ photons to be fired at the SiPM. Comparing this to figures such as the figures 4.19a, 4.50, it is clear that the saturation effects are occurring between 10% and 20%, clearly before 50%, where we expect SiPM saturation to occur from simulation. It is with this comparison to simulation that the conclusion is drawn that the dominant cause of saturation effects in the SND@LHC is not coming from SiPM saturation. Figure 4.50 shows all the SiPMs QDC saturation curve in the 1-bar illumination regime (highly representative of passing muon events) with the independent variable shown as the equivalent number of incident photons upon the SiPM. The purpose of overlaying all the saturation curves without a legend is not to show the individual SiPM saturation curve, but to show that all SiPMs are clearly losing linearity in their QDC response far before the simulation result of figure 4.49 predicts. The simulation shows that linearity is lost slightly around the point of $20 \times 10^3$ photons whereas the SiPMs of the PCB record an almost entire loss of linearity by $8 \times 10^3$ photons. The comparison to the simulation doesn't provide a definitive explanation for why the QDC linearity is lost this early, but it does narrow down the causes. ADC clipping is an unlikely candidate cause because the channels in figure 4.49 don't show a sharp change in the QDC response. The next potential cause of the problem is the linearity of the TOFPET ASIC. The property has been studied in references [112, 139, 140] which report that the TOFPETs linear energy response holds throughout the nominal dynamic range of 1500 pC.

[11] PDE$\geq$ 50% at 450 nm at 4.5$V$ over-voltage[88].

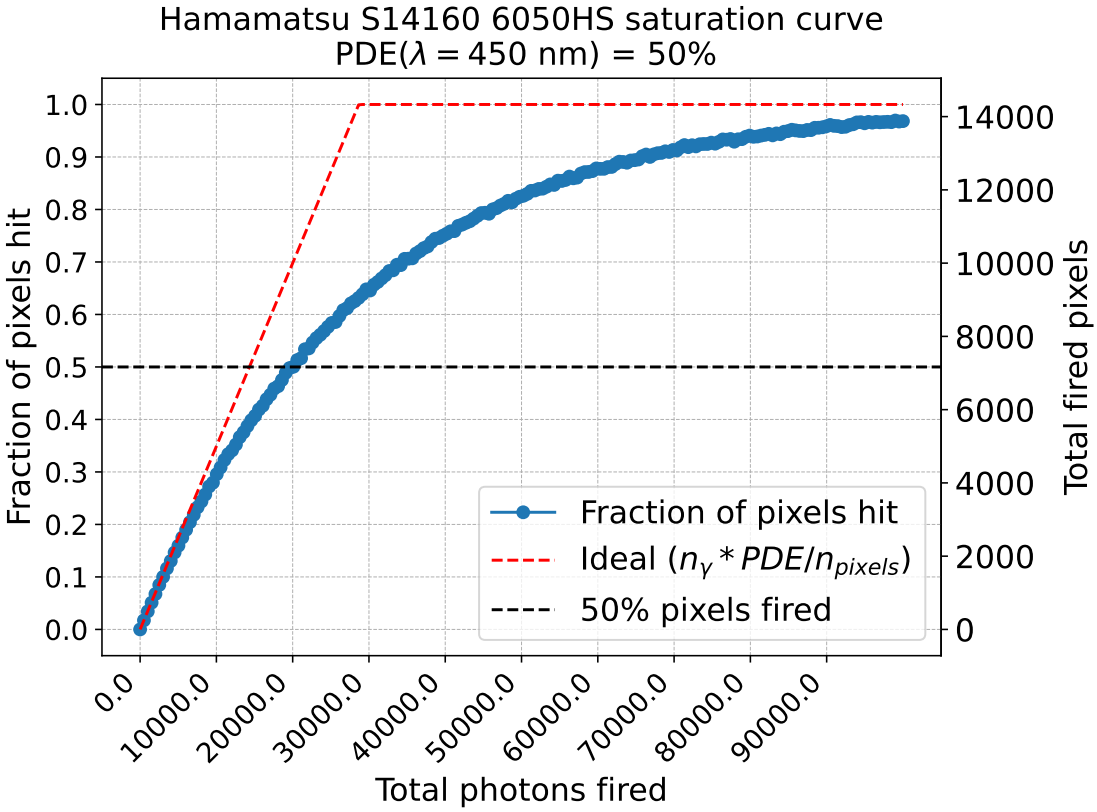



Figure 4.49: Results of a toy Monte Carlo simulation of the Hamamatsu S14160-6050HS to evaluate when SiPM saturation effects become visible.

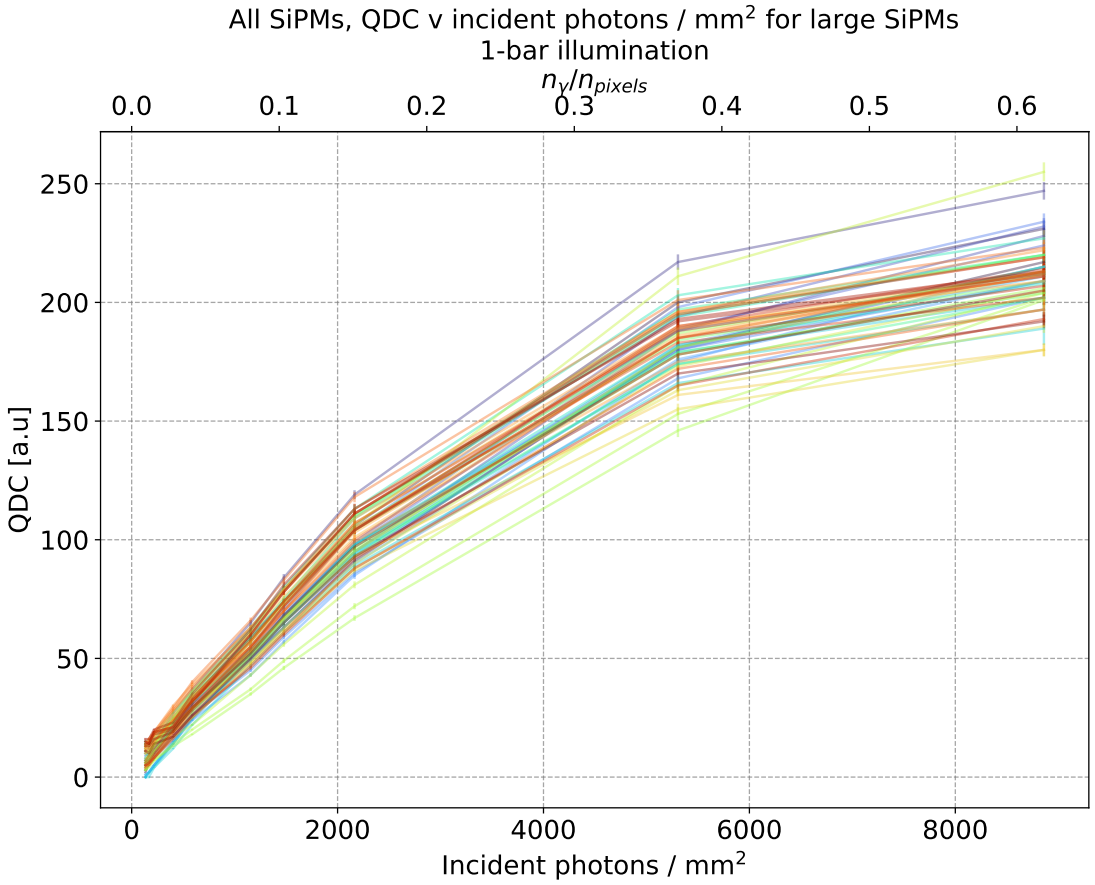



Figure 4.50: The QDC response curve in the 1-bar illumination regime for all the large SiPMs on the PCB. The upper x-axis displays the independent variable as number of incident photons divided by the total number of pixels on the SiPM.

Investigating the time-walk effect with the laser data serves as a good gauge of comparing with TI18 data. As explained in section 4.2, the time-walk is studied by forming a set of time versus QDC data points, where each data point is gathered from one run with the laser at a fixed intensity. It was observed that when the time-walk was compared between the available illumination modes, the firing of more than one SiPM greatly affected the measured time of the data, this is illustrated for one SiPM in figure 4.18. A strong argument behind this behaviour, which is also observed for other SiPMs throughout the PCB, is a transient voltage drop across the shared bias voltage. This transient voltage drop reduces the electric field in the silicon and slows down the avalanche process of all SiPMs detecting a photon. Figure 4.18 shows a roughly 2 ns timing delay induced between the single SiPM illumination regime and any multi-SiPM regime. The voltage drop hypothesis could be tested by monitoring the voltage on the shared bias line when illuminating multiple SiPMs. Cross-talk between the SiPM lines could also play a role here. The hypothesised reason behind why there is no increase in the timing offset displayed in figure 4.18 when greater numbers of SiPMs are illuminated comes from discussion with an electrical engineer in the SND@LHC collaboration. The hypothesis is that the TOFPET2 ASIC uses a feature known as a multiplexor. When multiple channels connected to a TOFPET2 ASIC register data and require use functionality in the ASIC that is not provided to each channel, additional logic is required from the ASIC to prioritise the order with which the data from channels is passed to the shared electronics. This is just

a hypothesis but the important conclusion for the collaboration is to be aware that this feature is present.

For most effective comparison between laser data and TI18, the single-SiPM illumination data was not considered important in this regard. To compare the degree of time-walk between cases one can devise a way to quantify the degree of time-walk: the difference in time measured between two QDC values, $QDC_{high}$ and $QDC_{low}$. Here these values are chosen to be 100 and 5 respectively. Figure 4.51 compares the time-walk observed from the laser studies, with the time-walk observed in TI18 for a SiPM at the same position on the PCB[12].

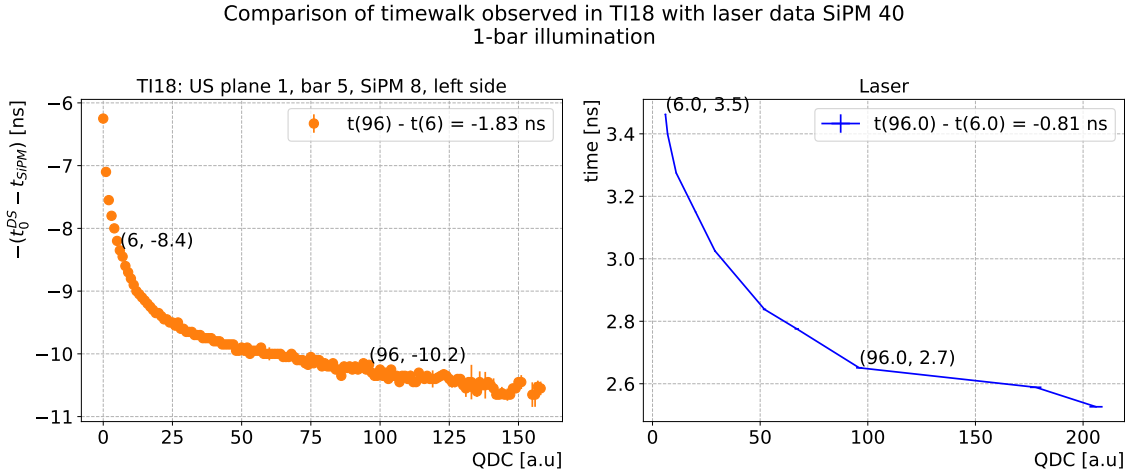



Figure 4.51: The time-walk observed in TI18, on the left, is multiplied by $-1$ in order to display the same inverse-exponential trend. As only the relative timing changes are of interest, this factor of $-1$ has no impact on the results.

Investigating how the degree of time-walk observed compares between laser and TI18 across the PCB for the 1-bar illumination is shown in figure 4.52 where it is clear that there is a substantially larger degree of time-walk observed with the laser measurements. One mechanism is provided to explain this: lower thresholds set in the calibration stage for the laser measurements would decrease the extent of the time-walk effect because the differing rise times of signals will induce a smaller timing difference when the threshold is lower. For this to be the case, the physics environment in the lab in EPFL would have to be less noisy than the TI18 tunnel.

[12]The SND@LHC contains 20 identical PCBs and the degree of time-walk should be invariant among all 20

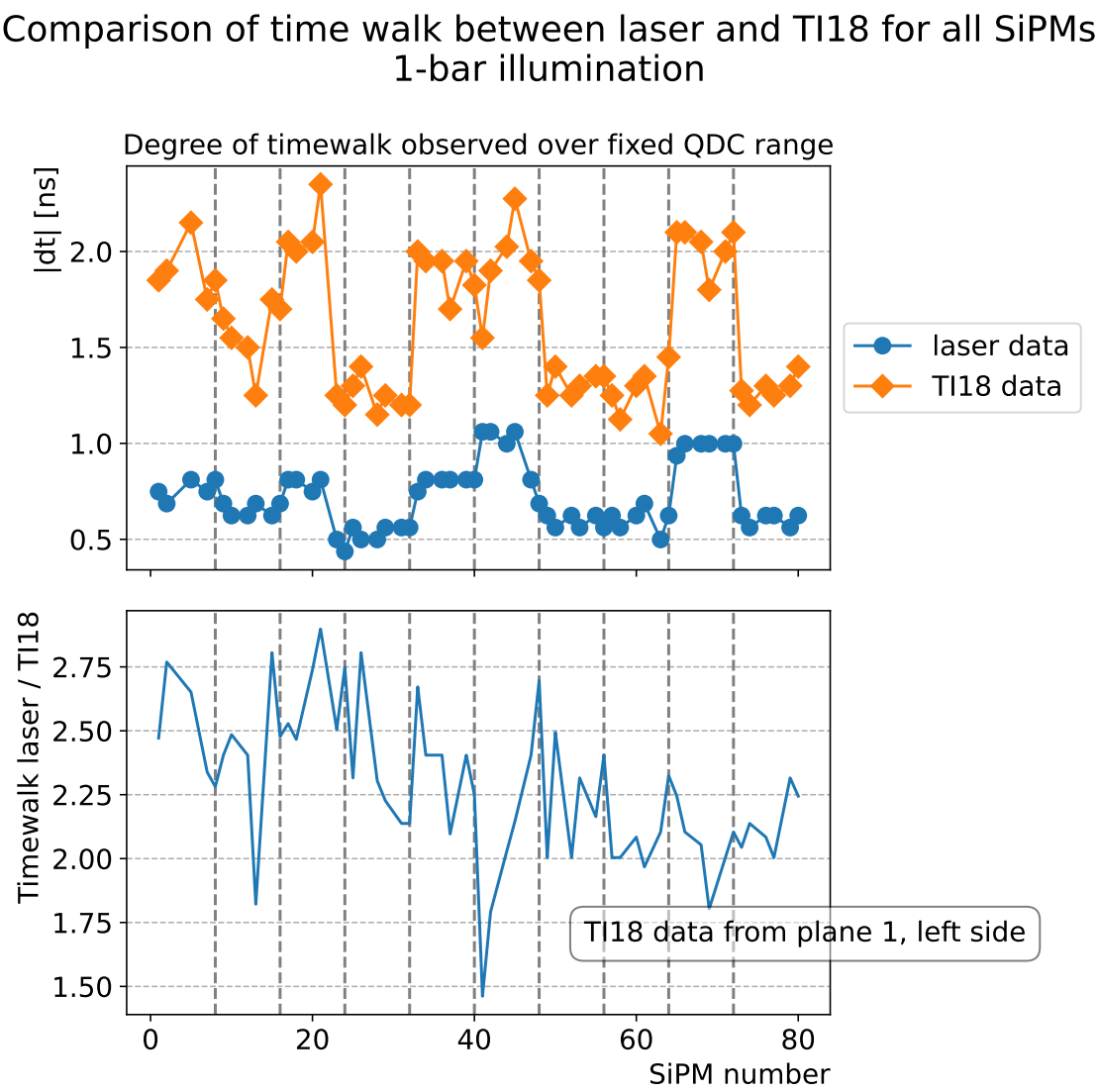



Figure 4.52: Degree of time-walk observed for SiPMs on the PCB compared with SiPMs in the same position of the plane 1, left side PCB of the full detector in TI18. The dashed vertical lines in each subplot distinguish the groupings of eight SiPMs coupled to the same scintillator.

A further area of study with the laser measurements was to investigate the characteristics of the data recorded by the small area SiPMs. The small SiPMs are thus far conspicuous in their absence from this thesis due to their almost complete lack of recorded data in the passing muon selected events. One expects the rate of small SiPM firing in the muon selected events to be far lower due to the lower gain (see section 3.2.3), but in the case of interactions such a muon deep-inelastic-scattering or the shower following a neutrino interaction, there can be very high energy depositions in the scintillator and one would expect the small SiPMs to fire. Time-walk plots and SiPM response curves of the small SiPMs could be made in the same fashion as for the large SiPMs, showing that the small SiPMs were in-fact registering data, but the multi-SiPM runs showed that the small SiPMs were never registering data in the same event as the large SiPMs[13], as shown in figure 4.53. Knowing that the small SiPMs are active, but not in the same event as the large

[13]As the statistics box in figure 4.53 shows, there were zero events with 100% of the illuminated SiPMs firing.

SiPMs poses the question of what the relative delay between the small and large channels is.

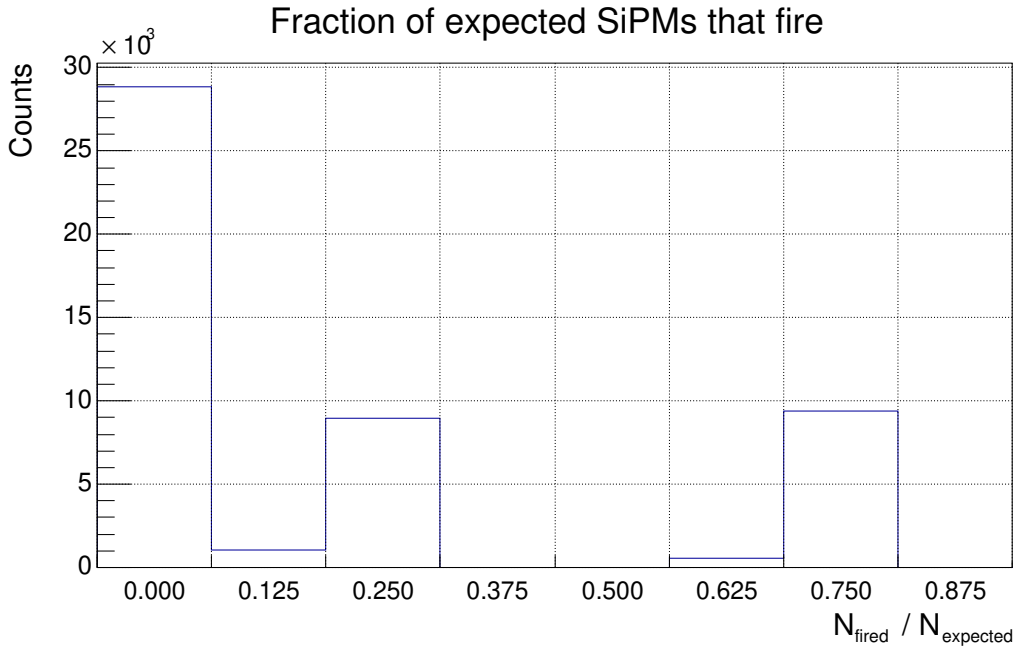



Figure 4.53: Fraction of expected SiPMs that fire in each event for a 1-bar illumination run. All of the expected SiPMs never fire in the same event but the peaks at 25% and 75% correspond respectively to: both small SiPMs firing and the six large SiPMs firing in the same event.

To study this delay, the event in which $\geq$ 50% of the expected SiPMs fire is defined as the start event and from there the next 100 events are scanned to look for the missing small SiPMs. If an expected small SiPM is found, the timing delay and QDC are recorded and the small SiPM is considered to be found. Then by examining this correlation between delay and the QDC recorded, the delay of the small SiPM is extracted as the number of clock cycles after the start event until the maximum of the QDC distribution is observed.

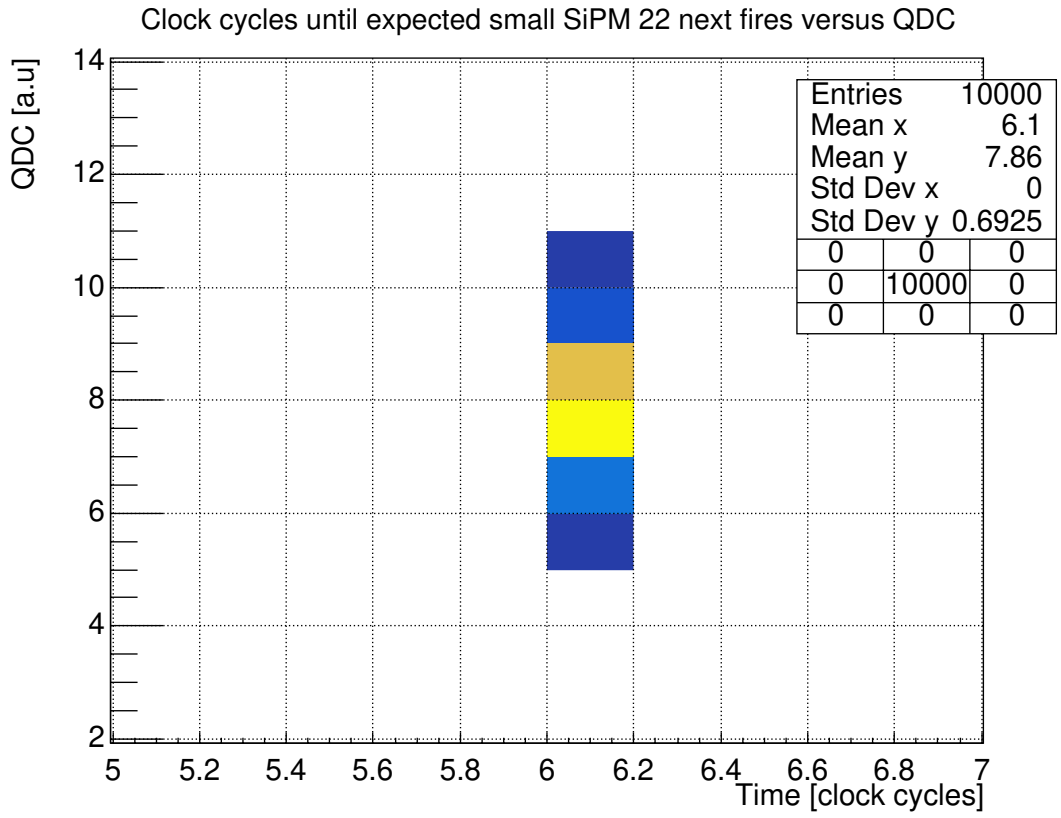
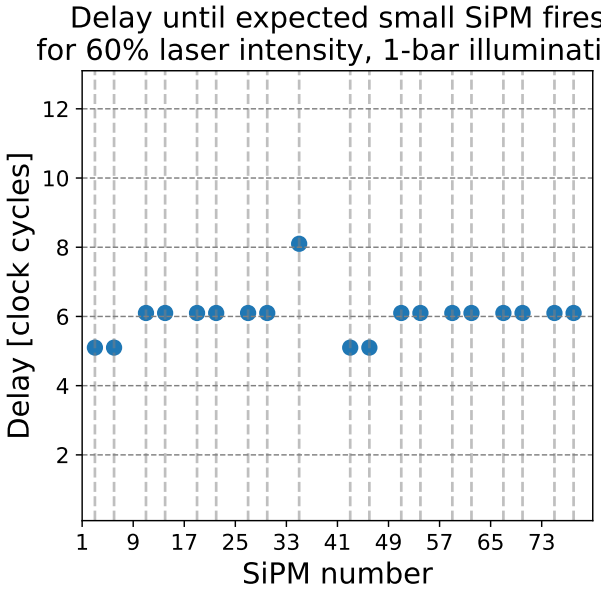



Figure 4.54: Correlation between the delay in clock cycles of the start event and the small SiPM registering data, with the registered QDC.

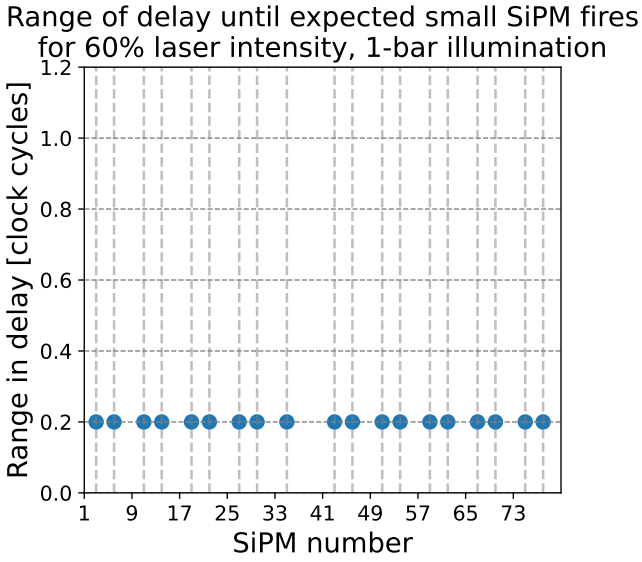
Extracting the delay of each small SiPM for a set illumination mode and a set laser intensity allows one to examine how this delay varies across the PCB, an example of which is shown in figure 4.55a. The delay is determined by first projecting the QDC–delay histogram onto the QDC (ordinate) axis and identifying the mode of this distribution. This modal QDC value defines a narrow slice - one bin wide - along the QDC axis. A timing distribution is then obtained by projecting the histogram onto the abscissa axis, using only entries from the modal QDC bin. The delay is defined as the mode of this resulting timing distribution. This figure also shows how well defined this delay can be: 100% of the data recorded by this small SiPM is within 0.2 clock cycles, which is 1.25 ns. At the highest intensity of the multi-SiPM runs, 60%, the minimal size of this delay is shared by all active small SiPMs in the system, as shown in figure 4.55b. The range in the delay is measured using similar steps to how the modal delay is determined but with one difference: once producing a timing distribution only containing hits from the modal QDC bin, the range is defined as the difference between the low bin-edge where entries first appear (6 clock cycles in figure 4.55a) and the high bin-edge of the last bin to record entries in this timing projection (6.2 clock cycles in figure 4.55a). The uniformity and minimality of the delay shown here may inspire confidence in the future inclusion of the small SiPMs in the US system, however these features are not shared at lower laser intensities of 20% and 40% as is shown in appendix A.14.

The reason for this delay can also be explained by the reasons given for the induced timing delay when moving from a single SiPM illumination to a multi-SiPM illumination: voltage drop and cross-talk. The cross-talk becomes a larger issue when it is from a large channel induced to a smaller channel due in-part to the

ratio of the sensitive areas, $A_{\text{large}}/A_{\text{small}} = 4$; however, the two types of SiPM do not have the same pixel pitch and geometric fill factors. The geometric fill factors of the large-area and small-area SiPMs are 74% and 31%, respectively; the pixel pitches of the large-area and small-area SiPMs are 50µm and 10µm, respectively. These values are not sufficient to make a definitive statement on the scale of the cross-talk as other factors such as over-voltage, gain and electronic response must be considered.




(a) Small SiPM delays for 1-bar illumination and 60% laser intensity.




(b) Small SiPM delay ranges for 1-bar illumination and 60% laser intensity.

One point in figure 4.55a is that a single small area SiPM was not functioning on the PCB used for the laser measurements, hence the single data point at SiPM number 35 without its partner small SiPM for that group which should be at number 38. The majority of the delays observed are identical but with exceptions. The delays of the small SiPMs are further studied at the end of the next chapter with TI18 neutrino events but the small SiPMs are not a primary area of concern of this thesis. The study of their behaviour is here to inform future studies seeking to improve an energy calibration of the Upstream Muon system.

# Chapter 5

# Spatial measurements using the aligned Upstream system

With the timing response of the scintillators aligned in the previous chapter, this chapter explains how a digitisation model in the simulation is constructed that allows for a comparison to the real data. Following this, the time resolution of the US can be translated into a spatial measurement which begins to bring the US to a point where it can actively contribute to the physics analyses of the experiment.

## 5.1 Method

### 5.1.1 Simulation implementation

Chapter 4 dealt with how to correct for the time-walk effect and align the timing response of all SiPMs in the US, equation 4.1.13 describes the timing components of the average time recorded by the SiPMs coupled to one side of a scintillator, $dt_{\text{side}}^{\text{ToF}}$, and can be written out relative to the DS system reference time, without a signal time-of-flight correction, as:

$$dt_{\text{side}} = \frac{1}{N_{\text{side}}} \sum_{\text{SiPM}}^{N_{\text{side}}} t_0^{\text{DS}} - t_{\text{SiPM}}^{\text{TW}} - d_{\text{SiPM}} \tag{5.1.1}$$

We know from equation 4.1.12 the alignment parameter is defined such that:

$$d_{\text{SiPM}} = E\left[t_0^{\text{DS}} - (t_{\text{SiPM}}^{\text{TW}} \pm t_{\text{SiPM}}^{\gamma})\right] \tag{5.1.2}$$

Where $E[X]$ denotes the expectation value of quantity $X$, $t_{\text{SiPM}}^{\gamma}$ is the scintillation photon time-of-flight from the energy deposition point to the SiPM first defined in equation 4.1.8 and the sign is positive (negative) for SiPMs on the left (right). Putting equations 5.1.1 and 5.1.2 together demonstrates that the only remaining timing component comes from the photon time-of-flight, given by the effective signal speed and the distance travelled by the signal, $t_{\text{side}}^{\text{ToF}} = (\sum_i^N t_i^{\text{TW ToF}} - d_i)/N$. Thus, if one also accounts for the photon time-of-flight in the scintillator, the side time distribution should be centred at zero, as was demonstrated in section 4.1.8. This alignment leads to a model that can be implemented in the SND@LHC simulation framework, as the effective signal speeds are measured in the previous chapter and the distance travelled by the signal is a known quantity in the simulation.

The measured time resolutions are not sufficient to capture the observed timing behaviour of the SiPMs in Upstream Muon System of SND@LHC. As discussed in

section 4.3.3 and section 4.3.6, the large tails in SiPM timing data and the bar-side timing data create difficulty in fitting the observed data with a function. Due to this, the time resolution was extracted as FWHM/2.355, which approximates a Gaussian, the large non-Gaussian tails mean that if the time resolution were implemented in the simulation with a simple Gaussian, there would be a large discrepancy between simulation and data. To resolve this, the scintillation photon ToF corrected and aligned bar-side time distributions, $dt^{\text{ToF}}_{\text{side}}$, defined in equation 4.1.13, measured in bins of $x-$position are used to create look-up histograms for the time resolution when digitising the simulation. An example of the bar-side aligned time in bins of $x-$position for both sides are shown in the two sub-plots of figure 5.1. Each aligned side-time distribution for a given bin in $x-$position is normalised to unit integral, approximating a PDF. When digitising the time response of the US system, the relevant normalised aligned side-time distribution is fetched and a value from it is randomly chosen according to the PDF. This data-driven approach encapsulates the exact timing behaviour observed in all scintillators such as scintillator-specific optical coupling imperfections, scintillator imperfections such as scratches or large scale scatterers as well as the residual time-walk effect and the edge effect observed around $x = 0$ in figure 5.1a which is discussed in detail in section 5.2.2.

Having used these look-up tables to smear the photon time-of-flight in the simulation, the resulting aligned side-time distributions in simulation are cross-checked to ensure the implementation is successful, the equivalent plot of figure 5.1a following the digitisation of passing muon simulation data is shown in figure 5.2.

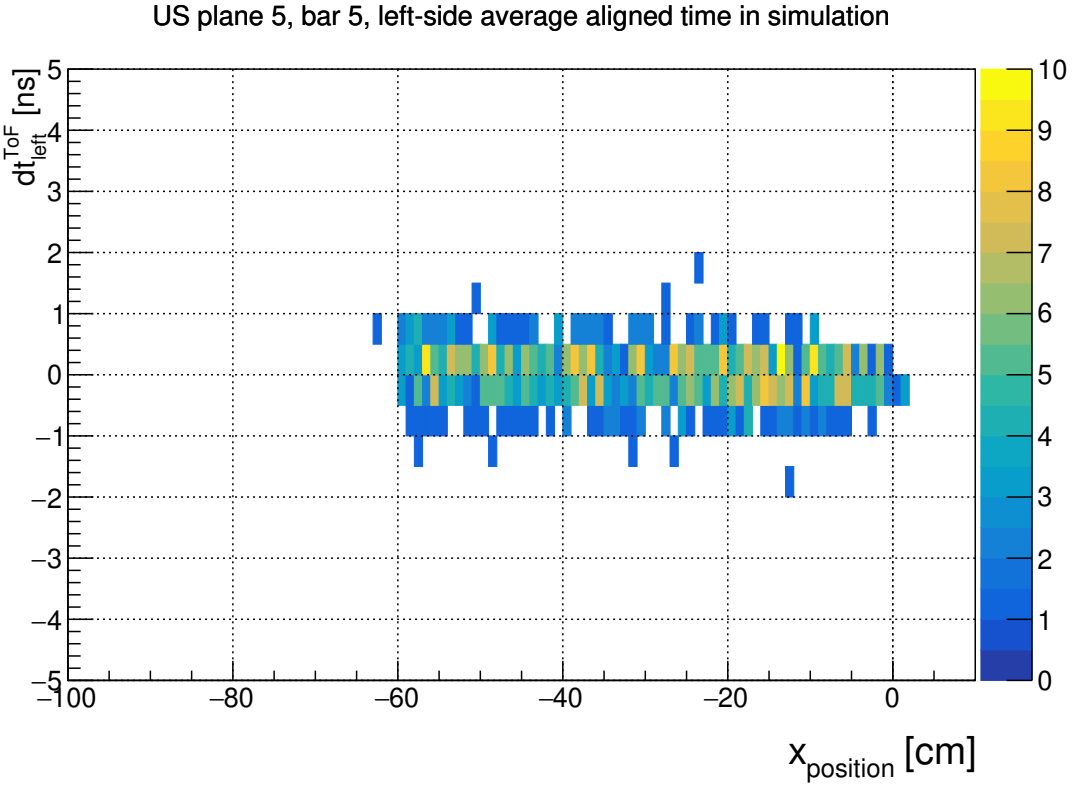
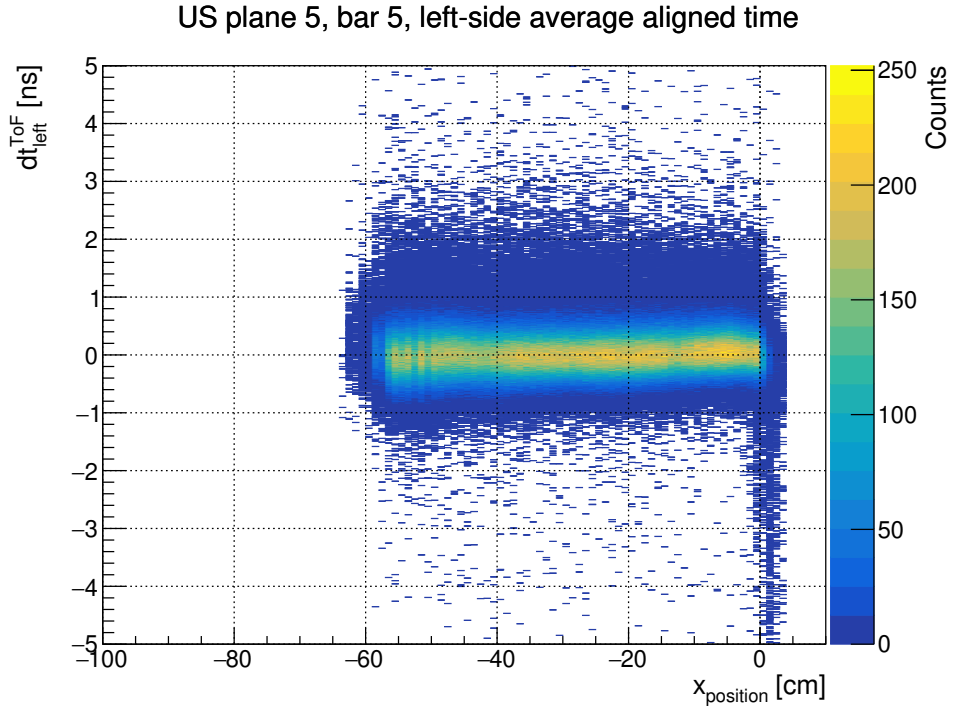
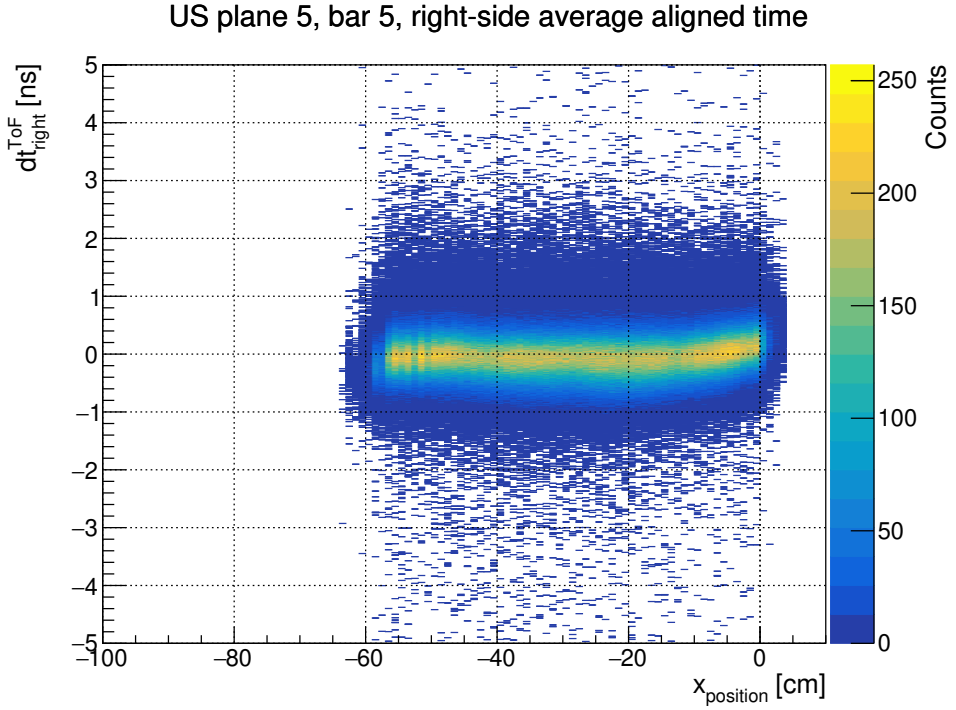



Figure 5.2: Aligned bar-side time distribution following digitisation of passing muon simulation data in bins of $x-$position for the left-side of the fifth scintillator in the fifth plane of the US system. The full sample size of passing muon simulation is used here.

While the look-up tables are used to accurately represent the time resolution




(a) Scintillation photon ToF corrected and aligned bar-side time, $dt_{\mathrm{L}}^{\mathrm{ToF}}$, distribution from selected passing muon events in bins of $x-$position for the left-side of the fifth scintillator in the fifth plane of the US system.




(b) Scintillation photon ToF corrected and aligned bar-side time, $dt_{\mathrm{R}}^{\mathrm{ToF}}$, distribution from selected passing muon events in bins of $x-$position for the left-side of the fifth scintillator in the fifth plane of the US system.

Figure 5.1: Scintillation photon ToF corrected and aligned bar-side times in bins of $x-$position for left (right) SiPMs on the top (bottom) sub-figure. The sub-figure plots are taken from the same scintillator.

of a bar-side measured in the data, there are few statistics per bar-side in the passing muon simulation, as is clear by the sample size of figure 5.2, meaning that the large tail at $x \approx 0\,\text{cm}$ shown in the experimental data of figure 5.1a is not visible in the simulated analysis because of its small effect as a fraction of the total statistics. A final but important step in the simulation digitisation implementation is a digitisation threshold: a lower limit on the energy deposition in the scintillator that should be reached for one to expect that the detector registers the energy deposition as a fired scintillator. The laboratory measurements to experimentally measure this threshold have not been performed so the setting of this threshold is done simply by varying this threshold in simulation and inspecting which threshold value gives the most realistic results. The value is set such that both sides of the scintillator, accounting for signal attenuation, must receive at least 600 keV. The analysis leading to this decision is outlined subsequently in section 5.2.1.

### 5.1.2 Spatial measurements in the US system

The aligned average time measured by a bar-side is multiplied by the average signal speed for that bar-side to produce a position in reference to the bar-centre and given a superscript $m$ to denote the position is in reference to the bar-centre[1]. This value is then translated into the physics frame of reference using equations 5.1.3 and 5.1.4, then referred to as $x_{\text{side}}$, with left and right being denoted as $L$ and $R$ respectively, as shown in figure 5.3. Henceforth the spatial measurement produced from one side of a scintillator, $x_{\text{side}}$, may be referred to as a one-sided spatial measurement. With both one-sided spatial measurements made, the average of these two positions corresponds to what is henceforth referred to as the $x-$barycentre, $x_B$, for this scintillator and is stated in equation 5.1.5. The difference between $x_L$ and $x_R$, $x_L - x_R$, gives a handle on the energy deposition width in $x$, henceforth referred to as $\lambda_x$ and stated in equation 5.1.6. Each firing scintillator in a given plane of the US provides independent measurements of $x_B$ and $\lambda_x$. An assumption in this model is that the earliest arriving photons set the time on each side. This is reflected in figure 5.3 by the vectors $x_L$ and $x_R$ pointing from $x = 0\,\text{cm}$ to the left and right edges of the generic energy deposition (red shaded circle) respectively.

$$x_L = x_m + c_L dt_L \tag{5.1.3}$$
$$x_R = x_m - c_R dt_R \tag{5.1.4}$$
$$x_B = \frac{1}{2}(x_L + x_R) \tag{5.1.5}$$
$$\lambda_x = (x_L - x_R) \tag{5.1.6}$$

The uncertainties on the one-sided spatial measurements and the $x-$barycentre are expressed as follows:

$$\sigma^2_{x_{\text{side}}} = dt^2_{\text{side}}\sigma^2_{c_{\text{side}}} + c^2_{\text{side}}\sigma^2_{dt_{\text{side}}}, \quad \sigma_{x_m} \approx 0 \tag{5.1.7}$$
$$\sigma_{x_{\text{B}}} = \frac{1}{2}\sqrt{\sigma^2_{x_{\text{L}}} + \sigma^2_{x_{\text{R}}}} \tag{5.1.8}$$

The uncertainty on $x_m$ is neglected from equation 5.1.7 because the civil engineering survey quotes the alignment to be within 0.5 mm for all detector components

[1]The reference point in $x$ is an artefact of the time-walk correction process and is explained in subsection 4.1.4.

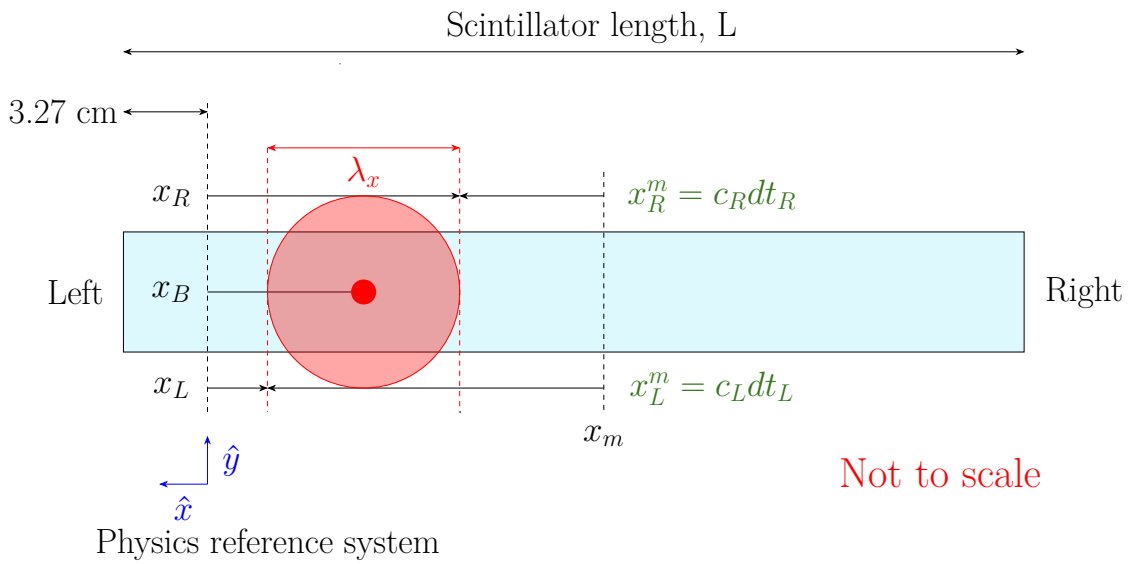



Figure 5.3: Schematic of the model used for producing spatial measurements in the US. The bar-centre, $x_m$, is the reference point to which the aligned times are produced. The shaded red circle represents a generic energy deposition in the scintillator where the left- and right-edges of this energy deposition create the photons which set the time on those respective sides.

[85]. Depending on the analysis goal, the set of measurements provided by a plane can be handled differently, such as by selecting the bar with the highest QDC, or by weighting the measurement from each bar, where the weight, $w_{\text{bar}}$, is given by the fraction of the total QDC in the plane that is recorded by the bar in question, $\text{QDC}_{\text{bar}}/\text{QDC}_{\text{plane}}$.

A key point in this model is that the resolution of either one-sided spatial measurement includes the resolution on the event-wise reference time, $\sigma_{t_0^{\text{DS}}}$. When measuring the resolution of $x-$barycentre, the component of resolution from the event-wise reference time cancels due to the different signs for left- and right-side spatial measurement calculations, shown in equation 5.1.3, 5.1.4. In the measurement of the resolution on $\lambda_x$ however, the resolution component from the event-wise reference time does not cancel, an effect which will be referred to in section 5.2.2.

In the $y-$dimension, only the QDC measured by each bar can be used to give a handle on $y-$position and the width of energy deposition in the $y-$dimension. Each bar provides a measurement of the energy loss in that bar of the plane[2]. Thus, one can use the fired scintillators in a plane to provide an estimate on the centre of the energy deposition in the $y-$dimension by QDC-weighting the $y-$positions of the scintillators, referred to as the $y-$barycentre, $y_B$:

$$y_B = \frac{1}{\text{QDC}_{\text{plane}}} \sum_{\text{bar}} \text{QDC}_{\text{bar}} \times y_{\text{bar}} \tag{5.1.9}$$

Where $\text{QDC}_{\text{bar}}$ is the total QDC measured in a scintillator by the SiPMs on each side, $\text{QDC}_{\text{bar}} = \sum_{\text{SiPM}} \text{QDC}_{\text{SiPM}}$. $\text{QDC}_{\text{plane}}$ is the total QDC measured in the plane by its constituent bars, $\text{QDC}_{\text{plane}} = \sum_{\text{bar}} \text{QDC}_{\text{bar}}$. The uncertainty in the $y-$barycentre is determined by propagating the uncertainties of equation 5.1.9:

[2]Energy loss in a scintillator scales with QDC and should be proportional within a certain range, however a calibration was not in place at the time of this thesis.

$$\sigma^2_{y_B} = \sum_{\text{bar}} w^2_{\text{bar}} \sigma^2_y$$

Where $w_{\text{bar}} = \text{QDC}_{\text{bar}}/\text{QDC}_{\text{plane}}$. The uncertainty the $y-$position measurement from each firing scintillator, $\sigma_y$, is treated as a uniform distribution of width 6 cm, the scintillator width in $y$: $\sigma_y = (6/\sqrt{12})\,\text{cm} = \sqrt{3}\,\text{cm}$:

$$\sigma_{y_B} = \sqrt{3 \sum_{\text{bar}} w^2_{\text{bar}}}\,\text{cm} \tag{5.1.10}$$

Then to produce a measure of the energy deposition width in $y$, $\lambda_y$, the difference in $y-$position between the highest bar and lowest bar is quoted:

$$\lambda_y = \Delta(y_{\text{max}},\, y_{\text{min}}) \tag{5.1.11}$$

A schematic of how $y_B$ and $\lambda_y$ are modelled is shown in figure 5.4 with the shaded red circle representing a generic energy deposition in a plane of scintillators.

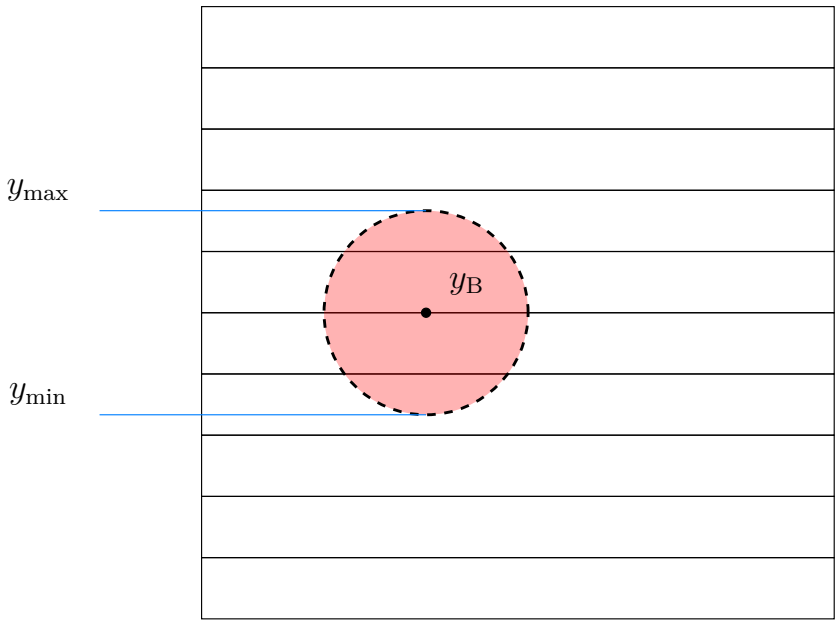



Figure 5.4: Illustration of four bars in the plane of the SND@LHC US system firing when traversed by a particle shower, illustrating how the $y-$barycentre and $\lambda_y$ values are measured.

With the model and algorithms in place, validation studies with selected muon events were performed. As mentioned in section 3.3, the majority of physics events in the experiment are prompt muons produced at IP1 which provide an ideal environment for commissioning work. Applying the same muon selection criteria as table 4.1, extrapolating the fitted DS track to the $z-$position of each plane of the US system to extract the expected track position, $x_{\text{track}}$, then evaluating the residual between the US spatial measurement and $x_{\text{track}}$ provides a handle on the spatial resolution of the US.

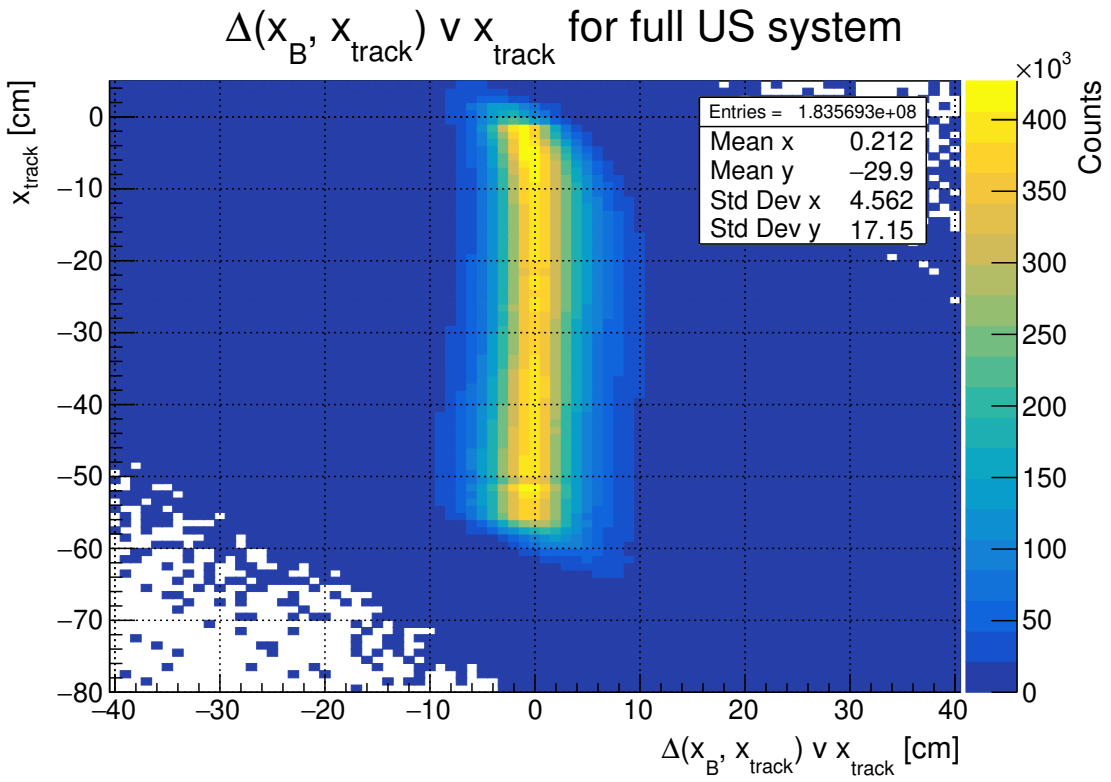



Figure 5.5: Residual between x-barycentre, $x_B$, and extrapolated track position, $x_{\text{track}}$, as a function of extrapolated track position for all scintillators in the US overlaid. Measured with data from run 5999.

Figure 5.5 shows this residual using the $x-$barycentre, $\Delta(x_B,\, x_{\text{track}})$, the residual can equally be computed using either one-sided spatial measurements, as will be investigated in section 5.2.2:

$$\Delta(x_i,\, x_{\text{track}}) = x_i - x_{\text{track}}, \qquad i \in \{\text{L}, \text{R}, \text{B}\} \tag{5.1.12}$$

The majority of the range of $x_{\text{track}}$ shows that the $x-$barycentre and track position are in good agreement, besides the large $x_{\text{track}}$ region. The causes for this divergence in agreement in the high $x_{\text{track}}$ region are discussed subsequently in section 5.2.2. A measurement of the $x-$barycentre resolution across the entire $x_{\text{track}}$ range can be quoted as the FWHM of the $\Delta(x_B,\, x_{\text{track}})$ distribution by projecting the two-dimensional onto the abscissa axis, this gives a FWHM of 6.8 cm. An analogous plot for the energy deposition width in $y$, as well as other measurable quantities, will also be discussed. The same plots as these are also investigated in the simulated data, which will also be discussed in this chapter.

### 5.1.3 Angular measurements

The spatial resolution of the SciFi tracker is far superior than that of the US due to the fine granularity of the fibres [141]. This makes the SciFi an ideal subsystem for a spatial measurements. However, the probability that a neutrino interaction is in a particular wall of the five target walls is a constant; 20% of neutrino interactions will take place in the final interaction wall, where only one spatial measurement in each of the $xz$ and $yz$ projections can be made with the SciFi tracker. A further 20% can have two spatial measurements per projection. For this reason, understanding

how the US system can work with the SciFi tracker to produce spatial information is important in maximising the physics capacity of SND@LHC. Angular measurements combine both systems allowing for the capacity of the subsystems' cooperation to be studied. Spatial measurements in the SciFi are easily determined due to the fine granularity in both $x$ and $y$ [85]. The SiPMs reading out vertically aligned fibre mats are referred to as vertically aligned SiPMs while SiPMs reading out horizontally aligned fibre mats are referred to as horizontally aligned SiPMs. A measurement of position in each projection is extracted as the median of the distribution of firing SiPM position, an example of the horizontal and vertical positions extracted from the SciFi is in figure 5.6. The uncertainty on this median hit position in each projection is taken as the standard error of the distribution.

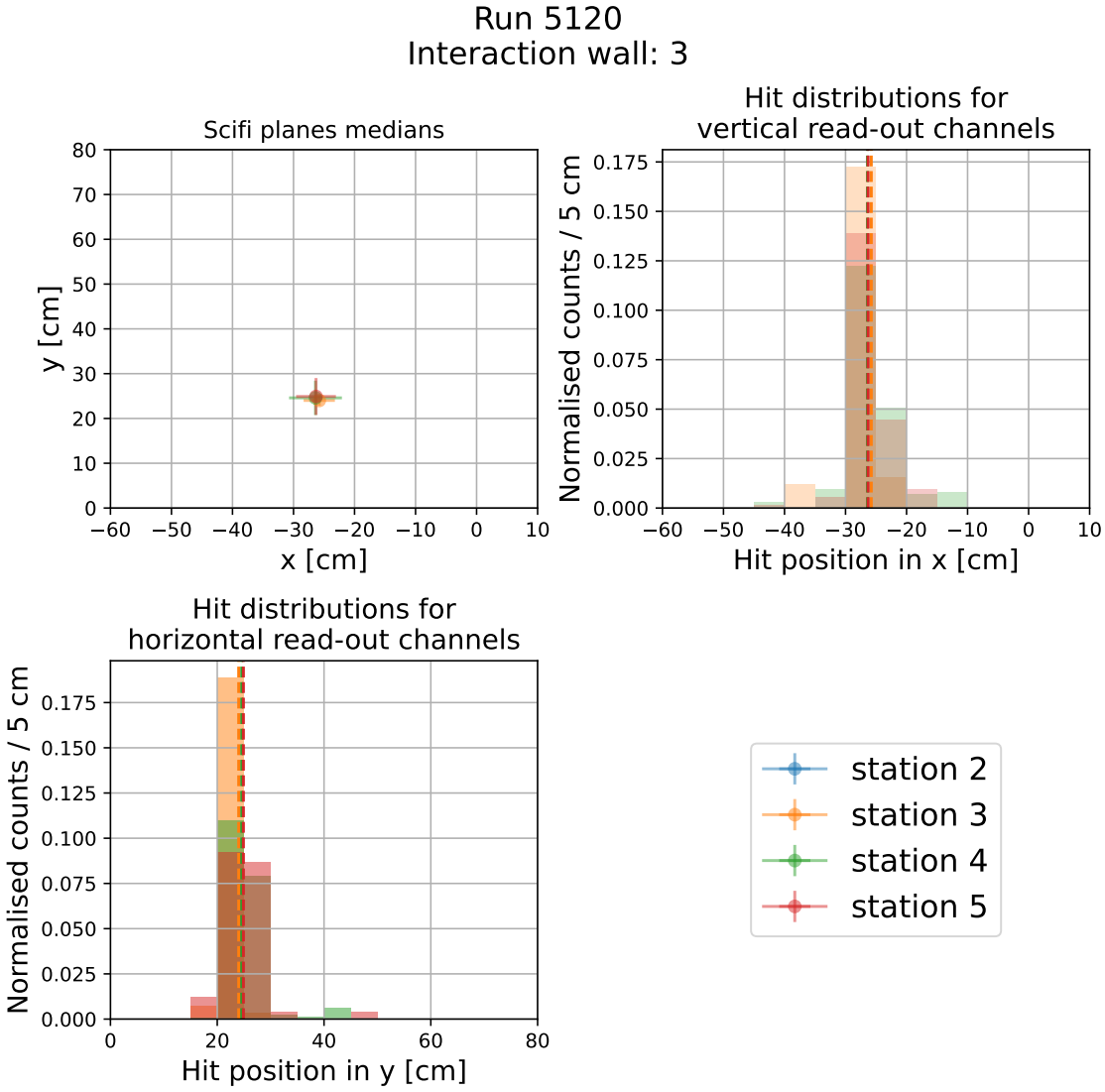



Figure 5.6: Distribution of SciFi firing SiPM positions for vertically aligned SiPM on the left and horizontally aligned SiPM on the right. The median value per plane for each projection is taken as the hit position. The specific data shown here is from the selected $\nu_\mu$ CCDIS event in run 5120.

A linear fit is performed using the combined set of SciFi tracker and US system spatial measurements with a $\chi^2$-minimisation method. The $\chi^2$-minimisation method here is important given the stronger spatial resolution of the SciFi. For

each projection, the function to be minimised is:

$$\chi^2(m, c) = \sum_i \frac{(x_i - (mz_i + c))^2}{\sigma_i}$$

The best fit values of $m$ and $c$ for each projection is then found by taking partial derivatives of $\chi^2$ with respect to each fit parameter:

$$\frac{\partial \chi^2}{\partial m} = 0, \quad \frac{\partial \chi^2}{\partial c} = 0$$

Thus, the better spatial resolution of the SciFi tracker is encoded in the angular measurements. Having performed the linear fit, the direction vectors in both projections, $xz$ and $yz$, can be defined:

$$\vec{d} = \frac{1}{\sqrt{\left(m_x^2 + m_y^2 + 1\right)}} (m_x,\, m_y,\, 1) = (d_x,\, d_y,\, 1) \tag{5.1.13}$$

The direction vectors then allow for the angular measurement to be computed. The polar angle, $\theta$, and the azimuthal angle, $\phi_X$, are computed as shown below in equations 5.1.14 and 5.1.15:

$$\theta = \arctan\left(\sqrt{d_x^2 + d_y^2}\right) \tag{5.1.14}$$

$$\phi = \arctan\left(d_y,\, d_x\right) \tag{5.1.15}$$

Where $\phi$ is computed using the two-component inverse tangent function to ensure the correct sign on the result. The angles computed using the spatial measurements of the SciFi and US systems can then be compared to the same angles measured with the track formed in the DS. The differences between these measured angles are defined in equations 5.1.16, 5.1.17:

$$\Delta\left(\theta_{\text{SF+US}},\, \theta_{\text{DS}}\right) = \theta_{\text{SF+US}} - \theta_{\text{DS}} \tag{5.1.16}$$

$$\Delta\left(\phi_{\text{SF+US}},\, \phi_{\text{DS}}\right) = \phi_{\text{SF+US}} - \phi_{\text{DS}} \tag{5.1.17}$$

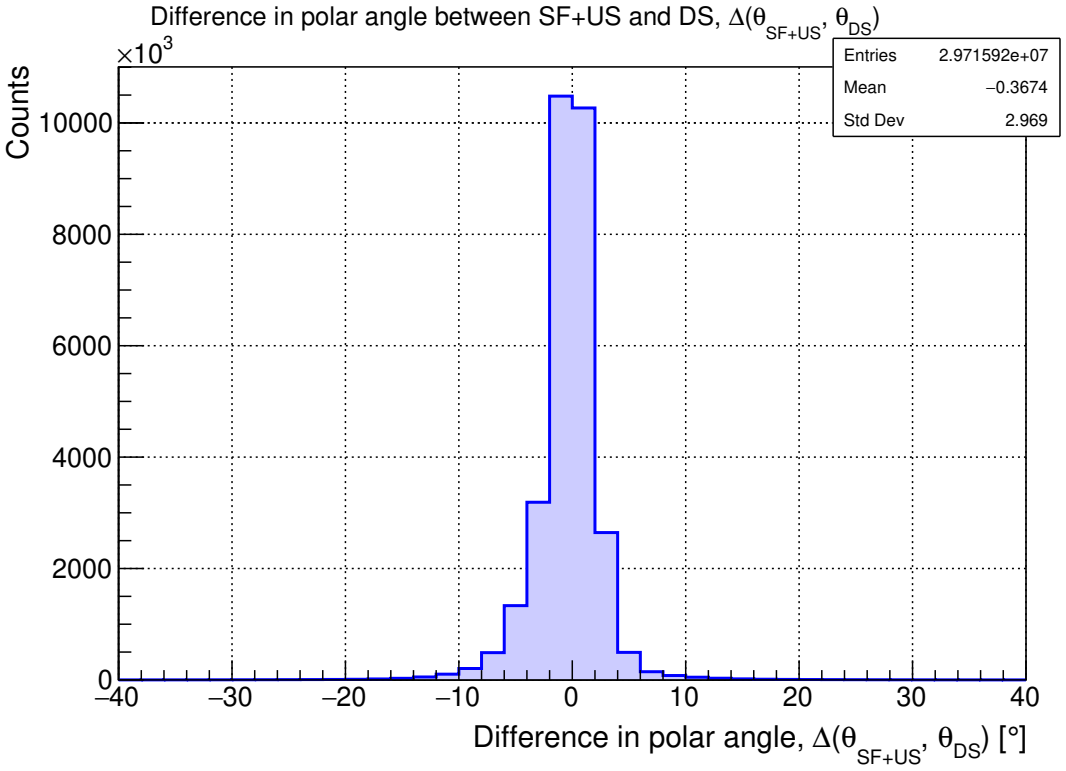



Figure 5.7: Distribution of the polar angle, $\theta$, residual measurement between the SciFi + US system (SF+US) and the muon system (DS) for muon events.

Figure 5.7 shows that the SciFi and US composite system is able to produce polar angles in strong agreement with the DS muon system. The accuracy of the composite system is strongly dependent on the spatial resolution of the SciFi. It is established in the collaboration that muon tracking can be performed with the SciFi tracker so figure 5.7 acts mostly to check that the inclusion of US system data does not spoil the ability of the SciFi tracker to produce track fits. While it is expected that the inclusion of the US in muon tracking does not

### 5.1.4 Kinematic studies of the high purity selection of muon neutrino charged current events

Section 3.5.2 outlines the work by the collaboration in identifying a high purity (low signal efficiency, high background rejection efficiency) sample of muon neutrino CCDIS events. These selected events provide an opportunity to study the US in an environment different to that of passing muons. Namely, how the US responds in cases of higher fired SiPM multiplicity and higher energy deposition. The results of studying several quantities are discussed in the following sections.

The feasibility of determining angular measurements is also studied with these selected $\nu_\mu$ CCDIS events. No run contains more than one selected $\nu_\mu$ CCDIS event so in the discussion of these events, the events will be referred by the run number they are in. Only a subset of these events were of sufficient quality to be included in the collaboration paper in PRL [125]. The whole high purity set of events will be referred to as the $\nu_\mu$ CCDIS events while the subset of these used in the PRL publication are referred to as the PRL selected events. The additional selection criterion which distinguish the PRL events is that to be included in PRL the interaction must occur

in either interaction wall number 3 or 4, whereas the whole high purity $\nu_\mu$ CCDIS include events from walls 2, 3, 4 and 5. Which target wall the interaction occurs in is estimated using an algorithm developed by the collaboration [142]. The reason for these additional criteria is described in section 3.5.2.

The US spatial measurements are performed as outlined in section 5.1.2, with the $x-$barycentre in a plane being quoted as the $x-$barycentre taken from the bar with the largest recorded QDC in that plane. The reason for this is to minimise the influence of the energy deposition from the muon on the spatial measurements in the US system, if the $x-$barycentre were quoted as an average of the measurements from all bars in the plane, then scintillators traversed by the muon would bias the result. As one scintillator is used for the $x-$barycentre measurement, the uncertainty on the central value is given by equation 5.1.8 while the uncertainty for the $y-$barycentre is given by equation 5.1.10.

Together the SciFi and US measurements with uncertainties can be used measure the shower direction as described in section 5.1.3. The fit performed with the SciFi and US spatial measurements will be referred to as the SF+US track. An example of a fitted $\nu_\mu$ CCDIS event is shown in figure 5.8 with the muon track also shown. In the case of studying the $\nu_\mu$ CCDIS events, an extra criterion based on the physics can be introduced for fitting the shower trajectory in the SciFi and US. Since the DS track and shower must originate from the same point, the DS track and shower track intersect at the $z-$position of the interaction wall, $z_{\text{int}}$. This reduces the degrees of freedom in the linear fit by one, allowing the constant term in the linear fit of the SF+US track to be expressed in terms of the shower fit gradient:

$$f_\mu(z_{\text{int}}) = f_{\text{SF+US}}(z_{\text{int}})$$
$$c_{\text{SF+US}} = f_\mu(z_{\text{int}}) + m_{\text{SF+US}} z_{\text{int}}$$

Where $f_\mu$ and $f_{\text{SF+US}}$ are the linear functions representing the muon track and shower respectively in either the horizontal or vertical axes as a function of $z-$position.

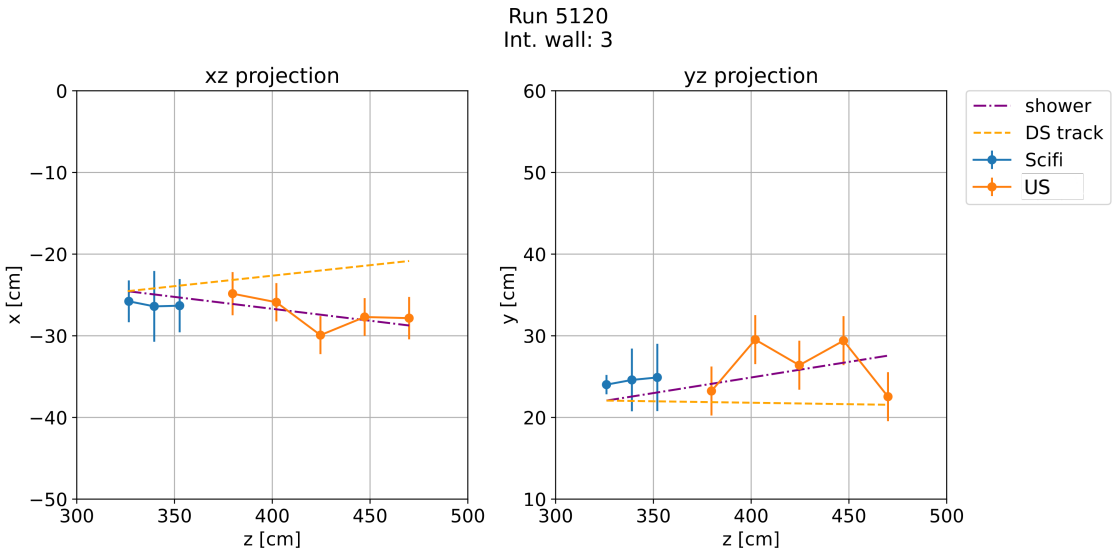



Figure 5.8: $\nu_\mu$ CCDIS selected event from run 5120 with SciFi positions, US barycentres and reconstructed muon track shown together. The uncertainties in $x-$position are determined using equation 5.1.8 while the uncertainties in $y-$position are determined using equation 5.1.10.

With the track fitting performed, the angular measurements outlined in the previous subsection can be used and compared to simulation. The neutrino simulation data, prepared as outlined in section 3.4, is subject to the same selection criteria as were used to prepare the simulation dataset used in the $\nu_\mu$ CCDIS article [125]. These criteria are detailed in section 3.5.2 tables 3.11, 3.12.

## 5.2 Results

### 5.2.1 Digitisation threshold study

The end of section 5.1.1 describes the digitisation threshold - the threshold energy that should be deposited in the scintillator to expect the US to record the data in reality. The value for this is estimated by studying how the digitisation threshold impacts observables in the system. From the scintillator material one expects a minimum ionising particle to deposit on average 1.8 MeV $cm^{-1}$. The end of section 2.6 discusses how to expected energy loss in a US muon system scintillator changes for the muons expected in SND@LHC relative to truly minimum ionising particles. The mean energy loss in 1 cm of Eljen EJ200 by the muons reaching is found in simulation to be 2.6 MeV. Since the QDC distributions, such as those shown for example as the dependent variable in figures 4.7, 4.9, clearly show a peak, attributable to the most probable muon energy at SND@LHC, the effective digitisation threshold in data must be less than the expected energy deposition by the muons arriving at SND@LHC, therefore the Bethe-Bloch expectation can be used to inform the digitisation threshold study.

The trialled values of the digitisation threshold vary from 800 keV to 2 MeV in steps of 400 keV. To understand the impact of the digitisation threshold on the simulation, the evolution of selected measurable values with respect to the digitisation threshold are studied using the simulated passing muon events meeting the selection criteria of table 4.1. The measured means and full width half maxima of $\Delta(x_{\mathrm{L}}, x_{\mathrm{track}})$ for the studied digitisation thresholds are shown in figure 5.9. Where $\Delta(x_{\mathrm{L}}, x_{\mathrm{track}})$ is defined in equation 5.1.12 as the difference between the left-sided spatial measurement, $x_{\mathrm{L}}$, and the expected position from the fitted DS track, $x_{\mathrm{track}}$. Given that the expected energy deposition by a muon in the scintillator is approximately 1.8 MeV, it is expected that once the digitisation threshold approaches 1.8 MeV, the effect on the $\Delta(x_{\mathrm{L}}, x_{\mathrm{track}})$ amplifies, as is observed for the mean value of $\Delta(x_{\mathrm{L}}, x_{\mathrm{track}})$ in all planes.

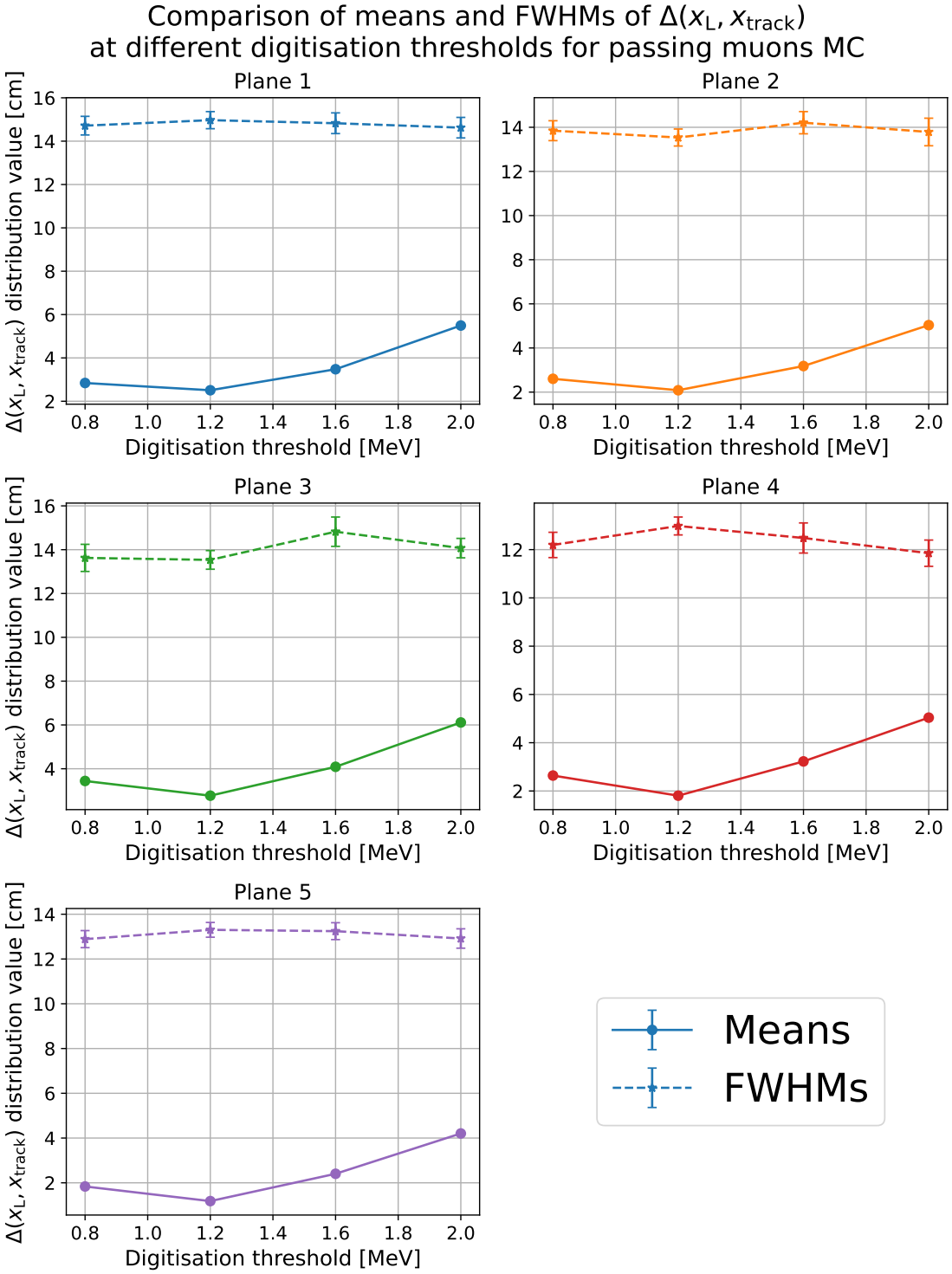



Figure 5.9: Impact of varying the digitisation threshold on the $\Delta\left(x_{\text{left}},\, x_{\text{track}}\right)$ measured with simulated passing muon data for all planes of the US system.

As expected from the Bethe-Bloch equation, once the digitisation threshold exceeds 1.8 MeV, the number of firing scintillators which meet the digitisation threshold drops significantly, shown in figure 5.10.

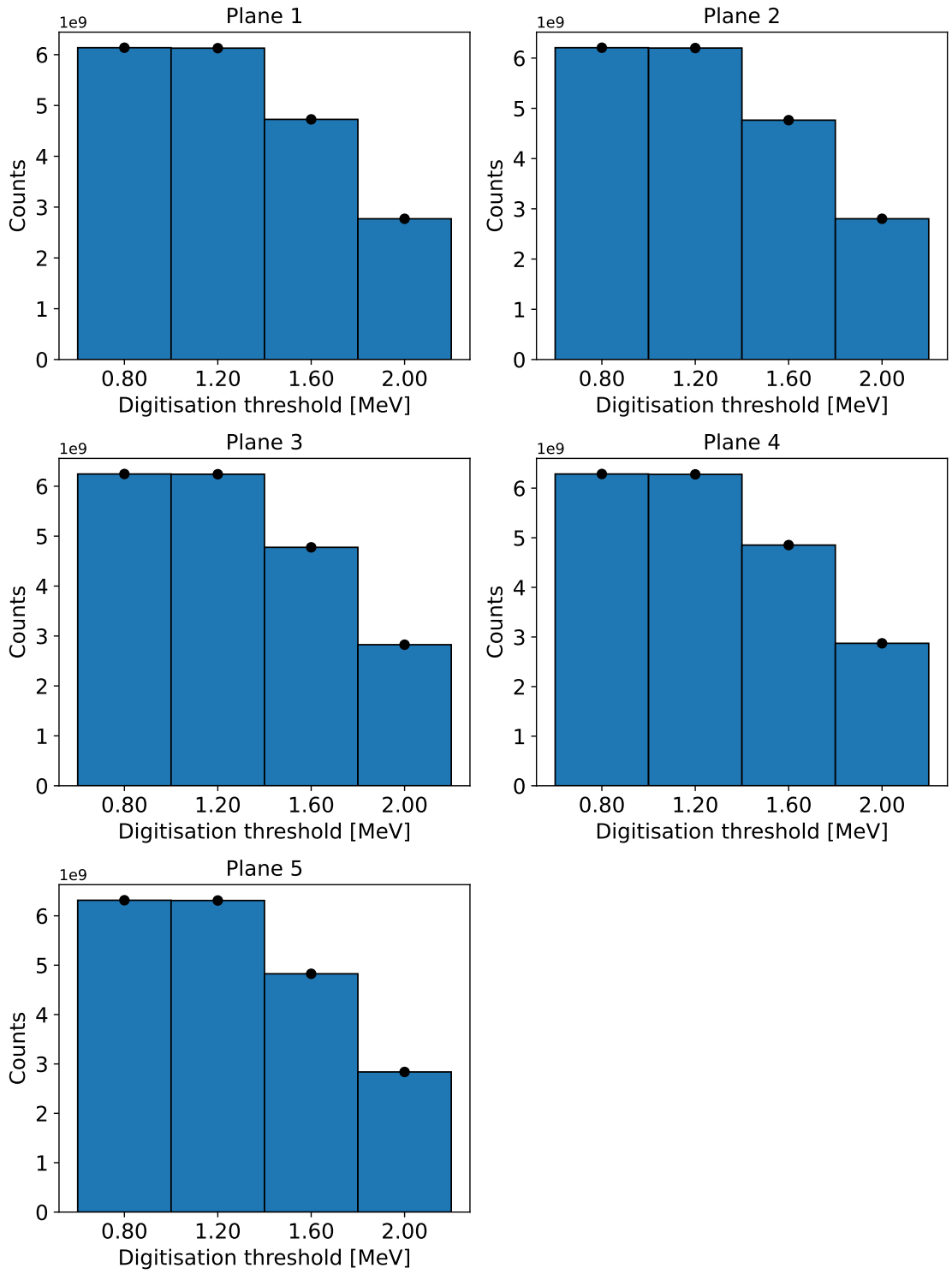



Figure 5.10: Number of valid hits passing the selection criteria for passing muons for the different digitisation thresholds studied.

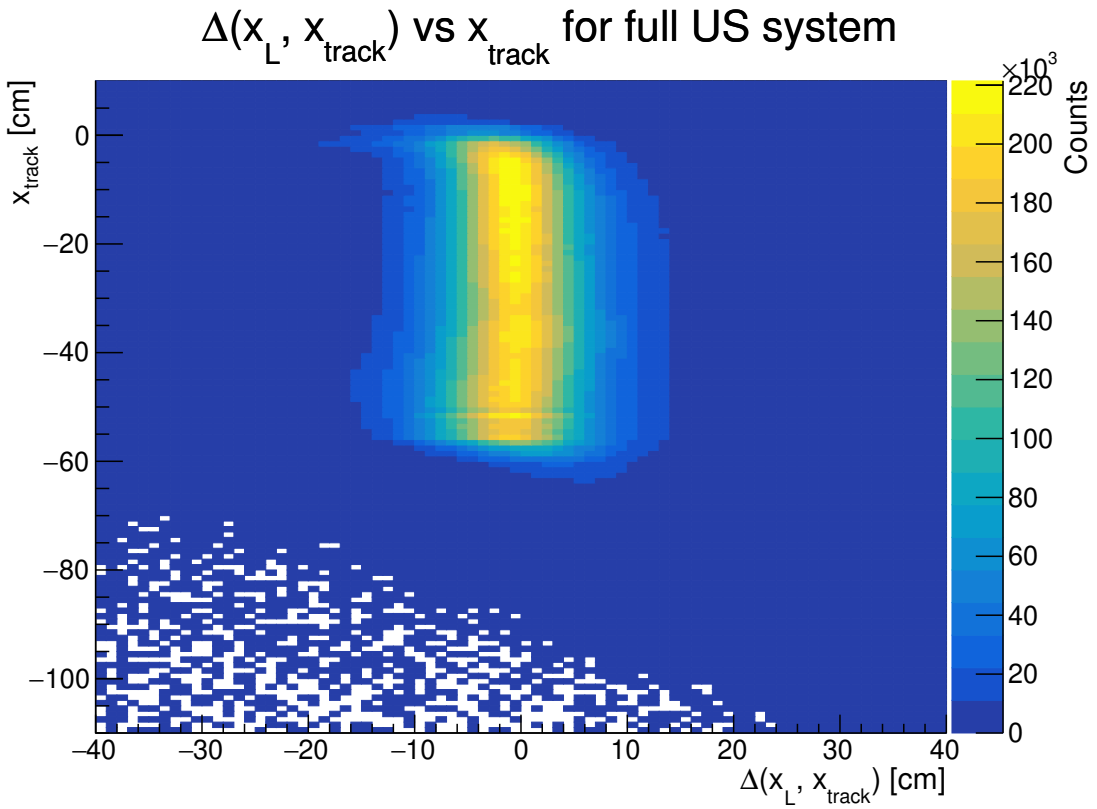
From these plots, it can be concluded that the digitisation threshold begins to impact the analysis at 1.6 MeV and there is no discernible difference between 800 keV and 1.2 MeV. A value of 1.2 MeV is selected.

### 5.2.2 Spatial measurement validation

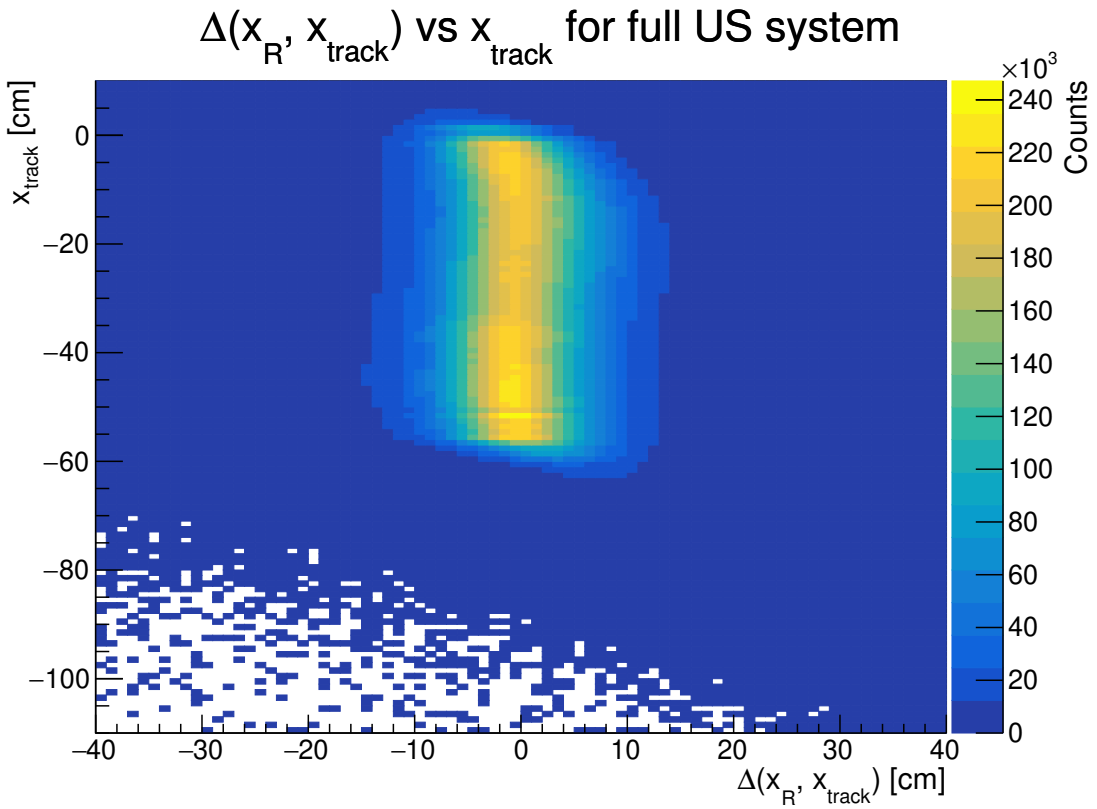
In this section, validation of the spatial measurement algorithms are provided using selected passing muons events in the experimental data as well as using the passing muon simulation data digitised using protocols described in section 5.1.1: the normalised look-up histograms of scintillator-side time in bins of $x-$position.

#### Horizontal dimension

The validation stages of the spatial measurement algorithms were undertaken with passing muon data: real data from TI18 satisfying the passing muon selection criteria of table 4.1, as well as through the simulated passing muon data satisfying the same selection criteria. Figure 5.5 shows one such validation plot for all bars of the US; the figure shows the residual between $x-$barycentre and the expected muon position, $x_{\text{track}}$. While the validation plot broadly shows the success of the algorithm, there is a clear breakdown in agreement at the left-side of the US system (high values of $x_{\text{track}}$) particularly for the left-side but also for the right-side to a lesser degree. This is an edge effect of the detector system stemming from signal arriving close to the SiPM read-out on the large $x-$position side.




(a) Residual between position measured by left-side SiPMs and expected muon position.




(b) Residual between position measured by right-side SiPMs and expected muon position.

Figure 5.11: The residual in position measured between left-side SiPMs (right-side SiPMs) and expected muon position from the track fitted in the DS on the top (bottom) sub-plot. All scintillators throughout the US are studied to form these histograms. The data are taken from run 5408 with the selection criteria from table 4.1

Before discussing this edge effect, the quantities in the plots are repeated to make clear what quantities are being discussed. $x_{\mathrm{L/R}}$ here is the distance between the $x = 0$ and the left/right edge of the energy deposition, as is shown schematically in figure 5.3. It is clear from sub-figure 5.11a that the divergence in residual, $\Delta(x_{\mathrm{side}}, x_{\mathrm{track}})$, at the left-side of the system (large $x_{\mathrm{track}}$) is stronger for the left-side of the experiment than for the right-side of the experiment, shown in sub-figure 5.11b. The manifestation of this divergence $\Delta(\mathrm{x_L}, \mathrm{x_{track}}) < 0$ is such that $\mathrm{x_L}$ must be smaller than $x_{\mathrm{track}}$. The left-side spatial measurement, $x_{\mathrm{L}}$, is determined in equation 5.1.3 as $x_{\mathrm{L}} = x_m + c_{\mathrm{L}} dt_{\mathrm{L}}$. The centre of the bar, $x_{\mathrm{m}}$, and the signal speed values, $c_{\mathrm{L}}$, have no spatial dependence: the only possible source is the value of $dt_{\mathrm{L}}$ being smaller than what would reproduce $x_{\mathrm{track}}$. Now the measurement of $dt_{\mathrm{left}}$ is recapped to understand what this means for the SiPM timing measurements themselves. The value of $dt_{\mathrm{left}}$ is the average of the aligned times on the left, where the aligned time of a SiPM is computed as $dt^{\mathrm{TW}}_{\mathrm{SiPM}} = t^{\mathrm{DS}}_0 - t^{\mathrm{TW}}_{\mathrm{SiPM}} - d_{\mathrm{SiPM}}$. The dependence of $t^{\mathrm{DS}}_0$ on $x-$position of a muon track is studied and shown to be negligible in appendix A.22. When the reconstructed muon track passes close to the SiPMs, the distribution of $t^{\mathrm{DS}}_0 - t^{\mathrm{TW}}_{\mathrm{SiPM}}$ possesses a large negative tail, due to large values of $t_{\mathrm{SiPM}}$, clearly visible in plots of $t^{\mathrm{DS}}_0 - t_{\mathrm{SiPM}}$ against $x-$position (with or without time-walk correction applied) such as figure 4.5. This can be explained through optical arguments: with increasing photon incidence angle at the bar-end with respect to the end-face normal vector, Fresnel reflection causes a larger fraction of the light to be reflected rather than transmitted to the SiPM, with total internal reflection occurring above at critical angle. If light is reflected rather than reaching the SiPM, then the QDC recorded by the SiPM will be reduced, which increases the time-walk of the SiPM due to a greater time being required until sufficient photons have arrived to reach the fixed threshold of the front-end electronics. Very low QDCs reduce the effectiveness of the time-walk correction, as was discussed at the end of section 4.1.4, leaving the value of $dt^{\mathrm{TW\ ToF}}_{\mathrm{SiPM}}$ too small, thereby causing the observed consequence of $x_{\mathrm{left}}$ being too small relative to $x_{\mathrm{track}}$.

The optics of this system were simulated separately from the main experimental software using the refractive indices of the scintillator (1.58) and air (1.00) or Wacker Silgel (1.40 at the sodium D-line of 589 nm [90])[3] while assuming that the light produced by scintillation is isotropic and thus in no dominant polarisation state [143]. The Fresnel equations for the reflection coefficient of s- and p-polarised light are [13]:

$$R_s = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t}$$
$$R_p = \frac{n_1 \cos\theta_t - n_2 \cos\theta_i}{n_1 \cos\theta_t + n_2 \cos\theta_i}$$

Where $\theta_i$ is the angle of incidence and $\theta_t$ is the angle of transmission which is determined with Snell's law. The assumption that light is in no dominant polarisation state is achieved by averaging the reflectivity coefficients for s- and p-polarisation. The refractive index of the silgel is not documented for 425 nm wavelength light so an estimate of the refractive index can be made using the Cauchy equation [144]:

$$n(\lambda\,[\mu m]) = A + \frac{B}{\lambda^2\,[\mu m]} + \frac{C}{\lambda^4\,[\mu m]} \tag{5.2.1}$$

[3]Each plane of the US has silgel on only one side, as outlined in table 3.1.

Where $A$, $B$ and $C$ are material dependent dispersion constants. Values of these constants are taken from similar materials to estimate the silgel refractive index for light of wavelength 425 nm, these values are documented and motivated in appendix A.17.

The optics simulation results in figure 5.12 highlight that the losses from reflectivity and then total internal reflection increase rapidly as the critical angle is approached. This information can be used in conjunction with the geometry of the system to return to the original problem: can reflection losses explain the divergence of the residual $\Delta(x_{\mathrm{L}}, x_{\mathrm{track}})$. To answer this question the incidence angle of photons upon SiPMs can be translated into a value of $x-$position in the SND@LHC physics reference frame:

$$x_{\mathrm{track}}(h,\, \theta_i) = x_{\mathrm{US\ left}} - h \tan\left(\frac{\pi}{2} - \theta_i\right) \tag{5.2.2}$$

The left-edge of US scintillators, $x_{\mathrm{US\ left}}$, are aligned to an $x-$value in the SND@LHC physics reference frame of $x_{\mathrm{US\ left}} = 3.27 \pm 0.14\,\mathrm{cm}$ as measured by the engineering survey at CERN. The conversion of photon incidence angle into an $x-$position is also dependent on the vertical distance that the photon must travel to reach the SiPM. This variable is denoted as $h$ in equation 5.2.2. The geometry underlying the second half of this equation is illustrated in figure 5.13, the remaining algebra (the factor of $-1$ and offset of $x_{\mathrm{US\ left}}$) is to produce the SND@LHC physics frame. The results in figure 5.12 use $h = 5\,\mathrm{cm}$ which indicates when reflection losses are predicted to begin for the upper- and lower-most SiPMs coupled to a scintillator-side. The optics simulation predicts such losses on scintillator-sides with silgel to begin when $x_{\mathrm{track}} \gtrsim 1\,\mathrm{cm}$ and losses on bar-sides without silgel to begin when $x_{\mathrm{track}} \gtrsim -3\,\mathrm{cm}$.

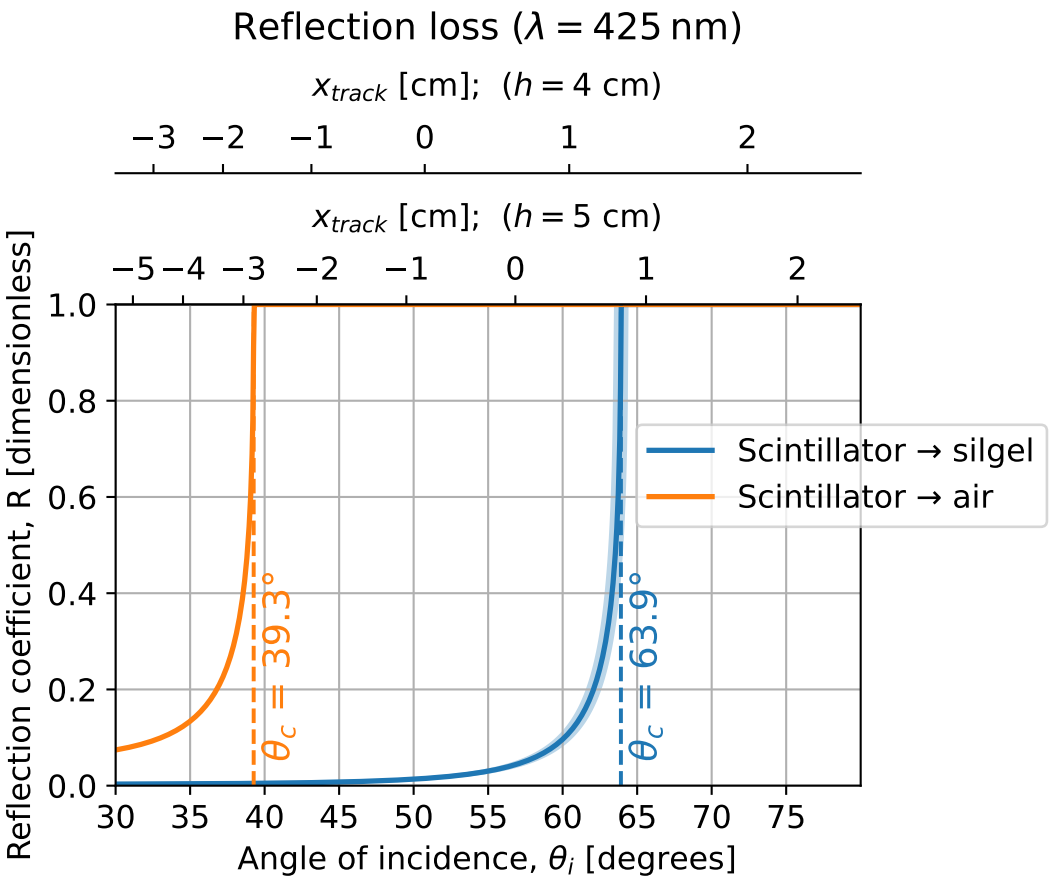



Figure 5.12: Expected reflection coefficient from the Fresnel equations averaged between s- and p-polarised light for the scintillator-silgel and scintillator-air barrier. The upper abscissa axes use equation 5.2.2 to translate the photon incidence angle into the $x_{\text{track}}$ value in the SND@LHC physics reference system that forms such an angle when signal photons travel two different vertical distances, as is outlined in figure 5.13.

The case for bar-sides without silgel is supported by the two QDC versus $x-$position correlation plots in figure 5.14. The observation that the data in figure 5.14b, SiPM 5 in figure 5.13, show a greater population of counts for $x_{\text{track}} > -3\,\text{cm}$ than the data in figure 5.14a, SiPM 1 in figure 5.13, is to be expected. At $x_{\text{track}} > -3\,\text{cm}$, a large fraction of the light reaching the $y-$position of SiPM 1 will be reflected due to the greater incidence angle, whereas more light that reaches the $y-$position of SiPM 5 will transmit through to the SiPM.

The case for bar-sides with silgel, however, is not well supported in the data. Many SiPMs coupled to the scintillator via silgel show QDC versus $x-$position correlation plots highly similar in nature to those of SiPMs coupled to scintillator-sides without silgel. Two example QDC versus $x-$position correlation plots to demonstrate this are shown in figure 5.14. This indicates the presence of an air-gap between the scintillator and silgel, negating the ability of the silgel to reduce reflection losses. The hypothesis of the presence of an air-gap could be simulated to conclude how an air-gap will impact the losses from reflection, however, it is not paramount in answering the original question of the breakdown in agreement of the residual between left-sided spatial measurement and expected track position, $\Delta(x_{\text{L}}, x_{\text{track}})$.

Now turning to the comparison of data and simulation using the passing muon

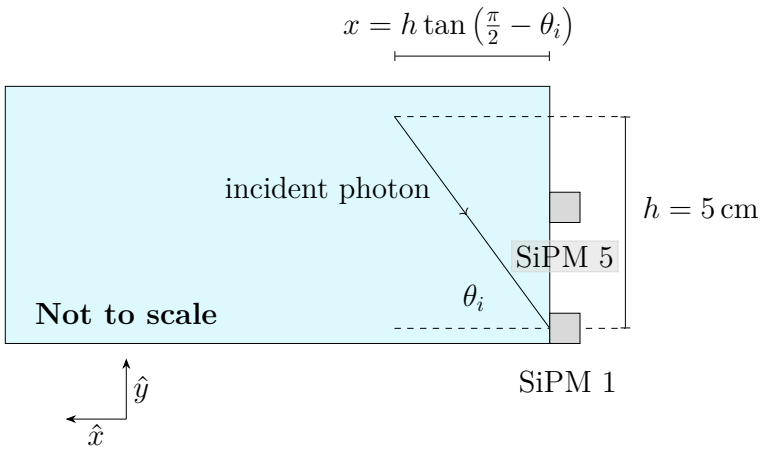
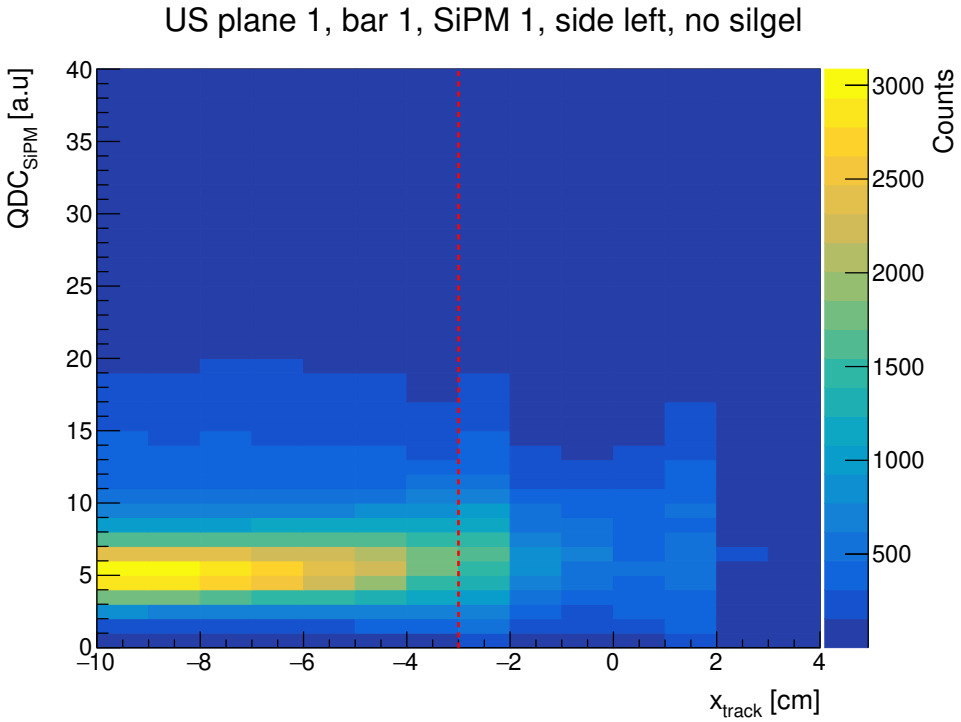
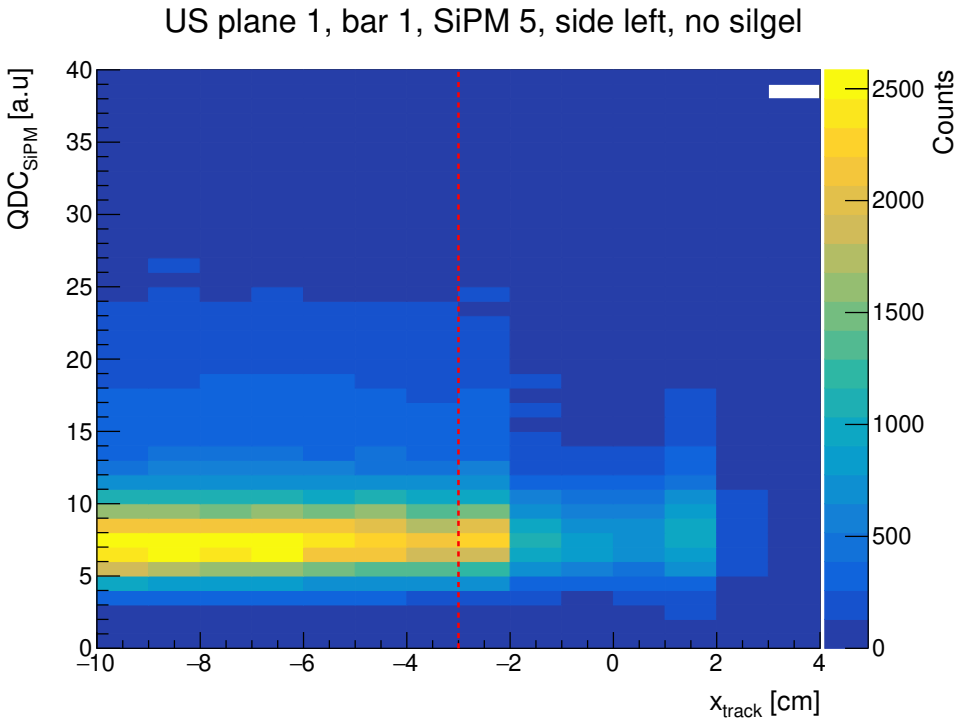



Figure 5.13: Schematic of a US scintillator with scintillation light travelling most of the height of the scintillator, $h$, to reach the SiPMs furthest away from the light creation point.

simulation data prepared as outlined in section 3.4. The simulated data also undergoes the selection criteria of table 4.1. Figure 5.15 shows how the measurement of the residual between $x-$barycentre and expected track position, $x-$track, compare to one another. The data-sets are all normalised to unit integral for clarity of comparison. From the above discussion of the edge-effects at the left-side of the US, around $x_{\text{track}} \approx 0$, the acceptance on the $x-$axis is truncated to only include scintillator hits such that $x_{\text{track}} < -2\,\text{cm}$. The same data-simulation comparison without this acceptance restriction criteria is shown in appendix A.18. Overall, the comparison between data and simulation makes it clear that the algorithm for measuring the $x-$barycentre produces comparable results for experimental data and simulation. There is, however, disagreement between the tails regardless of the lookup histograms from data employed to replicate the timing behaviour as accurately as possible. Efforts to further improve the comparison will be detailed following the subsequent experimental data to simulation comparison.




(a) SiPM 1 is at the bottom of the $y-$range of the bar, this SiPM is the case which is illustrated in figure 5.13 and figure 5.12 showing that at $x-$positions of $\sim -3$ cm the Fresnel losses and TIR losses are very large.




(b) SiPM 5 is in the centre of the $y-$range of the bar, meaning that only tracks very close in $x-$position are able to create scintillation photons impinging upon the SiPM with an angle approaching the critical angle.

Figure 5.14: Comparison of the QDC versus track position histograms for two SiPMs in the same bar. There is no silgel used on the left-side of this plane, meaning that the optical interface is between the scintillator and air. The dotted red line at $x_{\text{track}} = -3\,\text{cm}$ is where TIR is predicted to dominate.

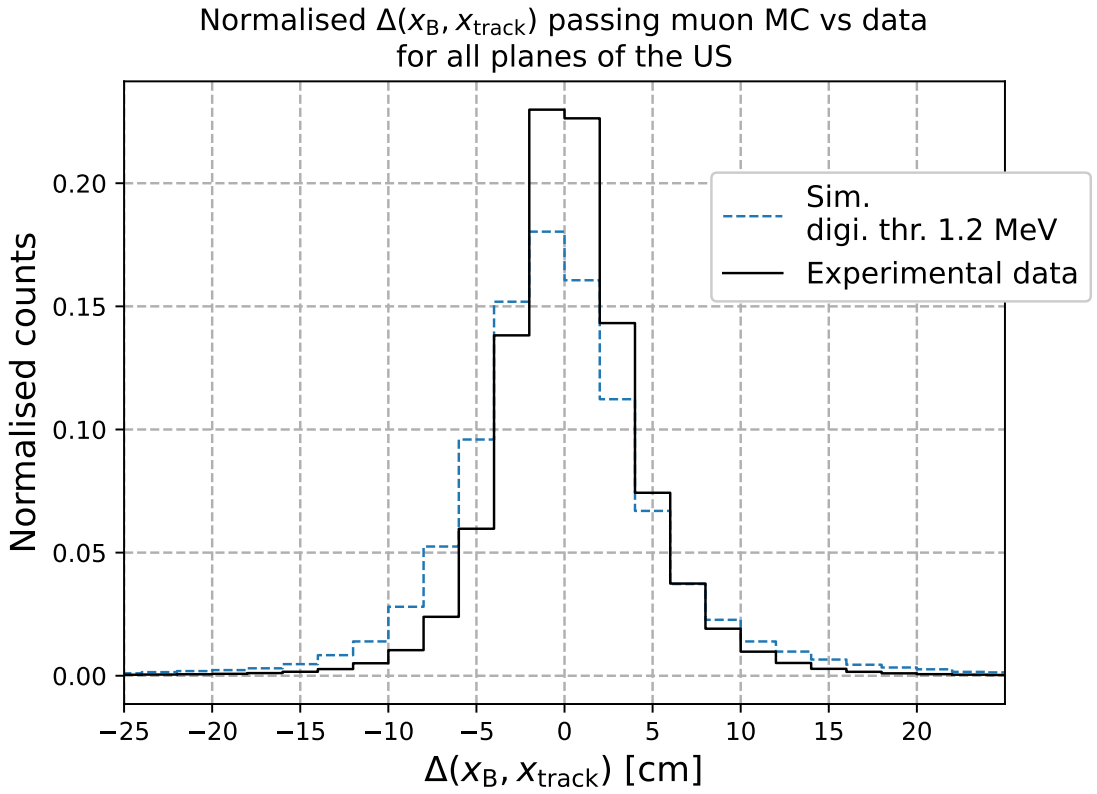



Figure 5.15: Comparison of the residual between $x-$barycentre and $x-$track for experimental data from run 5999 and passing muon simulation data at different digitisation thresholds. Both experimental data and simulation have the selection criteria of table 4.1 applied.

Equation 5.1.6 and figure 5.3 explain how an estimate of the lateral energy deposition width can be made. This is another quantity that is compared between simulation and data to evaluate the efficacy of the simulation implementation of the aligned US timing system. Before investigating the comparison of $\lambda_x$ in data and simulation, one area of investigation for tuning the analysis is studying how the distribution of time-walk corrected and aligned times measured on a scintillator-side is used to produce a spatial measurement. By default, the average of the corrected and aligned times measured by a set of SiPMs on a scintillator-side is used to produce a one-sided spatial measurement following equations 5.1.3, 5.1.4. However, the set of measured times may be sorted and interpolated to allow for different deciles of the set of timing measurements to be used as the time measurement from a scintillator-side. While the SiPMs themselves are all aligned to the DS system event-wise reference time, $t_0^{\mathrm{DS}}$, as described in section 4.1.7, residual timing offsets between a SiPM and $t_0^{\mathrm{DS}}$ may still be sufficient to induce a spatial offset in $\Delta\left(x_{\mathrm{side}}, x_{\mathrm{track}}\right)$. How the decile chosen impacts the agreement between simulation and data is shown for each side in figure 5.16 using the residuals between spatial measurements and expected position from the fitted track in the DS muon system.

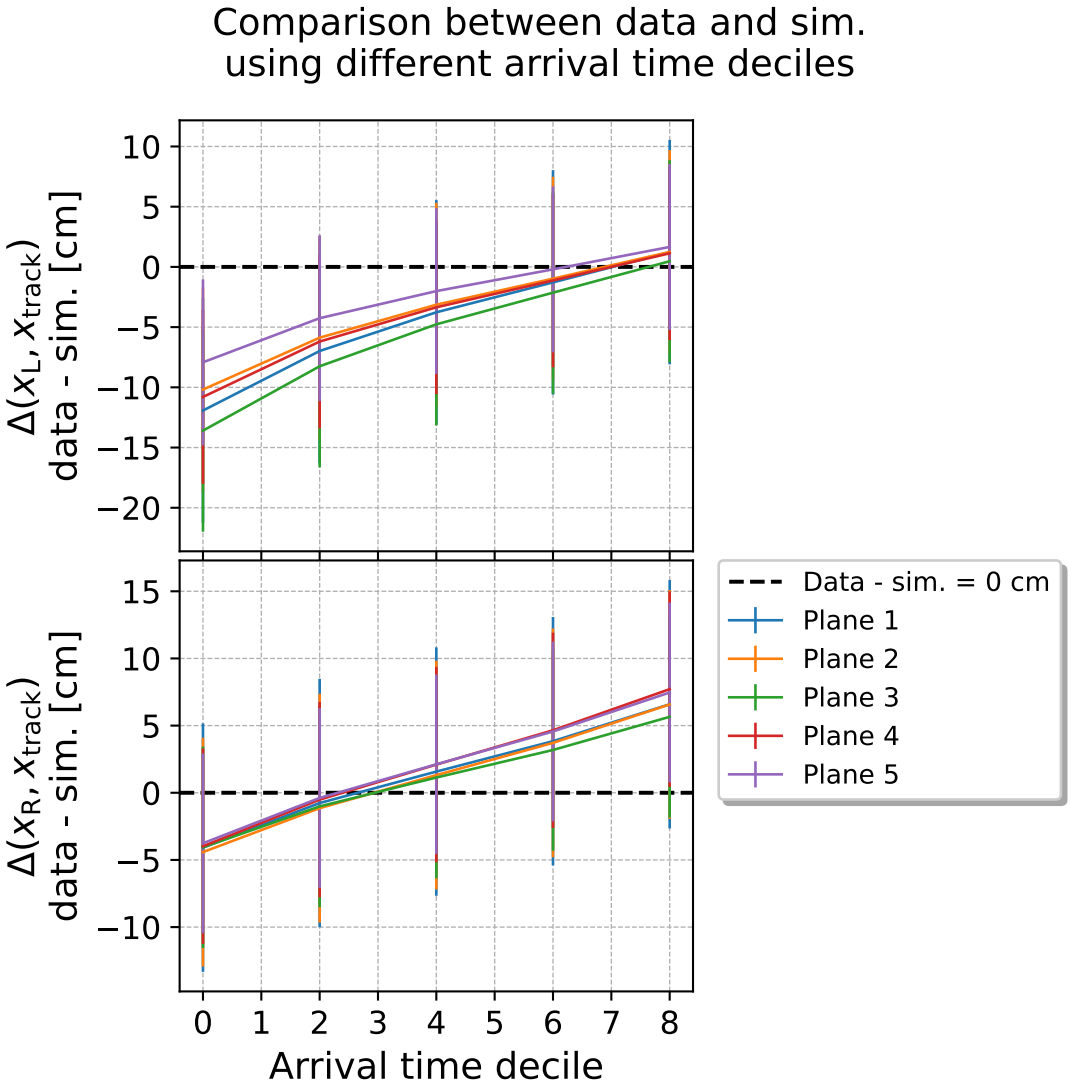



Figure 5.16: Data-simulation comparison for the residual between the scintillator-side spatial measurements and the expected position from the muon track fit in the DS muon system. The left-side residual comparison, $\Delta\left(x_{\mathrm{L}}, x_{\mathrm{track}}\right)$, in the upper sub-figure and the right-side residual comparison, $\Delta\left(x_{\mathrm{R}}, x_{\mathrm{track}}\right)$, in the upper sub-figure.

Both sub-figures show a monotonic trend between the data-simulation agreement and the decile of scintillator-side timing distribution used to create the spatial measurement for that side. The data-simulation agreement is strongest when the sixth decile is used for the left-side timing distribution and when the third decile is used for the right-side timing distribution. Using these observations in the construction of a $\lambda_x$ histogram will maximise the agreement between data and simulation. The criterion that the acceptance on the x−axis is truncated to only include scintillator hits such that $x_{\mathrm{track}} < -2\,\mathrm{cm}$ is also used applied to strongly suppress the discussed left-side edge-effect. The result of this is shown for the first plane of the US system in figure 5.17. Appendix A.20 shows these comparisons for all planes of the US muon system where all planes share the feature that the simulated data distribution features a larger tail in the positive $\lambda_x$ side. The reason for this excess in simulation at large values of $\lambda_x$ is not clear. From the underlying physics, as the events are selected to be muons, near minimum ionising particles at SND@LHC, it is expected that the energy deposition in a scintillator is point-like, thus $\lambda_x$ is around zero with

smearing from the resolution of the reference time relative scintillator side-time, $dt_{\text{side}}$, defined at the start of section 4.1.8 in equation 5.1.1 which is why negative values are observed.

It is pertinent here to recall that the resolution of the event-wise reference time, $t_0^{\text{DS}}$, is present in the resolution of the spatial measurements, the importance of this is discussed here. When the distribution of $\lambda_x$ is compared to the distribution of $x_{\text{B}}$, it is clear that the width of $\lambda_x$ is far greater than the width of $x_{\text{B}}$. The reason for this can be explained by returning to equation 5.1.1 describing the corrected and aligned scintillator-side time relative to the event-wise reference time, and equations 5.1.3, 5.1.4 for using the timing information to produce spatial measurements. When the side-time equations 5.1.1 are substituted into the spatial measurements equations 5.1.3 and 5.1.4, it can be shown that the component from the event-wise reference time is strongly suppressed when computing the $x-$barycentre and does not cancel when computing the energy deposition width, $\lambda_x$. This propagates through to the variance expressions for $x_{\text{B}}$ and $\lambda_x$, this is explicitly shown in appendix A.24.

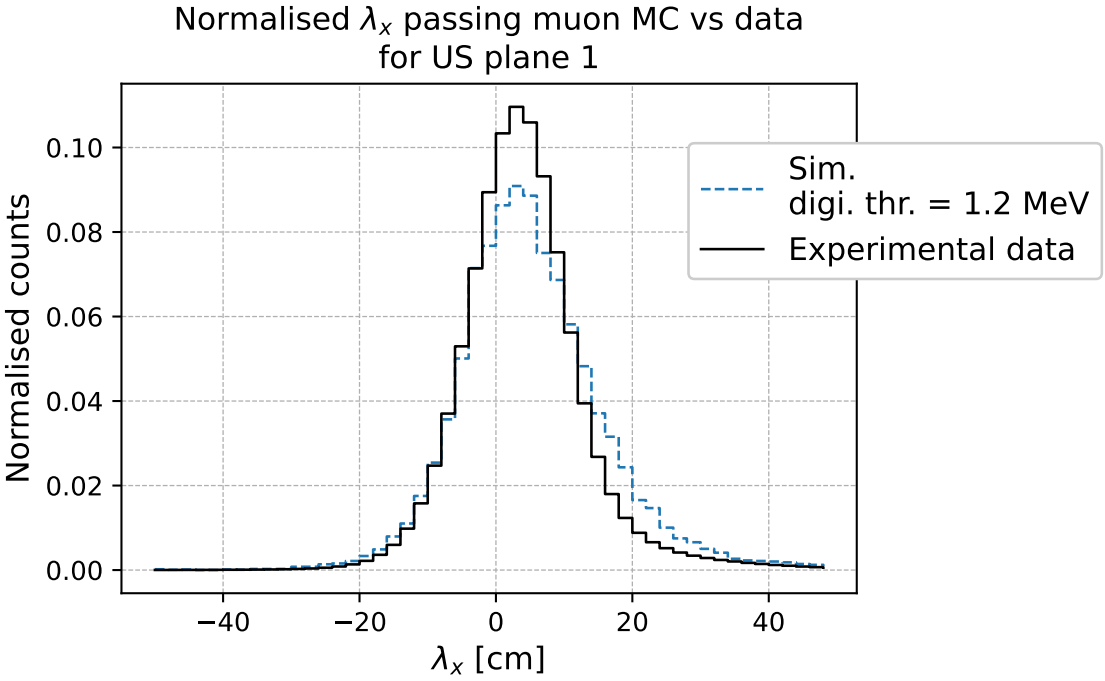



Figure 5.17: Comparison of the measured $\lambda_x$ distributions for experimental data from run 5999 and passing muon simulation data . Both experimental data and simulation have the selection criteria of table 4.1 applied to select passing muon events. The criterion of $x_{\text{track}} < -2\,\text{cm}$ is applied to minimise the discussed edge-effect on the left-side.

### Vertical dimension

Turning to the measurement of the $y-$barycentre: figure 5.18 shows there is a well aligned residual between $y-$barycentre and the track position, which validates the use of the algorithm for determining the $y-$barycentre.

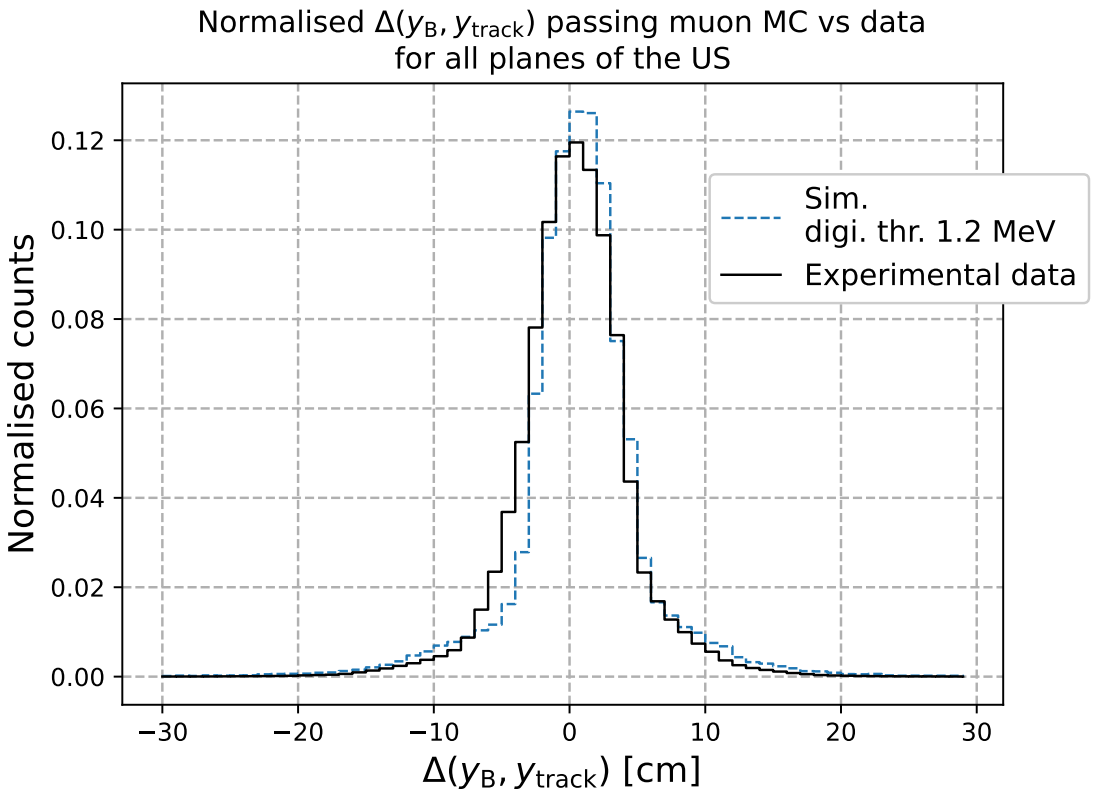



Figure 5.18: Residual in y-position measured between the barycentre and the extrapolated muon track position for all planes in the US for both TI18 data, from run 5999, and from the digitised passing muon simulation using a digitisation threshold of 1.2 MeV.

A quantity closely tied to the $y-$barycentre is the fired bar multiplicity in each plane which can be studied in simulation and data. Figure 5.19 shows a comparison of the fired bar multiplicity between simulation and data for selected passing muon events.

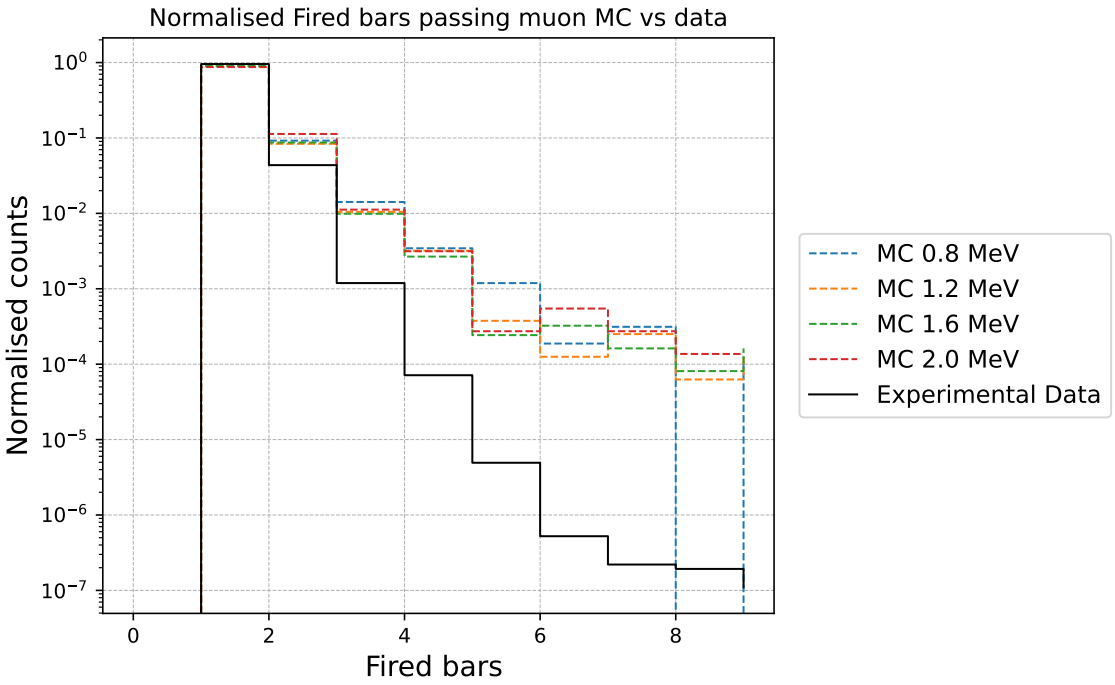



Figure 5.19: Comparison of fired bar multiplicity in plane 1 of the US system for passing muon events from data and simulation (with various digitisation thresholds shown in the legend). A logarithmic scale on the ordinate axis is used to highlight the discrepancy in the tail.

Figure 5.19 shows that the simulation has a tail in the fired bar multiplicity consistently greater over the four digitisation thresholds than what is observed with experimental data. This discrepancy is relatively minor and the underlying causes are not clear. Since the multiplicity is clearly independent on the digitisation threshold employed.

The final observable studied between between simulation and data is the energy deposition width in the $y-$dimension, $\lambda_y$, determined as the difference in position between the uppermost and lowermost firing bars, stated in equation 5.1.11. The comparison between simulation and data for selected passing muon events is shown in figure 5.20. On a linear ordinate axis, the agreement for $\lambda_y$ between data and simulation is strong, however it is still clear that the tails have a disagreement that would be more clear with a logarithmic scale.

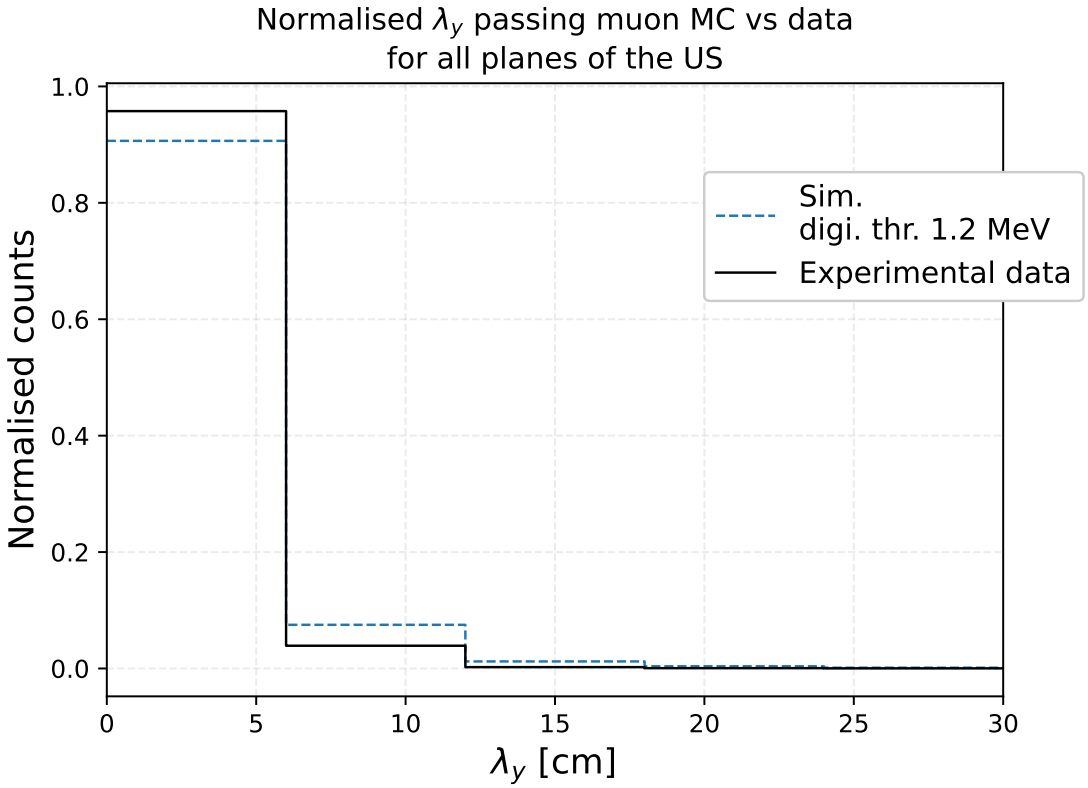



Figure 5.20: Comparison of data from run 5999 and simulation $\lambda_y$ distributions for all planes of the US.

Overall, all of the spatial measurements in both the horizontal and vertical dimensions show strong agreement between simulation and data, however there is room for improvement. The quantity showing the poorest agreement between data and simulation is the energy deposition width in the $x-$axis, $\lambda_x$.

### 5.2.3 Data quality of the $\nu_\mu$ CC events

Section 5.1.4 outlines how the data from the Scifi and US are put together with the reconstructed muon track to study the shower direction measurement prospects. When the data in each plane of the US are investigated, several features are present. Upon producing the spatial measurements for the US data (equations 5.1.3, 5.1.4, 5.1.9) for the $\nu_\mu$ CCDIS events, several bars produced $x-$barycentres far outwith the acceptance of the detector. Investigating the causes of these outlier hits showed two underlying commonalities: low numbers of SiPMs on either side firing, and unphysical[4] times on either side. The firing scintillators which fail either of these criteria will be referred to as outlier hits.

[4]Times which are unphysical are called so because the time produces a spatial measurement that is outwith the acceptance of the detector, thus they cannot be used to give an appropriate description of the physics.

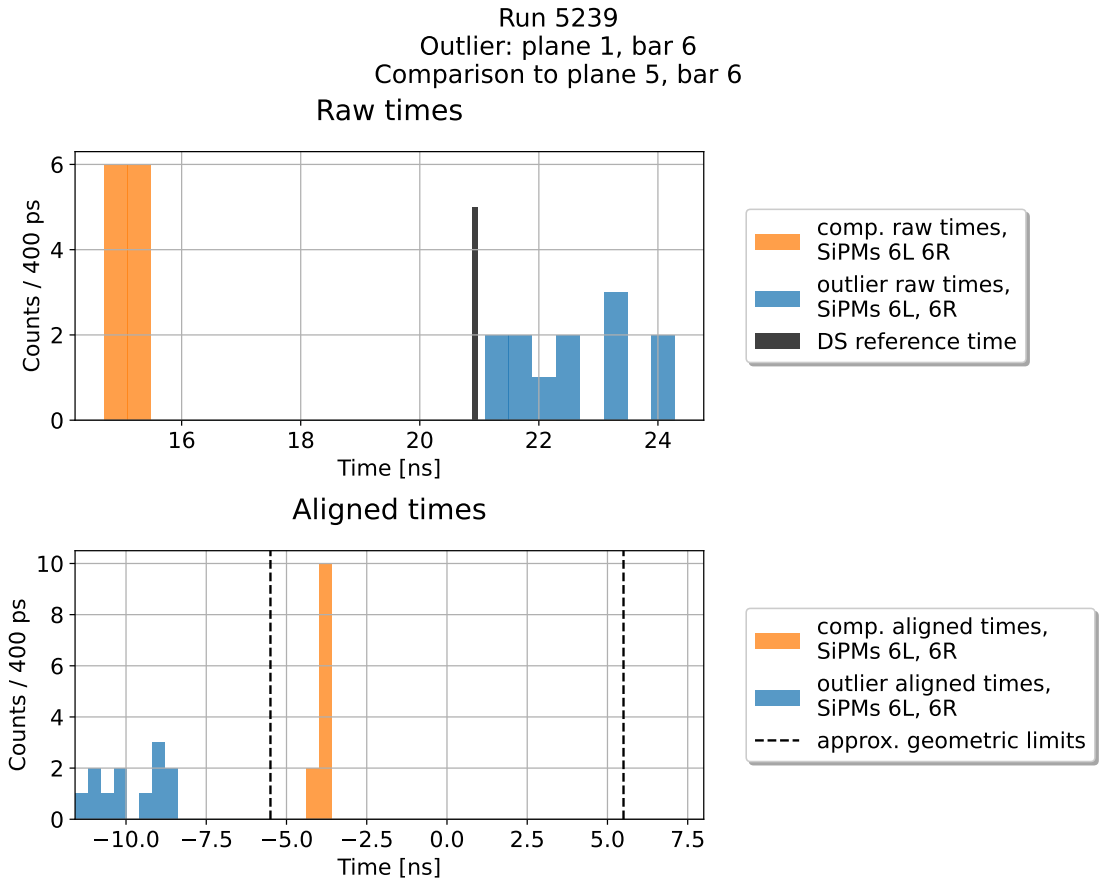



Figure 5.21: Timing data taken from $\nu_\mu$ CC event in run 5239. Top: raw SiPM times directly from the offline format data-file with no corrections applied whatsoever. Bottom: aligned SiPM times, with both time-walk correction and alignment parameter applied. The geometric limits in the lower sub-figure, explained in the main body, provide loose bounds on the possible times that would give physical results. The number of SiPMs firing on the left- and right-side of the scintillator are shown in the legend as SiPMs $N$L $M$R. A binning of 400 ps is chosen to reflect the typical time-walk corrected SiPM time resolution.

In figure 5.21, a comparison is made between a scintillator that gives an unphysical spatial measurement in $x$, the outlier, and a scintillator from the same event which yields an appropriate measurement, the comparison ('comp.'). The upper subplot shows the wholly unaltered times ('raw times') and the lower shows the times with the time-walk and time alignment corrections applied. Geometric limits on the aligned times can be produced by firstly recalling that an aligned time of 0 ns corresponds to light produced in the centre of the scintillator travelling half the bar length, $L/2 = 41.25$ cm, to the SiPMs. Therefore, times corresponding to light travelling more than $L/2$ infer energy depositions outside the sensitive area of the detector:

$$t_{\text{max}} = (L/2)/c_{\text{scint}}$$

Where $c_{\text{scint}}$ is the average time-walk corrected effective signal speed measured in the US system. These stringent geometric constraints are loosened to accept times which infer energy depositions coming from $L/2$ beyond either side of the sensitive area of the detector. The reason for this is purely to allow for some outliers to be plotted in the event displays such as figure 5.8 to see if the outliers are producing

positions which loosely agree with the other detector planes. Figure 5.21 shows that the raw times of the outlier hit are delayed relative to the comparison hit and the DS reference time which result in aligned times outwith the loose geometric constraints shown in the aligned times of the lower subplot. Physically this could be produced by the charged decay products of excited nuclei decaying after the neutrino interaction products have traversed the detector.

One further issue with the data quality of some $\nu_\mu$ CCDIS events arises in the number of fired SiPMs on a scintillator. As one sees in figure 5.22, only one large SiPM fires on the left side of the scintillator while zero SiPMs fire on the right hand side.

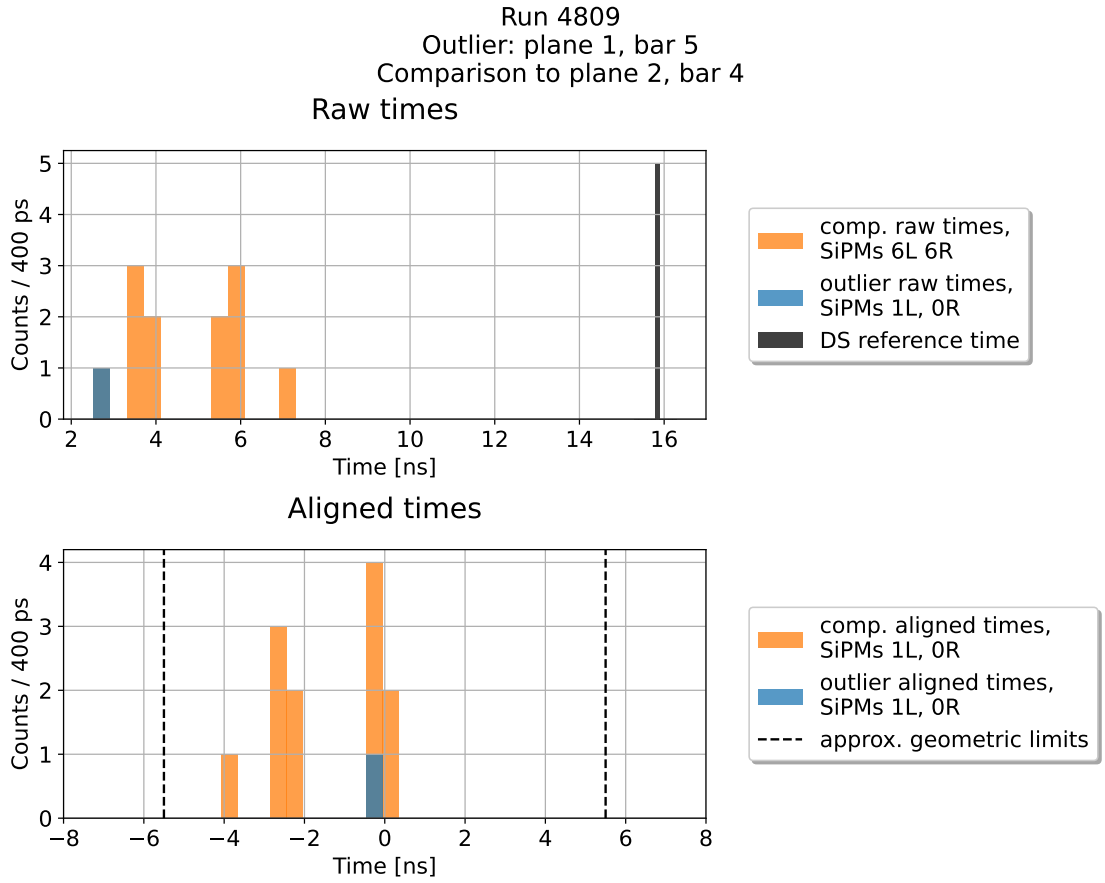



Figure 5.22: Timing data taken from $\nu_\mu$ CC event in run 4809. Identical plotting style to figure 5.21.

One could argue that this hit is in-fact signal as the chronology of the raw times make sense, the comparison times are in the subsequent plane, and accordingly the comparison times are delayed by $\approx 3$ ns which corresponds to about the distance in $z$ between planes of the US. Also the aligned times are well within the approximate geometric limits. Over the 16 total selected $\nu_\mu$ CCDIS events, comprising a total of 140 firing scintillators, 22 scintillators fail to have greater than three SiPMs on both sides firing, which is equivalent to $\approx 15.7\%$, shown in figure 5.23.

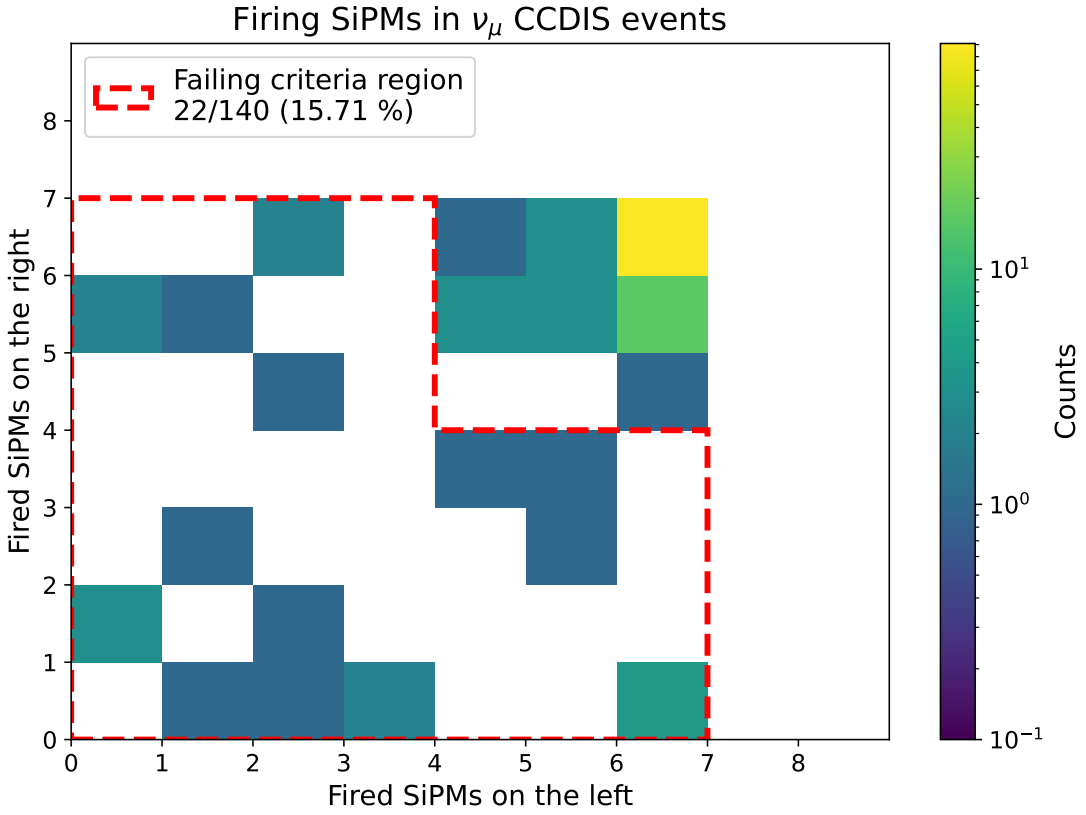



Figure 5.23: Two-dimensional histogram showing the number of SiPMs firing on the left and right side of scintillators for all 140 firing scintillators over the 16 selected $\nu_\mu$ CCDIS events.

An inspection of the correlation of firing SiPMs on left and right sides of the scintillator shows that one very rarely expects so few SiPMs to fire. The probability for events passing the muon selection criteria (table 4.1) yet failing to have at least four SiPMs firing on both sides can be extracted from figure 5.24 to be 1.37 %. Clearly, the selected $\nu_\mu$ CCDIS events have a far greater probability of failing this criteria on the number of fired SiPMs. The causes of this are not clear but the most obvious differences between these two classes of event are the energy deposited and the expected number of firing scintillators per plane. However, since there is no reason why one would expect so few firing SiPMs on either side for high energy deposition $\nu_\mu$ CCDIS events, these scintillators are not considered signal in this study of the $\nu_\mu$ CCDIS events.

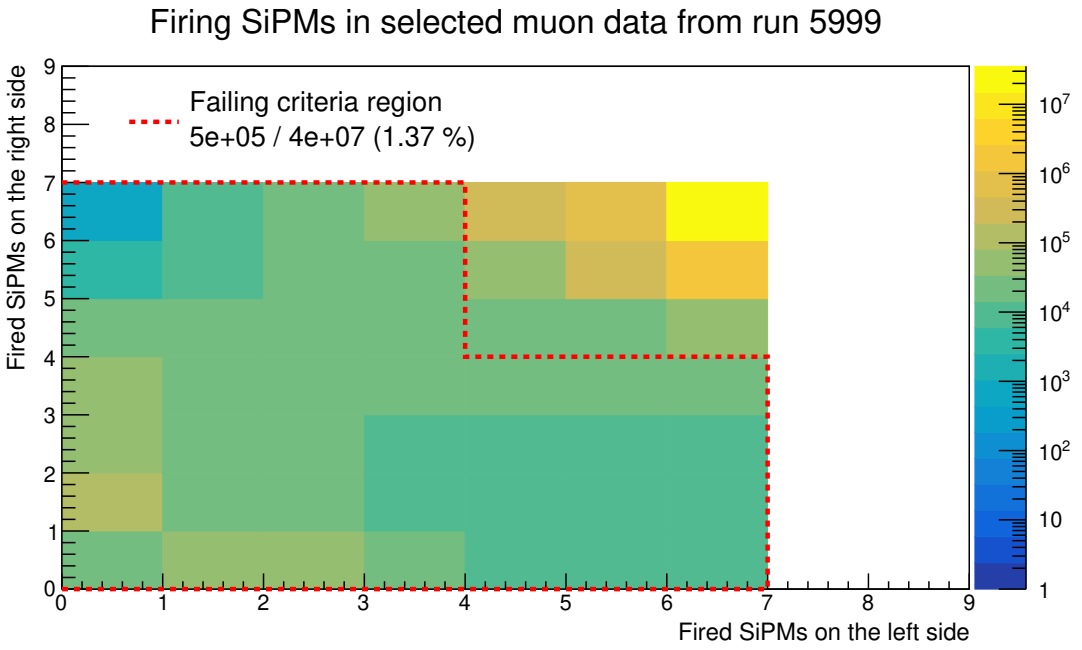



Figure 5.24: Events passing the muon selection criteria from run 5999. Correlation of the number of SiPMs firing on each side of a scintillator, for scintillators in the first plane of the US. The red dashed rectangle contains the region of the plot where events fail to have at least four SiPMs firing on both the left and right sides.

The choice of the minimum number of SiPMs required to fire on both sides is arbitrary: regardless of the choice, the selected $\nu_\mu$ CCDIS data meets the criteria at a greatly lower rate than the selected muon data. As the energy deposition is the most fundamental difference expected between passing muons and $\nu_\mu$ CCDIS events, the number of firing SiPMs can be studied for firing scintillators in passing muon events where there is a large energy deposition in a plane of scintillators. This same probability is evaluated in selected muon events where there is a QDC value of greater than 100 and also at 500 units, the results for these criteria are shown in table 5.1 and the two plots, identical in fashion to figure 5.24, are in appendix A.23.

| QDC in plane threshold | Hits failing criteria (%) |
|---|---|
| 0 | 1.37 |
| 100 | 3.41 |
| 500 | 5.46 |

Table 5.1: How the percentage of hits failing to meet the criteria of at least four SiPMs firing on both sides of the scintillator changes for three applied criteria on the QDC measured in the plane of scintillators containing the hit under study.

Table 5.1 shows that as the criteria on the QDC measured in the plane of scintillators containing the hit under study increases, so does the percentage of firing scintillators failing to meet the criteria of at least four SiPMs on both sides of the scintillator registering data. However the correlation between these variables is not sufficient to explain the behaviour observed in the $\nu_\mu$ CCDIS events. The most serious consequence of this observed behaviour is in the pending SND@LHC physics analyses relying on an energy calibration. Currently, an energy calibration of the experiment is in development [142]. How to deal with the fired US system scintillators with low numbers of firing SiPMs will present a non-trivial challenge.

These two issues: delayed timing response and low SiPM numbers are the two additional data quality checks that are implemented in the study of $\nu_\mu$ CCDIS events. The application of these two criteria removes all of the scintillators producing $x-$barycentres outside the acceptance of the US.

| **Delayed timing** | **SiPM multiplicity** |
|---|---|
| Average aligned time of both sides inside geometric limits | $\geq 4$ SiPMs on both sides firing |

Table 5.2: Additional data quality checks imposed upon the selected $\nu_\mu$ CCDIS selected events for the shower direction analysis.

Overall, 25.7% of the firing scintillators in the 16 selected $\nu_\mu$ CCDIS events[5] (140 firing scintillators) either report outlying timing measurements when measured as producing $x-$barycentres more than $L/2$ beyond either side of the scintillator acceptance or the scintillator does not meet the criteria on the number of firing SiPMs. It is the opinion of the author that there are clearly effects occurring in these events that are not yet understood by the collaboration. In the case of the number of firing SiPMs, solving this problem will be an important step towards a thorough energy calibration of the detector as firing scintillators failing this criterion will introduce a bias in the energy calibration without further action. The firing scintillators presenting delayed timing can present different issues down the line depending on the underlying physics: regarding an energy measurement, the hits with delayed timing can only cause issues if the hits are actually caused by coincidental physics processes, which is unlikely however the probability of this is not studied in this thesis.

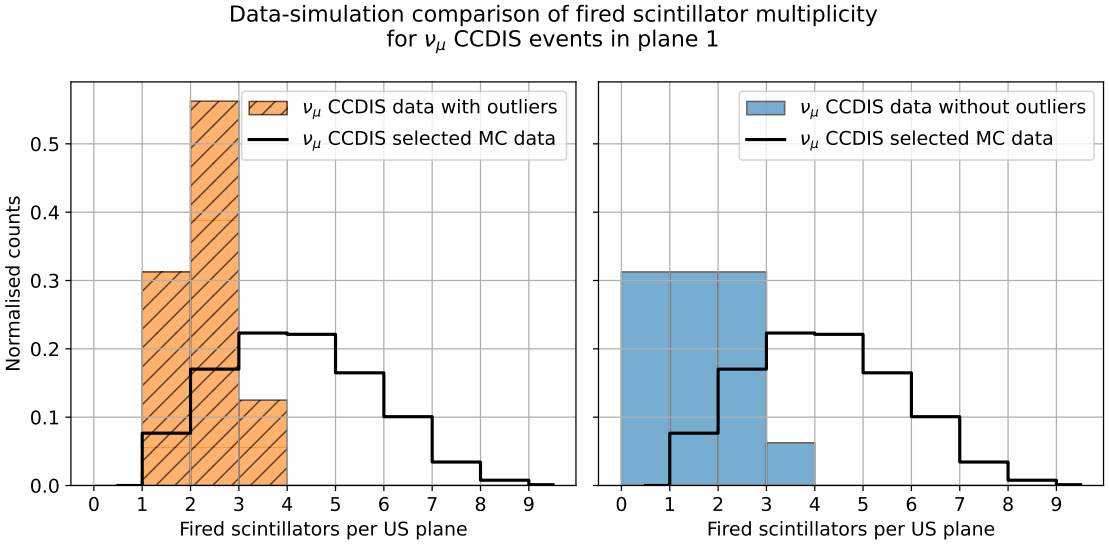



Figure 5.25: Multiplicity of firing scintillators in the first plane of the US muon system. Neutrino simulation data passing the $\nu_\mu$ CCDIS selection criteria is shown as well as the data events satisfying the $\nu_\mu$ CCDIS selection criteria split into two cases: including firing scintillators tagged as outliers (left) and not including those scintillators (right). The three data-sets are normalised to unit integral.

[5]A reminder that this is not identical to the events in the PRL article, the difference between these is stated in the opening of section 5.1.4.

The first key point regarding figure 5.25 is that all three datasets are normalised to unit integral for ease of comparison. Clearly, whether including outlier hits or not, the distribution expected from simulation is not reproduced in data, with a particular issue from the high multiplicity region. In the first plane of the US, simulation predicts that $\approx 53\%$ of events should have four or more planes firing whereas this is never observed in the US system. Why this large discrepancy exists is not clear. When the multiplicity of firing scintillators in later planes of the US are compared to simulation in this fashion, the agreement between data and simulation improves. The equivalent plot to figure 5.25 for the final plane of the US muon system is shown in appendix A.25.

The outlier hits could be in part caused by cross-talk on the traces of the PCB. Section 4.3.6 shows that cross-talk is measured in the aligned timing measurements of SiPMs coupled to the same scintillator and that the covariance measurement is required to well reproduce the observed time resolution of a scintillator-side. If cross-talk is creating some outlier hits observed in the selected $\nu_\mu$ CCDIS data then one effect that could be observed is a lower distribution in the QDC measured by outlier hits as opposed to the hits not flagged as outliers. Figure 5.26 shows the distribution of QDC measured in scintillators for hits flagged as outliers and those hits not flagged as outliers. Here is it recalled from the description of the $y$-barycentre in section 5.1.2 that the QDC measured by a scintillator is simply the sum of the QDC measurements by the SiPMs coupled to the scintillator, $\mathrm{QDC}_{\mathrm{bar}} = \sum_{\mathrm{SiPM}} \mathrm{QDC}_{\mathrm{SiPM}}$.

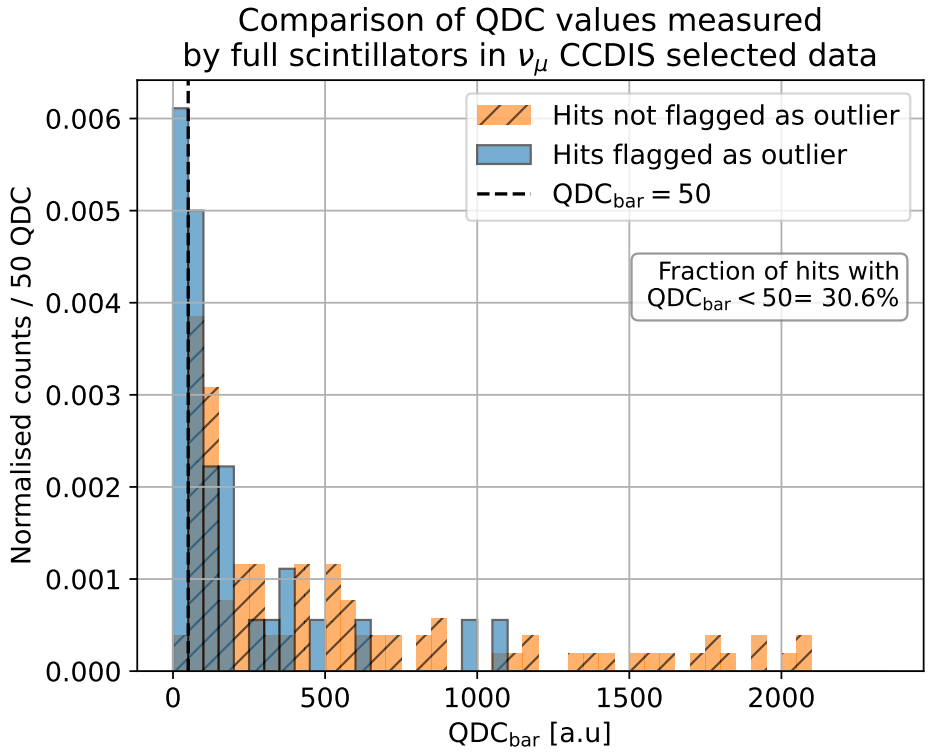



Figure 5.26: Distribution of QDC measured by a scintillator for the scintillators not flagged as outliers compared to the same distribution for scintillators flagged as outlier hits. Both distributions are normalised to unit integral to compare the shape more easily.

The fine binning at low values of $\mathrm{QDC}_{\mathrm{bar}}$ highlights the the fact that the outlier hits have an excess of $\mathrm{QDC}_{\mathrm{bar}}$ in the very lowest bin, less than 50 QDC. The outlier

hits $QDC_{bar}$ distribution does also extend to larger values but not to the same range shown by the hits not flagged as outlier. The hypothesis that cross-talk between traces of the US PCB is responsible for the outlier hits cannot be concluded from figure 5.26 but the excess at low values of $QDC_{bar}$ provides incentivise for further study. To more accurately study this hypothesis requires laboratory work. Injecting pulses of known voltages to SiPM channels and recording, with an oscilloscope, the raw waveform from neighbouring channels would allow for a conclusive and accurate understanding of how cross-talk impacts signals in the US muon system. Injecting pulses with voltages that would be expected at certain energy depositions in scintillators would allow for the cross-talk to be studied in conditions very similar to the $\nu_\mu$ CCDIS events in data. As discussed in section 2.4, the physics goals of SND@LHC require measuring the incident electron neutrino energy spectrum, since the radiation length in tungsten, the target material of SND@LHC, is approximately 0.35 cm [78], a poor energy resolution in the US system of the experiment may not pose a serious problem but it should be considered for future studies.

With these data quality cuts imposed, the event displays and fits such as figure 5.8 are produced and a study of the angular distributions in data and simulation can be made. The process of constructing direction vectors and then measurements of the polar and azimuthal angles are outlined in section 5.1.3 with equations 5.1.13, and 5.1.14 and 5.1.15 respectively. The resulting distributions for the measured polar angle in data and simulation are shown in figure 5.27.

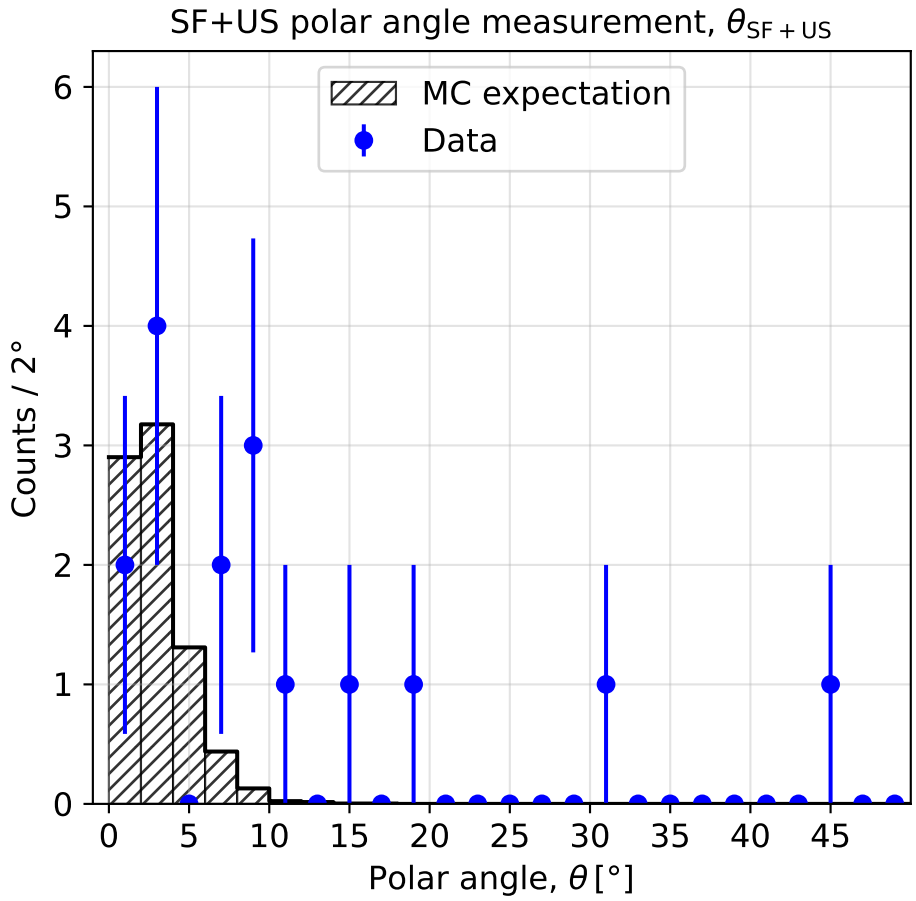



Figure 5.27: Measured polar angle from SciFi and US track, $\theta_{\mathrm{SF+US}}$, in the $\nu_\mu$ CCDIS events for data and simulation. The simulation dataset is scaled to the number of observed events in experimental data.

While the low statistics of $\nu_\mu$ CCDIS events in experimental data and the observed data quality issues restrict the conclusions that can be drawn from the data-simulation comparison in $\nu_\mu$ CCDIS interactions, the polar angle comparison, shown in figure 5.27, clearly shows a poor agreement. Continuing this analysis as the statistics of $\nu_\mu$ CCDIS events collected by the experiment increase may help to elucidate the underlying causes, however, without addressing the cause of the poor data quality addressed earlier in this subsection, deeper studying the data collected by the US system in high energy deposition events is only of limited utility. The data-simulation comparison plot for the measured azimuthal angle is not shown. As the azimuthal angular range is $[-180°, 180°]$ and there are only 16 events in data, there is insufficient information to be gained through the comparison as the data is too diffuse to show discrepancy from the expected uniform distribution.

# Chapter 6

# Multivariate analysis application of US spatial measurements

This chapter focusses on a physics analysis application of the spatial measurements achieved as a result of the time alignment of the US muon system. Specifically, the use of the spatial measurements in improving the muon identification efficiency. Section 3.5.3 provides important context for this chapter by outlining the publication by SND@LHC reporting the analysis of neutrino interactions which do not leave a muon in the final state, $\nu 0\mu$ events. That section reports that at the end of the analysis the largest remaining source of background comes from $\nu_\mu$ CCDIS events where the muon is not reconstructed. As explained in section 4.1.1, for a muon track to be reconstructed in the experiment, there must be hits left in three of the four detector planes of the muon system for both $xz$ and $yz$ projections, this means that final state muons exiting the acceptance of the detector before the 3rd station of the DS muon system are non-reconstructible and thus liable for background in both the $\nu_e$ and thus also the $\nu 0\mu$ analysis. A muon identification regime extended to the US utilising the spatial measurements described in chapter 5 could be used to reduce the background from unidentified muon final state events.

The input of the author to the simulated data is outlined in the beginning of chapter 5, where the digitisation of the simulation data is performed such that the digitised times are in accordance with the real timing data following time-walk correction and alignment. Then, the existing code for applying $\nu 0\mu$ selection criteria was used which the author did not help to create. The data resulting from the application of the selection criteria was then analysed solely by the author with two exceptions. The first exception is the use of an algorithm for filtering the data collected by the SciFi, and secondly using the filtered SciFi data to determine the interaction wall in the target: these algorithms were developed by a colleague in the collaboration. This data is used to develop a Gradient Boosted Decision Tree (BDT) in order to evaluate the probability of a given event containing a muon in the final-state. The chapter begins with an introduction to how Gradient Boosted Decision Trees operate, are trained, optimised and analysed before outlining the specific methods and results of the data analysis.

## 6.1 Gradient Boosted Decision Trees

The broad concept of Gradient Boosted Decision Trees is to use a set of weak learners (trees) where each sequential tree aims to correct the errors of the previous tree [145, 146]:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) \; + \; \eta\rho_m h(\mathbf{x}; \mathbf{a}_m) \tag{6.1.1}$$

Where $F_m(\mathbf{x})$ is the model at iteration $m$ for the observation set $\mathbf{x}$; $\rho_m$ is a scalar factor determined for each $m$ which weights the importance of the weak learner; $h(\mathbf{x}; \mathbf{a}_m)$ is the latest weak learner evaluated with parameter set $\mathbf{a}_m$ and $\eta$ is the learning rate which will be discussed later in this section. The values produced by $F_m(\mathbf{x})$ are mapped to a value in $[0, 1]$ through the sigmoid function [146]:

$$P(y = 1|\mathbf{x}) = \sigma(F_m(\mathbf{x})) = \frac{1}{1 + e^{-F_m(\mathbf{x})}} \tag{6.1.2}$$

This is the probability assigned to the event by the model that it is of a class $y = 1$. Each new learner in the ensemble model is trained to predict the error of the model. In binary classification, one can express the probability of observing an output $y$ given probability $p$:

$$P(y|p) = p^y(1 - p)^{1-y}$$

The log-likelihood, $\log(\mathcal{L})$, of observing the set of $N$ measurements, $\{y_i\}$, can be written as:

$$\mathcal{L} = \prod_i^N p_i^{y_i}(1 - p_i)^{1-y_i}$$
$$\log(\mathcal{L}) = \sum_i^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

The training of the $m^{th}$ learner in the ensemble can then be written as [145]:

$$h_m(\mathbf{x}) = \arg\min_h \sum_{i=1}^N \left( -\frac{\partial L(y_i, F_{m-1}(x_i))}{\partial F_{m-1}(x_i)} - h(x_i) \right)^2 \tag{6.1.3}$$

The partial derivative in the sum corresponds to the $n$-dimensional vector of residuals where each component is the residual for a feature of the model. Thus, 6.1.3 is finding the function $h_m(\mathbf{x})$ which best corrects for the remaining residuals after $m-1$ learners. The fact that each successive learner to the ensemble is trained on the residuals of the existing model gives rise to the term *gradient boosting* with the final set of weak learners being referred to as the BDT.

BDT implementations contain *hyper-parameters* which affect how the model trains. Several of these hyper-parameters are important in the application in this thesis. Firstly, *learning rate*, $\eta$ is a scalar applied to all weak learners in the model affecting the impact that a single learner can have. $\eta$ is shown in equation 6.1.1 and is useful to limit over-fitting. The number of trees allowed in the full ensemble model can be capped with the hyper-parameter *n-estimators*; *max-depth* caps the maximum number of decision splits between start and leaf nodes[1]; *min-child-weight* sets a lower limit on the number of events which are required in each leaf node.

[1] A leaf node is a final node of a tree where a prediction is made.

Finally *scale-pos-weight* is the ratio of the two classes in the dataset which is very important for datasets as unbalanced as what is dealt with especially in the $\nu 0\mu$ study in this chapter.

The BDT in this thesis is a binary classifier: it identifies events as either signal or background. The results of the BDT can then be categorised into a two-dimensional square matrix of truth values and predicted values, referred to as the *confusion matrix*. The coefficients of the confusion matrix contain the number of events in that category. The efficacy of the BDT can be defined through several metrics arising from the proportions of these coefficients; namely accuracy, precision, recall and background rejection efficiency (BRE):

$$\text{accuracy} = \frac{TP + TN}{TP + FP} \tag{6.1.4}$$

$$\text{precision} = \frac{TP}{TP + FP} \tag{6.1.5}$$

$$\text{recall} = \frac{TP}{TP + FN} \tag{6.1.6}$$

$$\text{BRE} = \frac{TN}{TN + FP} \tag{6.1.7}$$

Where $TP$, $TN$, $FP$ and $FN$ are respectively the correctly identified positive cases (true positive, $TP$), the correctly identified negative cases (true negative, $TN$), falsely identified positive cases (false positive, $FP$) and the falsely identified negative cases (false negative, $FN$) - these acronyms will be used henceforth for brevity. A model optimised on maximum precision will penalise the $FP$ rate, while optimising for maximum recall will penalise the $FN$ rate. In the case of optimising the signal sensitivity for a very imbalanced dataset, there is a trade-off between these two values to be made. For such cases, the harmonic mean of precision and recall is often employed, this is called the F1-score:

$$F1 = \frac{2PR}{P + R} \tag{6.1.8}$$

Where $P$ and $R$ are precision and recall respectively. The use of the harmonic mean to average precision and recall emphasises that both precision and recall must have strong scores. The use of the arithmetic mean[2] could cause the optimal $F1-$score to be found in cases where one of precision or recall is very strong and compensates for the other metric being very poor. With these metrics evaluated, there are many other standard studies that are used in machine learning to evaluate the performance of the model, such as the *receiver-operating-characteristic curve* (ROC curve) and *learning curve* which are used in this chapter to evaluate the model performance.

BDTs using data with such an imbalanced dataset benefit from training with *Stratified K-fold cross-validation* (K-fold cv). This training method is performed by splitting the dataset into $K$ subsets, where the proportion of signal and background in the whole dataset is retained in each of the $K$ subsets (stratification), then $K-1$ of these subsets are used for training with the final subset used for validation. This process is repeated $K$ times with the overall training scores being taken as the average of the $K$ repetitions.

[2]Where the arithmetic mean is simply: $\bar{x} = 1/N \sum_i^N x_i$

Following the optimisation of hyper-parameters and cross-validation, the ensemble of weak learners is then finally trained once more on the full dataset using the optimised hyper-parameters to produce the final ensemble of weak learners. The binary classifier BDT returns a probability estimate that an event belongs to the signal class, known as the signal classification threshold. By default, the signal classification threshold is set to 50%, however, this need not be fixed at 0.5. Studying the impact of altering the signal classification threshold on $TPR$ and $FPR$ is precisely what the ROC curve performs. The ROC curve highlights the trade-off in a binary classifier between true-positive rate ($TPR$) and false-positive rate ($FPR$). Where these values are defined as:

$$TPR = \frac{TP}{TP + FN} \tag{6.1.9}$$

$$FPR = \frac{FP}{FP + TN} \tag{6.1.10}$$

Guessing the result of flipping a coin is a good comparison thought experiment when contextualising the ROC curve of a binary classifier. Consider a coin flip repeated many times with tails considered the desired signal. Regardless of the probability, $p$, with which an observer guesses signal, the accuracy and precision of the observer is invariant and equal to 50%, however the underlying $TPR$ and $FPR$ vary[3]. For this reason, the random chance of a coin flip is a good base-line comparison for a binary classifier. The signal classification threshold can also be optimised with respect to a figure of merit, in this chapter the signal classification threshold is optimised with respect to the F1-score because losing true signal or allowing background to be classified as signal should both be avoided in a similar fashion. Once deployed in the analysis, a BDT is instanced with the expected features and its optimised hyper-parameters. For a given event, the values of these features are passed to the BDT which then returns the estimated signal probability that is used to decide whether to consider the event as signal or not.

BDTs have become somewhat ubiquitous in experimental particle physics over the last decade due to the high quality of open source software such as `XGBoost` [147] and `SciKit Learn` [148]. BDTs also have a high degree of model interpretability, which is important for being able to rationalise the conclusion the model has made with the physics of the system.

## 6.2 $\nu 0\mu$ BDT application

Section 3.5.3 reviews the publication by the SND@LHC collaboration observing neutrino events which do not possess a muon in the final state, referred to as $\nu 0\mu$ events. This study is performed with the full neutrino simulation dataset created as outlined in section 3.4. The neutrino simulation data includes neutrino interactions in the SND@LHC detector from all flavours. The selection criteria applied to the simulation data are not identical to the criteria listed in table 3.13 applied in the SND@LHC publication observing neutrinos without a muon in the final-state. The criteria applied to the simulated data are listed in table 6.1 with the impact of each criteria on the simulation data statistics shown in figure 6.1 and the breakdown by

[3]This can be shown by substituting the signal probability, $p$, and the true signal fraction of 50% into the expressions for accuracy (equation 6.1.4) and precision (equation 6.1.5) and the expressions for $TPR$ and $FPR$.

event-type following the application of all selection criteria shown in figure 6.2. In this chapter, these criteria will be referred to as the $\nu 0\mu$ criteria. Here the neutrino and anti-neutrino interaction modes considered signal and background are recalled: signal modes are all (anti-)neutrino flavour neutral current interactions, electron (anti-)neutrino charged current interaction and tau (anti-)neutrino charged current modes without a final state muon. Background is therefore muon (anti-)neutrino charged current interactions and tau (anti-)neutrino charged current modes where the tau lepton decays to a final state including a muon. It is also recalled that since SND@LHC operates without a strong magnetic field, neutrino and anti-neutrino charged current modes cannot be distinguished by the charge of the outgoing lepton. The discussion of the $\nu 0\mu$ analysis in section 3.5.3 states that after the criteria are applied, the remaining expected background is 0.3 events from $\nu_\mu$ CCDIS and $1.5 \times 10^{-2}$ events from neutral hadron interactions. Neutral hadron simulation data is not included in the development of the BDT.

Criterion A of table 6.1 reduces the fiducial volume of the SciFi detector in the $xy$ plane while criterion B simply removes events where the lowest two bars of the first two US planes fire. The reason for criterion B is due to the poor veto efficiency in this low$-y$ region as outlined as a reason for the added third plane of the veto system in figure 3.11 of section 3.2.5. Criterion C then simply removes any events with activity in the Veto system. Criterion E ensures the continuity of the SciFi tracker such that all planes following the interaction fire. Criterion F requires that the first two US planes fire. Criterion G requires that there are at least 35 SiPMs in the SciFi firing and criterion H requires that there is a total of 700 QDC measured in the US system[4]. Finally, to efficiently remove tracks containing a final state muon, events containing a sufficient number of firing DS muon system planes for a track to be fitted, three (or four in the $xz$-projection case), are removed from the analysis.

| Criterion label | Criterion name |
|---|---|
| A | SF fiducial criterion |
| B | US fiducial criterion |
| C | Veto criterion |
| E | SF continuity |
| F | First two US planes |
| G | Min SF hits |
| H | US QDC criterion |
| I | DS veto criterion |

Table 6.1: Criteria applied to the neutrino simulation data. With criteria labels corresponding to the bin labels of figure 6.1.

[4]700 is the value used for simulation data while 600 is the value used in data.

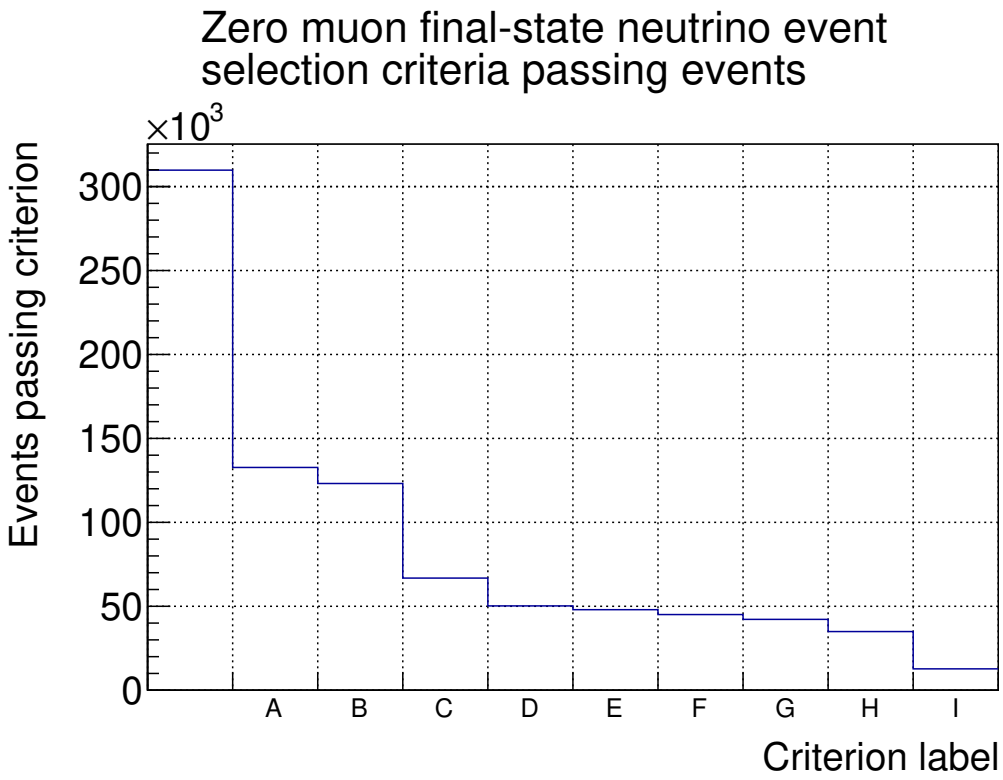



Figure 6.1: Impact of criteria (A-I) on the number of simulation events surviving the zero muon final-state neutrino event selection criteria. The bin labels correspond to the criteria as mapped in table 6.1.

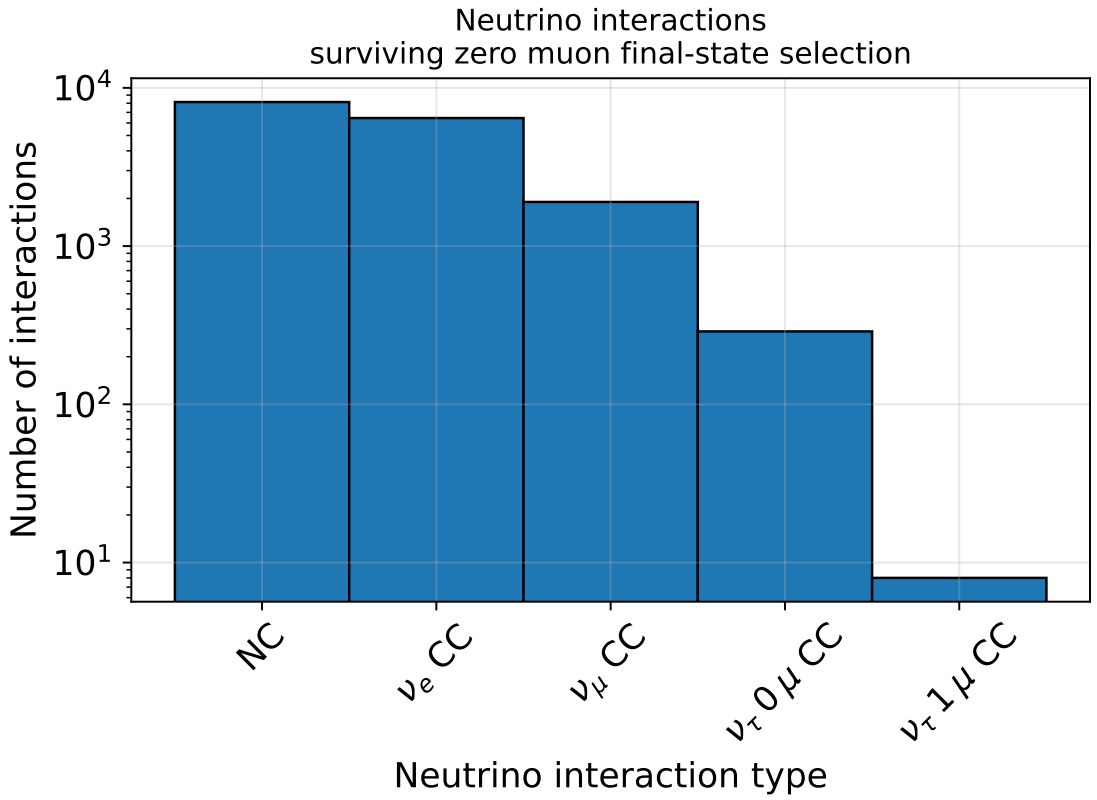


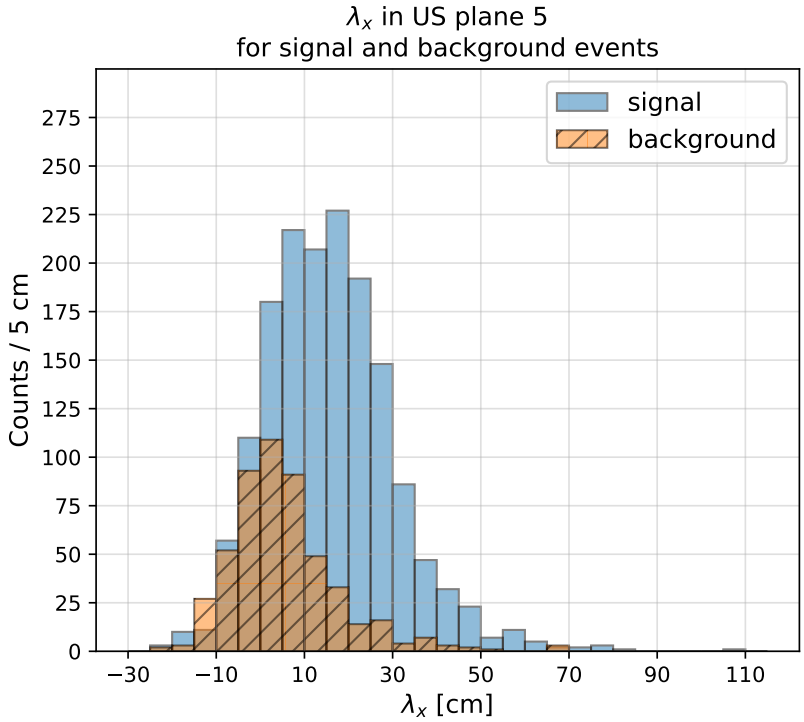

Figure 6.2: Distribution of neutrino simulation species and interaction mode following the application of cuts listed in table 6.1. This is the whole input dataset to the training of the BDT.

With the selection criteria applied, the production of the data utilised in the BDT is performed. Firstly, the interaction wall of the event is tagged using the algorithm developed by colleagues in the collaboration, used in both the $\nu_\mu$ CCDIS study [125] and the $\nu 0\mu$ study [126]. In all stations of the SciFi, the median hit position horizontally and vertically is computed using the same method as is described in section 5.1.4. Following this, the horizontal and vertical spatial measurements in the US system are computed. Now, a point to be recalled from section 3.3.6 is the use of a clustering algorithm using the firing scintillators in the DS muon system prior to the track fitting by GENFIT. This clustering algorithm is also used here: for events with firing scintillators in the DS, the clustering algorithms are used to produce spatial measurements in the DS. In the case that an event contains hits in the last plane of the US system or first two stations of the DS muon system, the most downstream two spatial measurements in each projection of the event are used to define a straight line[5]. This straight line can then be extrapolated upstream towards the interaction wall where the residual between the straight line and the spatial measurements in each plane of the US muon system and SciFi are recorded. This muon identification strategy is only extended to the final plane of the US muon system because the further upstream planes are more likely to contain scintillators firing due to the hadronic shower, which would pollute the information passed to the BDT. There may be cluster combinatorics in the two planes of the DS muon system, these are handled by defining the most likely pair of clusters to represent a non-reconstructible muon as the pair of clusters whose line minimises the resid-

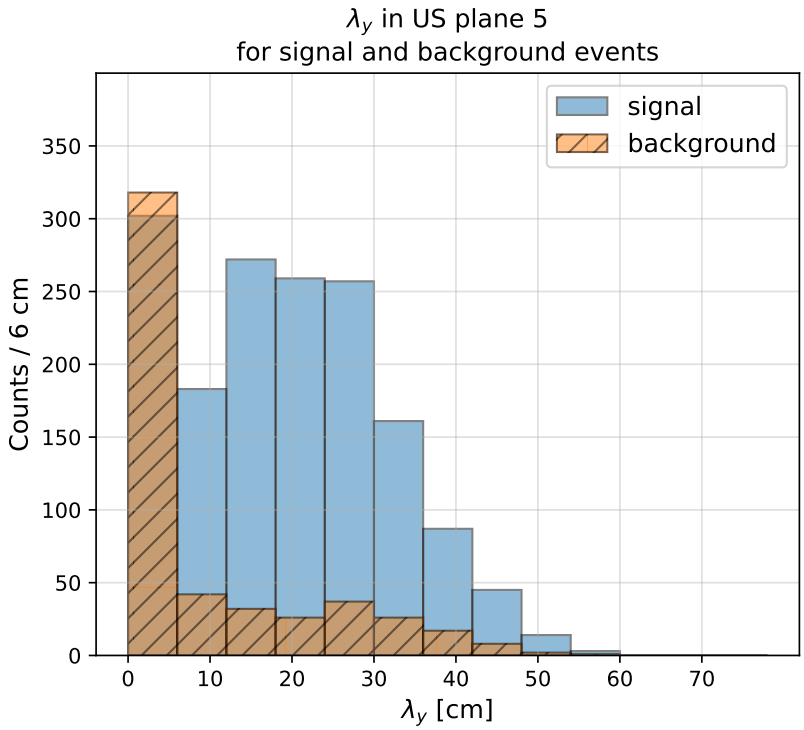
[5]Since this is just two data points in a projection, the straight line is defined, not fitted.

ual to the SciFi station immediately following the tagged interaction wall. Finally, the truth values must also be extracted from the simulation data as BDTs are a supervised learning method.

With the data extracted from the neutrino simulation, the signal and background distributions of several key variables are studied. Figure 6.3 shows how the $\lambda_x$ and $\lambda_y$ values for the final plane of the US system are distributed for signal and background.




(a) Horizontal energy deposition width, $\lambda_x$, measured in the fifth plane of the US system.




(b) Vertical energy deposition width, $\lambda_y$, measured in the fifth plane of the US system.

Figure 6.3: Measured energy horizontal and vertical deposition width in the fifth US plane using the neutrino simulation data which passes the $\nu 0\mu$ selection criteria of table 6.1.

Due to the construction of $\lambda_y$, defined in equation 5.1.11, and the granularity of

the US system[6], events with a single bar firing in a plane will record a $\lambda_y$ value of 0. Whereas horizontally the measurement of position and energy deposition width, $\lambda_x$, is continuous with its precision limited by the time resolution. The coarser granularity in $y-$position provides an observable with cleaner signal separation since there is no smearing by the time resolution. The computed $\lambda_x$ and $\lambda_y$ values for the US planes both show increasing separation power against muon final-state neutrino interactions as the plane number increases. One metric for measuring the separation power of a discriminant, $y$, referred to as the TMVA separation is outlined in section 3.1.10 of reference [149], and defined by the integral:

$$\langle S^2 \rangle = \frac{1}{2} \int \frac{(\,\hat{y}_S(y) - \hat{y}_B(y)\,)^2}{\hat{y}_S(y) + \hat{y}_B(y)} dy \tag{6.2.1}$$

Where $\hat{y}_S(y)$ and $\hat{y}_B(y)$ are the signal and background PDFs. The equation is such that PDFs with zero overlap with produce a separation, $\langle S^2 \rangle$, of one, and PDFs of identical shape will produce a value of zero. Computing this value for $\lambda_x$ and $\lambda_y$ at each plane of the US is shown below in figure 6.4. While the peak separation power of $\lambda_x$ is not in the final plane, it is clear that the later planes provide a greater separation power. The energy deposition width in either transverse direction in any plane will not provide a useless discriminator between signal and background, but each the energy deposition widths in the later planes can contribute to the multivariate method.

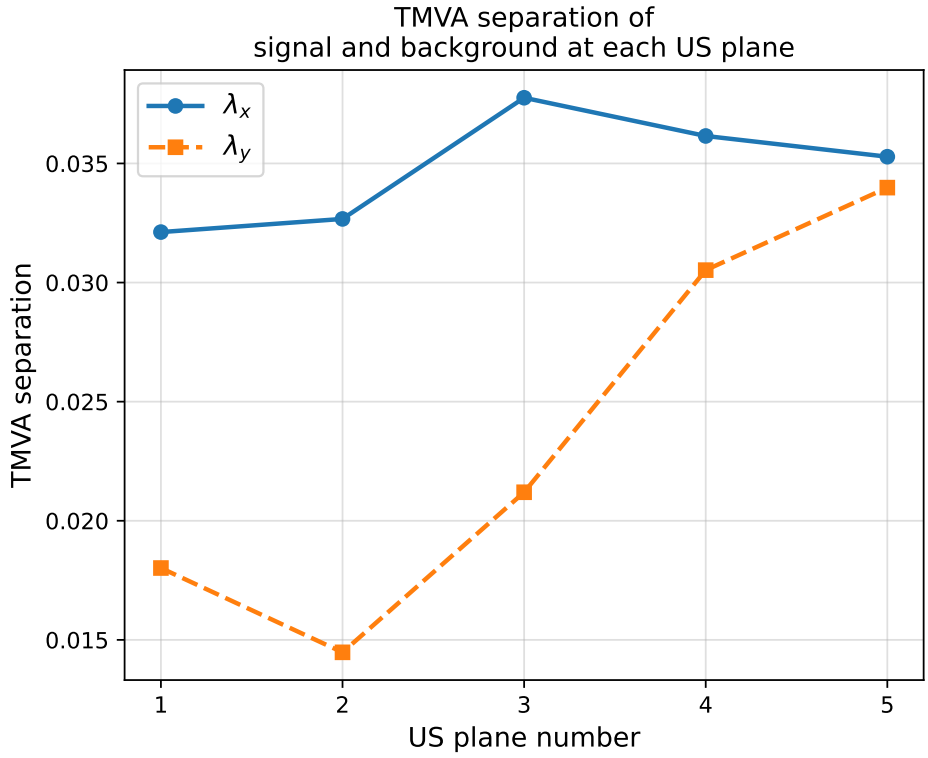



Figure 6.4: TMVA separation score for $\lambda_x$ and $\lambda_y$ measured in all five planes of the US system.

As explained earlier, the case where two planes of the DS muon system fire presents the ideal analysis case for separating signal from background, and in such

[6]A reminder that the US scintillators are 6 cm long in the $y-$dimension.

cases, straight lines can be produced in each projection to be extrapolated upstream in the detector. SciFi and US residuals are all of some use for distinguishing $\nu 0\mu$ events. The SciFi station immediately following the tagged interaction target wall provides the most accurate estimate of the neutrino interaction point, therefore spatial measurements in the DS muon system or late planes of the US should produce a straight line with a small residual to the SciFi spatial measurement immediately following the tagged interaction wall in both projections. Furthermore, the residual in each projection can be combined by adding the two residuals in quadrature, as shown in equation 6.2.2. This quantity is referred to as the resultant residual in the SciFi, $ds_{\mathrm{SF}}$.

$$ds_{\mathrm{SF}} = \sqrt{dx_{\mathrm{SF}}^2 + dy_{\mathrm{SF}}^2} \tag{6.2.2}$$

Having computed $ds_{\mathrm{SF}}$, figure 6.5 shows the signal and background distributions for $ds_{\mathrm{SF}}^2$ computed for the ideal background events which contain two firing planes in the DS system. It is clear from figure 6.5 that $ds_{\mathrm{SF}}$ is a powerful discriminating variable against events with two firing planes in the DS. However this subcategory of event only accounts for 30% of the events with a final-state muon.

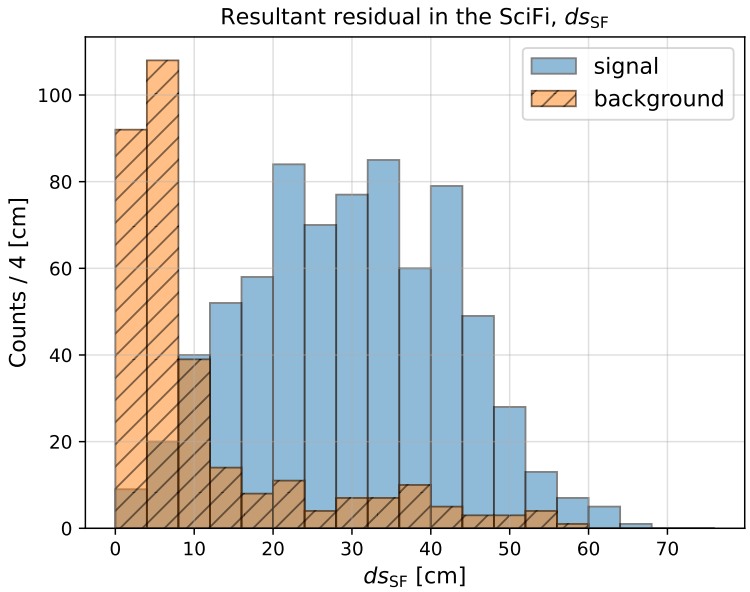



Figure 6.5: Resultant residual in the SciFi computed according to equation 6.2.2 for events containing two firing planes in the DS.

Multivariate methods provide a path to using all spatial measurements in the US as a means of improving the observation significance of $\nu 0\mu$ final-state neutrino interactions. The choice of multivariate method was the gradient boosted decision tree, BDT, in the XGBoost library [147]. All the features passed to the BDT are found in appendix A.26. The BDT was trained on the neutrino simulation data, described in section 3.4, that passes the $\nu 0\mu$ criteria of table 6.1 using a boolean flag indicating the presence of a muon as the target value. The balance of events following the selection criteria of table 6.1 is highly weighted towards signal events, the ratio of signal to background, $S : B \approx 7.8$ with the dataset is rather limited in statistics at $\sim 16,664$ events. The following hyper-parameters of the constructed BDT are optimised by the trial and error method constituent in the

SciKit Learn framework referred to as a *grid-search*. The grid-search uses a user-defined set of hyper-parameters to be optimised and a set of values for each hyper-parameter to be trialled. This information for the BDT of this chapter is shown in table 6.2. The grid-search creates a set of all permutations for the hyper-parameters to be trialled, then for each permutation instances a BDT with that permutation of hyper-parameters and performs $K-$fold cv, with $K = 10$, and performs a threshold optimisation in each of the $K$ folds used, where the $F1-$score is used as the metric of interest for threshold optimisation. This strategy produces the set of hyper-parameters which are capable of producing a model with the highest $F1-$score including after threshold optimisation. This operation, for a hyper-parameters set $\theta$ = (n-estimators, learning rate, max depth, min child weight), can be encoded as follows:

$$\text{CV score}(\theta) = \frac{1}{K}\sum_{k=1}^{K}\max_{t\in[0,1]} F1_k(t|\theta) \tag{6.2.3}$$

Where $t$ is the classification threshold value varied through [0, 1]; $F1_k(t|\theta)$ is the $F1-$score of the $k^{th}$ fold with classification threshold $t$ and fixed hyper-parameter set $\theta$; $\max_{t\in[0,1]}$ reflects that the threshold value producing the best $F1-$score is used and $1/K\sum_{k=1}^{K}$ shows that the $F1-$score is averaged over the $K$ folds. Thus, with equation 6.2.3 evaluated for each hyper-parameter permutation, the best permutation is simply taken as the one with the highest resulting CV score.

| Hyper-parameter | Values searched | Final value |
|---|---|---|
| *n−estimators* | 100, 200 | 200 |
| *learning rate* | 0.02, 0.03 | 0.03 |
| *max depth* | 1, 2, 3 | 3 |
| *min child weight* | 5, 10, 20, 40 | 5 |

Table 6.2: Table of the optimised hyper-parameter, the trialled values in the grid search method of `SciKit Learn` and the final value in the optimised BDT.

Next, the final model is trained using the optimised hyper-parameters and its optimal threshold is evaluated. Figure 6.6 shows how the $F1-$score, precision, recall and background rejection efficiency vary as the threshold for signal classification is altered.

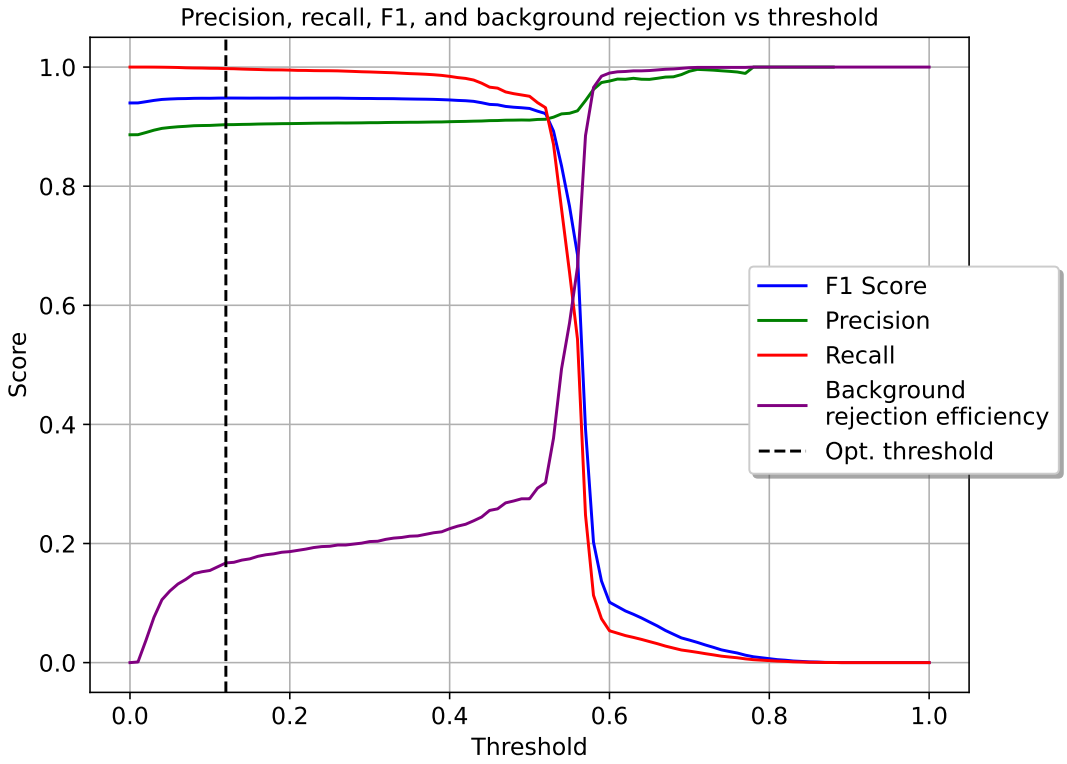



Figure 6.6: Impact of signal classification threshold on the $F1-$score, precision, recall and BRE scores. The threshold is set to optimise the $F1-$score.

With the signal classification threshold optimised for $F1-$score, the final confusion matrix of the BDT can be produced and is shown in figure 6.7. It is clear in figure 6.7 that the BDT is very efficient at retaining signal, reflected by the very strong recall curve in figure 6.6. Only a modest 19% of the background events are rejected with the BDT, however a key point of the BDT is its lightweight in analysis. The BDT is constructed in $0.340 \pm 0.009$s and classifies an event in $0.980 \pm 0.028\,\mu$s. It should be considered more akin to an additional, advanced selection criterion than a discrete analysis.

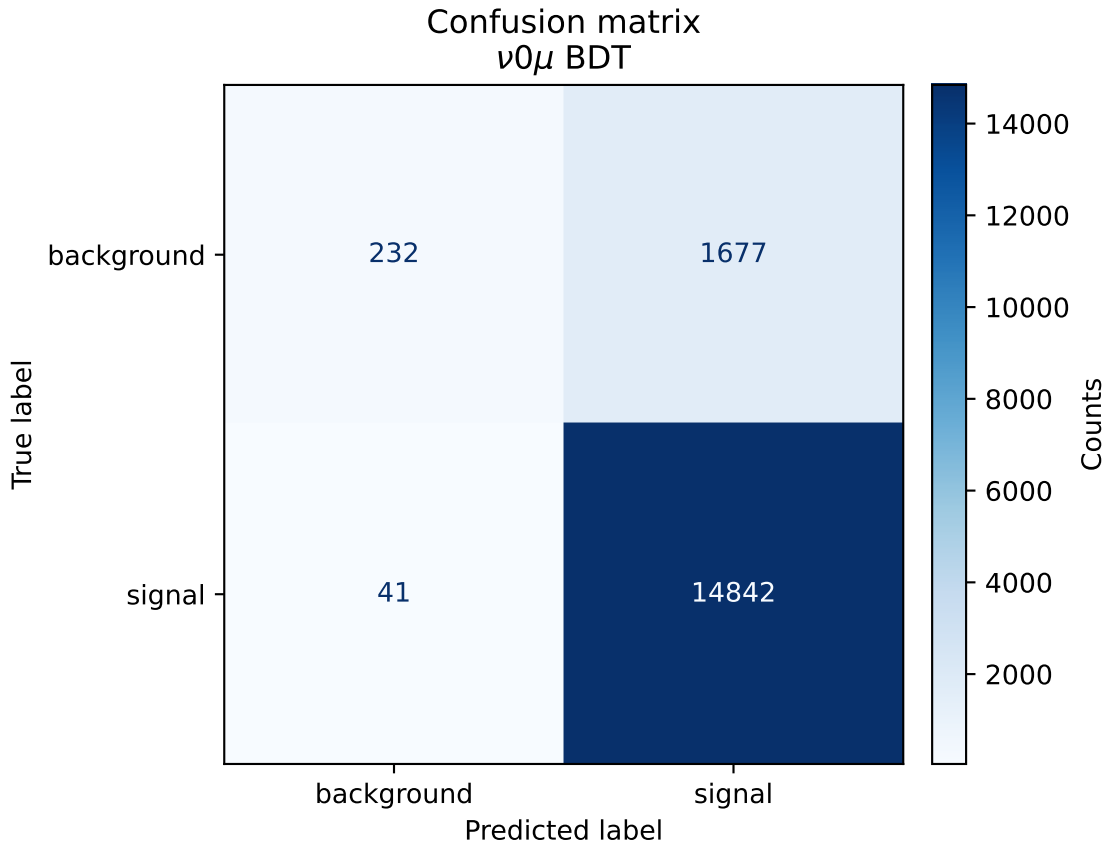



Figure 6.7: Confusion matrix obtained with the optimised BDT used for identifying events containing a final-state muon.

With a trained model in hand, the impact of the model upon the $\nu 0\mu$ analysis can be quantified by use of existing analysis infrastructure. This is done with simulated data and real events isolated by the collaboration. Neutrino and neutral hadron simulation data which survives the $\nu 0\mu$ selection criteria were used, only negligible passing muon simulation survives these criteria and thus is not necessary to be included in the observation significance.

The following description detailing the determination of an observation significance was not done by the author. The following method is the same as what is outlined in the description $\nu 0\mu$ analysis in section 3.5.3, some more details regarding the measurement of an observation significance are given in this section. The discriminating variable used in the significance observation is the measurement of the density of SciFi hits identical to that variable defined in the $\nu_\mu$ CCDIS analysis described in section 3.5.2. The choice of discriminating variable was made by the collaboration and the algorithm was designed by colleagues in the collaboration with no input whatever from the author. Simulation yields are scaled to the expected yields from the data set collected by the collaboration between 2022 and 2023, corresponding to an integrated luminosity of 68.6 $\text{fb}^{-1}$, this is the time period used in the published analysis of $\nu 0\mu$ events by the collaboration [126]. Lower limits of the discriminating variable are iterated over, with the yields at each lower limit determined by integrating the bins above this lower limit. The observation significance for the $\nu 0\mu$ study and for the $\nu_e$ CC channel alone are determined using methods outlined in reference [150] section 3. The $p$-value for observing a measured number of events in the signal region, $n_{\text{on}}$, and a measured number of events in the control

region, $n_{\text{off}}$[7]:

$$p = B_I(\rho, n_{\text{on}}, 1 + n_{\text{off}})/B_C(n_{\text{on}}, 1 + n_{\text{off}})$$

Where $B_I$ and $B_C$ are the incomplete and complete Beta functions respectively; $\rho = 1/(1+\tau)$, with $\tau$ the ratio of the control region mean to the signal region mean $\tau = \mu_{\text{off}}/\mu_{\text{on}}$. This $p$-value can then be converted to a corresponding $Z$-value using the inverse cumulative distribution function of the normal distribution:

$$Z = \Phi^{-1}(1 - p)$$

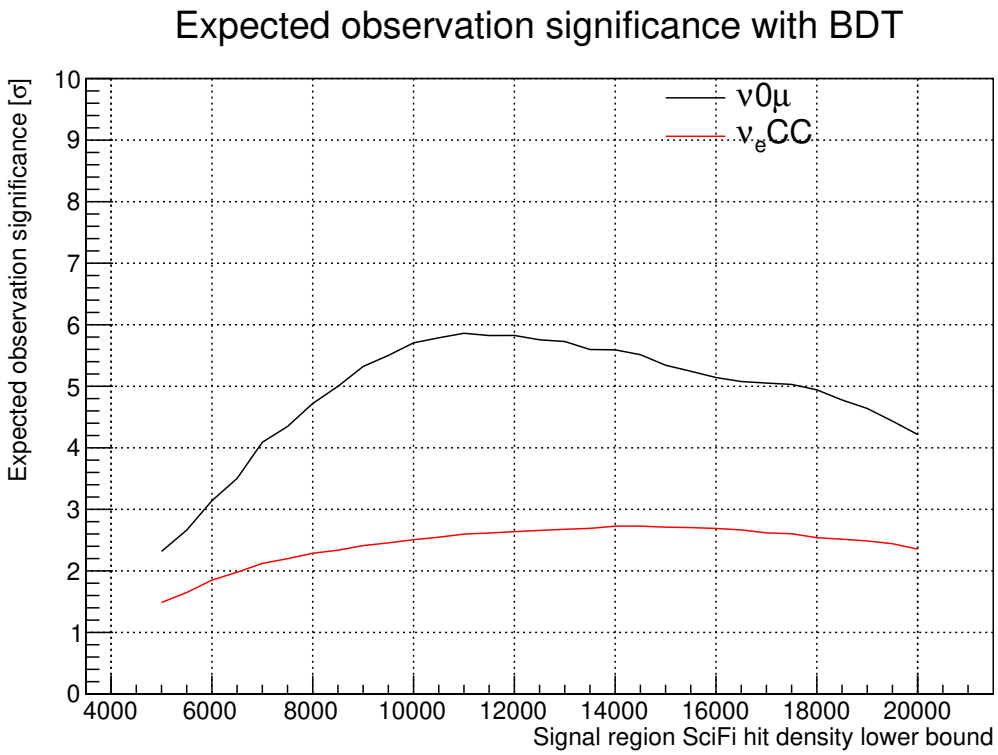
The impact of the BDT on the $\nu 0\mu$ observation significance is shown in figure 6.8b. The collaboration study into the zero muon final-state neutrino events uses a SciFi hit density requirement of $11 \times 10^3$, figure 6.8b shows that at this value of the discriminating variable, the use of the BDT reduces the observation significance of signal by $\approx 0.1\sigma$. The reason for this is that the BDT reduces too much signal relative to the reduction in background, figure 6.9a shows the reduction in $\nu 0\mu$ signal events while figure 6.9b shows the reduction in $\nu_\mu$ CC events. The confusion matrix of figure 6.7 can be used with equation 6.1.7 to demonstrate the background rejection efficiency within the neutrino simulation data is BRE $= 232/(232 + 1677) = 12.2\%$, which is shown to be too low a background rejection efficiency for the signal loss incurred by the BDT, as reflected by the negative change to the significance when the BDT is applied to the simulation data in the $\nu 0\mu$ study.

The motivation for this BDT study was that the spatial measurements in the late planes of the US system could be used to improve neutrino identification studies. To what extent the Upstream Muon System acts in either the background rejection or signal loss is not trivial to extricate due to the multivariate nature of the study. The study acts to investigate the utility of the Upstream Muon System to work in accordance with its neighbouring systems. While the first attempt at a method is unsuccessful in improving the observation significance of the $\nu 0\mu$ channel, the method is promising in extending the muon identification abilities of SND@LHC, as demonstrated by the 12.2% $\nu_\mu$ CC background rejection efficiency of the BDT. The data production and the BDT application provide a platform for future analyses taking advantage of the spatial measurement infrastructure provided by this thesis work. Future work could look to include the QDC measurements in the BDT: scintillators in the downstream planes of the US traversed only by muons would be expected to show a low QDC value relative to US planes of SciFi stations containing the highly ionising final-state products.

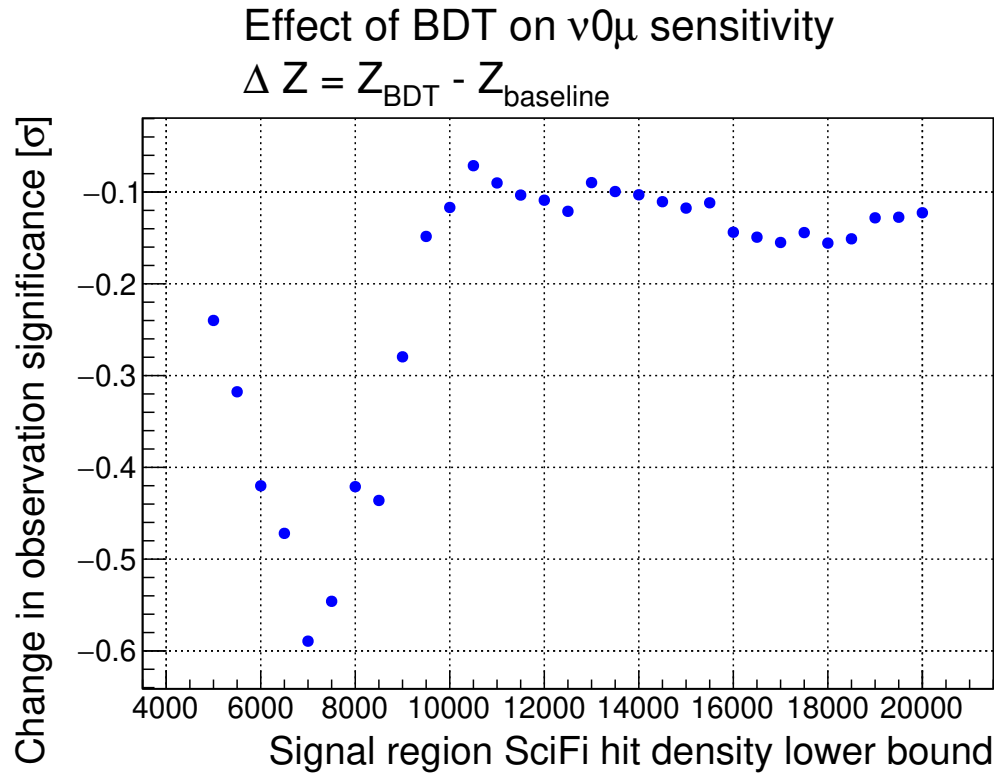
Another limitation of this study comes from the distribution of where the muons exit the acceptance of the detector. Criterion I of table 6.1 already requires that the third station or fourth vertical plane[8] do not fire, but where the final-state muon leaves the detector acceptance is important context in the background rejection capability of the BDT. The exit point of the final-state muon can be studied in simulation data and defined as when the $x-$position or $y-$position of the final-state muon track first exits the detector acceptance. Figure 6.10 shows the distribution for this quantity with the $z-$positions of the end of the SciFi, final US plane and

[7]The "on", "off" nomenclature stems from the rooting of this statistical method in gamma ray astronomy where photons can be detected when the telescope is directed at the source (on-source photons), or away from the source (off-source photons) [151].

[8]A reminder that the DS muon system has three stations comprising a plane of horizontally aligned scintillators and a plane of vertically aligned scintillators, then a final station comprised solely of a plane with vertically aligned scintillators.

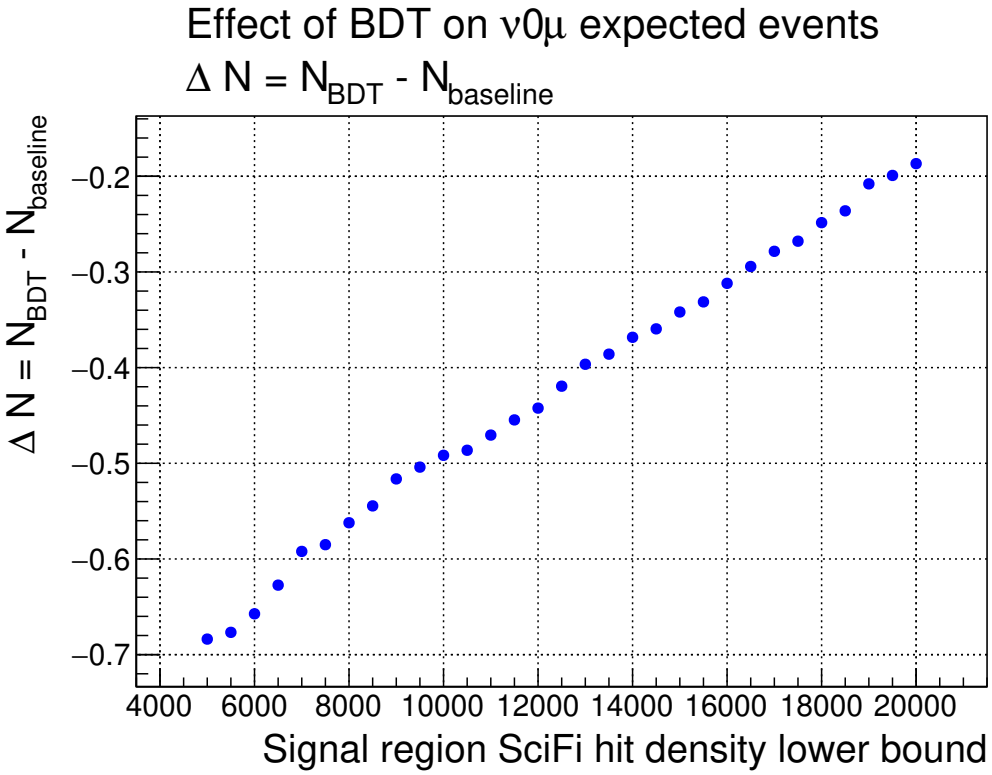



(a) The observation significance curves for the $\nu 0\mu$ study and $\nu_e$ channels with the BDT applied. The impact of the BDT on the $\nu_e$ observation is not considered.

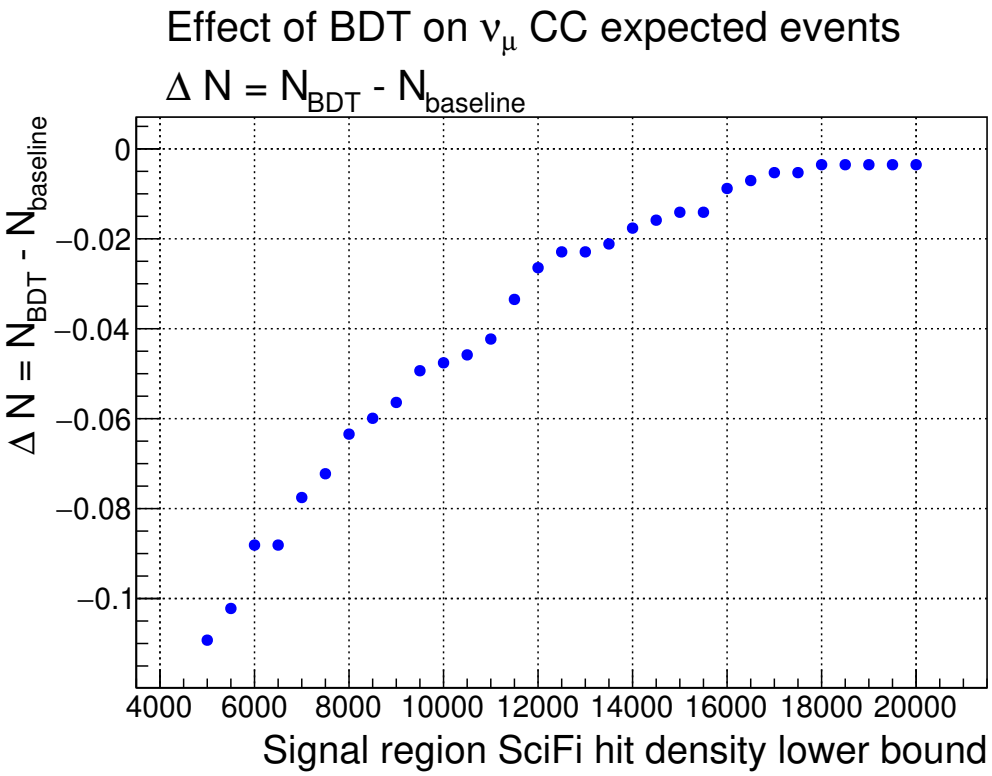



(b) The impact of the BDT on the $\nu 0\mu$ observation significance as a function of the SciFi hit density lower bound.

Figure 6.8




(a) Impact of the BDT on the expected number of $\nu 0\mu$ events in an integrated luminosity of 68.6 fb$^{-1}$.




(b) Impact of the BDT on the expected number of $\nu_\mu$ CC events in an integrated luminosity of 68.6 fb$^{-1}$.

Figure 6.9

third DS station shown as black dashed lines with their labels shown adjacent to the line. Also written on figure 6.10 is the fraction of final-state muon tracks which exit the detector acceptance before the final plane of the US muon system, shown to be 32% and therefore this fraction of the $\nu_\mu$ CC background is ineligible to be removed by the BDT presented in this study as hits in the final US plane or beyond are required.

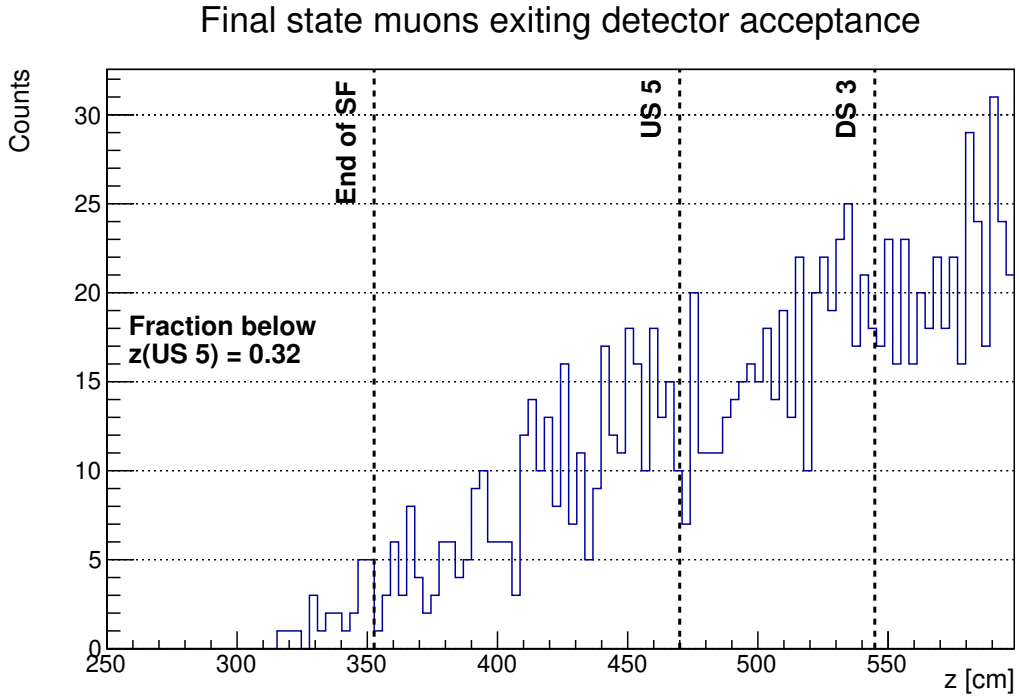



Figure 6.10: Distribution of the $z-$position at which the final-state muon exits the detector in either the $xz$ or $yz$ projection. The $z-$positions of the end of the SciFi, the final plane of the US and the third station of the DS systems are shown.

# Chapter 7

# Conclusion

The thesis begins with a review of the relevant supporting physics in chapter 2. It introduces the underlying principles of the Standard Model of particle physics, followed by a brief history of neutrino physics. This section contextualises the phenomena that place neutrinos at the heart of modern high-energy physics: from neutrino flavour oscillations providing a solution to the then long-standing Solar neutrino problem, through to how the existence of three neutrino flavours leads to two possible neutrino mass hierarchies and allows for CP violation in the neutrino sector. The interactions of neutrinos with matter are then discussed with particular attention paid to deep inelastic scattering, the scattering mechanism of paramount interest to the SND@LHC experiment. Neutrino physics at the LHC are then discussed, introducing the origins of the neutrino flux expected at SND@LHC, such as the decays of pions and kaons being responsible for the muon neutrino and anti-neutrino flux at SND@LHC. Finally, the physics goals of SND@LHC, the primary of which being a measurement of the gluon parton distribution function at Bjorken-$x$ values $\approx 10^{-6}$ and measuring the inclusive neutrino scattering cross-section at neutrino energies $\mathcal{O}(100\,\text{GeV})$ to $\mathcal{O}(\text{TeV})$.

Following the installation of the SND@LHC detector, the commissioning work on the US began in order for the subsystem to contribute as effectively as possible to the physics studies of the experiment. Chapter 4 deals with the initial steps taken from the collected experimental data, to modelling and correcting the observed time-walk effect then performing a system-wide time-alignment of the US system. Firstly, the unwanted time-walk effect had to be dealt with through a multi-stage method of developing selection criteria for passing muons, measuring the effective signal speed in the system, and then producing a functional description of the effect in order to statistically minimise the time-walk. The correction of the time-walk effect yields tangible systemic results best shown by the improved single-SiPM time resolution. Due to the construction of an event-wise reference time in either the DS muon system or the SciFi system, SiPM channels are able to be aligned to one-another through an alignment to the subsystem used to construct the event-wise reference time. The application of the so-called fine alignment brings all SiPMs of the US system into alignment, allowing for the full timing capabilities of a scintillator to be realised. The measurement of the time resolution on the side of a scintillator, read-out by six SiPMs, showed the presence of covariance between the timing measurements of SiPMs. The presence of the covariance was directly measured in data and found to be required to account for the greater than expected scintillator-side-time resolution. Following the time-walk correction and the alignment of the SiPMs, the typical resolution of a scintillator-side is found to be 302 ps while the typical resolution of

a full scintillator is 245 ps.

The final studies of this chapter describe self-contained laser studies performed in a laboratory at École Polytechnique Fédérale de Lausanne (EPFL). The purpose of these measurements was to study the electronic response of the SiPMs in the US PCB without the presence of a scintillator and with a known incident light. Having calibrated the laser, two variables were manipulated to study the response of the SiPMs on the PCB: the intensity of light incident upon the PCB and the number of firing neighbouring SiPMs. Collecting data manipulating these variables over a period of two weeks allowed the time-walk effect and QDC saturation curves to be studied in the cases where different numbers of neighbouring SiPMs were illuminated by light. The data collected in this controlled environment allowed for conclusions to be argued about the US PCB that are not reachable in the TI18 detector: namely that a transient voltage drop over the shared SiPM bias line causes a reduced electric field over the SiPMs which induces a time delay. For this conclusion to be thoroughly tested, further laboratory measurements would be required using an oscilloscope to read the voltage over the bias line. A further result of the laser measurements was the measured delay of the small-area SiPMs. It was observed that the two small-area SiPMs per SiPM group never fire in the same event as the six large-area SiPMs. The delay between an event where the majority of illuminated large-area SiPMs fire to the event where one or both small-area SiPMs fire was studied and shown to be relatively constant across the small-area SiPMs. This observation provides a foundation for further studies of the small-area SiPMs. If their raison d'être, providing an extended QDC dynamic range to the US system, were to become critical in optimising physics measurements of the experiment, the collaboration should look to this thesis and the laser data collected in Lausanne.

With the time-walk correction and alignment in-place, the beginning of chapter 5 outlines how the alignment process of chapter 4 naturally leads to a method of implementing the timing behaviour of the US system into the simulation framework of the SND@LHC experiment. While the time resolution is extracted using the FWHM, this value does not capture the large, non-Gaussian tails and so for an accurate reproduction of the timing behaviour in the simulation, look-up histograms were used in the process of digitising the timing behaviour of the US system. Furthermore, the time alignment of the US system then allowed for horizontal spatial measurements to be produced using the effective signal speed values measured also in chapter 4, while vertical spatial measurements can be produced using the relative QDC measurements of the scintillators in each plane of the US system. These spatial measurements were validated using selected passing muon events, the dominant event type for the experiment. Studying the horizontal and vertical spatial residuals between the spatial measurements in the US system and the expected position from the track fitted in the DS muon system show that both left- and right-sides of all scintillators in the US system produce positions that agree with the position expected by the fitted track in the DS. The spatial measurements in passing muon data were also compared to those collected with passing muon simulation to validate the simulation implementation of the spatial measurement infrastructure. Horizontally, the left- and right-side positions were used together to produce either a central position of the energy deposition, referred to as the $x-$barycentre; or the energy deposition width, $\lambda_x$. All the horizontal spatial measurements are shown in the schematic of figure 5.3 while the vertical measurement are shown in figure 5.4. The spatial measurements open an entirely unexplored set of analysis opportunities for SND@LHC, the first of these to be studied in this thesis is how the spatial measure-

ments in the US can be combined with spatial measurements in the SciFi tracker to perform angular measurements. Passing muon events were used to investigate the angular agreement between tracks fitted with the DS muon system and the SciFi plus US composite system.

Chapter 5 also presents an investigation of the SiPM data in the set of selected $\nu_\mu$ CCDIS events isolated by the collaboration in an analysis outlined in section 3.5.2. The investigation in this chapter draws out two data quality issues: firing scintillators with significantly delayed timing data, and firing scintillators with very few SiPMs on either side. Overall, 25.7% of the firing scintillators in the selected $\nu_\mu$ CCDIS events contain either of these two data quality issues. The former of these problems presents issues when using the timing data to perform a spatial measurement but also the chronology of the detector is out of order, i.e. SiPMs in the US system registering times later than SiPMs in the DS muon system. The issue of low firing SiPM multiplicity was compared to what was observed in passing muon data. This highlighted a far greater rate of low firing SiPM multiplicity in the selected $\nu_\mu$ CCDIS events than what was observed for selected passing muon events. It was also observed that this rate of low firing SiPM multiplicity increases when there is a greater total QDC value measured in a plane of scintillators, thus a US PCB. However, even with selected passing muon events producing very high QDC values in a plane, the rate of low SiPM multiplicity does not reflect what is measured in the selected $\nu_\mu$ CCDIS events. It is stressed in this section of the thesis that this will present a challenge when the collaboration intends to use the US muon system for calorimetry and the issue should be studied more in the near future. Angular measurements were also performed using the selected $\nu_\mu$ CCDIS events including measuring the scattering angle between the tracks formed in the SciFi plus US composite system and the DS muon system. While the resulting angular measurements show promise, the comparison to simulation shows an excess number of events with a greater polar angle than expected from simulation. Similarly for the measured scattering angle, there is an excess number of events with a greater angle. The impact of the data quality issues on the resulting angular measurements is not clear but the issues must be tackled in order to achieve the best performance from the detector system.

The final chapter presenting analysis, chapter 6, builds upon all the work in the preceding chapters to use the Upstream Muon System of SND@LHC to improve the observation significance of neutrino interactions without a muon in the final-state. The opening section of the chapter begins by outlining the underlying principles of Gradient Boosted Decision Trees and the implementation of this in the XGBoost library including the hyper-parameters that can be optimised in the model. Following this, the selection criteria used to produce a signal-dominant simulation data-set, where signal is neutrino events without a muon in the final-state, were outlined along with the signal and background distributions of key observables used in the BDT such as the energy deposition widths in the final plane of the US system. The BDT event classification probability was then optimised with the goal of maximally improving the observation significance of the $\nu 0\mu$ analysis. This model provides a strong proof of concept that the spatial measurement infrastructure created in this thesis can yield results in the broader physics goals of the experiment. It is unsuccessful in improving the observation significance of the $\nu 0\mu$ study with the observables used in the BDT as the observation significance reduced by $0.1\sigma$ following the BDT output being used as a final section criteria.

# List of Figures

2.1 The fermions and bosons of the Standard Model. With each particle is shown the spin, mass and charge. The neutrino mass limits here are not up to date. . . . 14
2.2 Continuous decay spectrum from 'Radium E' [22]. . . . 15
2.3 Energy spectrum of neutrinos produced by mechanisms [26] determined by the Standard Solar Model. . . . 16
2.4 Comparison of $^8$B solar $\nu_\mu + \nu_\tau$ flux vs $\nu_e$ flux measured by SNO and SK collaborations. The diagonal bands show the predicted all-flavour $^8$B solar neutrino flux from SNO and what is predicted by the SSM [30]. . . . 17
2.5 Measurements of the double-ratio of muon to electron atmospheric neutrino rates by various experiments. Central values depicted with a square refer to iron-based tracking detectors, while circular central values are water-based Cherenkov detectors [33]. . . . 18
2.6 The two possible mass hierarchies given the measurements of $\Delta m^2_{ij}$. The flavour sharing in each mass eigenstate is also depicted. Taken from [39]. . . . 19
2.7 Two-body interaction with mediator four-momentum $q$. . . . 21
2.8 Total neutrino and anti-neutrino per nucleon charged current cross sections (isoscalar target) divided by neutrino energy as a function of neutrino energy. The total (anti-)neutrino per nucleon cross section is shown as well as the three contributions: quasielastic scattering (dashed), resonance production (dot-dashed) and DIS (dashed). Taken from [53]. . . . 23
2.9 Experimental measurements of charged current neutrino interaction cross-section in the energy range $\mathcal{O}(1)$GeV - $\mathcal{O}(100)$GeV. The approximate energy range of LHC neutrinos passing through SND@LHC is shown in the magenta rectangle. Adapted from [54]. . . . 24
2.10 Charged current resonance production neutrino scattering where a $\Delta^+$ is produced before decaying back to a neutron ground state through the emission of a $\pi^+$ through the strong interaction [55]. The decay to $\pi^0 p^+$ is also allowed. . . . 25
2.11 Lowest-order Feynman diagram for deep inelastic neutrino scattering [56]. . . . 26
2.13 Charged-current tree-level diagram showing single charm production from a neutrino interaction. . . . 28
2.12 Two rare processes by which neutrino DIS can produce charmonium ($c\bar{c}$). . . . 29

2.14 Incident neutrino spectrum for the SND@LHC on the right column, FASER$\nu$ on the left column. The scale of the ordinate axis represents the number of neutrinos passing through the detector's cross-sectional area for 150 fb$^{-1}$ of integrated luminosity. The neutrino flux is shown for each flavour, split by production method and estimated using up to six different event generators depicted with unique line styles. Taken from [65]. . . . . 32
2.15 Correlation between the pseudo-rapidity of the charm meson, $\eta_{\text{meson}}$, and the electron neutrino, $\eta_{\text{neutrino}}$. Data obtained from simulations with POWHEG and PYTHIA8 from [66]. . . . . 33
2.16 Correlation of $x_1$, $x_2$, the proton momentum fractions carried by the scattering gluons involved in open charm production at the LHC in events resulting in a neutrino in the SND@LHC $\eta$-range. Taken from [66]. . . . . 34
2.17 Left: differential cross section for charm production at 13 TeV. Right: ratio of differential cross section to the reference differential cross section at 7 TeV, $4 < \eta < 4.5$. Taken from [66]. . . . . 34
2.18 Leptonic decay of a $D_s^-$ meson to a tau final state. . . . . 35
2.19 Normalised energy spectrum of final state NC DIS neutrino interactions and dark matter DIS interaction for three possible mediator masses, $m_V$. Taken from [66]. . . . . 36
2.20 Projected sensitivity of the SND@LHC experiment to the leptophobic $U(1)_B$ dark matter portal by scattering off protons. Elastic scattering is shown in dashed blue, and DIS scattering in solid blue. Taken from [66]. . . . . 37
2.21 Mass stopping power for muons, pions and protons in various materials, expressed as $-1/\rho\langle dE/dx\rangle$. Taken from [79]. . . . . 39
2.22 Initial energy spectrum of muons reaching SND@LHC measured from simulation and normalised to unit integral. . . . . 40

3.1 A schematic illustrating the position of both FASER and SND@LHC relative to ATLAS, and how the LHC magnets and rock removes the majority of charged particles and hadrons from impinging upon the experiments. Figure taken from reference [83]. . . . . 41
3.2 Layout of the SND@LHC detector. The coordinate systems of the $xy$ and $yz$ planes are shown in the top left and right corners respectively. Taken from [84]. . . . . 42
3.3 The SND@LHC detector. Particles traversing the experiment from left to right. Taken from [85]. . . . . 42
3.4 SND@LHC uses 250 $\mu$m diameter double-cladded polystyrene based scintillating fibres from Kuraray [86]. . . . . 43
3.5 Schematic displaying the five walls of the emulsion target. . . . . 43
3.6 Photograph and schematic of finished neutron shielding cold box. . . 44
3.7 . . . . . 46
3.8 Diagram illustrating total internal reflection (TIR) in a scintillator wrapped with Mylar foil, showing the critical angle condition and refractive indices. . . . . 47

3.9 A photograph of SND@LHC, where one can see the final two vertically aligned planes and the final horizontally aligned plane of the DS system. The vertically aligned planes have a smaller length in $x$ than the horizontally aligned planes, the last of which is labelled 'DS3H'. . 49
3.10 Schematic diagram of the veto bi-plane from all sides. . . . . 49
3.11 Positions of SciFi tracks extrapolated to the planes of the Veto system for events with less than 13 SiPMs firing in the Veto system. The first plane is shown on the left and the second plane on the right. Taken from [91]. . . . . 50
3.12 Energy level diagram of an organic molecule with $\pi$-electron structure. From [93]. . . . . 52
3.13 Schematic of a pn-junction powered in reverse bias with voltage, $V_R$. $E$ denotes the electric field within the depletion zone and $E_{\text{app}}$ denotes the electric field from the external potential. The direction of these fields indicate the direction that a test positive charge would accelerate. $W$ indicates the width of the depletion zone. Taken from [97]. . . . . 54
3.14 Depiction of how the current across the photodiode increases with the applied reverse bias. Taken from [96]. . . . . 55
3.15 Avalanche triggering probability of electrons and holes in a pixel. Taken from [96]. . . . . 55
3.16 The absorption coefficient of photons in silicon as a function of photon wavelength for different temperatures between 40 K and 415 K. Taken from [100]. . . . . 55
3.17 a.) n-on-p pixel structure, more sensitive in the red region of light. b.) p-on-n pixel structure, more sensitive in the blue, near-UV region of light. Figure taken from [96]. . . . . 56
3.18 Typical photon detection efficiency curve for the Hamamatsu S14160/S14161 series SiPMs. The nature of the data varies between SiPM type and even among SiPMs of the same type. This figure demonstrates the typical data for these series of SiPM, the peak PDE is quoted as 50% at $\Delta V = 2.7\,\text{V}$ for all SiPMs in this series [88]. . . . . 57
3.19 Equivalent circuit for a SiPM pixel and quenching resistor. The resistance of the pixel during an avalanche is denoted as $R_d$. Taken from [96] . . . . . 58
3.20 Depiction of left: prompt, middle: delayed and right: external optical cross-talk. Taken from [98]. . . . . 61
3.21 Data for a typical SiPM in the Hamamatsu S14160/S14161 series for how gain, OCT probability and PDE($\lambda = 450\,\text{nm}$) varies with the applied over-voltage. Figure taken from [88]. . . . . 62
3.22 Measurements of the DCR for various over-voltages with linear fits for a range of temperatures. This data is collected for a Hamamatsu S10362-11-050C. The Taken from [105]. . . . . 63
3.23 DCR measurements at a wide range of over-voltages (labelled $\text{V}_{\text{OV}}$ in the figure but referred to as $\Delta V$ in this thesis) for SiPMs irradiated to different levels. The SiPMs used in this study are from Hamamatsu (labelled HPK in the upper-left of the plot) with a pitch of 25 µm. Taken from [102]. . . . . 63
3.24 Illustration of how two signals of different signal shapes will be affected by the time-walk effect when a fixed threshold is applied. . . . 64

3.25 Photo of the SND@LHC FE board. The dashed red box shows the dual connector strips which connect to the SiPM PCB via a small interposer PCB. The solid red box shows the two TOFPET2 ASICs. Taken from [85]. . . . . 65
3.26 Photo of the SND@LHC DAQ board. The dotted rectangle on the left shows the four connector slots for reading in digitised SiPM data from the FE boards. The central dotted rectangle shows the Enclustra Mercury SA 1 module including the FPGA. On the top right a rectangle highlights the power socket and on the bottom right the Ethernet and TTCrx ports. Taken from [85]. . . . . 66
3.27 Schematic of the event building process [85]. Step 1 corresponds to the collection of hits into 25 ns events. Step 2 corresponds to the application of the two-stage noise filter. Taken from [85]. . . . . 67
3.28 Illustration of the difference between the SND@LHC time alignment before (A) and after (B) running the event builder synchronously with the LHC BST clock. Before, a noise hit could split a physics event into two separate recorded events. With the LHC BST clock, the expected start time of an event can remove this unwanted feature. 68
3.29 Simplified schematic of the global SND@LHC DAQ system [85]. . . . 70
3.30 Energy spectra of neutrons and neutral kaons interacting in the target system of SND@LHC shown for both with (blank) and without (shaded) accompanying hits in the Veto subsystem. Taken from [124] 73
3.31 DS tracks shown in the red data series, Monte Carlo simulation in blue. The dotted lines at 12 cm and 64 cm represent the lower and upper limits on $y$ considered for the muon flux measurement. Taken from [84]. . . . . 75
3.32 Distribution of the sum of SciFi hit density weights as measured with 100 GeV pions in the 2023 test beam campaign. The dotted vertical line is at $11 \times 10^3$. Taken from [126]. . . . . 80

4.1 Distribution of fitted track slopes in the $x$ and $y$ projections. Data from run 5408 with the only criteria applied being the requirement of a fitted track in the muon system. . . . . 83
4.2 Location of Q12 quadrupole magnet and the SND@LHC positioned in TI18. . . . . 84
4.3 Distribution of the average time value measured by the fired horizontal muon system scintillators used in track reconstruction. This distribution is taken from run 5408. . . . . 85
4.4 Residual between the $y$-position of the track extrapolated to the 5th US plane and the y-midpoint of a fired scintillator in the 5th US plane from run 5097. . . . . 87
4.5 2D histogram of $dt_{\mathrm{SiPM}}$ against the extrapolated DS $x-$value. The vertical red dashed lined indicate the fit range. Data from run 5408. . 88
4.6 Schematic of a scintillator illustrating what the fit range used to determine the effective signal speed in the scintillator is. The figure is not to scale. . . . . 89
4.7 QDC of a single SiPM correlated against $dt^{\mathrm{ToF}}_{\mathrm{SiPM}}$. The lower pane shows the ratio of the data to the fit for each QDC point. . . . . 91

4.8 Time resolution of a SiPM with signal time-of-flight correction, time-walk correction and its fine alignment parameter applied. Time resolution extracted according to equation 4.1.10. . . . . 92
4.9 Definition of the alignment parameter. Using a centrally positioned SiPM on the left side of the 5th bar of the 5th plane of the US system as an example. . . . . 93
4.10 Corrected and aligned SiPMs coupled to the same side of a scintillator in the with the modal times and standard deviations shown. . . . . 95
4.11 Average time measured by fired SiPMs on left side of scintillator. The time resolution is extracted according to equation 4.1.10. . . . . 96
4.12 Schematic of the experimental set-up inside the dark room at EPFL. The items in this schematic are not to scale. There are eight SiPMs visible to the laser: the one-bar illumination regime. . . . . 97
4.13 Schematic illustrating the positions of the two SiPMs used for calibrating the laser. The red Gaussian simplistically illustrates the Gaussian beam profile of the laser output. . . . . 98
4.14 Both figures here were not produced by the author but by the PhD student who collected the data at EPFL. . . . . 99
4.15 Plot of the ADC recorded by channel 30 for laser intensities between 0 and 100% following the correction of saturation effects using the ratio of ADC values between the two calibration channels. The linear fit is performed in the range [20%, 100%]. . . . . 100
4.16 Plot of ADC recorded by channel 30 and the calibration laser intensities following the linear regression defined in the legend. The small discontinuity around 20% is an artefact of the calibration process with negligible impact on the resulting analysis. . . . . 100
4.17 Time versus QDC for a single laser run for a single SiPM on the PCB. The unit "[cc]" refers to the fraction of the 6.25 ns clock-cycle. . . . . 102
4.18 Comparison of the time-walk behaviour observed for a given SiPM in the different illumination regimes investigated. . . . . 103
4.19 Comparison of the different illumination modes for a large area SiPM (top) and small area SiPM (bottom). . . . . 104
4.20 Effect of selection requirements applied on events for selecting muons 105
4.21 Distribution of the fitted track $\chi^2_\nu$. . . . . 105
4.22 Distribution of measured effective signal speeds with no time walk correction applied for run 5408. Error bars correspond to the estimated systematic uncertainty. . . . . 107
4.23 Mean centred correlation matrix of the measured effective signal speeds in the five equal-length segments throughout the scintillators. By definition of equation 4.3.1, matrix is symmetric and the main diagonal values are all unity. . . . . 108
4.24 Run stability plots of the effective signal speed before time-walk correction. Overlaid histograms of the quantity on the top subplot and a channel-wise plot for the quantity on the lower plot. . . . . 109
4.25 Zoom-in on run stability plot for uncorrected $c_{\mathrm{SiPM}}$ values on the left of plane 3 in the Upstream muon system. . . . . 110
4.26 Effect on the measured effective signal speed when applying the time-walk correction function. . . . . 111
4.27 QDC value for a SiPM correlated against the x-position. . . . . 112

4.28 Zoom-in of $dt^{\text{ToF}}_{\text{SiPM}}$ versus $x-$position for SiPM 5, bar 5, plane 5 of the US zoomed-in around the MPV peak to illustrate what impact on time can be induced by a small QDC change around the most probable QDC value. . . . . 113
4.29 Run stability plots of the time-walk corrected effective signal speed. Overlaid histograms of the quantity on the top subplot and a channel-wise plot for the quantity on the lower plot. . . . . 114
4.30 Zoom in on run stability plot for time-walk corrected effective signal speed values in the US. Channels 320 to 399 correspond to the left side of the 3rd plane of the US. . . . . 115
4.31 The effect on time resolution by applying the time walk correction parameters for run 5408. . . . . 116
4.32 Measured time resolutions after the application of a time-walk correction for US plane 5, bar 5, SiPM 5 with data from run 5999. The time resolution at $L_\mu$ and $x_{\text{left}} - 5$ is measured and the ratio shown in the top right. . . . . 118
4.33 Distribution and graph showing how the measured time-walk corrected time resolution changes over a distance of approximately 50.6 cm for all the SiPMs in the US muon system. . . . . 119
4.34 Distribution of $\chi^2_\nu$ values using only statistical uncertainties from the time-walk fits of the US SiPM data from run 5999. . . . . 120
4.35 Distributions of the RMS residual for the time-walk fit on data from run 5999. . . . . 121
4.36 Distribution of $\chi^2_\nu$ using the quadratic sum of statistical uncertainty and numerically optimised estimate of the systematic uncertainty from the time-walk fits of the US SiPM data from run 5999. . . . . 123
4.37 Distribution of measured $\sigma_{\text{sys}}$ using MLE. . . . . 124
4.38 Determined DS alignment parameters from run 5999 for the US muon system. These alignment parameters are appropriate for use on any data using the LHC synchronous time alignment. . . . . 125
4.39 The timing value mean truncated to within $\pm 1\sigma$ measured for all US channels after applying the time-walk, signal time-of-flight corrections and alignment parameters for run 5999. . . . . 126
4.40 The estimated timing component coming from the trace-line measurements supply an explanation for the observed inverse-u shape of the alignment parameters for a given PCB. . . . . 128
4.41 Average aligned time distribution for all the SiPMs on the left side of the 5th scintillator in the 5th plane. The FWHM extracted directly from the data and $\sigma_t^{\text{bar-side}}$ is measured by removing the resolution component from the event-wise reference time. . . . . 129
4.42 Measured bar-side time resolutions for all scintillators in the . The green vertical lines indicate the borders between planes of the . . . . . 130
4.43 Correlation histogram between the aligned times of two SiPMs reading out the same side of a scintillator in the 3rd plane of the . . . . . 131
4.44 Scatter plot of the truncated covariance measured for SiPM pairs coupled to the same side of a US bar. . . . . 132
4.45 Residual in bar-side time resolution between what is directly measured and what is calculated from SiPM resolutions and covariances. . . . . 134

4.46 Measured time resolution of full scintillators of the US system with data taken from run 5999. . . . . 135
4.47 Residual (measured - calculated) full scintillator time resolution measurement for all scintillators in the US system. . . . . 136
4.48 For each bootstrap (sample with replacement for the measured residual time resolutions), the mean residual is determined and these bootstrapped means are filled into this histogram. . . . . 137
4.49 Results of a toy Monte Carlo simulation of the Hamamatsu S14160-6050HS to evaluate when SiPM saturation effects become visible. . . 139
4.50 The QDC response curve in the 1-bar illumination regime for all the large SiPMs on the PCB. The upper x-axis displays the independent variable as number of incident photons divided by the total number of pixels on the SiPM. . . . . 140
4.51 The time-walk observed in TI18, on the left, is multiplied by $-1$ in order to display the same inverse-exponential trend. As only the relative timing changes are of interest, this factor of $-1$ has no impact on the results. . . . . 141
4.52 Degree of time-walk observed for SiPMs on the PCB compared with SiPMs in the same position of the plane 1, left side PCB of the full detector in TI18. The dashed vertical lines in each subplot distinguish the groupings of eight SiPMs coupled to the same scintillator. . . . . 142
4.53 Fraction of expected SiPMs that fire in each event for a 1-bar illumination run. All of the expected SiPMs never fire in the same event but the peaks at 25% and 75% correspond respectively to: both small SiPMs firing and the six large SiPMs firing in the same event. . . . . 143
4.54 Correlation between the delay in clock cycles of the start event and the small SiPM registering data, with the registered QDC. . . . . 144

5.2 Aligned bar-side time distribution following digitisation of passing muon simulation data in bins of $x-$position for the left-side of the fifth scintillator in the fifth plane of the US system. The full sample size of passing muon simulation is used here. . . . . 147
5.1 Scintillation photon ToF corrected and aligned bar-side times in bins of $x-$position for left (right) SiPMs on the top (bottom) sub-figure. The sub-figure plots are taken from the same scintillator. . . . . 148
5.3 Schematic of the model used for producing spatial measurements in the US. The bar-centre, $x_m$, is the reference point to which the aligned times are produced. The shaded red circle represents a generic energy deposition in the scintillator where the left- and right-edges of this energy deposition create the photons which set the time on those respective sides. . . . . 150
5.4 Illustration of four bars in the plane of the SND@LHC US system firing when traversed by a particle shower, illustrating how the $y-$barycentre and $\lambda_y$ values are measured. . . . . 151
5.5 Residual between x-barycentre, $x_B$, and extrapolated track position, $x_{\text{track}}$, as a function of extrapolated track position for all scintillators in the US overlaid. Measured with data from run 5999. . . . . 152

5.6 Distribution of SciFi firing SiPM positions for vertically aligned SiPM on the left and horizontally aligned SiPM on the right. The median value per plane for each projection is taken as the hit position. The specific data shown here is from the selected $\nu_\mu$ CCDIS event in run 5120. . . . . 153
5.7 Distribution of the polar angle, $\theta$, residual measurement between the SciFi + US system (SF+US) and the muon system (DS) for muon events. . . . . 155
5.8 $\nu_\mu$ CCDIS selected event from run 5120 with SciFi positions, US barycentres and reconstructed muon track shown together. The uncertainties in $x-$position are determined using equation 5.1.8 while the uncertainties in $y-$position are determined using equation 5.1.10. 156
5.9 Impact of varying the digitisation threshold on the $\Delta(x_{\text{left}}, x_{\text{track}})$ measured with simulated passing muon data for all planes of the US system. . . . . 159
5.10 Number of valid hits passing the selection criteria for passing muons for the different digitisation thresholds studied. . . . . 160
5.11 The residual in position measured between left-side SiPMs (right-side SiPMs) and expected muon position from the track fitted in the DS on the top (bottom) sub-plot. All scintillators throughout the US are studied to form these histograms. The data are taken from run 5408 with the selection criteria from table 4.1 . . . . . 162
5.12 Expected reflection coefficient from the Fresnel equations averaged between s- and p-polarised light for the scintillator-silgel and scintillator-air barrier. The upper abscissa axes use equation 5.2.2 to translate the photon incidence angle into the $x_{\text{track}}$ value in the SND@LHC physics reference system that forms such an angle when signal photons travel two different vertical distances, as is outlined in figure 5.13. . . 165
5.13 Schematic of a US scintillator with scintillation light travelling most of the height of the scintillator, $h$, to reach the SiPMs furthest away from the light creation point. . . . . 166
5.14 Comparison of the QDC versus track position histograms for two SiPMs in the same bar. There is no silgel used on the left-side of this plane, meaning that the optical interface is between the scintillator and air. The dotted red line at $x_{\text{track}} = -3\,\text{cm}$ is where TIR is predicted to dominate. . . . . 167
5.15 Comparison of the residual between $x-$barycentre and $x-$track for experimental data from run 5999 and passing muon simulation data at different digitisation thresholds. Both experimental data and simulation have the selection criteria of table 4.1 applied. . . . . 168
5.16 Data-simulation comparison for the residual between the scintillator-side spatial measurements and the expected position from the muon track fit in the DS muon system. The left-side residual comparison, $\Delta(x_{\text{L}}, x_{\text{track}})$, in the upper sub-figure and the right-side residual comparison, $\Delta(x_{\text{R}}, x_{\text{track}})$, in the upper sub-figure. . . . . 169
5.17 Comparison of the measured $\lambda_x$ distributions for experimental data from run 5999 and passing muon simulation data . Both experimental data and simulation have the selection criteria of table 4.1 applied to select passing muon events. The criterion of $x_{\text{track}} < -2\,\text{cm}$ is applied to minimise the discussed edge-effect on the left-side. . . . . 170

5.18 Residual in y-position measured between the barycentre and the extrapolated muon track position for all planes in the US for both TI18 data, from run 5999, and from the digitised passing muon simulation using a digitisation threshold of 1.2 MeV. . . . . 171
5.19 Comparison of fired bar multiplicity in plane 1 of the US system for passing muon events from data and simulation (with various digitisation thresholds shown in the legend). A logarithmic scale on the ordinate axis is used to highlight the discrepancy in the tail. . . . . 172
5.20 Comparison of data from run 5999 and simulation $\lambda_y$ distributions for all planes of the US. . . . . 173
5.21 Timing data taken from $\nu_\mu$ CC event in run 5239. Top: raw SiPM times directly from the offline format data-file with no corrections applied whatsoever. Bottom: aligned SiPM times, with both time-walk correction and alignment parameter applied. The geometric limits in the lower sub-figure, explained in the main body, provide loose bounds on the possible times that would give physical results. The number of SiPMs firing on the left- and right-side of the scintillator are shown in the legend as SiPMs $N$L $M$R. A binning of 400 ps is chosen to reflect the typical time-walk corrected SiPM time resolution. 174
5.22 Timing data taken from $\nu_\mu$ CC event in run 4809. Identical plotting style to figure 5.21. . . . . 175
5.23 Two-dimensional histogram showing the number of SiPMs firing on the left and right side of scintillators for all 140 firing scintillators over the 16 selected $\nu_\mu$ CCDIS events. . . . . 176
5.24 Events passing the muon selection criteria from run 5999. Correlation of the number of SiPMs firing on each side of a scintillator, for scintillators in the first plane of the US. The red dashed rectangle contains the region of the plot where events fail to have at least four SiPMs firing on both the left and right sides. . . . . 177
5.25 Multiplicity of firing scintillators in the first plane of the US muon system. Neutrino simulation data passing the $\nu_\mu$ CCDIS selection criteria is shown as well as the data events satisfying the $\nu_\mu$ CCDIS selection criteria split into two cases: including firing scintillators tagged as outliers (left) and not including those scintillators (right). The three data-sets are normalised to unit integral. . . . . 178
5.26 Distribution of QDC measured by a scintillator for the scintillators not flagged as outliers compared to the same distribution for scintillators flagged as outlier hits. Both distributions are normalised to unit integral to compare the shape more easily. . . . . 179
5.27 Measured polar angle from SciFi and US track, $\theta_{\mathrm{SF+US}}$, in the $\nu_\mu$ CCDIS events for data and simulation. The simulation dataset is scaled to the number of observed events in experimental data. . . . . 181

6.1 Impact of criteria (A-I) on the number of simulation events surviving the zero muon final-state neutrino event selection criteria. The bin labels correspond to the criteria as mapped in table 6.1. . . . . 187
6.2 Distribution of neutrino simulation species and interaction mode following the application of cuts listed in table 6.1. This is the whole input dataset to the training of the BDT. . . . . 188

6.3 Measured energy horizontal and vertical deposition width in the fifth US plane using the neutrino simulation data which passes the $\nu 0\mu$ selection criteria of table 6.1. . . . . 190
6.4 TMVA separation score for $\lambda_x$ and $\lambda_y$ measured in all five planes of the US system. . . . . 191
6.5 Resultant residual in the SciFi computed according to equation 6.2.2 for events containing two firing planes in the DS. . . . . 192
6.6 Impact of signal classification threshold on the $F1-$score, precision, recall and BRE scores. The threshold is set to optimise the $F1-$score. 194
6.7 Confusion matrix obtained with the optimised BDT used for identifying events containing a final-state muon. . . . . 195
6.8 . . . . . 197
6.9 . . . . . 198
6.10 Distribution of the $z-$position at which the final-state muon exits the detector in either the $xz$ or $yz$ projection. The $z-$positions of the end of the SciFi, the final plane of the US and the third station of the DS systems are shown. . . . . 199

A.1 Energy deposition from muons in the simulated data. The thickness in $z$ of a scintillator is 1 cm. . . . . 229
A.3 Exploded schematic for the aluminium housing and installation structure for the US and DS muon system. Taken from [85]. . . . . 231
A.4 Data for a typical SiPM from the Hamamatsu S14160-1310PS/-3010PS/-6010PS models showing for how gain, OCT probability and PDE($\lambda =$ 450 nm) varies with the applied over-voltage. Figure taken from [101]. 232
A.5 Multiplicity of fired bars in each plane of the US muon system under the criteria of table 4.1. . . . . 233
A.6 The measured $\Lambda_{23}$ distribution for run 5999 following the application of the selection criteria in table 4.1. . . . . 234
A.7 Values of $\sigma_{\Lambda_{ij}}$ determined for either combination of $i$ and $j$ for the three target runs of SND@LHC used in this thesis. The secondary ordinate axis shows the translation of $\sigma_{\Lambda_{ij}}$ into a value for $\sigma_{t_0^{\mathrm{DS}}}$. The horizontal line corresponds to the mean of the six measurements. . . 235
A.8 Outlier in the uncorrected time resolution measurements . . . . . 236
A.9 US alignment parameters to the muon system, DS, for the 'new' time alignment. Produced using run 5408. . . . . 236
A.10 US alignment parameters to the muon system, DS, for the 'new+LHC synch' time alignment. Produced using run 5999. . . . . 237
A.11 Correlation between US SiPMs and muon system reference time for run 5408. . . . . 238
A.12 A 3x2 grid of SiPM response curves of different groups of SiPMs in various illumination modes, all showing clear saturation effects before the prediction from simulation in figure 4.49 of around 40%. . . . . 240
A.15 Simulated energy spectrum of muons passing through SND@LHC. . . 243
A.16 Caption . . . . . 244
A.17 Data-simulation comparison of $\lambda_x$ distributions for selected passing muon events in data and simulation, 1.2 MeV digitisation threshold applied. . . . . 245

A.19 Correlation plot of the event-wise reference time in the DS muon system against the $x-$position of a track fitted in the DS when extrapolated to the $z-$position of the final plane in the US muon system. 246
A.18 The fired bars for all planes of the US for the PRL event in run 5132. 247
A.20 The variable changed between these two correlation plots is the minimum QDC measured by the whole plane of scintillators of which a scintillator under study is a constituent. . . . . 248
A.21 Distribution of the sum (difference) of the left- and right-side effective signal speeds squared for all US system scintillators in the top (bottom) sub-figure. Data from from run 5999. . . . . 250
A.22 Multiplicity of firing scintillators in the fifth plane of the US muon system. Neutrino simulation data passing the $\nu_\mu$ CCDIS selection criteria is shown as well as the data events satisfying the $\nu_\mu$ CCDIS selection criteria split into two cases: including firing scintillators tagged as outliers (left) and not including those scintillators (right). The three data-sets are normalised to unit integral. . . . . 251

# List of Tables

2.1 Values for the squared differences between mass eigenstates. All values are taken from reference [11]. More up-to date values may exist but these are here to contextualise the two different possible mass hierarchies. . . . . 19

3.1 Tabulation of what side gel is placed on for each US plane . . . . . 47
3.2 Scintillator-SiPM configurations of veto and US subsystems. Note that a third Veto plane is added at the end of 2023 [91]. . . . . 51
3.3 The number of pixels in each type of SiPM, width of the pixels (pitch) and the geometric fill factor. Values taken from the respectively data sheets for the SiPM models, found at reference [88] for the large area SiPMs and reference [101] for the small area SiPMs. . . . . 59
3.4 The breakdown voltages are taken from the respectively data sheets for the SiPM models, found at reference [88] for the large area SiPMs and reference [101] for the small area SiPMs. . . . . 61
3.5 Estimated Hamamatsu S14160-6050HS SiPM performance parameters at the SND@LHC chosen over-voltage of $4.5\,V$. . . . . 62
3.6 As is clear, the misalignment reappeared for several days due to a technically issue. The uncovered time period from December to March 29 2023 is due to the LHC not running over these dates. . . . 69
3.7 Runs used in chapter 4 to study the timing behaviour in the experiment. 69
3.8 Tracking efficiency measured in data for both SciFi and DS, each using both available tracking algorithms. Adapted from [127]. . . . . 74
3.9 Measured muon flux in the SciFi and DS detectors with distinct fiducial areas. Adapted from [84]. . . . . 75
3.10 Measured muon flux in the SciFi and DS detectors using a common fiducial area. Adapted from [84]. . . . . 76
3.11 Fiducial volume selection criteria and the fraction events for data and neutrino simulation [125] which pass each subsequent criterion. . . . . 77
3.12 $\nu_\mu$ CCDIS identification selection criteria and the fraction of events in data and neutrino simulation passing each successive criterion. The final efficiencies are tabulated in the bottom two rows. Data taken from [125]. . . . . 77
3.13 Number of events passing each selection cut in the data, $\nu 0\mu$ simulation (signal) and $\nu 1\mu$ (background). . . . . 79

4.1 Selection criteria for events and fired scintillators applied to data. . . 86
4.2 Upper and lower parameter limits for fitting the time-walk function, equation 4.1.9. . . . . 90
4.3 Fixed parameters during laser data taking and their values. . . . . 101

4.4 Measured skew and kurtosis of the residual bar-side time resolution. These values are then used with the D'Agostino-Pearson test to produce a probability that the data comes from an underlying normal distribution. . . . 133
4.5 Measured skew and kurtosis of the residual full scintillator time resolution. These values are then used with the D'Agostino-Pearson test to produce a probability that the data comes from an underlying normal distribution. . . . 137

5.1 How the percentage of hits failing to meet the criteria of at least four SiPMs firing on both sides of the scintillator changes for three applied criteria on the QDC measured in the plane of scintillators containing the hit under study. . . . 177
5.2 Additional data quality checks imposed upon the selected $\nu_\mu$ CCDIS selected events for the shower direction analysis. . . . 178

6.1 Criteria applied to the neutrino simulation data. With criteria labels corresponding to the bin labels of figure 6.1. . . . 186
6.2 Table of the optimised hyper-parameter, the trialled values in the grid search method of `SciKit Learn` and the final value in the optimised BDT. . . . 193

A.1 Laser data runs for individually illuminated large SiPMs, multiple and single SiPM groups of eight were studied at these laser intensity points. . . . 238
A.2 Laser data runs for individually illuminated small SiPMs were studied at these laser intensity points. . . . 239
A.3 Illumination combinations used in 2-group and 3-group setups. . . . 239
A.4 Dispersion constants and estimated refractive index at 425 nm for silicone-based materials. Units of the dispersion constants $A$, $B$ and $C$ are [dimensionless], [$\mu m^2$] and [$\mu m^4$]. . . . 244
A.5 Features used to train the $\nu 0\mu$ BDT. . . . 251
A.6 Table of frequently used timing quantities to be used as an aid to the reader. . . . 252

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# Appendix A

# Appendices

## A.1 Energy deposition in the US from muons

Shown here is a histogram normalised to unit integral of the simulated energy loss from muons in the scintillator material of the US system, Eljen EJ200. The most probable value of the energy loss in the material is 2.36 MeV, which is approximately what one expects from 1 TeV muons travelling through carbon from the Bethe-Bloch curve of figure 2.21

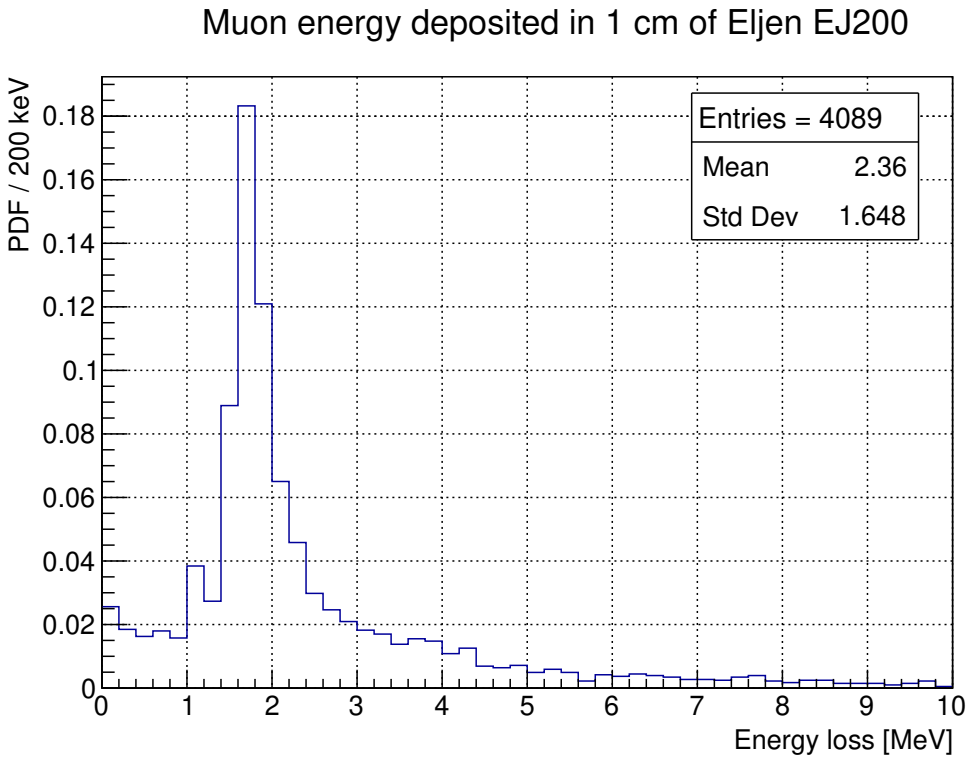
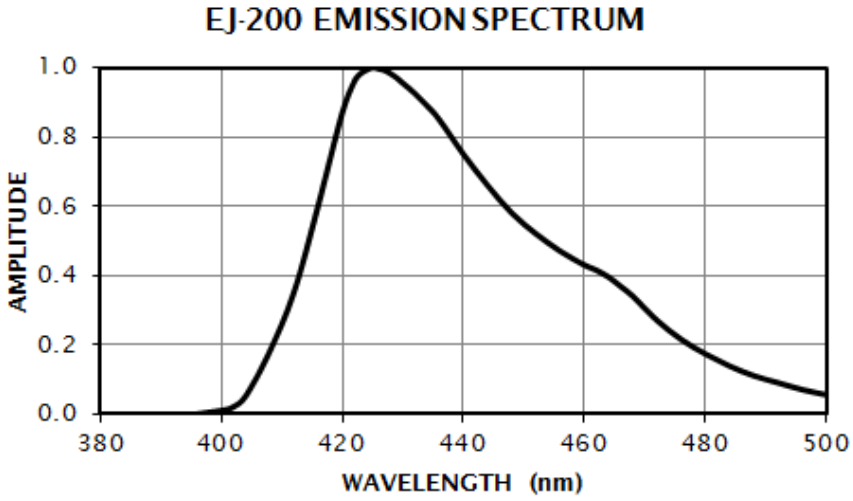



Figure A.1: Energy deposition from muons in the simulated data. The thickness in $z$ of a scintillator is 1 cm.

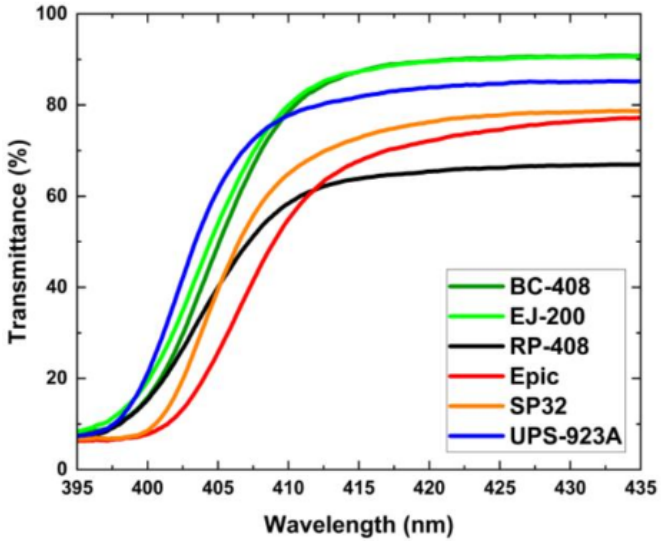
## A.2 Emission spectrum of EJ-200

Here the third criteria of an appropriate scintillator from reference [81] is validated: "The medium should be transparent to the wavelength of its own emission for good

light collection.”.




(a) Emission spectrum of EJ200, taken from [80].




(b) Transmission spectrum of various plastic scintillators including the Eljen EJ-200, taken from [152].

One sees clearly the light green line for EJ-200 in figure A.2b which exceeds 80% for wavelengths $\lambda \gtrsim 412.5$ nm. Thus, some light emitted by the scintillator will be in the wavelength, $\lambda$, range 400 nm$< \lambda <$ 412.5 nm where the transmittance is begins at $\approx$20% and rises rapidly to $\approx$80%. This region of the emission spectrum is not ideal, but does not act sufficiently against the rest of the emission spectrum, where the transmittance levels off around 90%.

## A.3 Aluminium housing structure for the US and DS

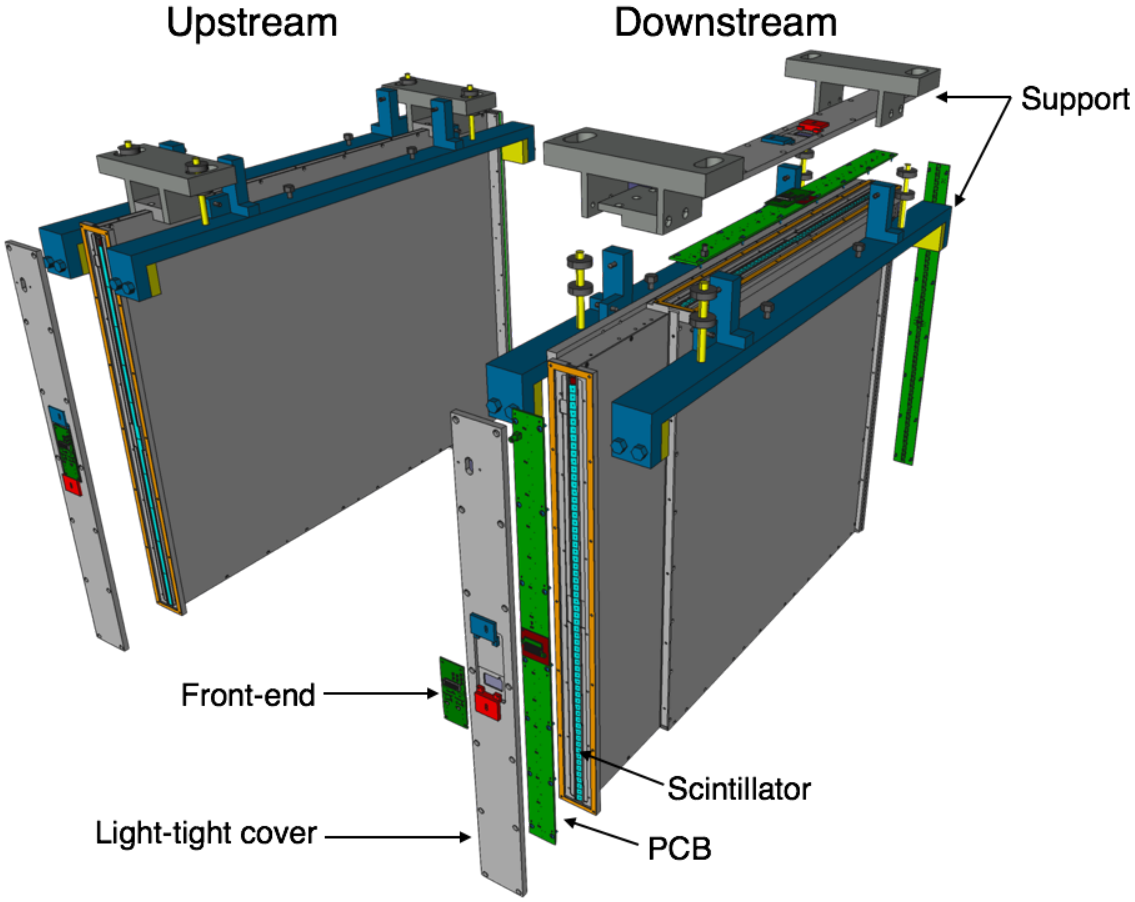



Figure A.3: Exploded schematic for the aluminium housing and installation structure for the US and DS muon system. Taken from [85].

The designs and implementations of the two housings and PCB fixings shown in figure A.3 are largely similar. The aluminium housing provides light-tightness through the light-tight cover screwed on over the PCB. In practice, to ensure light-tightness, the seams between the aluminium housing and the light-tight cover had to be taped over as the screws alone did not suffice. The support pieces are used to minimise any movement of the detector stations post-installation which maximises the effectiveness of the engineering survey used to create the geometry files used in analyses.

## A.4 Typical data informing how small SiPM parameters vary with over-voltage

Figure A.4 shows how the gain, OCT and PDE($\lambda = 450\,\text{nm}$) will vary with the applied over-voltage for a typical SiPM from the Hamamatsu S14160-1310PS/-3010PS/-6010PS models. The small area SiPMs used in the US muon system are the Hamamatsu S14160-3010PS type. These SiPM operate with an over-voltage of $10\,V$ in the US system so the approximate values may only be very roughly estimated. The specific values for these SiPMs at an over-voltage of $10\,V$ is not of great interest to this thesis.

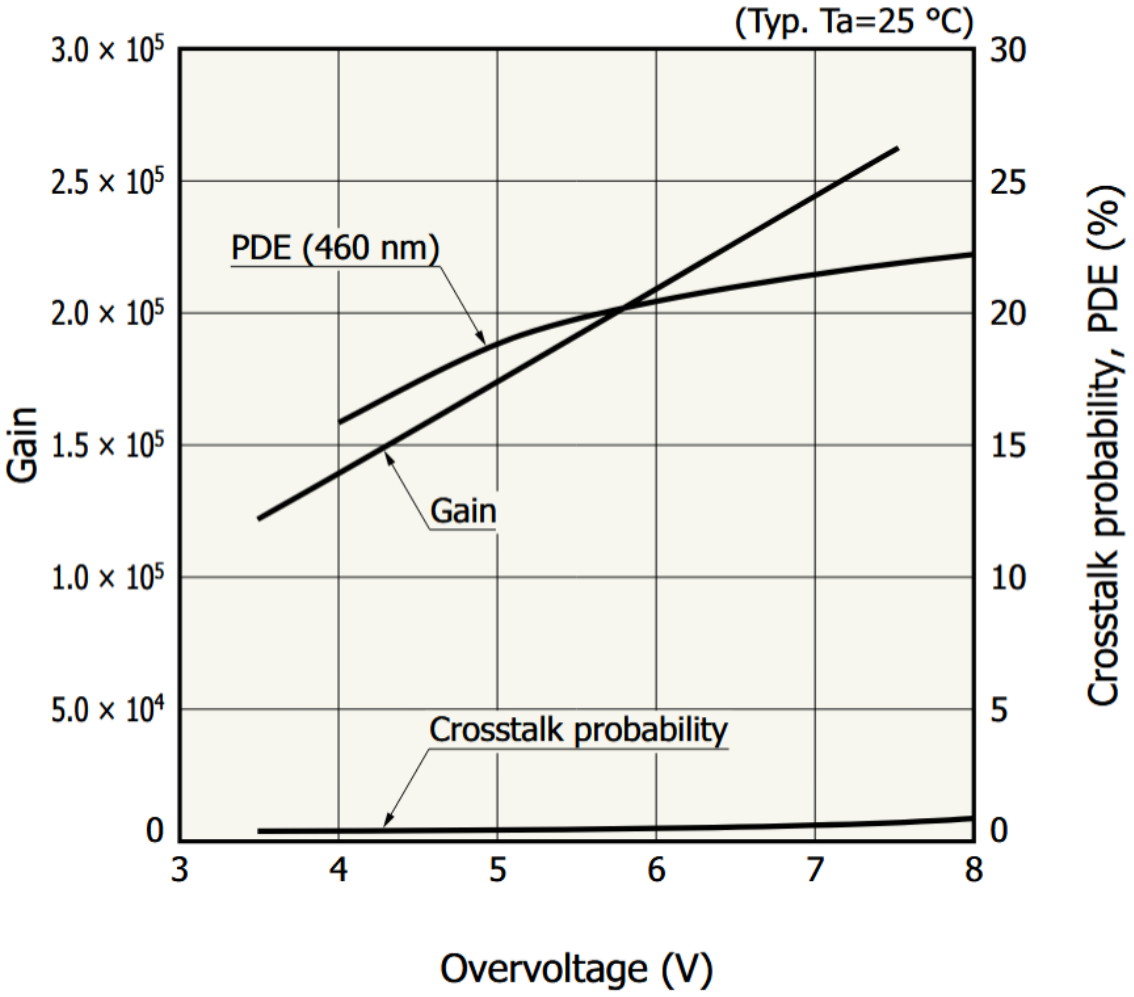



Figure A.4: Data for a typical SiPM from the Hamamatsu S14160-1310PS/-3010PS/-6010PS models showing for how gain, OCT probability and PDE($\lambda$ = 450 nm) varies with the applied over-voltage. Figure taken from [101].

## A.5 Fired US bar multiplicity in selected muon events

Here shown are histograms of the US firing bar multiplicity in each plane when the muon selection criteria of table 4.1 are applied. Clearly in the vast majority of cases, one bar only fires in each plane.

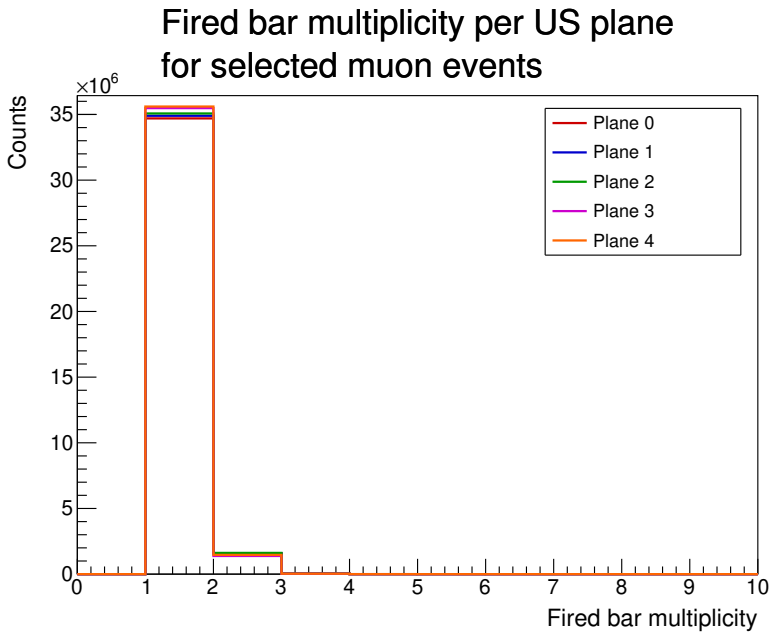



Figure A.5: Multiplicity of fired bars in each plane of the US muon system under the criteria of table 4.1.

## A.6 Evaluating the uncertainty on the muon system reference time

$t_0^{\mathrm{DS}}$ is defined as the average time recorded by the SiPMs coupled to horizontally aligned scintillators in the muon system. Each horizontally aligned scintillator in the muon system is read out by one SiPM on each side and there are three planes of horizontally aligned scintillators in the muon system.

$$t_0^{DS} = \frac{1}{3}\sum_{i}^{3} t_{\mathrm{DSH\ i}} \tag{A.6.1}$$

Where $t_{\mathrm{DSH\ i}}$ is the average time measured by the SiPMs attached to horizontally aligned scintillators in the DS muon system. The sum is over the planes of horizontally aligned scintillators in the track fit which must be equal to three. From equation A.6.1, an expression for the variance of $t_0^{\mathrm{DS}}$ can be written down:

$$\sigma^2_{t_0^{\mathrm{DS}}} = \left(\frac{1}{3}\right)^2 \sum_{i}^{3} \sigma^2_{t_{\mathrm{DSH\ i}}} \tag{A.6.2}$$

Then one can define $\Lambda_{ij}$ as the difference in average time measured by successive horizontally aligned planes of scintillators in the DS muon system:

$$\Lambda_{ij} = t_{\mathrm{DSH\ j}} - t_{\mathrm{DSH\ i}} \tag{A.6.3}$$

Where $j$ and $i$ denote the indices of two adjacent DS horizontal planes. The possible combinations for this are $\Lambda_{32}$ and $\Lambda_{21}$. The distributions of these $\Lambda_{ij}$ values are studied with the selection criteria of table 4.1 applied. Similarly to what was done above for $t_0^{\mathrm{DS}}$, an expression for the variance of $\Lambda_{ij}$ can be written down:

$$\sigma^2_{\Lambda_{ij}} = \sigma^2_{t_{\mathrm{DSH\ j}}} + \sigma^2_{t_{\mathrm{DSH\ i}}} \tag{A.6.4}$$

Values of $\sigma_{\Lambda_{ij}}$ are extracted from the distribution by approximating the underlying Gaussian width as FWHM/2.355 with the uncertainty on the distribution coming from the binning uncertainty. An example of the distribution of $\Lambda_{ij}$ is shown in figure A.6. As the FWHM is determined as the difference in time between the first bin above half the maximum value and the final bin above the half the maximum value:

$$\text{FWHM} = \text{first bin above } \frac{\max}{2} - \text{last bin above } \frac{\max}{2} \tag{A.6.5}$$

The systematic uncertainty on each of these components is given by $\pm\frac{\text{bin width}}{2}$. When this uncertainty is propagated through, the uncertainty on the FWHM from the binning, $\sigma^{\text{binning}}_{\text{FWHM}}$ is given by $\pm\,\text{bin width}/\sqrt{2}$. Then finally the uncertainty on the underlying Gaussian sigma:

$$\sigma^{\text{binning}}_{\text{FWHM}} = \frac{\text{bin width}}{\sqrt{2}}$$
$$\sigma^{\text{binning}}_{\text{FWHM/2.355}} = \frac{\text{bin width}}{2.355\sqrt{2}}$$

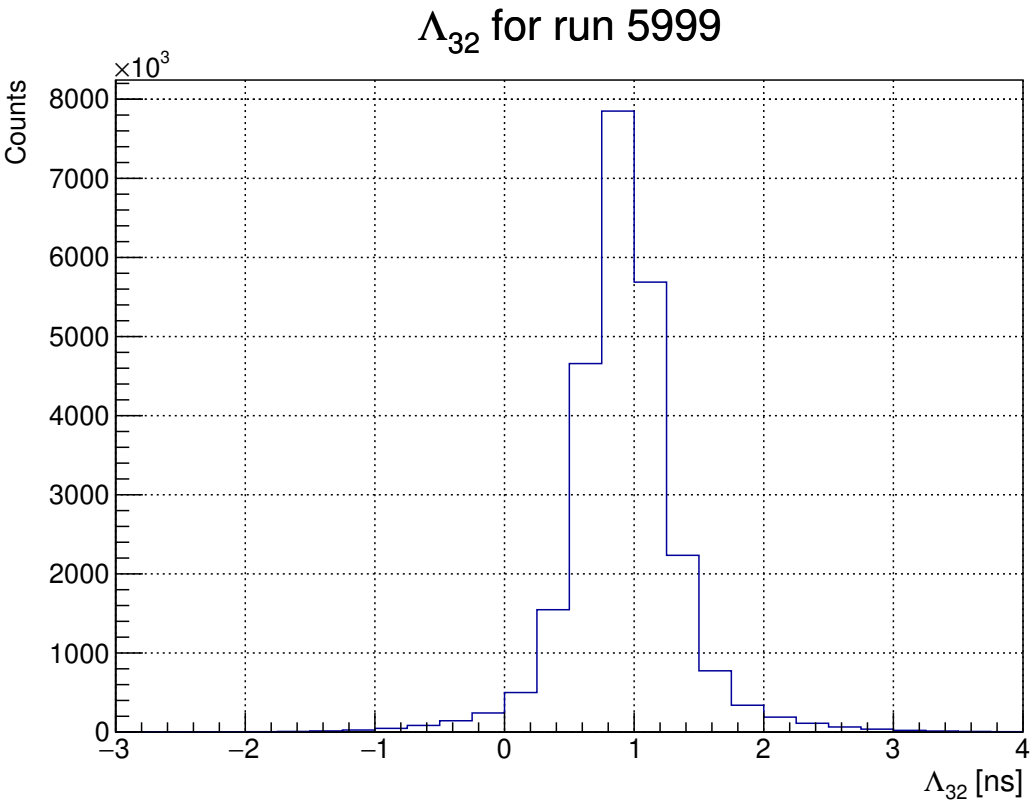



Figure A.6: The measured $\Lambda_{23}$ distribution for run 5999 following the application of the selection criteria in table 4.1.

Figure A.7 shows that the measured values of $\sigma_{\Lambda_{ij}}$ are compatible measured over the three target runs of SND@LHC in this thesis (5097, 5408, 5999).

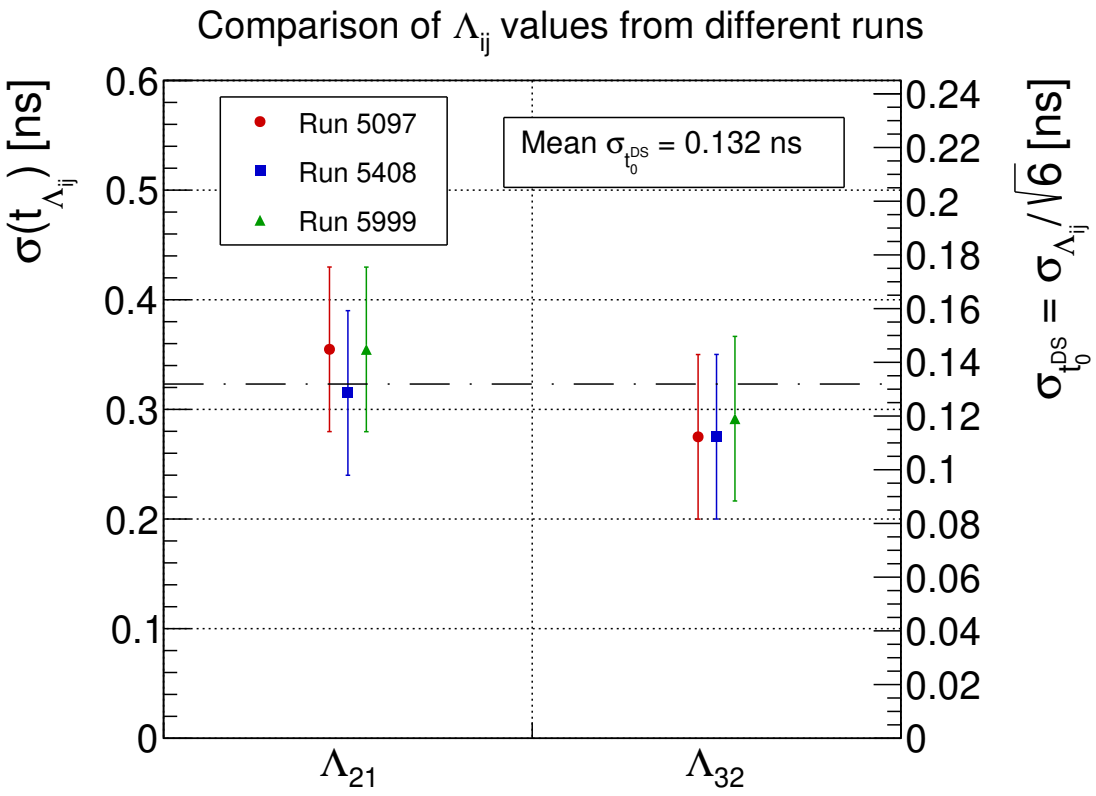



Figure A.7: Values of $\sigma_{\Lambda_{ij}}$ determined for either combination of $i$ and $j$ for the three target runs of SND@LHC used in this thesis. The secondary ordinate axis shows the translation of $\sigma_{\Lambda_{ij}}$ into a value for $\sigma_{t_0^{\mathrm{DS}}}$. The horizontal line corresponds to the mean of the six measurements.

This observed compatibility between $\Lambda_{21}$ and $\Lambda_{32}$ means that $\sigma_{t_{\mathrm{DS\ i}}}$ does not vary greatly for different planes of horizontally aligned scintillators, $i$: $\sigma_{t_{\mathrm{DSH\ i}}} = \sigma_{t_{\mathrm{DSH}}}\ \forall\, i$. Therefore, equation A.6.4 can be simplified to:

$$\sigma_{\Lambda_{ij}} = \sqrt{2} \times \sigma_{t_{\mathrm{DSH}}} \tag{A.6.6}$$

And equation A.6.2 can be simplified to:

$$\sigma_{t_0^{\mathrm{DS}}} = \frac{1}{\sqrt{3}} \sigma_{t_{\mathrm{DSH}}} \tag{A.6.7}$$

Equations A.6.6 and A.6.7 can give the final expression for the uncertainty on $t_0^{\mathrm{DS}}$:

$$\sigma_{t_0^{\mathrm{DS}}} = \frac{1}{\sqrt{6}} \sigma_{t_{\mathrm{DSH}}} \tag{A.6.8}$$

## A.7 Outlier uncorrected time resolution

The SiPM here records a very broad time distribution featuring a shoulder peak which is not present in other channels. The cause of its poor time resolution is not studied but the resolution is not so bad as to spoil the entire side of the scintillator.

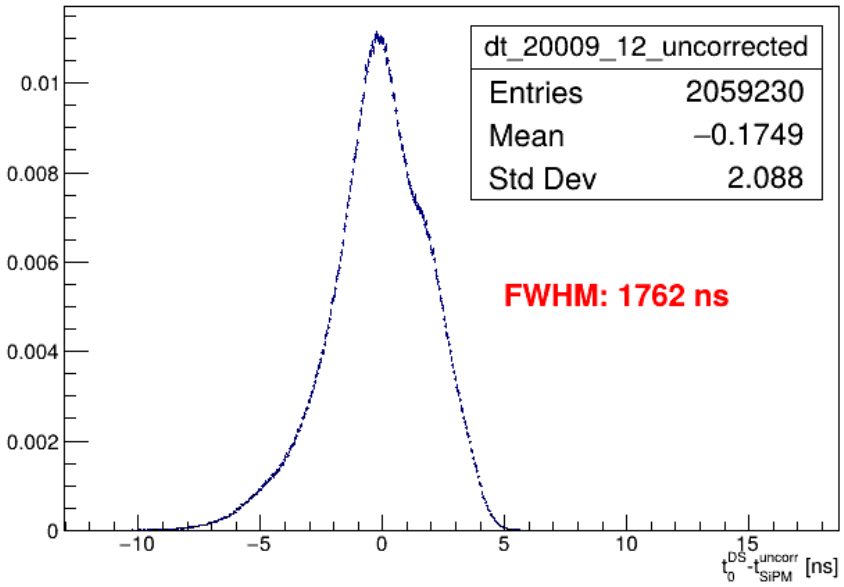



Figure A.8: Outlier in the uncorrected time resolution measurements

## A.8 All US alignment parameter sets

In the main body, the alignment parameters for the 'old' time alignment are shown. Here, the alignment parameters for the 'new' time alignment, produced with run 5408 and shown in figure A.9. Then, 'new+LHC synch' alignment parameters produced with run 5999 are given in figure A.10.

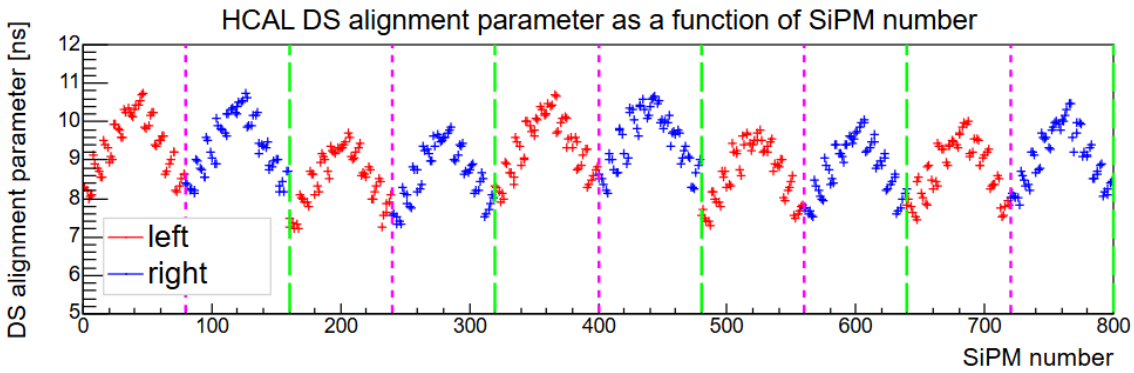



Figure A.9: US alignment parameters to the muon system, DS, for the 'new' time alignment. Produced using run 5408.

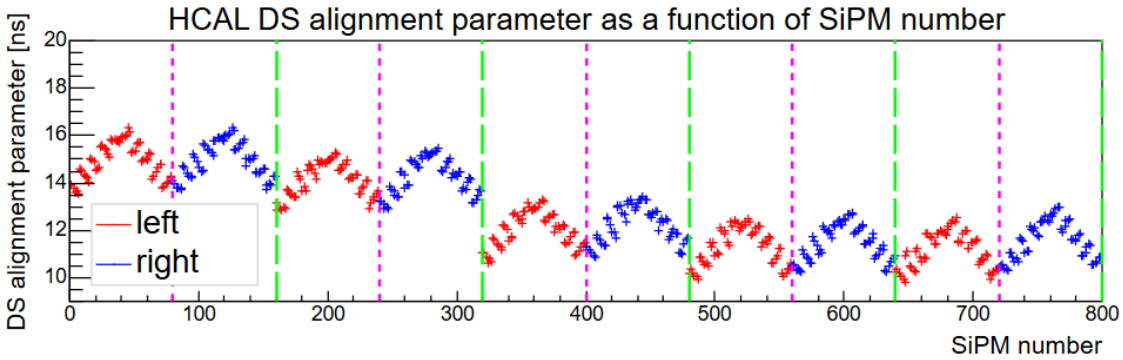



Figure A.10: US alignment parameters to the muon system, DS, for the 'new+LHC synch' time alignment. Produced using run 5999.

## A.9 Bar-side time resolution derivation

Returning now to summing $dt_{\text{SiPM}}^{\text{TW ToF}}$ in order to elucidate the time resolution of a bar-end. Equation 4.3.8 allows the following expression to be written down with the following statements: the uncertainty contribution from the alignment parameter is sufficiently small to be neglected; $\sigma_{t_0^{DS}}$ is independent of SiPM number.

$$y = dt_{\text{side}}^{\text{ToF}} = t_0^{DS} - \sum_{\text{SiPM}} \left( t_{\text{SiPM}}^{\text{TW ToF}} - d_{\text{SiPM}} \right) \tag{A.9.1}$$

$$\begin{aligned} \sigma_y^2 = & \left( \frac{\partial y}{\partial t_0^{DS}} \right)^2 \sigma_{t_0^{DS}}^2 \\ & + \sum_{i=1}^{N} \left[ \left( \frac{\partial y}{\partial t_i^{\text{TW ToF}}} \right)^2 \sigma_{t_i^{\text{TW ToF}}}^2 + \frac{\partial y}{\partial t_i^{\text{TW ToF}}} \frac{\partial y}{\partial t_0^{\text{DS}}} \sigma_{t_i^{\text{TW ToF}}, t_0^{\text{DS}}} \right] \\ & + \sum_{1 \le i < j \le N} \left( \frac{\partial y}{\partial t_i^{\text{TW ToF}}} \right) \left( \frac{\partial y}{\partial t_j^{\text{TW ToF}}} \right) \sigma_{t_i^{\text{TW ToF}}, t_j^{\text{TW ToF}}} \end{aligned} \tag{A.9.2}$$

Where $\sum_{1 \le i < j \le N}$ is the sum over the pair-wise combinations of same-side SiPMs, thus of length $\binom{N}{2}$. The partial derivatives required for equation A.9.2 can be calculated from equation 4.3.8 and the covariance coefficients are determined by 2D correlation plots between time-walk corrected SiPM timing values.

$$\frac{\partial y}{\partial t_0^{DS}} = 1, \quad \frac{\partial y}{\partial t_i^{\text{TW ToF}}} = -\frac{1}{N}, \ \forall i \tag{A.9.3}$$

Using the partial derivatives above in equation A.9.2,

$$\begin{aligned} \sigma_y^2 = & \sigma_{t_0^{DS}}^2 \\ & + \frac{1}{N^2} \left[ \sum_{i=1}^{N} \sigma_{t_i^{\text{TW ToF}}}^2 + 2 \sum_{1 \le i < j \le N} \sigma_{t_i^{\text{TW ToF}}, t_j^{\text{TW ToF}}} \right] \\ & - \frac{2}{N} \sum_{i}^{N} \sigma_{t_0^{\text{DS}}, t_i^{\text{TW ToF}}} \end{aligned}$$

The factors of 2 are introduced to account for both of the unordered pairs of SiPMs, $(i, j)$ and $(j, i)$, which have identical covariance. Which recovers the equation quoted in section 4.3.5.

## A.10 Correlation between US SiPMs and muon system reference time

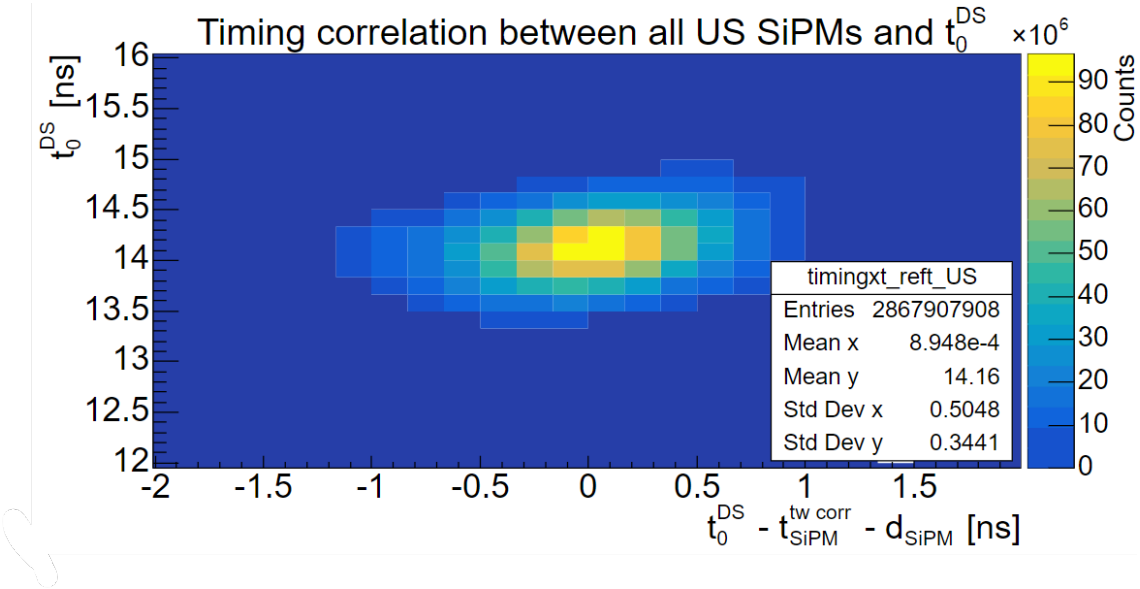



Figure A.11: Correlation between US SiPMs and muon system reference time for run 5408.

The covariance is measured to be 0.023 $ns^2$, this value is measured for each run, and is used for all US SiPMs.

## A.11 Laser intensity SiPM measuring points

| Uncalibrated relative intensity (%) | Calibrated relative intensity (%) | $n_\gamma$ / $mm^2$ |
|---|---|---|
| 8.5 | 8.3 | 3.7 |
| 8.7 | 8.5 | 4.6 |
| 9.0 | 8.8 | 6.0 |
| 9.5 | 9.9 | 11.2 |
| 10.0 | 11.0 | 16.3 |
| 12.0 | 14.4 | 32.1 |
| 14.0 | 16.3 | 41.1 |
| 20.0 | 20.4 | 60.1 |
| 40.0 | 40 | 147.4 |
| 60.0 | 60 | 246.1 |

Table A.1: Laser data runs for individually illuminated large SiPMs, multiple and single SiPM groups of eight were studied at these laser intensity points.

| Uncalibrated relative intensity (%) | Calibrated relative intensity (%) | $n_\gamma$ / mm$^2$ |
|---|---|---|
| 15.0 | 17.0 | 44.2 |
| 20.0 | 20.4 | 60.1 |
| 25.0 | 25.0 | 80.2 |
| 30.0 | 30.0 | 103.9 |
| 40.0 | 40.0 | 147.4 |
| 60.0 | 60.0 | 246.1 |
| 80.0 | 80.0 | 341.8 |

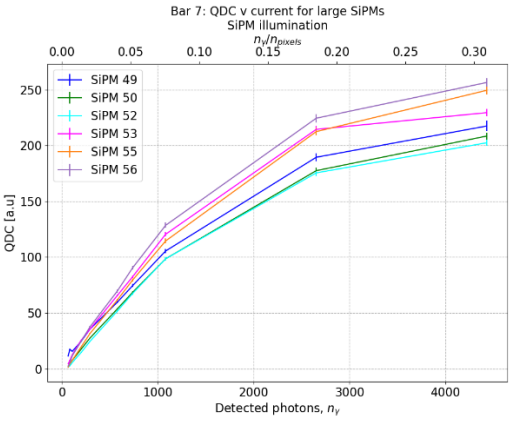
Table A.2: Laser data runs for individually illuminated small SiPMs were studied at these laser intensity points.

## A.12 Groups of bars illuminated in multi-bar illumination modes

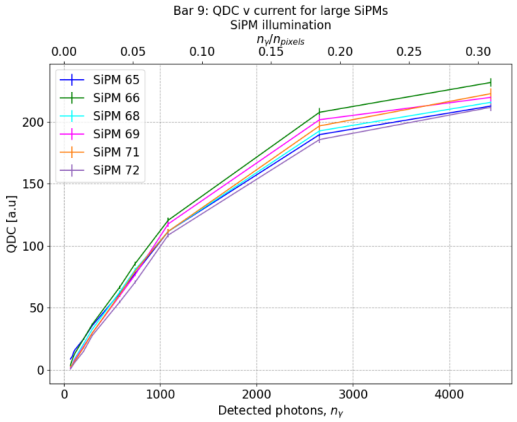
| **Illumination regime** | **Cominations of groups used** |
|---|---|
| 2-group | [1, 2], [3, 4], [5, 6],[7, 8], [9, 10] |
| 3-group | [2, 3, 4], [5, 6, 7] |

Table A.3: Illumination combinations used in 2-group and 3-group setups.

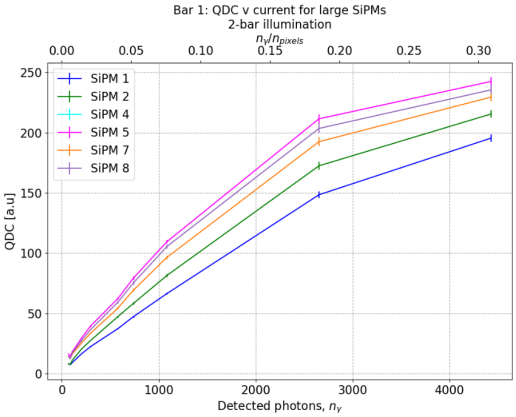
# A.13 Example SiPM response curves



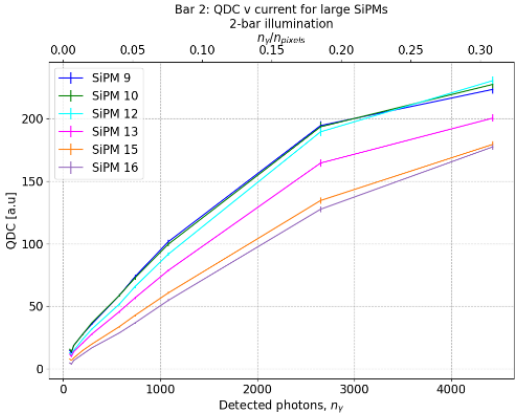

(a) QDC response curve for the SiPMs in bar 7 with a single SiPM illumination.



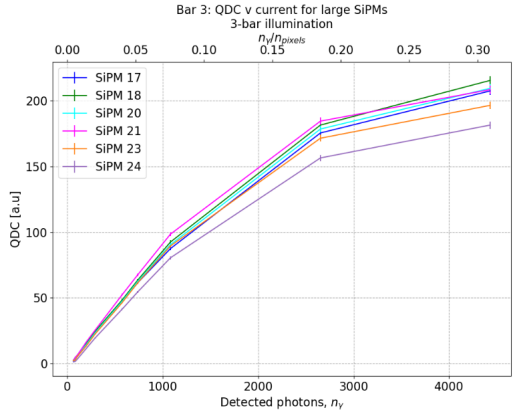

(b) QDC response curve for the SiPMs in bar 9 with a single SiPM illumination.



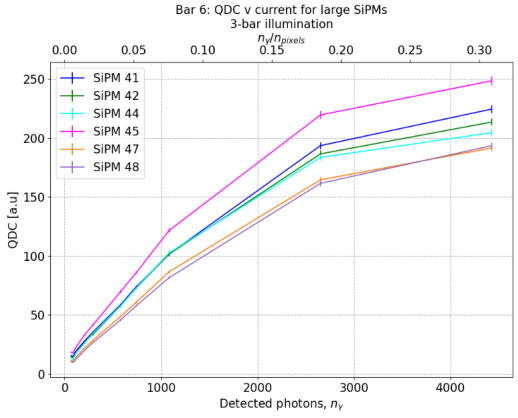

(c) QDC response curve for the SiPMs in bar 1 with a 2-bar illumination.




(d) QDC response curve for the SiPMs in bar 2 with a 2-bar illumination.




(e) QDC response curve for the SiPMs in bar 3 with a 3-bar illumination.

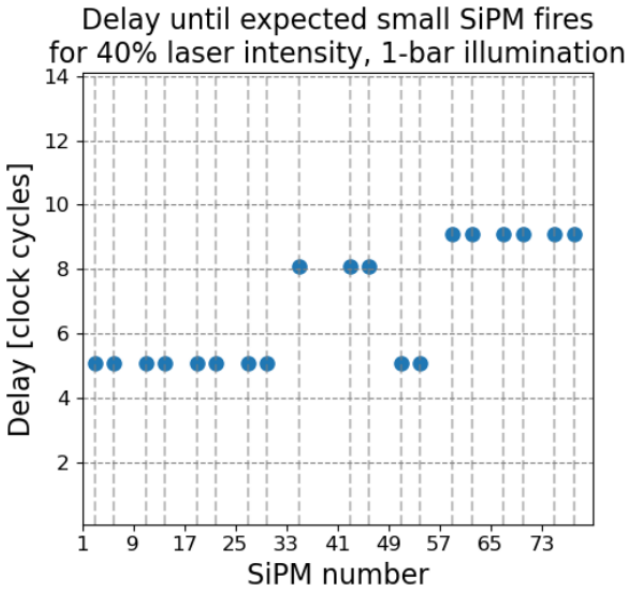



(f) QDC response curve for the SiPMs in bar 6 with a 3-bar illumination.

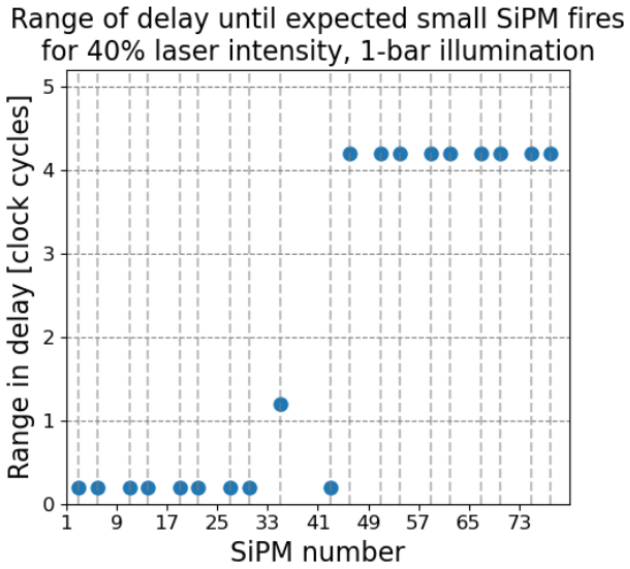
Figure A.12: A 3x2 grid of SiPM response curves of different groups of SiPMs in various illumination modes, all showing clear saturation effects before the prediction from simulation in figure 4.49 of around 40%.

## A.14 Example small SiPM delay - QDC correlations

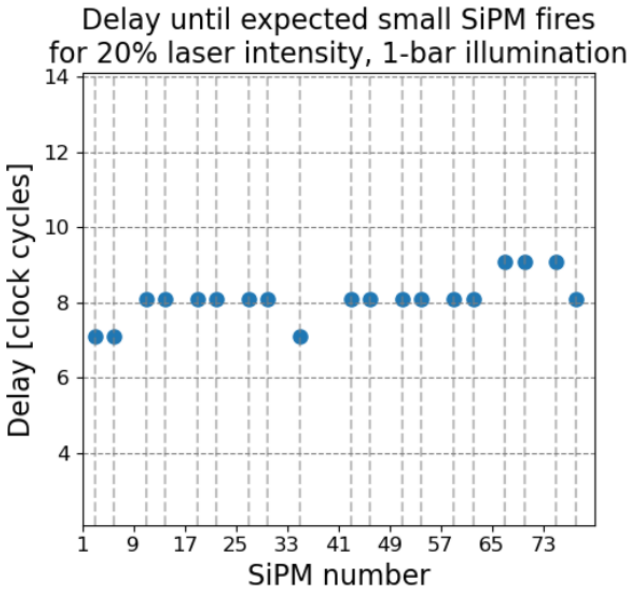
Here the small SiPM delays and the ranges of the delays are shown for laser intensities of 40% and 20%. Unlike for the intensity of 60% shown in the thesis main body, the range of the measured delays is not as minimal and uniform.



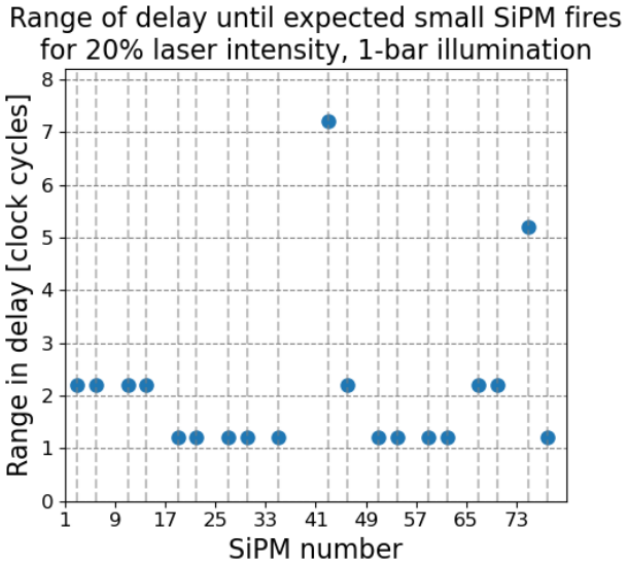

(a) Small SiPM delays for 1-bar illumination and 40% laser intensity.




(b) Small SiPM delay ranges for 1-bar illumination and 40% laser intensity.




(a) Small SiPM delays for 1-bar illumination and 20% laser intensity.




(b) Small SiPM delay ranges for 1-bar illumination and 20% laser intensity.

## A.15 Signal velocity from Maxwell's Equations

Equivalent derivations to this may be found, for example, in references [13, 153]. Beginning from Maxwell's third equation (Faraday's law); assuming no free charges in the dielectric ($\rho = 0$); no current in the dielectric ($\mathbf{J} = 0$) and that the absolute permeability and permittivity, $\epsilon$ and $\mu$, are constant throughout the dielectric:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

Where $\mathbf{E}$ is the electric field and $\mathbf{H}$ is the magnetising field. Taking the curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) \tag{A.15.1}$$

Ampère's Law, shown in equation A.15.2:

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} \tag{A.15.2}$$

Can then be substituted into equation A.15.1:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \tag{A.15.3}$$

The vector identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla (\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ in conjunction with the condition that there are no sources of electric field in the dielectric region, $\nabla \cdot \mathbf{E} = 0$, allows equation A.15.3 to be simplified:

$$\nabla^2 \mathbf{E} = \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \tag{A.15.4}$$

Equation A.15.4 allows for a ready comparison with the standard wave equation [13]. The signal velocity can be extracted by recalling that $\epsilon = \epsilon_0 \epsilon_r$, $\mu = \mu_0 \mu_r$, use of the identity $c = 1/\sqrt{\epsilon_0 \mu_0}$ and that the dielectric is non-magnetic ($\mu_r \approx 1$) which allows the signal velocity to be extracted as:

$$v = \frac{1}{\sqrt{\epsilon_r}} \tag{A.15.5}$$

Where equation A.15.5 returns the equation presented in the discussion of section 4.3.5.

## A.16 Energy spectrum of muons in SND@LHC

As explained in section 3.4, the $pp-$collisions at IP1 are simulated using DPM-JET3 and SIBYLL, the products of these collisions are then tracked to a scoring plane upstream of SND@LHC using FLUKA. The muons travelling through SND@LHC are then simulated using GEANT4. The energy spectrum of the muons reaching SND@LHC is extracted from the simulation and shown in figure A.15.

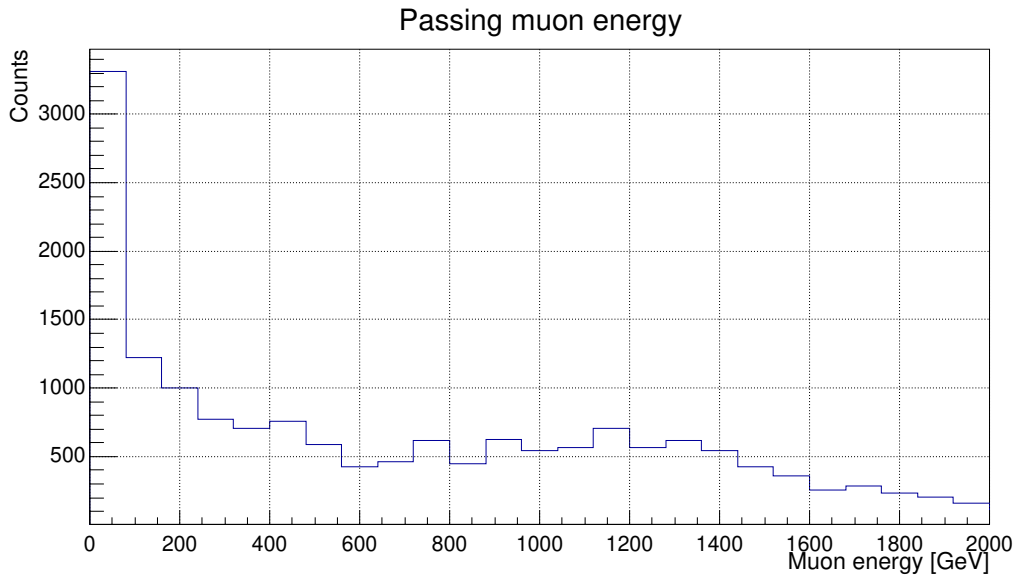



Figure A.15: Simulated energy spectrum of muons passing through SND@LHC.

## A.17 Variation in Wacker Silgel refractive index

To estimate the refractive index of Wacker SilGel 612 A/B [90] for light at a wavelength of $\lambda = 425\,\text{nm}$, the Cauchy equation is used with material dependent dispersion constants of similar materials. Table A.4 documents the materials used and their dispersion constants. The motivation for these three materials is that they are silicone based and all intended for optical coupling applications.

| **Dispersion Constants (A, B, C)** | $n(425\,\text{nm})$ | **Source Material** |
|---|---|---|
| $(1.398,\ 6.5 \times 10^{-3},\ 8 \times 10^{-5})$ | 1.44 | GE RTV615 (Sylgard-like) [154] |
| $(1.400,\ 7 \times 10^{-3},\ 1 \times 10^{-4})$ | 1.44 | Dow Corning Q2-3067 [155] |
| $(1.402,\ 7.5 \times 10^{-3},\ 1.2 \times 10^{-4})$ | 1.45 | Nusil LS-6257 [156] |

Table A.4: Dispersion constants and estimated refractive index at 425 nm for silicone-based materials. Units of the dispersion constants $A$, $B$ and $C$ are [dimensionless], [$\mu m^2$] and [$\mu m^4$].

## A.18 $x$-barycentre data-simulation comparison

Here shown in figure A.16 is the data-simulation comparison of the $\Delta\,(x_B,\ x_{\text{track}})$ for passing muon events. Both data and simulation isolate the events with the selection criteria of table 4.1. In section 5.2.2 a similar plot is presented in figure 5.15 where a criteria is placed on $x_{\text{track}}$ such that $x_{\text{track}} < -2\,\text{cm}$ to remove the edge effect reported in section 5.2.2. It is clear that this additional selection criteria has a minimal impact on the results.

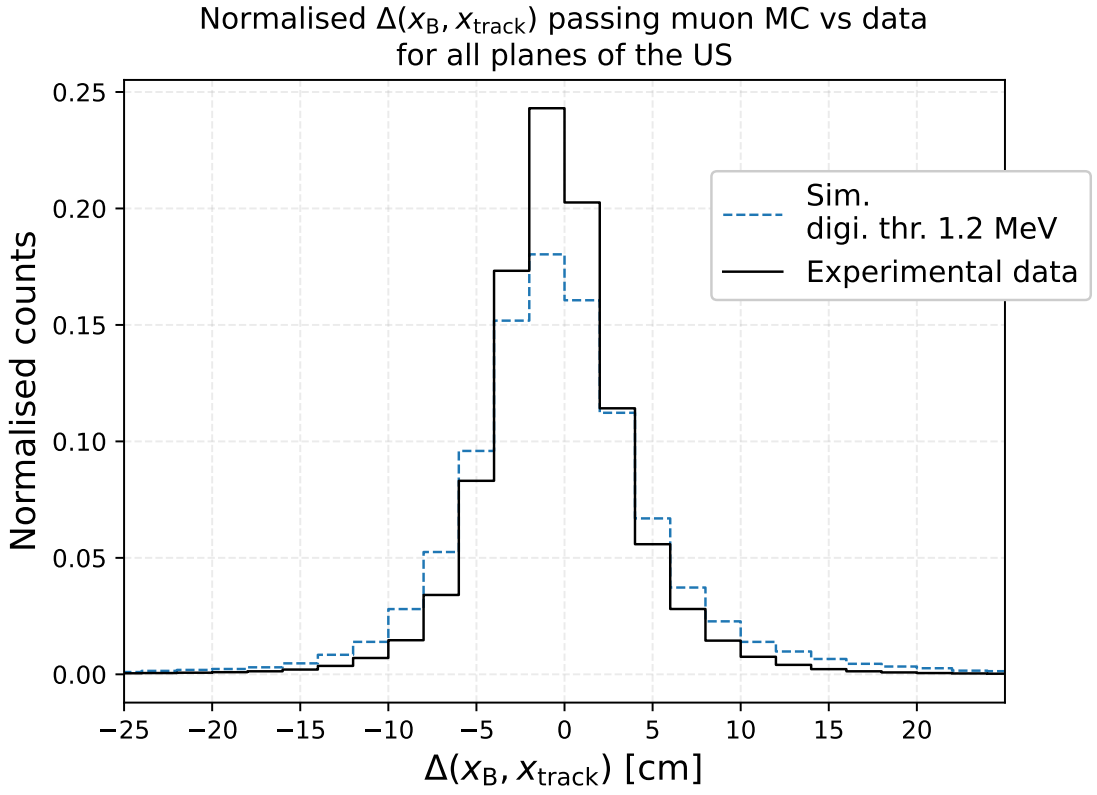



Figure A.16: Caption

# A.19

# A.20 Energy deposition $x-$width

Here shown is the comparison of the $\lambda_x$ distributions in data and simulation. The data is from run 5999 events passing the passing muon selection criteria of table 4.1, the simulation data is from the muon simulation data and also has the criteria of table 4.1 applied.

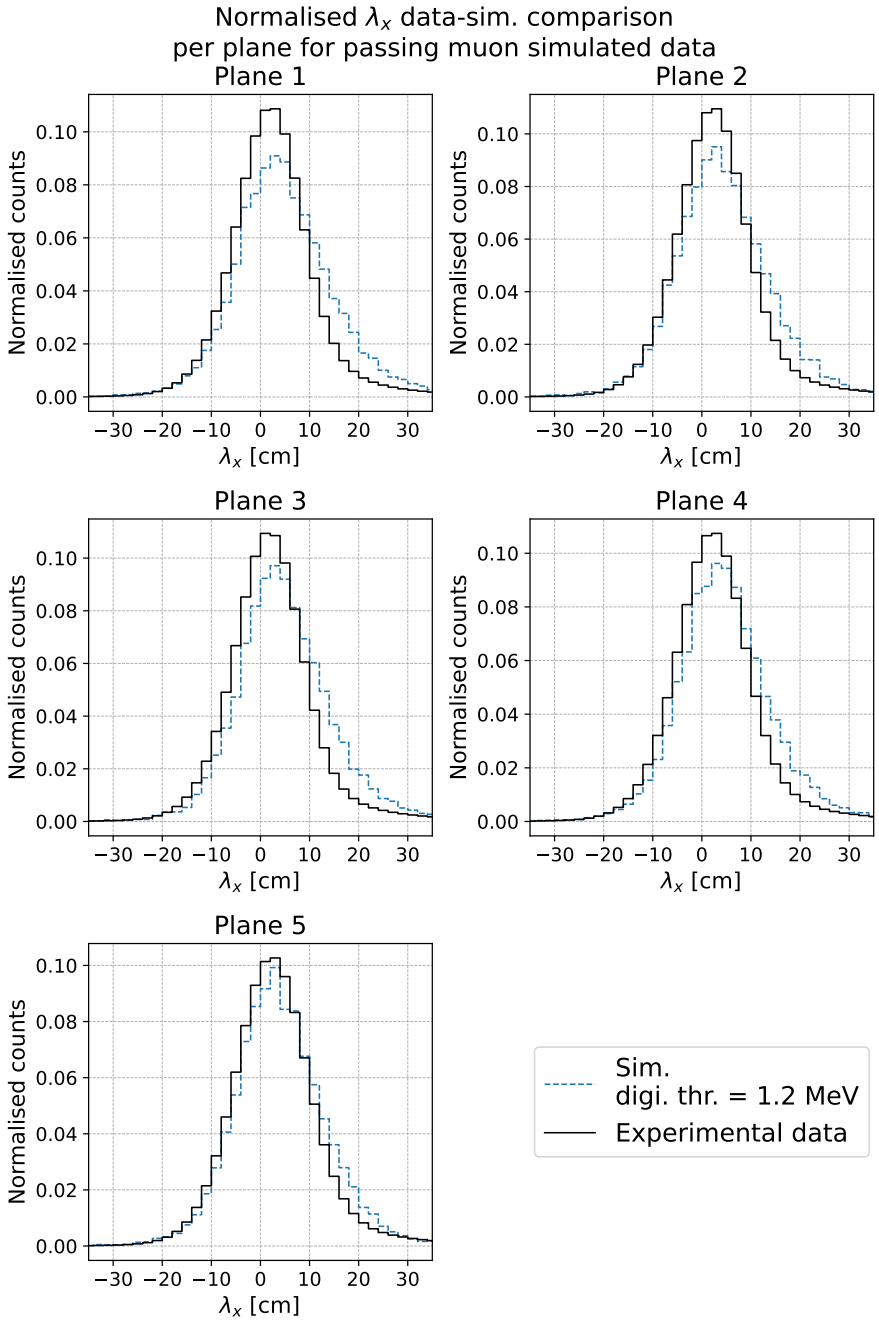



Figure A.17: Data-simulation comparison of $\lambda_x$ distributions for selected passing muon events in data and simulation, 1.2 MeV digitisation threshold applied.

## A.21 Run 5132 plane displays

## A.22 Spatial dependence of the DS system event-wise reference time

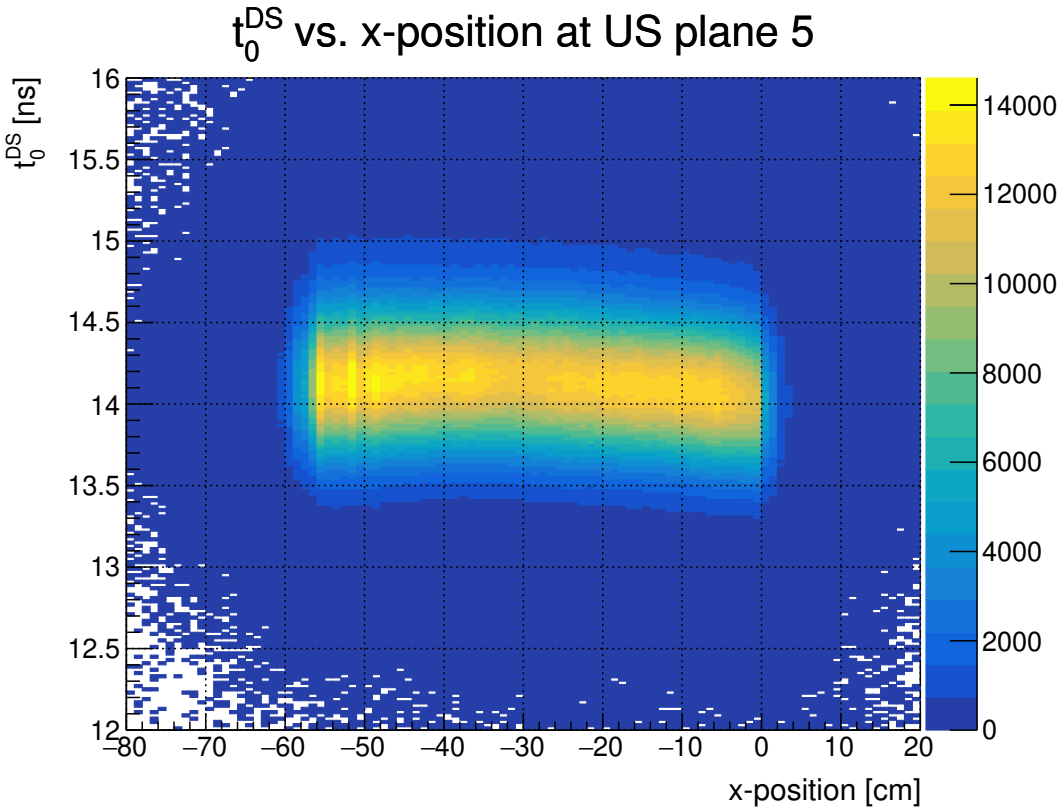



Figure A.19: Correlation plot of the event-wise reference time in the DS muon system against the $x-$position of a track fitted in the DS when extrapolated to the $z-$position of the final plane in the US muon system.

The value of the DS system event-wise reference time shows a slight curve when correlated with the DS track $x-$position when the track is extrapolated to the $z-$position of US plane 5. The dependence is so weak that it can effectively be neglected.

## A.23 Firing SiPMs in high QDC muon events

In the selected $\nu_\mu$ CCDIS events, 15.7% of the firing scintillators fail to meet the criteria of four or more SiPMs on both sides registering data. In the events selected as passing muons with the criteria of table 4.1, only 1.37% of events fail this criteria. The most fundamental difference between the passing muon events and $\nu_\mu$ CCDIS events is the energy deposited. By only studying the SiPMs firing rate for hits constituent in planes where the plane records a minimum QDC, the rate at which hits fail to have four or more SiPMs registering data increases, as reported in table 5.1, however not to a degree sufficient to explain the observation in the selected $\nu_\mu$ CCDIS events. Here the SiPM left and right correlation plots for passing muon events with two different values used for the minimum QDC values required in a plane for its constituent hits to be studied.

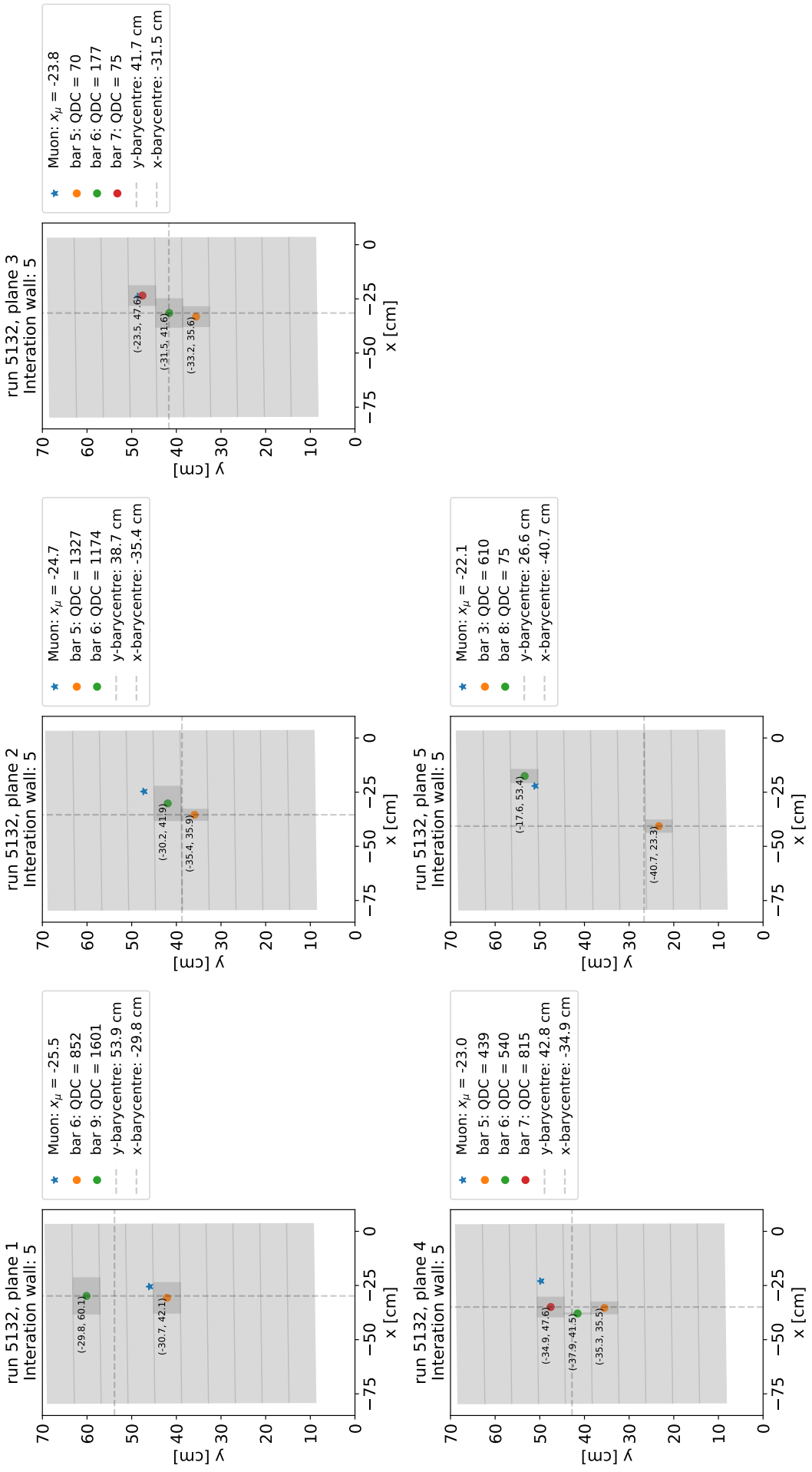
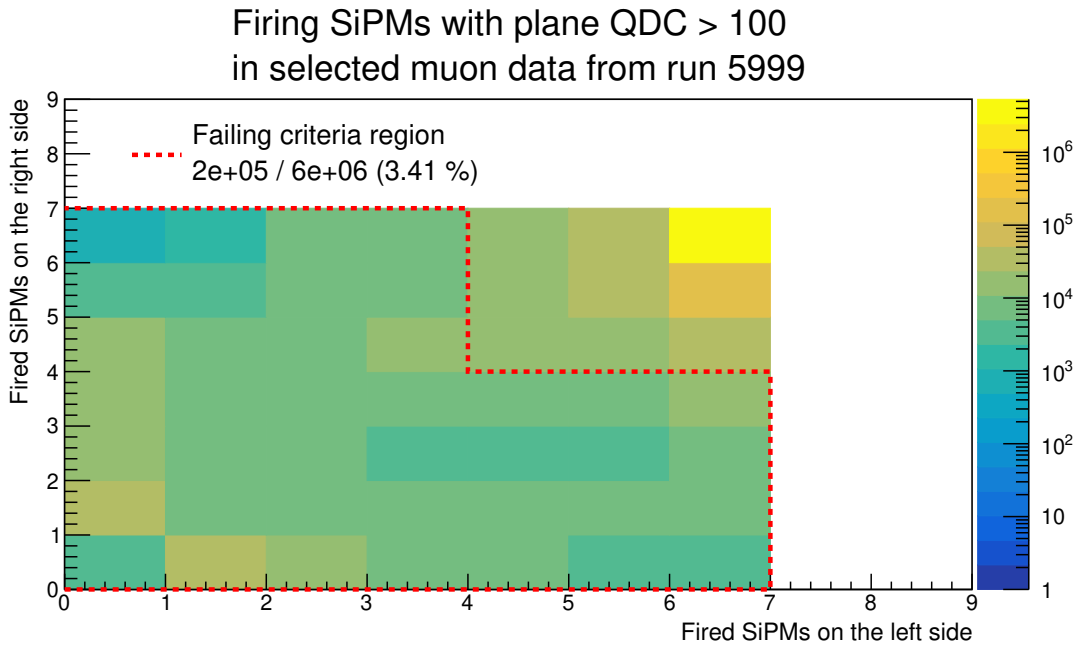
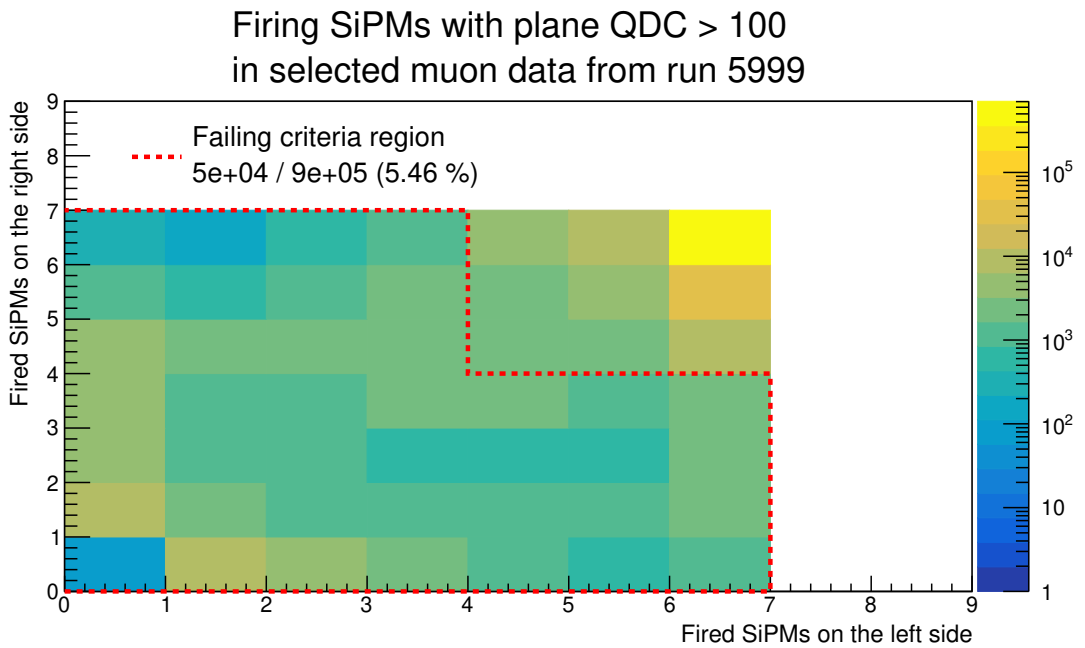



Figure A.18: The fired bars for all planes of the US for the PRL event in run 5132.

Figures A.20a and A.20b show how the percentage of firing scintillators failing to record four or more SiPMs on both sides change as the minimum required plane QDC criteria is set to 100 QDC and 500 QDC respectively.




(a) Residual between position measured by left-hand SiPMs and expected muon position




(b) Residual between position measured by right-hand SiPMs and expected muon position

Figure A.20: The variable changed between these two correlation plots is the minimum QDC measured by the whole plane of scintillators of which a scintillator under study is a constituent.

## A.24 Variance of spatial measurements

One observation of section 5.2.2 is that the variance observed for the energy deposition width, $\lambda_x = x_L - x_R$, is considerably greater than the variance of the

$x-$barycentre, $x_B = (x_L + x_R)/2$. When the scintillator-side time equations 5.1.1 and spatial measurement equations 5.1.3, 5.1.4 are combined, the spatial measurements can be expressed as below.

$$x_L = c_L \left( t_0^{\mathrm{DS}} - \frac{1}{N_L} \sum_i \left( t_i^{\mathrm{TW}} - d_i \right) \right) + x_m$$
$$x_R = x_m - c_R \left( t_0^{\mathrm{DS}} - \frac{1}{N_R} \sum_i \left( t_i^{\mathrm{TW}} - d_i \right) \right)$$

Where $N_L$ and $N_R$ are the number of firing SiPMs on the left- and right-sides respectively. Expressing $\lambda_x = x_L - x_R$ and $x_B = 1/2(x_L + x_R)$ using the above equations:

$$x_B = x_m + \frac{1}{2} \left( t_0^{\mathrm{DS}}(c_L - c_R) - \frac{c_L}{N_L} \sum_i \left( t_i^{TW} - d_i \right) + \frac{c_R}{N_R} \sum_i \left( t_i^{TW} - d_i \right) \right) \quad \text{(A.24.1)}$$
$$\lambda_x = t_0^{\mathrm{DS}} \left( c_L + c_R \right) - \frac{c_L}{N_L} \sum_i \left( t_i^{TW} - d_i \right) - \frac{c_R}{N_R} \sum_i \left( t_i^{TW} - d_i \right) \quad \text{(A.24.2)}$$

When computed the variance for these observables, the variance of the barcentre is negligible as explained in section 5.1.2. Consider the component involving the event-wise reference time, $t_0^{\mathrm{DS}}$, in equations A.24.1, A.24.2:

$$X = \alpha t_0^{\mathrm{DS}} \text{ where } \begin{cases} \alpha = \frac{1}{2}(c_L - c_R) \text{ for } x_{\mathrm{B}} \\ \alpha = (c_L + c_R) \text{ for } \lambda_x \end{cases}$$

The variance of this term can be written as:

$$\mathrm{Var}(X) = \langle \alpha^2 \rangle \mathrm{Var}(t_0^{\mathrm{DS}}) + \langle (t_0^{\mathrm{DS}})^2 \rangle \mathrm{Var}(\alpha)$$

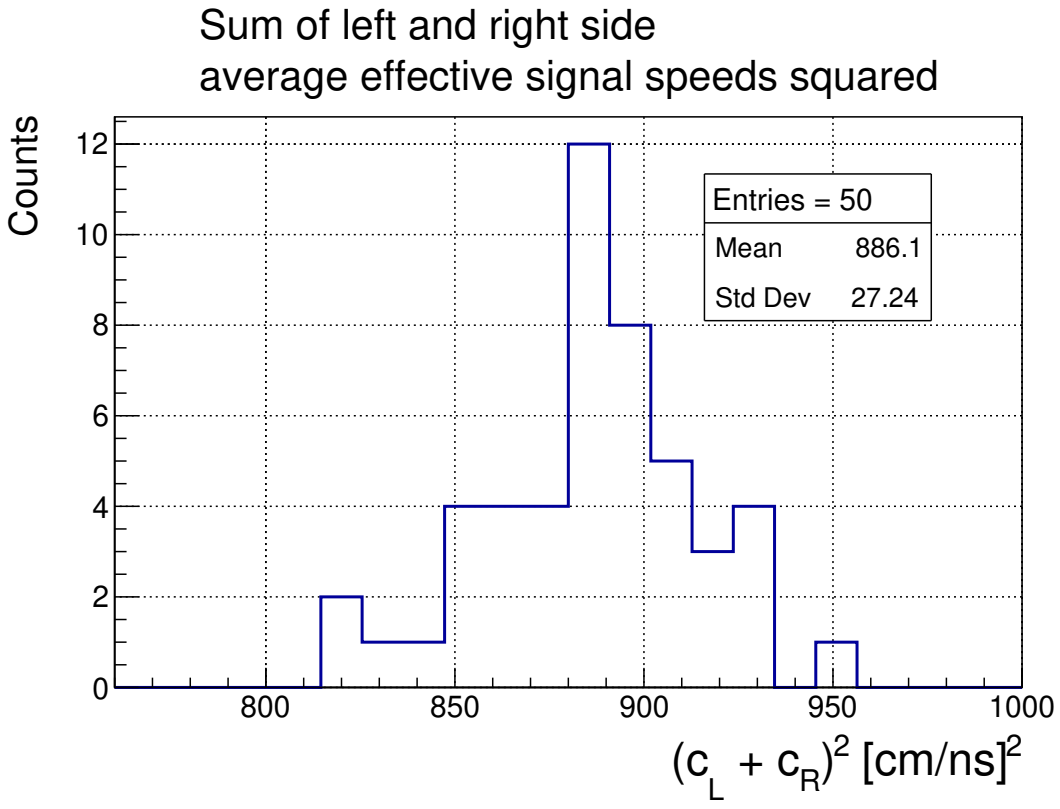
Where angled brackets indicate the mean of a value. While $c_L \neq c_R$, they are very similar in value:

$$\langle \alpha^2 \rangle = \begin{cases} 0.8\,(\mathrm{cm/ns})^2 \text{ for } x_{\mathrm{B}} \\ 886\,(\mathrm{cm/ns})^2 \text{ for } \lambda_x \end{cases}$$

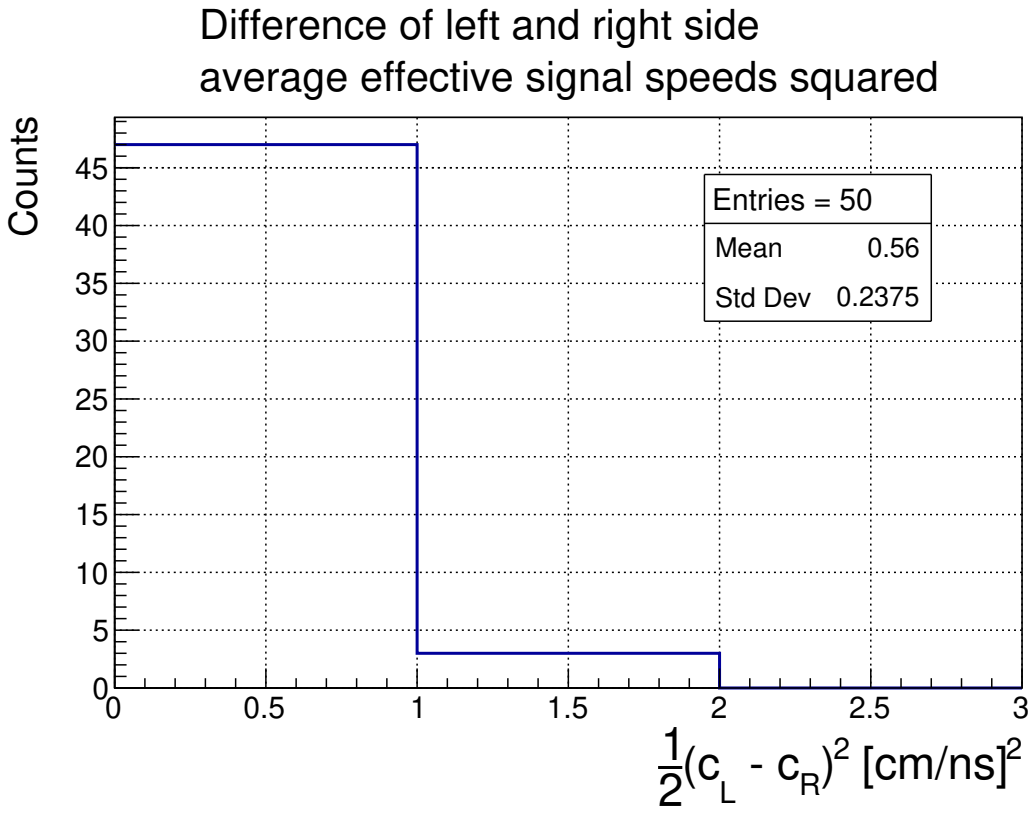
Where the distributions used to produce the values of $\langle \alpha^2 \rangle$ are shown below in figure Therefore the variance of $\alpha t_0^{\mathrm{DS}}$ is greatly suppressed for $x_{\mathrm{B}}$ as opposed to $\lambda_x$.

## A.25 Multiplicity comparison for $\nu_\mu$ CCDIS data

Figure 5.25 is used in section 5.2.3 to highlight the discrepancy between simulation and data when measuring the number of firing scintillators expected in the first plane of the US muon system for $\nu_\mu$ CCDIS events. This discrepancy is at its strongest in the first plane of the US system, with the agreement increasing as further downstream stations of the US are studied. Figure A.22 shows the comparison of firing scintillator multiplicity for the final plane of the US system, where the agreement is strong.




(a)




(b)

Figure A.21: Distribution of the sum (difference) of the left- and right-side effective signal speeds squared for all US system scintillators in the top (bottom) sub-figure. Data from from run 5999.

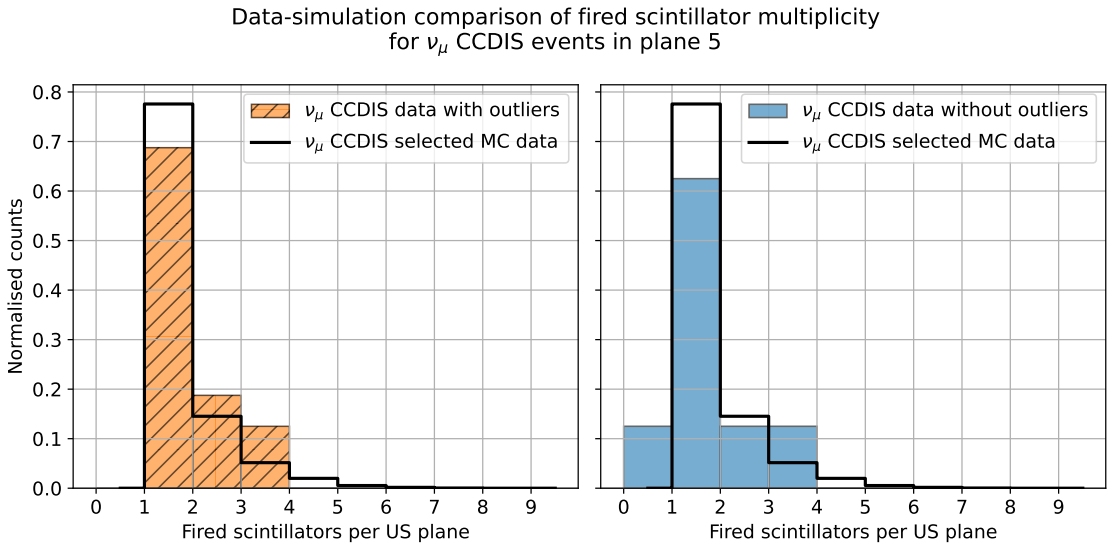



Figure A.22: Multiplicity of firing scintillators in the fifth plane of the US muon system. Neutrino simulation data passing the $\nu_\mu$ CCDIS selection criteria is shown as well as the data events satisfying the $\nu_\mu$ CCDIS selection criteria split into two cases: including firing scintillators tagged as outliers (left) and not including those scintillators (right). The three data-sets are normalised to unit integral.

## A.26 $\nu 0\mu$ BDT features

Table A.5 shows the features used to train the BDT to select $\nu 0\mu$ events and reject the remaining events containing a muon in the final-state.

$ds_{\mathrm{SF}}$ is defined in equation 6.2.2. While $dx$ and $dy$ are similarly defined as the residuals between the line formed by hit clusters in the detector and the spatial measurements in each plane of the US.

| Feature | Note |
|---|---|
| Scifi median $x-$position | |
| Scifi median $y-$position | |
| $ds_{\mathrm{SF}}$ | |
| $dx_{\mathrm{SF}}$ | |
| $dy_{\mathrm{SF}}$ | |
| $\lambda_x$ | For all planes |
| $dx$ | For all planes |
| $\lambda_y$ | For all planes |
| $dy$ | For all planes |

Table A.5: Features used to train the $\nu 0\mu$ BDT.

## Timing quantities

Here the frequently used timing quantities are summarised concisely for the reader. This is not an exhaustive list of any timing quantity used in the thesis, just a list of the more frequently used quantities.

It should be recalled that bar-side or full bar quantities all have the time-walk correction applied so the superscript TW is dropped due to redundancy.

| Quantity | Definition |
|---|---|
| $t_{\text{SiPM}}$ | The time measured by a SiPM. |
| $t_0^{\text{DS}}$ | DS event-wise reference time. Definition in equation 4.1.1. |
| $t_0^{\text{SF}}$ | SciFi event-wise reference time. Definition in equation 4.1.2. |
| $dt_{\text{SiPM}}$ | Given by $t_0^{\text{DS}} - t_{\text{SiPM}}$. |
| $dt_{\text{SiPM}}^{\text{SF}}$ | Given by $t_0^{\text{SF}} - t_{\text{SiPM}}$. |
| $X^{\text{TW}}$ | Quantity $X$ has the time-walk correction applied. Definition in equation 4.1.9. |
| $X^{\text{ToF}}$ | Quantity $X$ has the signal time-of-flight correction applied. Definition in equation 4.1.8. |
| $t_{\text{side}}$ | The average of time-walk corrected and aligned SiPMs on one side. Definition in equation 4.3.8. |
| $t_{\text{bar}}$ | The average of time-walk corrected and aligned SiPMs in the full scintillator. Definition in equation 4.3.8. |

Table A.6: Table of frequently used timing quantities to be used as an aid to the reader.